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Math 75: Introduction to Linear Algebra

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Included content from the original text:

- Sections 1.1-1.6
- Sections 2.1-2.6, and 2.9
- Sections 3.1-3.3
- Sections 4.1, 4.2, and 4.5
- Sections 5.1-5.4 and 5.6

From the list of included content, the following (and related content) has been omitted. Additional content may have been removed. The removal of this content is for instructional purposes and do not reflect the opinions of either Nicholson or Lyryx Learning Inc.

- content related to block multiplication, including Definition 2.10, Theorems 2.3.4 and 2.3.5, Example 2.3.12, and Theorem 3.1.5
- content related to directed graphs including Theorem 2.3.6
- content related to inverses of matrix transformations including Theorem 2.4.6 and Example 2.4.12
- content related to the Smith Normal Form including Theorem 2.5.3 and Example 2.5.4

- content related to the uniqueness of the reduced row echelon form including Theorem 2.5.4
- some content related to rotations, reflections and projections including including Example 2.6.7, Theorems 2.6.5 and 2.6.6, and Example 2.6.9
- content related to the determinant as a function (Theorem 3.1.6)
- content related to the laws of cosines including Theorem 4.2.2 and Definition 4.5
- content related to the cross product including Definition 4.8 and Theorem 4.2.5
- content related to decomposing a vector as the sum of orthogonal vectors including Theorems 5.3.4 and 5.3.6.

Chapter 1

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Systems of Linear Equations

1.1 Solutions and Elementary Operations

Practical problems in many fields of study—such as biology, business, chemistry, computer science, economics, electronics, engineering, physics and the social sciences—can often be reduced to solving a system of linear equations. Linear algebra arose from attempts to find systematic methods for solving these systems, so it is natural to begin this book by studying linear equations.

If a, b, and c are real numbers, the graph of an equation of the form

$$ax+by=c$$

is a straight line (if a and b are not both zero), so such an equation is called a *linear* equation in the variables x and y. However, it is often convenient to write the variables as $x_1, x_2, \dots, x_n$, particularly when more than two variables are involved. An equation of the form

$$a_1x_1+a_2x_2+\cdots+a_nx_n=b$$

is called a **linear equation** in the n variables $x_1, x_2, \dots, x_n$. Here $a_1, a_2, \dots, a_n$ denote real numbers (called the **coefficients** of $x_1, x_2, \dots, x_n$, respectively) and b is also a number (called the **constant term** of the equation). A finite collection of linear equations in the variables $x_1, x_2, \dots, x_n$ is called a **system of linear equations** in these variables. Hence,

$$2x_1-3x_2+5x_3=7$$

is a linear equation; the coefficients of x_1, x_2, and x_3 are 2, -3, and 5, and the constant term is 7. Note that each variable in a linear equation occurs to the first power only.

Given a linear equation $a_1x_1+a_2x_2+\cdots+a_nx_n=b$, a sequence $s_1, s_2, \dots, s_n$ of n numbers is called a **solution** to the equation if

$$a_1s_1+a_2s_2+\cdots+a_ns_n=b$$

that is, if the equation is satisfied when the substitutions $x_1=s_1, x_2=s_2, \dots, x_n=s_n$ are made. A sequence of numbers is called **a solution to a system** of equations if it is a solution to every equation in the system.

For example, $x=-2$, $y=5$, $z=0$ and $x=0$, $y=4$, $z=-1$ are both solutions to the system

$$\begin{aligned} x+y+z&=3\\ 2x+y+3z&=1 \end{aligned}$$

A system may have no solution at all, or it may have a unique solution, or it may have an infinite family of solutions. For instance, the system $x+y=2$, $x+y=3$ has no solution because the sum of two numbers cannot be 2 and 3 simultaneously. A system that has no solution is called **inconsistent**; a system with at least one solution is called **consistent**. The system in the following example has infinitely many solutions.

Example 1.1.1

Show that, for arbitrary values of s and t,

$$\begin{aligned} x_1 &= t - s + 1 \\ x_2 &= t + s + 2 \\ x_3 &= s \\ x_4 &= t \end{aligned}$$

is a solution to the system

$$\begin{array}{rcrcrcrcr} x_1 & - & 2x_2 & + & 3x_3 & + & x_4 & = & -3 \\ 2x_1 & - & x_2 & + & 3x_3 & - & x_4 & = & 0 \end{array}$$

Solution. Simply substitute these values of x_1, x_2, x_3, and x_4 in each equation.

$$\begin{aligned} x_1 - 2x_2 + 3x_3 + x_4 &= (t - s + 1) - 2(t + s + 2) + 3s + t = -3 \\ 2x_1 - x_2 + 3x_3 - x_4 &= 2(t - s + 1) - (t + s + 2) + 3s - t = 0 \end{aligned}$$

Because both equations are satisfied, it is a solution for all choices of s and t.

The quantities s and t in Example 1.1.1 are called **parameters**, and the set of solutions, described in this way, is said to be given in **parametric form** and is called the **general solution** to the system. It turns out that the solutions to *every* system of equations (if there *are* solutions) can be given in parametric form (that is, the variables x_1, x_2, $\dots$ are given in terms of new independent variables s, t, etc.). The following example shows how this happens in the simplest systems where only one equation is present.

Example 1.1.2

Describe all solutions to $3x - y + 2z = 6$ in parametric form.

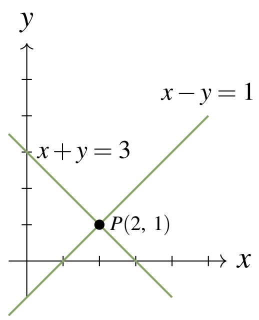
Solution. Solving the equation for y in terms of x and z, we get $y = 3x + 2z - 6$. If s and t are arbitrary then, setting $x = s$, $z = t$, we get solutions

$$\begin{aligned} x &= s \\ y &= 3s + 2t - 6 \qquad s \text{ and } t \text{ arbitrary} \\ z &= t \end{aligned}$$

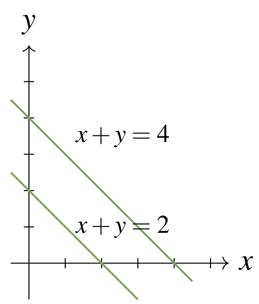
Of course we could have solved for x: $x = \frac{1}{3}(y - 2z + 6)$. Then, if we take $y = p$, $z = q$, the solutions are represented as follows:

$$\begin{array}{rcl} x & = & \frac{1}{3}(p - 2q + 6) \\ y & = & p \qquad\qquad\qquad p \text{ and } q \text{ arbitrary} \\ z & = & q \end{array}$$

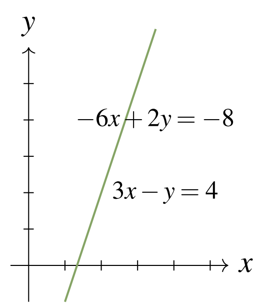
The same family of solutions can "look" quite different!



(a) Unique Solution
($x = 2, y = 1$)



(b) No Solution



(c) Infinitely many solutions
($x = t, y = 3t - 4$)

Figure 1.1.1

When only two variables are involved, the solutions to systems of linear equations can be described geometrically because the graph of a linear equation $ax + by = c$ is a straight line if a and b are not both zero. Moreover, a point $P(s, t)$ with coordinates s and t lies on the line if and only if $as + bt = c$—that is when $x = s$, $y = t$ is a solution to the equation. Hence the solutions to a *system* of linear equations correspond to the points $P(s, t)$ that lie on *all* the lines in question.

In particular, if the system consists of just one equation, there must be infinitely many solutions because there are infinitely many points on a line. If the system has two equations, there are three possibilities for the corresponding straight lines:

1. *The lines intersect at a single point. Then the system has a* unique solution *corresponding to that point.*
2. *The lines are parallel (and distinct) and so do not intersect. Then the system has* no solution.
3. *The lines are identical. Then the system has* infinitely many solutions*—one for each point on the (common) line.*

These three situations are illustrated in Figure 1.1.1. In each case the graphs of two specific lines are plotted and the corresponding equations are indicated. In the last case, the equations are $3x - y = 4$ and $-6x + 2y = -8$, which have identical graphs.

With three variables, the graph of an equation $ax + by + cz = d$ can be shown to be a plane (see Section 4.2) and so again provides a "picture" of the set of solutions. However, this graphical method has its limitations: When more than three variables are involved, no physical image of the graphs (called hyperplanes) is possible. It is necessary to turn to a more "algebraic" method of solution.

Before describing the method, we introduce a concept that simplifies the computations involved. Consider the following system

$$\begin{array}{rcrcrcrcr} 3x_1 & + & 2x_2 & - & x_3 & + & x_4 & = & -1 \\ 2x_1 & & & - & x_3 & + & 2x_4 & = & 0 \\ 3x_1 & + & x_2 & + & 2x_3 & + & 5x_4 & = & 2 \end{array}$$

of three equations in four variables. The array of numbers[1]

$$\left[\begin{array}{rrrr|r} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array}\right]$$

occurring in the system is called the **augmented matrix** of the system. Each row of the matrix consists of the coefficients of the variables (in order) from the corresponding equation, together with the constant

[1]A rectangular array of numbers is called a **matrix**. Matrices will be discussed in more detail in Chapter 2.

term. For clarity, the constants are separated by a vertical line. The augmented matrix is just a different way of describing the system of equations. The array of coefficients of the variables

$$\left[\begin{array}{rrrr} 3 & 2 & -1 & 1 \\ 2 & 0 & -1 & 2 \\ 3 & 1 & 2 & 5 \end{array}\right]$$

is called the **coefficient matrix** of the system and $\left[\begin{array}{r} -1 \\ 0 \\ 2 \end{array}\right]$ is called the **constant matrix** of the system.

Elementary Operations

The algebraic method for solving systems of linear equations is described as follows. Two such systems are said to be **equivalent** if they have the same set of solutions. A system is solved by writing a series of systems, one after the other, each equivalent to the previous system. Each of these systems has the same set of solutions as the original one; the aim is to end up with a system that is easy to solve. Each system in the series is obtained from the preceding system by a simple manipulation chosen so that it does not change the set of solutions.

As an illustration, we solve the system $x+2y=-2$, $2x+y=7$ in this manner. At each stage, the corresponding augmented matrix is displayed. The original system is

$$\begin{array}{rcr} x+2y & = & -2 \\ 2x+y & = & 7 \end{array} \qquad \left[\begin{array}{rr|r} 1 & 2 & -2 \\ 2 & 1 & 7 \end{array}\right]$$

First, subtract twice the first equation from the second. The resulting system is

$$\begin{array}{rcr} x+2y & = & -2 \\ -3y & = & 11 \end{array} \qquad \left[\begin{array}{rr|r} 1 & 2 & -2 \\ 0 & -3 & 11 \end{array}\right]$$

which is equivalent to the original (see Theorem 1.1.1). At this stage we obtain $y=-\frac{11}{3}$ by multiplying the second equation by $-\frac{1}{3}$. The result is the equivalent system

$$\begin{array}{rcr} x+2y & = & -2 \\ y & = & -\frac{11}{3} \end{array} \qquad \left[\begin{array}{rr|r} 1 & 2 & -2 \\ 0 & 1 & -\frac{11}{3} \end{array}\right]$$

Finally, we subtract twice the second equation from the first to get another equivalent system.

$$\begin{array}{rcr} x & = & \frac{16}{3} \\ y & = & -\frac{11}{3} \end{array} \qquad \left[\begin{array}{rr|r} 1 & 0 & \frac{16}{3} \\ 0 & 1 & -\frac{11}{3} \end{array}\right]$$

Now *this* system is easy to solve! And because it is equivalent to the original system, it provides the solution to that system.

Observe that, at each stage, a certain operation is performed on the system (and thus on the augmented matrix) to produce an equivalent system.

Definition 1.1 Elementary Operations

The following operations, called ***elementary operations****, can routinely be performed on systems of linear equations to produce equivalent systems.*

I. *Interchange two equations.*

II. *Multiply one equation by a* nonzero *number.*

III. *Add a multiple of one equation to a* different *equation.*

Theorem 1.1.1

Suppose that a sequence of elementary operations is performed on a system of linear equations. Then the resulting system has the same set of solutions as the original, so the two systems are equivalent.

The proof is given at the end of this section.

Elementary operations performed on a system of equations produce corresponding manipulations of the *rows* of the augmented matrix. Thus, multiplying a row of a matrix by a number k means multiplying *every entry* of the row by k. Adding one row to another row means adding *each entry* of that row to the corresponding entry of the other row. Subtracting two rows is done similarly. Note that we regard two rows as equal when corresponding entries are the same.

In hand calculations (and in computer programs) we manipulate the rows of the augmented matrix rather than the equations. For this reason we restate these elementary operations for matrices.

Definition 1.2 Elementary Row Operations

The following are called ***elementary row operations*** *on a matrix.*

I. *Interchange two rows.*

II. *Multiply one row by a* nonzero *number.*

III. *Add a multiple of one row to a* different *row.*

In the illustration above, a series of such operations led to a matrix of the form

$$\left[\begin{array}{cc|c} 1 & 0 & * \\ 0 & 1 & * \end{array}\right]$$

where the asterisks represent arbitrary numbers. In the case of three equations in three variables, the goal is to produce a matrix of the form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right]$$

This does not always happen, as we will see in the next section. Here is an example in which it does happen.

Example 1.1.3

Find all solutions to the following system of equations.

$$\begin{array}{rcr} 3x+4y+z &=& 1 \\ 2x+3y &=& 0 \\ 4x+3y-z &=& -2 \end{array}$$

Solution. The augmented matrix of the original system is

$$\left[\begin{array}{rrr|r} 3 & 4 & 1 & 1 \\ 2 & 3 & 0 & 0 \\ 4 & 3 & -1 & -2 \end{array}\right]$$

To create a 1 in the upper left corner we could multiply row 1 through by $\frac{1}{3}$. However, the 1 can be obtained without introducing fractions by subtracting row 2 from row 1. The result is

$$\left[\begin{array}{rrr|r} 1 & 1 & 1 & 1 \\ 2 & 3 & 0 & 0 \\ 4 & 3 & -1 & -2 \end{array}\right]$$

The upper left 1 is now used to "clean up" the first column, that is create zeros in the other positions in that column. First subtract 2 times row 1 from row 2 to obtain

$$\left[\begin{array}{rrr|r} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -2 \\ 4 & 3 & -1 & -2 \end{array}\right]$$

Next subtract 4 times row 1 from row 3. The result is

$$\left[\begin{array}{rrr|r} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -2 \\ 0 & -1 & -5 & -6 \end{array}\right]$$

This completes the work on column 1. We now use the 1 in the second position of the second row to clean up the second column by subtracting row 2 from row 1 and then adding row 2 to row 3. For convenience, both row operations are done in one step. The result is

$$\left[\begin{array}{rrr|r} 1 & 0 & 3 & 3 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -7 & -8 \end{array}\right]$$

Note that the last two manipulations *did not affect* the first column (the second row has a zero there), so our previous effort there has not been undermined. Finally we clean up the third column. Begin by multiplying row 3 by $-\frac{1}{7}$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 3 & 3 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & \frac{8}{7} \end{array}\right]$$

Now subtract 3 times row 3 from row 1, and then add 2 times row 3 to row 2 to get

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{3}{7} \\ 0 & 1 & 0 & \frac{2}{7} \\ 0 & 0 & 1 & \frac{8}{7} \end{array}\right]$$

The corresponding equations are $x = -\frac{3}{7}$, $y = \frac{2}{7}$, and $z = \frac{8}{7}$, which give the (unique) solution.

Every elementary row operation can be **reversed** by another elementary row operation of the same type (called its **inverse**). To see how, we look at types I, II, and III separately:

Type I *Interchanging two rows is reversed by interchanging them again.*

Type II *Multiplying a row by a nonzero number k is reversed by multiplying by $1/k$.*

Type III *Adding k times row p to a different row q is reversed by adding $-k$ times row p to row q (in the new matrix). Note that $p \neq q$ is essential here.*

To illustrate the Type III situation, suppose there are four rows in the original matrix, denoted R_1, R_2, R_3, and R_4, and that k times R_2 is added to R_3. Then the reverse operation adds $-k$ times R_2, to R_3. The following diagram illustrates the effect of doing the operation first and then the reverse:

$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{bmatrix} \to \begin{bmatrix} R_1 \\ R_2 \\ R_3 + kR_2 \\ R_4 \end{bmatrix} \to \begin{bmatrix} R_1 \\ R_2 \\ (R_3 + kR_2) - kR_2 \\ R_4 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{bmatrix}$$

The existence of inverses for elementary row operations and hence for elementary operations on a system of equations, gives:

Proof of Theorem 1.1.1. Suppose that a system of linear equations is transformed into a new system by a sequence of elementary operations. Then every solution of the original system is automatically a solution of the new system because adding equations, or multiplying an equation by a nonzero number, always results in a valid equation. In the same way, each solution of the new system must be a solution to the original system because the original system can be obtained from the new one by another series of elementary operations (the inverses of the originals). It follows that the original and new systems have the same solutions. This proves Theorem 1.1.1. □

Exercises for 1.1

Exercise 1.1.1 In each case verify that the following are solutions for all values of s and t.

a. $x = 19t - 35$
$y = 25 - 13t$
$z = t$

is a solution of

$$\begin{aligned} 2x + 3y + z &= 5 \\ 5x + 7y - 4z &= 0 \end{aligned}$$

b. $x_1 = 2s + 12t + 13$
$x_2 = s$
$x_3 = -s - 3t - 3$
$x_4 = t$

is a solution of

$$\begin{aligned} 2x_1 + 5x_2 + 9x_3 + 3x_4 &= -1 \\ x_1 + 2x_2 + 4x_3 &= 1 \end{aligned}$$

Exercise 1.1.2 Find all solutions to the following in parametric form in two ways.

a. $3x + y = 2$

b. $2x + 3y = 1$

c. $3x - y + 2z = 5$

d. $x - 2y + 5z = 1$

Exercise 1.1.3 Regarding $2x = 5$ as the equation $2x + 0y = 5$ in two variables, find all solutions in parametric form.

Exercise 1.1.4 Regarding $4x - 2y = 3$ as the equation $4x - 2y + 0z = 3$ in three variables, find all solutions in parametric form.

Exercise 1.1.5 Find all solutions to the general system $ax = b$ of one equation in one variable (a) when $a = 0$ and (b) when $a \neq 0$.

Exercise 1.1.6 Show that a system consisting of exactly one linear equation can have no solution, one solution, or infinitely many solutions. Give examples.

Exercise 1.1.7 Write the augmented matrix for each of the following systems of linear equations.

a. $\begin{aligned} x - 3y &= 5 \\ 2x + y &= 1 \end{aligned}$

b. $\begin{aligned} x + 2y &= 0 \\ y &= 1 \end{aligned}$

c. $\begin{aligned} x - y + z &= 2 \\ x - z &= 1 \\ y + 2x &= 0 \end{aligned}$

d. $\begin{aligned} x + y &= 1 \\ y + z &= 0 \\ z - x &= 2 \end{aligned}$

Exercise 1.1.8 Write a system of linear equations that has each of the following augmented matrices.

a. $\left[\begin{array}{rrr|r} 1 & -1 & 6 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & -1 & 0 & 1 \end{array}\right]$

b. $\left[\begin{array}{rrr|r} 2 & -1 & 0 & -1 \\ -3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 3 \end{array}\right]$

Exercise 1.1.9 Find the solution of each of the following systems of linear equations using augmented matrices.

a. $\begin{aligned} x - 3y &= 1 \\ 2x - 7y &= 3 \end{aligned}$

b. $\begin{aligned} x + 2y &= 1 \\ 3x + 4y &= -1 \end{aligned}$

c. $\begin{aligned} 2x + 3y &= -1 \\ 3x + 4y &= 2 \end{aligned}$

d. $\begin{aligned} 3x + 4y &= 1 \\ 4x + 5y &= -3 \end{aligned}$

Exercise 1.1.10 Find the solution of each of the following systems of linear equations using augmented matrices.

a. $\begin{aligned} x + y + 2z &= -1 \\ 2x + y + 3z &= 0 \\ -2y + z &= 2 \end{aligned}$

b. $\begin{aligned} 2x + y + z &= -1 \\ x + 2y + z &= 0 \\ 3x - 2z &= 5 \end{aligned}$

Exercise 1.1.11 Find all solutions (if any) of the following systems of linear equations.

a. $\begin{aligned} 3x - 2y &= 5 \\ -12x + 8y &= -20 \end{aligned}$

b. $\begin{aligned} 3x - 2y &= 5 \\ -12x + 8y &= 16 \end{aligned}$

Exercise 1.1.12 Show that the system

$$\left\{ \begin{array}{rcrcrcl} x & + & 2y & - & z & = & a \\ 2x & + & y & + & 3z & = & b \\ x & - & 4y & + & 9z & = & c \end{array} \right.$$

is inconsistent unless $c = 2b - 3a$.

Exercise 1.1.13 By examining the possible positions of lines in the plane, show that two equations in two variables can have zero, one, or infinitely many solutions.

Exercise 1.1.14 In each case either show that the statement is true, or give an example[2] showing it is false.

a. If a linear system has n variables and m equations, then the augmented matrix has n rows.

b. A consistent linear system must have infinitely many solutions.

c. If a row operation is done to a consistent linear system, the resulting system must be consistent.

d. If a series of row operations on a linear system results in an inconsistent system, the original system is inconsistent.

Exercise 1.1.15 Find a quadratic $a+bx+cx^2$ such that the graph of $y=a+bx+cx^2$ contains each of the points $(-1, 6)$, $(2, 0)$, and $(3, 2)$.

Exercise 1.1.16 Solve the system $\left\{\begin{array}{l} 3x+2y=5 \\ 7x+5y=1 \end{array}\right.$ by changing variables $\left\{\begin{array}{rl} x = & 5x' - 2y' \\ y = & -7x' + 3y' \end{array}\right.$ and solving the resulting equations for x' and y'.

Exercise 1.1.17 Find a, b, and c such that

$$\frac{x^2-x+3}{(x^2+2)(2x-1)} = \frac{ax+b}{x^2+2} + \frac{c}{2x-1}$$

[*Hint*: Multiply through by $(x^2+2)(2x-1)$ and equate coefficients of powers of x.]

Exercise 1.1.18 A zookeeper wants to give an animal 42 mg of vitamin A and 65 mg of vitamin D per day. He has two supplements: the first contains 10% vitamin A and 25% vitamin D; the second contains 20% vitamin A and 25% vitamin D. How much of each supplement should he give the animal each day?

Exercise 1.1.19 Workmen John and Joe earn a total of \$24.60 when John works 2 hours and Joe works 3 hours. If John works 3 hours and Joe works 2 hours, they get \$23.90. Find their hourly rates.

Exercise 1.1.20 A biologist wants to create a diet from fish and meal containing 183 grams of protein and 93 grams of carbohydrate per day. If fish contains 70% protein and 10% carbohydrate, and meal contains 30% protein and 60% carbohydrate, how much of each food is required each day?

1.2 Gaussian Elimination

The algebraic method introduced in the preceding section can be summarized as follows: Given a system of linear equations, use a sequence of elementary row operations to carry the augmented matrix to a "nice" matrix (meaning that the corresponding equations are easy to solve). In Example 1.1.3, this nice matrix took the form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right]$$

The following definitions identify the nice matrices that arise in this process.

[2] Such an example is called a **counterexample**. For example, if the statement is that "all philosophers have beards", the existence of a non-bearded philosopher would be a counterexample proving that the statement is false. This is discussed again in Appendix B.

Definition 1.3 Row-Echelon Form (Reduced)

A matrix is said to be in ***row-echelon form*** *(and will be called a* ***row-echelon matrix****) if it satisfies the following three conditions:*

1. *All* ***zero rows*** *(consisting entirely of zeros) are at the bottom.*
2. *The first nonzero entry from the left in each nonzero row is a* 1*, called the* ***leading 1*** *or* ***pivot*** *for that row.*
3. *Each leading* 1 *is to the right of all leading* 1*s in the rows above it.*

A row-echelon matrix is said to be in ***reduced row-echelon form*** *(and will be called a* ***reduced row-echelon matrix****) if, in addition, it satisfies the following condition:*

4. *Each leading* 1 *is the only nonzero entry in its column.*

The row-echelon matrices have a "staircase" form, as indicated by the following example (the asterisks indicate arbitrary numbers).

$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The leading 1s proceed "down and to the right" through the matrix. Entries above and to the right of the leading 1s are arbitrary, but all entries below and to the left of them are zero. Hence, a matrix in row-echelon form is in reduced form if, in addition, the entries directly above each leading 1 are all zero. Note that a matrix in row-echelon form can, with a few more row operations, be carried to reduced form (use row operations to create zeros above each leading one in succession, beginning from the right).

Example 1.2.1

The following matrices are in row-echelon form (for any choice of numbers in $*$-positions).

$$\begin{bmatrix} 1 & * & * \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

The following, on the other hand, are in reduced row-echelon form.

$$\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The choice of the positions for the leading 1s determines the (reduced) row-echelon form (apart from the numbers in $*$-positions).

The importance of row-echelon matrices comes from the following theorem.

Theorem 1.2.1

Every matrix can be brought to (reduced) row-echelon form by a sequence of elementary row operations.

In fact we can give a step-by-step procedure for actually finding a row-echelon matrix. Observe that while there are many sequences of row operations that will bring a matrix to row-echelon form, the one we use is systematic and is easy to program on a computer. Note that the algorithm deals with matrices in general, possibly with columns of zeros.

Gaussian[3] Algorithm[4]

Step 1. If the matrix consists entirely of zeros, stop—it is already in row-echelon form.

Step 2. Otherwise, find the first column from the left containing a nonzero entry (call it a), and move the row containing that entry to the top position.

Step 3. Now multiply the new top row by $1/a$ to create a leading 1.

Step 4. By subtracting multiples of that row from rows below it, make each entry below the leading 1 zero.

This completes the first row, and all further row operations are carried out on the remaining rows.

Step 5. Repeat steps 1–4 on the matrix consisting of the remaining rows.

The process stops when either no rows remain at step 5 or the remaining rows consist entirely of zeros.

Observe that the gaussian algorithm is recursive: When the first leading 1 has been obtained, the procedure is repeated on the remaining rows of the matrix. This makes the algorithm easy to use on a computer. Note that the solution to Example 1.1.3 did not use the gaussian algorithm as written because the first leading 1 was not created by dividing row 1 by 3. The reason for this is that it avoids fractions. However, the general pattern is clear: Create the leading 1s from left to right, using each of them in turn to create zeros below it. Here are two more examples.

Example 1.2.2

Solve the following system of equations.

$$\begin{array}{rcrcrcr} 3x & + & y & - & 4z & = & -1 \\ x & & & + & 10z & = & 5 \\ 4x & + & y & + & 6z & = & 1 \end{array}$$

[3]Carl Friedrich Gauss (1777–1855) ranks with Archimedes and Newton as one of the three greatest mathematicians of all time. He was a child prodigy and, at the age of 21, he gave the first proof that every polynomial has a complex root. In 1801 he published a timeless masterpiece, Disquisitiones Arithmeticae, in which he founded modern number theory. He went on to make ground-breaking contributions to nearly every branch of mathematics, often well before others rediscovered and published the results.

[4]The algorithm was known to the ancient Chinese.

Solution. The corresponding augmented matrix is

$$\left[\begin{array}{rrr|r} 3 & 1 & -4 & -1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

Create the first leading one by interchanging rows 1 and 2

$$\left[\begin{array}{rrr|r} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & -1 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

Now subtract 3 times row 1 from row 2, and subtract 4 times row 1 from row 3. The result is

$$\left[\begin{array}{rrr|r} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 1 & -34 & -19 \end{array}\right]$$

Now subtract row 2 from row 3 to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

This means that the following reduced system of equations

$$\begin{aligned} x \quad + 10z &= 5 \\ y - 34z &= -16 \\ 0 &= -3 \end{aligned}$$

is equivalent to the original system. In other words, the two have the same solutions. But this last system clearly has no solution (the last equation requires that x, y and z satisfy $0x+0y+0z=-3$, and no such numbers exist). Hence the original system has no solution.

Example 1.2.3

Solve the following system of equations.

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= 1 \\ 2x_1 - 4x_2 + x_3 \phantom{{}+3x_4} &= 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 &= 4 \end{aligned}$$

Solution. The augmented matrix is

$$\left[\begin{array}{rrrr|r} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right]$$

Subtracting twice row 1 from row 2 and subtracting row 1 from row 3 gives

$$\left[\begin{array}{rrrr|r} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right]$$

Now subtract row 2 from row 3 and multiply row 2 by $\frac{1}{3}$ to get

$$\left[\begin{array}{rrrr|r} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This is in row-echelon form, and we take it to reduced form by adding row 2 to row 1:

$$\left[\begin{array}{rrrr|r} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The corresponding reduced system of equations is

$$\begin{array}{rcrcl} x_1 - 2x_2 & & + & x_4 = & 2 \\ & & x_3 & - 2x_4 = & 1 \\ & & & 0 = & 0 \end{array}$$

The leading ones are in columns 1 and 3 here, so the corresponding variables x_1 and x_3 are called leading variables. Because the matrix is in reduced row-echelon form, these equations can be used to solve for the leading variables in terms of the nonleading variables x_2 and x_4. More precisely, in the present example we set $x_2 = s$ and $x_4 = t$ where s and t are arbitrary, so these equations become

$$x_1 - 2s + t = 2 \quad \text{and} \quad x_3 - 2t = 1$$

Finally the solutions are given by

$$\begin{aligned} x_1 &= 2 + 2s - t \\ x_2 &= s \\ x_3 &= 1 + 2t \\ x_4 &= t \end{aligned}$$

where s and t are arbitrary.

The solution of Example 1.2.3 is typical of the general case. To solve a linear system, the augmented matrix is carried to reduced row-echelon form, and the variables corresponding to the leading ones are called **leading variables**. Because the matrix is in reduced form, each leading variable occurs in exactly one equation, so that equation can be solved to give a formula for the leading variable in terms of the nonleading variables. It is customary to call the nonleading variables “free” variables, and to label them by new variables $s, t, \dots,$ called **parameters**. Hence, as in Example 1.2.3, every variable x_i is given by a formula in terms of the parameters s and t. Moreover, every choice of these parameters leads to a solution to the system, and every solution arises in this way. This procedure works in general, and has come to be called

Gaussian Elimination

To solve a system of linear equations proceed as follows:

1. *Carry the augmented matrix to a reduced row-echelon matrix using elementary row operations.*
2. *If a row* $\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 1 \end{array}\right]$ *occurs, the system is inconsistent.*
3. *Otherwise, assign the nonleading variables (if any) as parameters, and use the equations corresponding to the reduced row-echelon matrix to solve for the leading variables in terms of the parameters.*

There is a variant of this procedure, wherein the augmented matrix is carried only to row-echelon form. The nonleading variables are assigned as parameters as before. Then the last equation (corresponding to the row-echelon form) is used to solve for the last leading variable in terms of the parameters. This last leading variable is then substituted into all the preceding equations. Then, the second last equation yields the second last leading variable, which is also substituted back. The process continues to give the general solution. This procedure is called **back-substitution**. This procedure can be shown to be numerically more efficient and so is important when solving very large systems.[5]

Example 1.2.4

Find a condition on the numbers a, b, and c such that the following system of equations is consistent. When that condition is satisfied, find all solutions (in terms of a, b, and c).

$$\begin{aligned} x_1 + 3x_2 + x_3 &= a \\ -x_1 - 2x_2 + x_3 &= b \\ 3x_1 + 7x_2 - x_3 &= c \end{aligned}$$

Solution. We use gaussian elimination except that now the augmented matrix

$$\left[\begin{array}{rrr|c} 1 & 3 & 1 & a \\ -1 & -2 & 1 & b \\ 3 & 7 & -1 & c \end{array}\right]$$

has entries a, b, and c as well as known numbers. The first leading one is in place, so we create zeros below it in column 1:

$$\left[\begin{array}{rrr|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & -2 & -4 & c-3a \end{array}\right]$$

[5] With n equations where n is large, gaussian elimination requires roughly $n^3/2$ multiplications and divisions, whereas this number is roughly $n^3/3$ if back substitution is used.

The second leading 1 has appeared, so use it to create zeros in the rest of column 2:

$$\left[\begin{array}{rrr|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & c-a+2b \end{array}\right]$$

Now the whole solution depends on the number $c-a+2b = c-(a-2b)$. The last row corresponds to an equation $0 = c-(a-2b)$. If $c \neq a-2b$, there is *no* solution (just as in Example 1.2.2). Hence:

The system is consistent if and only if $c = a-2b$.

In this case the last matrix becomes

$$\left[\begin{array}{rrr|c} 1 & 0 & -5 & -2a-3b \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Thus, if $c = a-2b$, taking $x_3 = t$ where t is a parameter gives the solutions

$$x_1 = 5t-(2a+3b) \quad x_2 = (a+b)-2t \quad x_3 = t.$$

Rank

It can be proven that the *reduced* row-echelon form of a matrix A is uniquely determined by A. That is, no matter which series of row operations is used to carry A to a reduced row-echelon matrix, the result will always be the same matrix. (A proof is given at the end of Section 2.5.) By contrast, this is not true for row-echelon matrices: Different series of row operations can carry the same matrix A to *different* row-echelon matrices. Indeed, the matrix $A = \left[\begin{array}{rrr} 1 & -1 & 4 \\ 2 & -1 & 2 \end{array}\right]$ can be carried (by one row operation) to the row-echelon matrix $\left[\begin{array}{rrr} 1 & -1 & 4 \\ 0 & 1 & -6 \end{array}\right]$, and then by another row operation to the (reduced) row-echelon matrix $\left[\begin{array}{rrr} 1 & 0 & -2 \\ 0 & 1 & -6 \end{array}\right]$. However, it *is* true that the number r of leading 1s must be the same in each of these row-echelon matrices (this will be proved in Chapter 5). Hence, the number r depends only on A and not on the way in which A is carried to row-echelon form.

Definition 1.4 Rank of a Matrix

The ***rank*** *of matrix A is the number of leading 1s in any row-echelon matrix to which A can be carried by row operations.*

Example 1.2.5

Compute the rank of $A = \begin{bmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & -5 & 8 \end{bmatrix}$.

Solution. The reduction of A to row-echelon form is

$$A = \begin{bmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & -5 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 4 \\ 0 & -1 & 5 & -8 \\ 0 & 1 & -5 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 4 \\ 0 & 1 & -5 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Because this row-echelon matrix has two leading 1s, rank $A = 2$.

Suppose that rank $A = r$, where A is a matrix with m rows and n columns. Then $r \leq m$ because the leading 1s lie in different rows, and $r \leq n$ because the leading 1s lie in different columns. Moreover, the rank has a useful application to equations. Recall that a system of linear equations is called consistent if it has at least one solution.

Theorem 1.2.2

Suppose a system of m equations in n variables is ***consistent****, and that the rank of the augmented matrix is r.*

1. *The set of solutions involves exactly $n - r$ parameters.*
2. *If $r < n$, the system has infinitely many solutions.*
3. *If $r = n$, the system has a unique solution.*

Proof. The fact that the rank of the augmented matrix is r means there are exactly r leading variables, and hence exactly $n - r$ nonleading variables. These nonleading variables are all assigned as parameters in the gaussian algorithm, so the set of solutions involves exactly $n - r$ parameters. Hence if $r < n$, there is at least one parameter, and so infinitely many solutions. If $r = n$, there are no parameters and so a unique solution. □

Theorem 1.2.2 shows that, for any system of linear equations, exactly three possibilities exist:

1. *No solution. This occurs when a row* $\begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$ *occurs in the row-echelon form of the augmented matrix. This is the case where the system is inconsistent.*

. *Unique solution. This occurs when* every *variable is a leading variable (every variable column has a pivot).*

. *Infinitely many solutions. This occurs when the system is consistent and there is at least one nonleading variable, so at least one parameter is involved (at least one variable has no pivot and the right-hand side column has no pivot).*

Example 1.2.6

Suppose the matrix A in Example 1.2.5 is the augmented matrix of a system of $m = 3$ linear equations in $n = 3$ variables. As rank $A = r = 2$, the set of solutions will have $n - r = 1$ parameter. The reader can verify this fact directly.

Many important problems involve **linear inequalities** rather than **linear equations**. For example, a condition on the variables x and y might take the form of an inequality $2x - 5y \leq 4$ rather than an equality $2x - 5y = 4$. There is a technique (called the **simplex algorithm**) for finding solutions to a system of such inequalities that maximizes a function of the form $p = ax + by$ where a and b are fixed constants.

Exercises for 1.2

Exercise 1.2.1 Which of the following matrices are in reduced row-echelon form? Which are in row-echelon form?

a. $\left[\begin{array}{rrr} 1 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right]$

b. $\left[\begin{array}{rrrr} 2 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$

c. $\left[\begin{array}{rrrr} 1 & -2 & 3 & 5 \\ 0 & 0 & 0 & 1 \end{array}\right]$

d. $\left[\begin{array}{rrrrr} 1 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$

e. $\left[\begin{array}{rr} 1 & 1 \\ 0 & 1 \end{array}\right]$

f. $\left[\begin{array}{rrr} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{array}\right]$

Exercise 1.2.2 Carry each of the following matrices to reduced row-echelon form.

a. $\left[\begin{array}{rrrrrrr} 0 & -1 & 2 & 1 & 2 & 1 & -1 \\ 0 & 1 & -2 & 2 & 7 & 2 & 4 \\ 0 & -2 & 4 & 3 & 7 & 1 & 0 \\ 0 & 3 & -6 & 1 & 6 & 4 & 1 \end{array}\right]$

b. $\left[\begin{array}{rrrrrrr} 0 & -1 & 3 & 1 & 3 & 2 & 1 \\ 0 & -2 & 6 & 1 & -5 & 0 & -1 \\ 0 & 3 & -9 & 2 & 4 & 1 & -1 \\ 0 & 1 & -3 & -1 & 3 & 0 & 1 \end{array}\right]$

Exercise 1.2.3 The augmented matrix of a system of linear equations has been carried to the following by row operations. In each case solve the system.

a. $\left[\begin{array}{rrrrrr|r} 1 & 2 & 0 & 3 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

b. $\left[\begin{array}{rrrrrr|r} 1 & -2 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 5 & 0 & -3 & -1 \\ 0 & 0 & 0 & 0 & 1 & 6 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

c. $\left[\begin{array}{rrrrr|r} 1 & 2 & 1 & 3 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

d. $\left[\begin{array}{rrrrr|r} 1 & -1 & 2 & 4 & 6 & 2 \\ 0 & 1 & 2 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

Exercise 1.2.4 Find all solutions (if any) to each of the following systems of linear equations.

a. $\begin{aligned} x - 2y &= 1 \\ 4y - x &= -2 \end{aligned}$

b. $\begin{aligned} 3x - y &= 0 \\ 2x - 3y &= 1 \end{aligned}$

c. $\begin{aligned} 2x + y &= 5 \\ 3x + 2y &= 6 \end{aligned}$

d. $\begin{aligned} 3x - y &= 2 \\ 2y - 6x &= -4 \end{aligned}$

e. $\begin{aligned} 3x - y &= 4 \\ 2y - 6x &= 1 \end{aligned}$

f. $\begin{aligned} 2x - 3y &= 5 \\ 3y - 2x &= 2 \end{aligned}$

Exercise 1.2.5 Find all solutions (if any) to each of the following systems of linear equations.

a. $\begin{aligned} x + y + 2z &= 8 \\ 3x - y + z &= 0 \\ -x + 3y + 4z &= -4 \end{aligned}$

b. $\begin{aligned} -2x + 3y + 3z &= -9 \\ 3x - 4y + z &= 5 \\ -5x + 7y + 2z &= -14 \end{aligned}$

c. $\begin{aligned} x + y - z &= 10 \\ -x + 4y + 5z &= -5 \\ x + 6y + 3z &= 15 \end{aligned}$

d. $\begin{aligned} x + 2y - z &= 2 \\ 2x + 5y - 3z &= 1 \\ x + 4y - 3z &= 3 \end{aligned}$

e. $$\begin{aligned} 5x+y \phantom{{}-z} &= 2 \\ 3x-y+2z &= 1 \\ x+y-z &= 5 \end{aligned}$$

f. $$\begin{aligned} 3x-2y+z &= -2 \\ x-y+3z &= 5 \\ -x+y+z &= -1 \end{aligned}$$

g. $$\begin{aligned} x+y+z &= 2 \\ x \phantom{{}+y} + z &= 1 \\ 2x+5y+2z &= 7 \end{aligned}$$

h. $$\begin{aligned} x+2y-4z &= 10 \\ 2x-y+2z &= 5 \\ x+y-2z &= 7 \end{aligned}$$

Exercise 1.2.6 Express the last equation of each system as a sum of multiples of the first two equations. [*Hint*: Label the equations, use the gaussian algorithm.]

a. $$\begin{aligned} x_1+x_2+x_3 &= 1 \\ 2x_1-x_2+3x_3 &= 3 \\ x_1-2x_2+2x_3 &= 2 \end{aligned}$$

b. $$\begin{aligned} x_1+2x_2-3x_3 &= -3 \\ x_1+3x_2-5x_3 &= 5 \\ x_1-2x_2+5x_3 &= -35 \end{aligned}$$

Exercise 1.2.7 Find all solutions to the following systems.

a. $$\begin{aligned} 3x_1+8x_2-3x_3-14x_4 &= 2 \\ 2x_1+3x_2-x_3-2x_4 &= 1 \\ x_1-2x_2+x_3+10x_4 &= 0 \\ x_1+5x_2-2x_3-12x_4 &= 1 \end{aligned}$$

b. $$\begin{aligned} x_1-x_2+x_3-x_4 &= 0 \\ -x_1+x_2+x_3+x_4 &= 0 \\ x_1+x_2-x_3+x_4 &= 0 \\ x_1+x_2+x_3+x_4 &= 0 \end{aligned}$$

c. $$\begin{aligned} x_1-x_2+x_3-2x_4 &= 1 \\ -x_1+x_2+x_3+x_4 &= -1 \\ -x_1+2x_2+3x_3-x_4 &= 2 \\ x_1-x_2+2x_3+x_4 &= 1 \end{aligned}$$

d. $$\begin{aligned} x_1+x_2+2x_3-x_4 &= 4 \\ 3x_2-x_3+4x_4 &= 2 \\ x_1+2x_2-3x_3+5x_4 &= 0 \\ x_1+x_2-5x_3+6x_4 &= -3 \end{aligned}$$

Exercise 1.2.8 In each of the following, find (if possible) conditions on a and b such that the system has no solution, one solution, and infinitely many solutions.

a. $$\begin{aligned} x-2y &= 1 \\ ax+by &= 5 \end{aligned}$$

b. $$\begin{aligned} x+by &= -1 \\ ax+2y &= 5 \end{aligned}$$

c. $$\begin{aligned} x-by &= -1 \\ x+ay &= 3 \end{aligned}$$

d. $$\begin{aligned} ax+y &= 1 \\ 2x+y &= b \end{aligned}$$

Exercise 1.2.9 In each of the following, find (if possible) conditions on a, b, and c such that the system has no solution, one solution, or infinitely many solutions.

a. $$\begin{aligned} 3x+y-z &= a \\ x-y+2z &= b \\ 5x+3y-4z &= c \end{aligned}$$

b. $$\begin{aligned} 2x+y-z &= a \\ 2y+3z &= b \\ x \phantom{{}+2y} -z &= c \end{aligned}$$

c. $$\begin{aligned} -x+3y+2z &= -8 \\ x \phantom{{}+3y} +z &= 2 \\ 3x+3y+az &= b \end{aligned}$$

d. $$\begin{aligned} x+ay &= 0 \\ y+bz &= 0 \\ z+cx &= 0 \end{aligned}$$

e. $$\begin{aligned} 3x-y+2z &= 3 \\ x+y-z &= 2 \\ 2x-2y+3z &= b \end{aligned}$$

f. $$\begin{aligned} x+ay-z &= 1 \\ -x+(a-2)y+z &= -1 \\ 2x+2y+(a-2)z &= 1 \end{aligned}$$

Exercise 1.2.10 Find the rank of each of the matrices in Exercise 1.2.1.

Exercise 1.2.11 Find the rank of each of the following matrices.

a. $\left[\begin{array}{rrr} 1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{array}\right]$

b. $\left[\begin{array}{rrr} -2 & 3 & 3 \\ 3 & -4 & 1 \\ -5 & 7 & 2 \end{array}\right]$

c. $\left[\begin{array}{rrrr} 1 & 1 & -1 & 3 \\ -1 & 4 & 5 & -2 \\ 1 & 6 & 3 & 4 \end{array}\right]$

d. $\left[\begin{array}{rrrr} 3 & -2 & 1 & -2 \\ 1 & -1 & 3 & 5 \\ -1 & 1 & 1 & -1 \end{array}\right]$

e. $\left[\begin{array}{cccc} 1 & 2 & -1 & 0 \\ 0 & a & 1-a & a^2+1 \\ 1 & 2-a & -1 & -2a^2 \end{array}\right]$

f. $\left[\begin{array}{cccc} 1 & 1 & 2 & a^2 \\ 1 & 1-a & 2 & 0 \\ 2 & 2-a & 6-a & 4 \end{array}\right]$

Exercise 1.2.12 Consider a system of linear equations with augmented matrix A and coefficient matrix C. In each case either prove the statement or give an example showing that it is false.

a. If there is more than one solution, A has a row of zeros.

b. If A has a row of zeros, there is more than one solution.

c. If there is no solution, the reduced row-echelon form of C has a row of zeros.

d. If the row-echelon form of C has a row of zeros, there is no solution.

e. There is no system that is inconsistent for every choice of constants.

f. If the system is consistent for some choice of constants, it is consistent for every choice of constants.

Now assume that the augmented matrix A has 3 rows and 5 columns.

g. If the system is consistent, there is more than one solution.

h. The rank of A is at most 3.

i. If rank $A = 3$, the system is consistent.

j. If rank $C = 3$, the system is consistent.

Exercise 1.2.13 Find a sequence of row operations carrying

$$\left[\begin{array}{ccc} b_1+c_1 & b_2+c_2 & b_3+c_3 \\ c_1+a_1 & c_2+a_2 & c_3+a_3 \\ a_1+b_1 & a_2+b_2 & a_3+b_3 \end{array}\right] \text{ to } \left[\begin{array}{ccc} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{array}\right]$$

Exercise 1.2.14 In each case, show that the reduced row-echelon form is as given.

a. $\left[\begin{array}{ccc} p & 0 & a \\ b & 0 & 0 \\ q & c & r \end{array}\right]$ with $abc \neq 0$; $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$

b. $\left[\begin{array}{ccc} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{array}\right]$ where $c \neq a$ or $b \neq a$; $\left[\begin{array}{ccc} 1 & 0 & * \\ 0 & 1 & * \\ 0 & 0 & 0 \end{array}\right]$

Exercise 1.2.15 Show that $\left\{\begin{array}{rl} az+\ by+\ cz &= 0 \\ a_1x+b_1y+c_1z &= 0 \end{array}\right.$ always has a solution other than $x = 0$, $y = 0$, $z = 0$.

Exercise 1.2.16 Find the circle $x^2+y^2+ax+by+c = 0$ passing through the following points.

a. $(-2,\ 1)$, $(5,\ 0)$, and $(4,\ 1)$

b. $(1,\ 1)$, $(5,\ -3)$, and $(-3,\ -3)$

Exercise 1.2.17 Three Nissans, two Fords, and four Chevrolets can be rented for \$106 per day. At the same rates two Nissans, four Fords, and three Chevrolets cost \$107 per day, whereas four Nissans, three Fords, and two Chevrolets cost \$102 per day. Find the rental rates for all three kinds of cars.

Exercise 1.2.18 A school has three clubs and each student is required to belong to exactly one club. One year the students switched club membership as follows:
Club A. $\frac{4}{10}$ remain in A, $\frac{1}{10}$ switch to B, $\frac{5}{10}$ switch to C.
Club B. $\frac{7}{10}$ remain in B, $\frac{2}{10}$ switch to A, $\frac{1}{10}$ switch to C.
Club C. $\frac{6}{10}$ remain in C, $\frac{2}{10}$ switch to A, $\frac{2}{10}$ switch to B.

If the fraction of the student population in each club is unchanged, find each of these fractions.

Exercise 1.2.19 Given points $(p_1,\ q_1)$, $(p_2,\ q_2)$, and $(p_3,\ q_3)$ in the plane with p_1, p_2, and p_3 distinct, show that they lie on some curve with equation $y = a+bx+cx^2$. [*Hint*: Solve for a, b, and c.]

Exercise 1.2.20 The scores of three players in a tournament have been lost. The only information available is the total of the scores for players 1 and 2, the total for players 2 and 3, and the total for players 3 and 1.

a. Show that the individual scores can be rediscovered.

b. Is this possible with four players (knowing the totals for players 1 and 2, 2 and 3, 3 and 4, and 4 and 1)?

Exercise 1.2.21 A boy finds \$1.05 in dimes, nickels, and pennies. If there are 17 coins in all, how many coins of each type can he have?

Exercise 1.2.22 If a consistent system has more variables than equations, show that it has infinitely many solutions. [*Hint*: Use Theorem 1.2.2.]

1.3 Homogeneous Equations

A system of equations in the variables $x_1, x_2, \ldots, x_n$ is called **homogeneous** if all the constant terms are zero—that is, if each equation of the system has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$$

Clearly $x_1 = 0, x_2 = 0, \ldots, x_n = 0$ is a solution to such a system; it is called the **trivial solution**. Any solution in which at least one variable has a nonzero value is called a **nontrivial solution**. Our chief goal in this section is to give a useful condition for a homogeneous system to have nontrivial solutions. The following example is instructive.

Example 1.3.1

Show that the following homogeneous system has nontrivial solutions.

$$\begin{aligned} x_1 - x_2 + 2x_3 - x_4 &= 0 \\ 2x_1 + 2x_2 \qquad + x_4 &= 0 \\ 3x_1 + x_2 + 2x_3 - x_4 &= 0 \end{aligned}$$

Solution. The reduction of the augmented matrix to reduced row-echelon form is outlined below.

$$\left[\begin{array}{rrrr|r} 1 & -1 & 2 & -1 & 0 \\ 2 & 2 & 0 & 1 & 0 \\ 3 & 1 & 2 & -1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrr|r} 1 & -1 & 2 & -1 & 0 \\ 0 & 4 & -4 & 3 & 0 \\ 0 & 4 & -4 & 2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrr|r} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

The leading variables are x_1, x_2, and x_4, so x_3 is assigned as a parameter—say $x_3 = t$. Then the general solution is $x_1 = -t$, $x_2 = t$, $x_3 = t$, $x_4 = 0$. Hence, taking $t = 1$ (say), we get a nontrivial solution: $x_1 = -1$, $x_2 = 1$, $x_3 = 1$, $x_4 = 0$.

The existence of a nontrivial solution in Example 1.3.1 is ensured by the presence of a parameter in the solution. This is due to the fact that there is a *nonleading* variable (x_3 in this case). But there *must* be a nonleading variable here because there are four variables and only three equations (and hence at *most* three leading variables). This discussion generalizes to a proof of the following fundamental theorem.

Theorem 1.3.1

If a homogeneous system of linear equations has more variables than equations, then it has a nontrivial solution (in fact, infinitely many).

Proof. Suppose there are m equations in n variables where $n > m$, and let R denote the reduced row-echelon form of the augmented matrix. If there are r leading variables, there are $n - r$ nonleading variables, and so $n - r$ parameters. Hence, it suffices to show that $r < n$. But $r \leq m$ because R has r leading 1s and m rows, and $m < n$ by hypothesis. So $r \leq m < n$, which gives $r < n$. □

Note that the converse of Theorem 1.3.1 is not true: if a homogeneous system has nontrivial solutions, it need not have more variables than equations (the system $x_1 + x_2 = 0$, $2x_1 + 2x_2 = 0$ has nontrivial solutions but $m = 2 = n$.)

Theorem 1.3.1 is very useful in applications. The next example provides an illustration from geometry.

Example 1.3.2

We call the graph of an equation $ax^2 + bxy + cy^2 + dx + ey + f = 0$ a **conic** if the numbers a, b, and c are not all zero. Show that there is at least one conic through any five points in the plane that are not all on a line.

Solution. Let the coordinates of the five points be (p_1, q_1), (p_2, q_2), (p_3, q_3), (p_4, q_4), and (p_5, q_5). The graph of $ax^2 + bxy + cy^2 + dx + ey + f = 0$ passes through (p_i, q_i) if

$$ap_i^2 + bp_iq_i + cq_i^2 + dp_i + eq_i + f = 0$$

This gives five equations, one for each i, linear in the six variables a, b, c, d, e, and f. Hence, there is a nontrivial solution by Theorem 1.3.1. If $a = b = c = 0$, the five points all lie on the line with equation $dx + ey + f = 0$, contrary to assumption. Hence, one of a, b, c is nonzero.

Linear Combinations and Basic Solutions

As for rows, two columns are regarded as **equal** if they have the same number of entries and corresponding entries are the same. Let $\mathbf{x}$ and $\mathbf{y}$ be columns with the same number of entries. As for elementary row operations, their **sum** $\mathbf{x} + \mathbf{y}$ is obtained by adding corresponding entries and, if k is a number, the **scalar product** $k\mathbf{x}$ is defined by multiplying each entry of $\mathbf{x}$ by k. More precisely:

$$\text{If } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \text{ then } \mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix} \text{ and } k\mathbf{x} = \begin{bmatrix} kx_1 \\ kx_2 \\ \vdots \\ kx_n \end{bmatrix}.$$

A sum of scalar multiples of several columns is called a **linear combination** of these columns. For example, $s\mathbf{x} + t\mathbf{y}$ is a linear combination of $\mathbf{x}$ and $\mathbf{y}$ for any choice of numbers s and t.

Example 1.3.3

If $\mathbf{x} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ then $2\mathbf{x} + 5\mathbf{y} = \begin{bmatrix} 6 \\ -4 \end{bmatrix} + \begin{bmatrix} -5 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Example 1.3.4

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$. If $\mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, determine whether $\mathbf{v}$ and $\mathbf{w}$ are linear combinations of $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$.

Solution. For $\mathbf{v}$, we must determine whether numbers r, s, and t exist such that $\mathbf{v} = r\mathbf{x} + s\mathbf{y} + t\mathbf{z}$, that is, whether

$$\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} = r\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} r+2s+3t \\ s+t \\ r+t \end{bmatrix}$$

Equating corresponding entries gives a system of linear equations $r+2s+3t = 0$, $s+t = -1$, and $r+t = 2$ for r, s, and t. By gaussian elimination, the solution is $r = 2-k$, $s = -1-k$, and $t = k$ where k is a parameter. Taking $k = 0$, we see that $\mathbf{v} = 2\mathbf{x} - \mathbf{y}$ is a linear combination of $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$.

Turning to $\mathbf{w}$, we again look for r, s, and t such that $\mathbf{w} = r\mathbf{x} + s\mathbf{y} + t\mathbf{z}$; that is,

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = r\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} r+2s+3t \\ s+t \\ r+t \end{bmatrix}$$

leading to equations $r+2s+3t = 1$, $s+t = 1$, and $r+t = 1$ for real numbers r, s, and t. But this time there is *no* solution as the reader can verify, so $\mathbf{w}$ is *not* a linear combination of $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$.

Our interest in linear combinations comes from the fact that they provide one of the best ways to describe the general solution of a homogeneous system of linear equations. When solving such a system with n variables $x_1, x_2, \dots, x_n$, write the variables as a column[6] matrix: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$. The trivial solution is denoted $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. As an illustration, the general solution in Example 1.3.1 is $x_1 = -t$, $x_2 = t$, $x_3 = t$, and $x_4 = 0$, where t is a parameter, and we would now express this by saying that the general solution is $\mathbf{x} = \begin{bmatrix} -t \\ t \\ t \\ 0 \end{bmatrix}$, where t is arbitrary.

Now let $\mathbf{x}$ and $\mathbf{y}$ be two solutions to a homogeneous system with n variables. Then any linear combination $s\mathbf{x} + t\mathbf{y}$ of these solutions turns out to be again a solution to the system. More generally:

Any linear combination of solutions to a homogeneous system is again a solution. (1.1)

In fact, suppose that a typical equation in the system is $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$, and suppose that $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ are solutions. Then $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$ and $a_1y_1 + a_2y_2 + \cdots + a_ny_n = 0$.

[6]The reason for using columns will be apparent later.

Hence $s\mathbf{x}+t\mathbf{y} = \begin{bmatrix} sx_1+ty_1 \\ sx_2+ty_2 \\ \vdots \\ sx_n+ty_n \end{bmatrix}$ is also a solution because

$$\begin{aligned} a_1(sx_1+ty_1)+a_2(sx_2+ty_2)+\cdots+a_n(sx_n+ty_n) \\ &= [a_1(sx_1)+a_2(sx_2)+\cdots+a_n(sx_n)]+[a_1(ty_1)+a_2(ty_2)+\cdots+a_n(ty_n)] \\ &= s(a_1x_1+a_2x_2+\cdots+a_nx_n)+t(a_1y_1+a_2y_2+\cdots+a_ny_n) \\ &= s(0)+t(0) \\ &= 0 \end{aligned}$$

A similar argument shows that Statement 1.1 is true for linear combinations of more than two solutions.

The remarkable thing is that *every* solution to a homogeneous system is a linear combination of certain particular solutions and, in fact, these solutions are easily computed using the gaussian algorithm. Here is an example.

Example 1.3.5

Solve the homogeneous system with coefficient matrix

$$A = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -3 & 6 & 1 & 0 \\ -2 & 4 & 4 & -2 \end{bmatrix}$$

Solution. The reduction of the augmented matrix to reduced form is

$$\left[\begin{array}{rrrr|r} 1 & -2 & 3 & -2 & 0 \\ -3 & 6 & 1 & 0 & 0 \\ -2 & 4 & 4 & -2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrr|r} 1 & -2 & 0 & -\frac{1}{5} & 0 \\ 0 & 0 & 1 & -\frac{3}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

so the solutions are $x_1 = 2s+\frac{1}{5}t$, $x_2 = s$, $x_3 = \frac{3}{5}t$, and $x_4 = t$ by gaussian elimination. Hence we can write the general solution $\mathbf{x}$ in the matrix form

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s+\frac{1}{5}t \\ s \\ \frac{3}{5}t \\ t \end{bmatrix} = s\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix} = s\mathbf{x}_1 + t\mathbf{x}_2.$$

Here $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$ are particular solutions determined by the gaussian algorithm.

The solutions $\mathbf{x}_1$ and $\mathbf{x}_2$ in Example 1.3.5 are denoted as follows:

Definition 1.5 Basic Solutions

The gaussian algorithm systematically produces solutions to any homogeneous linear system, called ***basic solutions****, one for every parameter.*

Moreover, the algorithm gives a routine way to express *every* solution as a linear combination of basic solutions as in Example 1.3.5, where the general solution $\mathbf{x}$ becomes

$$\mathbf{x} = s\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix} = s\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{5}t\begin{bmatrix} 1 \\ 0 \\ 3 \\ 5 \end{bmatrix}$$

Hence by introducing a new parameter $r = t/5$ we can multiply the original basic solution $\mathbf{x}_2$ by 5 and so eliminate fractions. For this reason:

Convention:
Any nonzero scalar multiple of a basic solution will still be called a basic solution.

In the same way, the gaussian algorithm produces basic solutions to *every* homogeneous system, one for each parameter (there are *no* basic solutions if the system has only the trivial solution). Moreover every solution is given by the algorithm as a linear combination of these basic solutions (as in Example 1.3.5). If A has rank r, Theorem 1.2.2 shows that there are exactly $n-r$ parameters, and so $n-r$ basic solutions. This proves:

Theorem 1.3.2

Let A be an $m \times n$ matrix of rank r, and consider the homogeneous system in n variables with A as coefficient matrix. Then:

1. *The system has exactly $n-r$ basic solutions, one for each parameter.*
2. *Every solution is a linear combination of these basic solutions.*

Example 1.3.6

Find basic solutions of the homogeneous system with coefficient matrix A, and express every solution as a linear combination of the basic solutions, where

$$A = \begin{bmatrix} 1 & -3 & 0 & 2 & 2 \\ -2 & 6 & 1 & 2 & -5 \\ 3 & -9 & -1 & 0 & 7 \\ -3 & 9 & 2 & 6 & -8 \end{bmatrix}$$

Solution. The reduction of the augmented matrix to reduced row-echelon form is

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 0 & 2 & 2 & 0 \\ -2 & 6 & 1 & 2 & -5 & 0 \\ 3 & -9 & -1 & 0 & 7 & 0 \\ -3 & 9 & 2 & 6 & -8 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrrr|r} 1 & -3 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 6 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

so the general solution is $x_1 = 3r - 2s - 2t$, $x_2 = r$, $x_3 = -6s + t$, $x_4 = s$, and $x_5 = t$ where r, s, and t are parameters. In matrix form this is

$$\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{array}\right] = \left[\begin{array}{c} 3r - 2s - 2t \\ r \\ -6s + t \\ s \\ t \end{array}\right] = r\left[\begin{array}{r} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{array}\right] + s\left[\begin{array}{r} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{array}\right] + t\left[\begin{array}{r} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{array}\right]$$

Hence basic solutions are

$$\mathbf{x}_1 = \left[\begin{array}{r} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{array}\right], \quad \mathbf{x}_2 = \left[\begin{array}{r} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{array}\right], \quad \mathbf{x}_3 = \left[\begin{array}{r} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{array}\right]$$

Exercises for 1.3

Exercise 1.3.1 Consider the following statements about a system of linear equations with augmented matrix A. In each case either prove the statement or give an example for which it is false.

a. If the system is homogeneous, every solution is trivial.

b. If the system has a nontrivial solution, it cannot be homogeneous.

c. If there exists a trivial solution, the system is homogeneous.

d. If the system is consistent, it must be homogeneous.

Now assume that the system is homogeneous.

e. If there exists a nontrivial solution, there is no trivial solution.

f. If there exists a solution, there are infinitely many solutions.

g. If there exist nontrivial solutions, the row-echelon form of A has a row of zeros.

h. If the row-echelon form of A has a row of zeros, there exist nontrivial solutions.

i. If a row operation is applied to the system, the new system is also homogeneous.

Exercise 1.3.2 In each of the following, find all values of a for which the system has nontrivial solutions, and determine all solutions in each case.

a. $$\begin{array}{rcl} x - 2y + z &=& 0 \\ x + ay - 3z &=& 0 \\ -x + 6y - 5z &=& 0 \end{array}$$

b. $$\begin{array}{rcl} x + 2y + z &=& 0 \\ x + 3y + 6z &=& 0 \\ 2x + 3y + az &=& 0 \end{array}$$

c. $$\begin{array}{rcl} x + y - z &=& 0 \\ ay - z &=& 0 \\ x + y + az &=& 0 \end{array}$$

d. $$\begin{array}{rcl} ax + y + z &=& 0 \\ x + y - z &=& 0 \\ x + y + az &=& 0 \end{array}$$

Exercise 1.3.3 Let $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, and $\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. In each case, either write $\mathbf{v}$ as a linear combination of $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$, or show that it is not such a linear combination.

a. $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}$

b. $\mathbf{v} = \begin{bmatrix} 4 \\ 3 \\ -4 \end{bmatrix}$

c. $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

d. $\mathbf{v} = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$

Exercise 1.3.4 In each case, either express $\mathbf{y}$ as a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$, or show that it is not such a linear combination. Here:

$$\mathbf{a}_1 = \begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \quad \text{and } \mathbf{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

a. $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix}$

b. $\mathbf{y} = \begin{bmatrix} -1 \\ 9 \\ 2 \\ 6 \end{bmatrix}$

Exercise 1.3.5 For each of the following homogeneous systems, find a set of basic solutions and express the general solution as a linear combination of these basic solutions.

a. $$\begin{aligned} x_1 + 2x_2 - x_3 + 2x_4 + x_5 &= 0 \\ x_1 + 2x_2 + 2x_3 + x_5 &= 0 \\ 2x_1 + 4x_2 - 2x_3 + 3x_4 + x_5 &= 0 \end{aligned}$$

b. $$\begin{aligned} x_1 + 2x_2 - x_3 + x_4 + x_5 &= 0 \\ -x_1 - 2x_2 + 2x_3 + x_5 &= 0 \\ -x_1 - 2x_2 + 3x_3 + x_4 + 3x_5 &= 0 \end{aligned}$$

c. $$\begin{aligned} x_1 + x_2 - x_3 + 2x_4 + x_5 &= 0 \\ x_1 + 2x_2 - x_3 + x_4 + x_5 &= 0 \\ 2x_1 + 3x_2 - x_3 + 2x_4 + x_5 &= 0 \\ 4x_1 + 5x_2 - 2x_3 + 5x_4 + 2x_5 &= 0 \end{aligned}$$

d. $$\begin{aligned} x_1 + x_2 - 2x_3 - 2x_4 + 2x_5 &= 0 \\ 2x_1 + 2x_2 - 4x_3 - 4x_4 + x_5 &= 0 \\ x_1 - x_2 + 2x_3 + 4x_4 + x_5 &= 0 \\ -2x_1 - 4x_2 + 8x_3 + 10x_4 + x_5 &= 0 \end{aligned}$$

Exercise 1.3.6

a. Does Theorem 1.3.1 imply that the system $\left\{ \begin{aligned} -z + 3y &= 0 \\ 2x - 6y &= 0 \end{aligned} \right.$ has nontrivial solutions? Explain.

b. Show that the converse to Theorem 1.3.1 is not true. That is, show that the existence of nontrivial solutions does *not* imply that there are more variables than equations.

Exercise 1.3.7 In each case determine how many solutions (and how many parameters) are possible for a homogeneous system of four linear equations in six variables with augmented matrix A. Assume that A has nonzero entries. Give all possibilities.

a. Rank $A = 2$.

b. Rank $A = 1$.

c. A has a row of zeros.

d. The row-echelon form of A has a row of zeros.

Exercise 1.3.8 The graph of an equation $ax + by + cz = 0$ is a plane through the origin (provided that not all of a, b, and c are zero). Use Theorem 1.3.1 to show that two planes through the origin have a point in common other than the origin $(0,\ 0,\ 0)$.

Exercise 1.3.9

a. Show that there is a line through any pair of points in the plane. [*Hint*: Every line has equation $ax + by + c = 0$, where a, b, and c are not all zero.]

b. Generalize and show that there is a plane $ax + by + cz + d = 0$ through any three points in space.

Exercise 1.3.10 The graph of

$$a(x^2 + y^2) + bx + cy + d = 0$$

is a circle if $a \neq 0$. Show that there is a circle through any three points in the plane that are not all on a line.

Exercise 1.3.11 Consider a homogeneous system of linear equations in n variables, and suppose that the augmented matrix has rank r. Show that the system has nontrivial solutions if and only if $n > r$.

Exercise 1.3.12 If a consistent (possibly nonhomogeneous) system of linear equations has more variables than equations, prove that it has more than one solution.

1.4 An Application to Network Flow

There are many types of problems that concern a network of conductors along which some sort of flow is observed. Examples of these include an irrigation network and a network of streets or freeways. There are often points in the system at which a net flow either enters or leaves the system. The basic principle behind the analysis of such systems is that the total flow into the system must equal the total flow out. In fact, we apply this principle at every junction in the system.

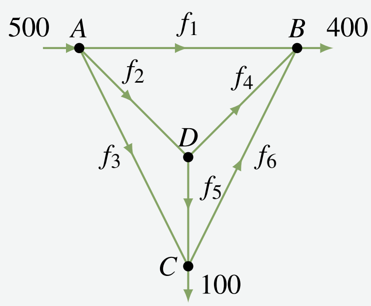
Junction Rule

At each of the junctions in the network, the total flow into that junction must equal the total flow out.

This requirement gives a linear equation relating the flows in conductors emanating from the junction.

Example 1.4.1

A network of one-way streets is shown in the accompanying diagram. The rate of flow of cars into intersection A is 500 cars per hour, and 400 and 100 cars per hour emerge from B and C, respectively. Find the possible flows along each street.



Solution. Suppose the flows along the streets are f_1, f_2, f_3, f_4, f_5, and f_6 cars per hour in the directions shown. Then, equating the flow in with the flow out at each intersection, we get

Intersection A	$500 = f_1 + f_2 + f_3$
Intersection B	$f_1 + f_4 + f_6 = 400$
Intersection C	$f_3 + f_5 = f_6 + 100$
Intersection D	$f_2 = f_4 + f_5$

These give four equations in the six variables $f_1, f_2, \dots, f_6$.

$$\begin{array}{rcrcrcrcrcrcl} f_1 & + & f_2 & + & f_3 & & & & & & & = & 500 \\ f_1 & & & & & + & f_4 & & & + & f_6 & = & 400 \\ & & & & f_3 & & & + & f_5 & - & f_6 & = & 100 \\ & & f_2 & & & - & f_4 & - & f_5 & & & = & 0 \end{array}$$

The reduction of the augmented matrix is

$$\left[\begin{array}{rrrrrr|r} 1 & 1 & 1 & 0 & 0 & 0 & 500 \\ 1 & 0 & 0 & 1 & 0 & 1 & 400 \\ 0 & 0 & 1 & 0 & 1 & -1 & 100 \\ 0 & 1 & 0 & -1 & -1 & 0 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrrrr|r} 1 & 0 & 0 & 1 & 0 & 1 & 400 \\ 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Hence, when we use f_4, f_5, and f_6 as parameters, the general solution is

$$f_1 = 400 - f_4 - f_6 \qquad f_2 = f_4 + f_5 \qquad f_3 = 100 - f_5 + f_6$$

This gives all solutions to the system of equations and hence all the possible flows.

Of course, not all these solutions may be acceptable in the real situation. For example, the flows $f_1, f_2, \dots, f_6$ are all *positive* in the present context (if one came out negative, it would mean traffic flowed in the opposite direction). This imposes constraints on the flows: $f_1 \geq 0$ and $f_3 \geq 0$ become

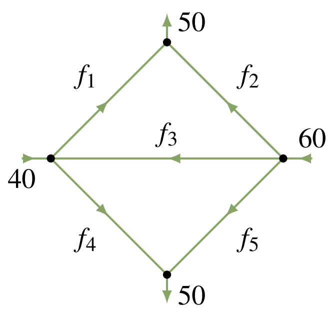
$$f_4 + f_6 \leq 400 \qquad f_5 - f_6 \leq 100$$

Further constraints might be imposed by insisting on maximum values on the flow in each street.

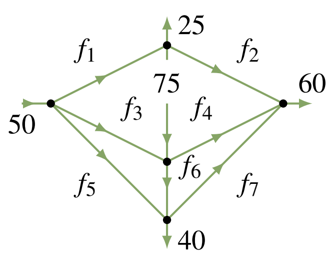
Exercises for 1.4

Exercise 1.4.1 Find the possible flows in each of the following networks of pipes.

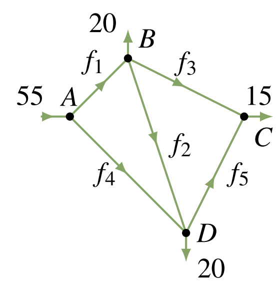
a.



b.



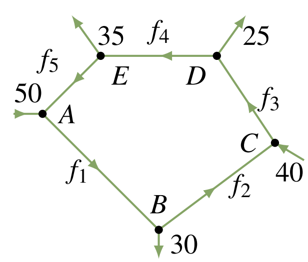
Exercise 1.4.2 A proposed network of irrigation canals is described in the accompanying diagram. At peak demand, the flows at interchanges A, B, C, and D are as shown.



a. Find the possible flows.

b. If canal BC is closed, what range of flow on AD must be maintained so that no canal carries a flow of more than 30?

Exercise 1.4.3 A traffic circle has five one-way streets, and vehicles enter and leave as shown in the accompanying diagram.



a. Compute the possible flows.

b. Which road has the heaviest flow?

1.5 An Application to Electrical Networks[7]

In an electrical network it is often necessary to find the current in amperes (A) flowing in various parts of the network. These networks usually contain resistors that retard the current. The resistors are indicated by a symbol ( ), and the resistance is measured in ohms (Ω). Also, the current is increased at various points by voltage sources (for example, a battery). The voltage of these sources is measured in volts (V), and they are represented by the symbol ( ). We assume these voltage sources have no resistance. The flow of current is governed by the following principles.

Ohm's Law

The current I and the voltage drop V across a resistance R are related by the equation $V = RI$.

Kirchhoff's Laws

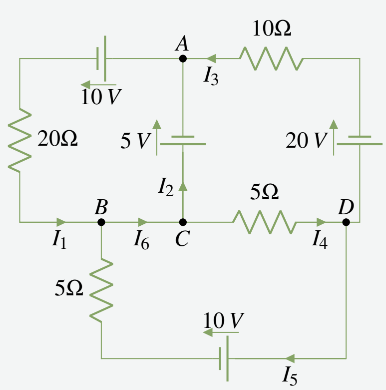
1. *(Junction Rule) The current flow into a junction equals the current flow out of that junction.*
2. *(Circuit Rule) The algebraic sum of the voltage drops (due to resistances) around any closed circuit of the network must equal the sum of the voltage increases around the circuit.*

When applying rule 2, select a direction (clockwise or counterclockwise) around the closed circuit and then consider all voltages and currents positive when in this direction and negative when in the opposite direction. This is why the term *algebraic sum* is used in rule 2. Here is an example.

Example 1.5.1

Find the various currents in the circuit shown.

Solution.



First apply the junction rule at junctions A, B, C, and D to obtain

$$\begin{array}{ll} \text{Junction } A & I_1 = I_2 + I_3 \\ \text{Junction } B & I_6 = I_1 + I_5 \\ \text{Junction } C & I_2 + I_4 = I_6 \\ \text{Junction } D & I_3 + I_5 = I_4 \end{array}$$

Note that these equations are not independent (in fact, the third is an easy consequence of the other three). Next, the circuit rule insists that the sum of the voltage increases (due to the sources) around a closed circuit must equal the sum of the voltage drops (due to resistances). By Ohm's law, the voltage loss across a resistance R (in the direction of the current I) is RI. Going counterclockwise around three closed circuits yields

[7] This section is independent of Section 1.4

$$\begin{array}{lrl} \text{Upper left} & 10+\quad 5 &= 20I_1 \\ \text{Upper right} & -5+\quad 20 &= 10I_3+5I_4 \\ \text{Lower} & -10 &= -5I_5-5I_4 \end{array}$$

Hence, disregarding the redundant equation obtained at junction C, we have six equations in the six unknowns $I_1, \dots, I_6$. The solution is

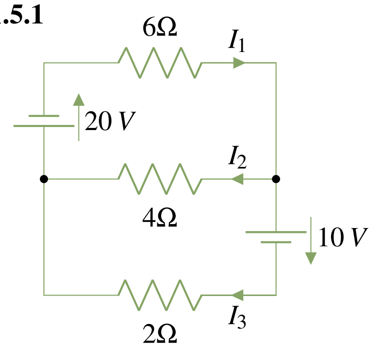
$$\begin{array}{ll} I_1 = \frac{15}{20} & I_4 = \frac{28}{20} \\ I_2 = \frac{-1}{20} & I_5 = \frac{12}{20} \\ I_3 = \frac{16}{20} & I_6 = \frac{27}{20} \end{array}$$

The fact that I_2 is negative means, of course, that this current is in the opposite direction, with a magnitude of $\frac{1}{20}$ amperes.

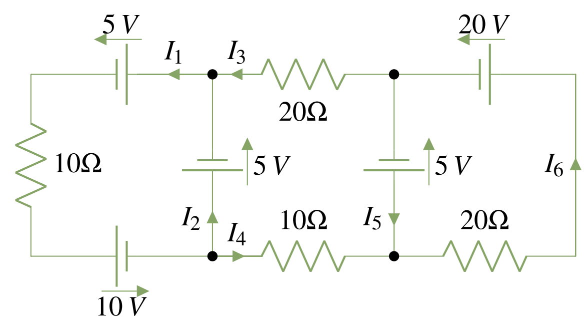
Exercises for 1.5

In Exercises 1 to 4, find the currents in the circuits.

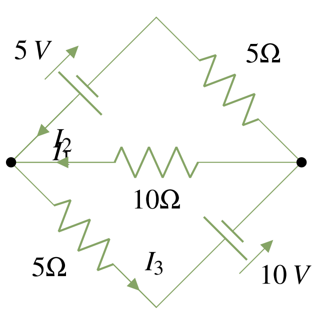
Exercise 1.5.1



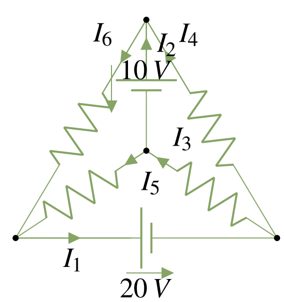
Exercise 1.5.3



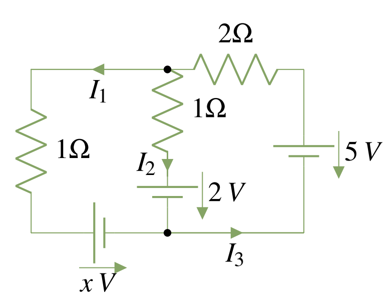
Exercise 1.5.2



Exercise 1.5.4 All resistances are 10Ω.



Exercise 1.5.5
Find the voltage x such that the current $I_1 = 0$.



1.6 An Application to Chemical Reactions

When a chemical reaction takes place a number of molecules combine to produce new molecules. Hence, when hydrogen H_2 and oxygen O_2 molecules combine, the result is water H_2O. We express this as

$$H_2 + O_2 \to H_2O$$

Individual atoms are neither created nor destroyed, so the number of hydrogen and oxygen atoms going into the reaction must equal the number coming out (in the form of water). In this case the reaction is said to be *balanced*. Note that each hydrogen molecule H_2 consists of two atoms as does each oxygen molecule O_2, while a water molecule H_2O consists of two hydrogen atoms and one oxygen atom. In the above reaction, this requires that twice as many hydrogen molecules enter the reaction; we express this as follows:

$$2H_2 + O_2 \to 2H_2O$$

This is now balanced because there are 4 hydrogen atoms and 2 oxygen atoms on each side of the reaction.

Example 1.6.1

Balance the following reaction for burning octane C_8H_{18} in oxygen O_2:

$$C_8H_{18} + O_2 \to CO_2 + H_2O$$

where CO_2 represents carbon dioxide. We must find positive integers x, y, z, and w such that

$$xC_8H_{18} + yO_2 \to zCO_2 + wH_2O$$

Equating the number of carbon, hydrogen, and oxygen atoms on each side gives $8x = z$, $18x = 2w$ and $2y = 2z + w$, respectively. These can be written as a homogeneous linear system

$$\begin{array}{rcrcrcrcl} 8x & & & - & z & & & = & 0 \\ 18x & & & & & - & 2w & = & 0 \\ & & 2y & - & 2z & - & w & = & 0 \end{array}$$

which can be solved by gaussian elimination. In larger systems this is necessary but, in such a simple situation, it is easier to solve directly. Set $w = t$, so that $x = \frac{1}{9}t$, $z = \frac{8}{9}t$, $2y = \frac{16}{9}t + t = \frac{25}{9}t$. But x, y, z, and w must be positive integers, so the smallest value of t that eliminates fractions is 18. Hence, $x = 2$, $y = 25$, $z = 16$, and $w = 18$, and the balanced reaction is

$$2C_8H_{18} + 25O_2 \to 16CO_2 + 18H_2O$$

The reader can verify that this is indeed balanced.

It is worth noting that this problem introduces a new element into the theory of linear equations: the insistence that the solution must consist of positive integers.

Exercises for 1.6

In each case balance the chemical reaction.

Exercise 1.6.1 $CH_4 + O_2 \to CO_2 + H_2O$. This is the burning of methane CH_4.

Exercise 1.6.2 $NH_3 + CuO \to N_2 + Cu + H_2O$. Here NH_3 is ammonia, CuO is copper oxide, Cu is copper, and N_2 is nitrogen.

Exercise 1.6.3 $CO_2 + H_2O \to C_6H_{12}O_6 + O_2$. This is called the photosynthesis reaction—$C_6H_{12}O_6$ is glucose.

Exercise 1.6.4 $Pb(N_3)_2 + Cr(MnO_4)_2 \to Cr_2O_3 + MnO_2 + Pb_3O_4 + NO$.

Supplementary Exercises for Chapter 1

Exercise 1.1 We show in Chapter 4 that the graph of an equation $ax+by+cz=d$ is a plane in space when not all of a, b, and c are zero.

a. By examining the possible positions of planes in space, show that three equations in three variables can have zero, one, or infinitely many solutions.

b. Can two equations in three variables have a unique solution? Give reasons for your answer.

Exercise 1.2 Find all solutions to the following systems of linear equations.

a. $$\begin{aligned} x_1 + x_2 + x_3 - x_4 &= 3 \\ 3x_1 + 5x_2 - 2x_3 + x_4 &= 1 \\ -3x_1 - 7x_2 + 7x_3 - 5x_4 &= 7 \\ x_1 + 3x_2 - 4x_3 + 3x_4 &= -5 \end{aligned}$$

b. $$\begin{aligned} x_1 + 4x_2 - x_3 + x_4 &= 2 \\ 3x_1 + 2x_2 + x_3 + 2x_4 &= 5 \\ x_1 - 6x_2 + 3x_3 &= 1 \\ x_1 + 14x_2 - 5x_3 + 2x_4 &= 3 \end{aligned}$$

Exercise 1.3 In each case find (if possible) conditions on a, b, and c such that the system has zero, one, or infinitely many solutions.

a. $$\begin{aligned} x + 2y - 4z &= 4 \\ 3x - y + 13z &= 2 \\ 4x + y + a^2z &= a+3 \end{aligned}$$

b. $$\begin{aligned} x + y + 3z &= a \\ ax + y + 5z &= 4 \\ x + ay + 4z &= a \end{aligned}$$

Exercise 1.4 Show that any two rows of a matrix can be interchanged by elementary row transformations of the other two types.

Exercise 1.5 If $ad \neq bc$, show that $\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$ has reduced row-echelon form $\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$.

Exercise 1.6 Find a, b, and c so that the system

$$\begin{aligned} x + ay + cz &= 0 \\ bx + cy - 3z &= 1 \\ ax + 2y + bz &= 5 \end{aligned}$$

has the solution $x=3$, $y=-1$, $z=2$.

Exercise 1.7 Solve the system

$$\begin{aligned} x + 2y + 2z &= -3 \\ 2x + y + z &= -4 \\ x - y + iz &= i \end{aligned}$$

where $i^2=-1$. [See Appendix A.]

Exercise 1.8 Show that the *real* system

$$\left\{\begin{aligned} x + y + z &= 5 \\ 2x - y - z &= 1 \\ -3x + 2y + 2z &= 0 \end{aligned}\right.$$

has a *complex* solution: $x=2$, $y=i$, $z=3-i$ where $i^2=-1$. Explain. What happens when such a real system has a unique solution?

Exercise 1.9 A man is ordered by his doctor to take 5 units of vitamin A, 13 units of vitamin B, and 23 units of vitamin C each day. Three brands of vitamin pills are available, and the number of units of each vitamin per pill are shown in the accompanying table.

	Vitamin		
Brand	**A**	**B**	**C**
1	1	2	4
2	1	1	3
3	0	1	1

a. Find all combinations of pills that provide exactly the required amount of vitamins (no partial pills allowed).

b. If brands 1, 2, and 3 cost 3¢, 2¢, and 5¢ per pill, respectively, find the least expensive treatment.

Exercise 1.10 A restaurant owner plans to use x tables seating 4, y tables seating 6, and z tables seating 8, for a total of 20 tables. When fully occupied, the tables seat 108 customers. If only half of the x tables, half of the y tables, and one-fourth of the z tables are used, each fully occupied, then 46 customers will be seated. Find x, y, and z.

Exercise 1.11

a. Show that a matrix with two rows and two columns that is in reduced row-echelon form must have one of the following forms:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & * \\ 0 & 0 \end{bmatrix}$$

[*Hint*: The leading 1 in the first row must be in column 1 or 2 or not exist.]

b. List the seven reduced row-echelon forms for matrices with two rows and three columns.

c. List the four reduced row-echelon forms for matrices with three rows and two columns.

Exercise 1.12 An amusement park charges \$7 for adults, \$2 for youths, and \$0.50 for children. If 150 people enter and pay a total of \$100, find the numbers of adults, youths, and children. [*Hint*: These numbers are nonnegative *integers*.]

Exercise 1.13 Solve the following system of equations for x and y.

$$\begin{aligned} x^2 + xy - y^2 &= 1 \\ 2x^2 - xy + 3y^2 &= 13 \\ x^2 + 3xy + 2y^2 &= 0 \end{aligned}$$

[*Hint*: These equations are linear in the new variables $x_1 = x^2$, $x_2 = xy$, and $x_3 = y^2$.]

Chapter 2

Matrix Algebra

In the study of systems of linear equations in Chapter 1, we found it convenient to manipulate the augmented matrix of the system. Our aim was to reduce it to row-echelon form (using elementary row operations) and hence to write down all solutions to the system. In the present chapter we consider matrices for their own sake. While some of the motivation comes from linear equations, it turns out that matrices can be multiplied and added and so form an algebraic system somewhat analogous to the real numbers. This "matrix algebra" is useful in ways that are quite different from the study of linear equations. For example, the geometrical transformations obtained by rotating the euclidean plane about the origin can be viewed as multiplications by certain 2×2 matrices. These "matrix transformations" are an important tool in geometry and, in turn, the geometry provides a "picture" of the matrices. Furthermore, matrix algebra has many other applications, some of which will be explored in this chapter. This subject is quite old and was first studied systematically in 1858 by Arthur Cayley.[1]

2.1 Matrix Addition, Scalar Multiplication, and Transposition

A rectangular array of numbers is called a **matrix** (the plural is **matrices**), and the numbers are called the **entries** of the matrix. Matrices are usually denoted by uppercase letters: A, B, C, and so on. Hence,

$$A = \left[\begin{array}{rrr} 1 & 2 & -1 \\ 0 & 5 & 6 \end{array} \right] \quad B = \left[\begin{array}{rr} 1 & -1 \\ 0 & 2 \end{array} \right] \quad C = \left[\begin{array}{r} 1 \\ 3 \\ 2 \end{array} \right]$$

are matrices. Clearly matrices come in various shapes depending on the number of **rows** and **columns**. For example, the matrix A shown has 2 rows and 3 columns. In general, a matrix with m rows and n columns is referred to as an $\boldsymbol{m \times n}$ ***matrix*** or as having **size** $\boldsymbol{m \times n}$. Thus matrices A, B, and C above have sizes 2×3, 2×2, and 3×1, respectively. A matrix of size $1 \times n$ is called a **row matrix**, whereas one of size $m \times 1$ is called a **column matrix**. Matrices of size $n \times n$ for some n are called **square** matrices.

Each entry of a matrix is identified by the row and column in which it lies. The rows are numbered from the top down, and the columns are numbered from left to right. Then the $\boldsymbol{(i,\ j)}$**-entry** of a matrix is the number lying simultaneously in row i and column j. For example,

$$\text{The } (1,\ 2)\text{-entry of } \left[\begin{array}{rr} 1 & -1 \\ 0 & 1 \end{array} \right] \text{ is } -1.$$

$$\text{The } (2,\ 3)\text{-entry of } \left[\begin{array}{rrr} 1 & 2 & -1 \\ 0 & 5 & 6 \end{array} \right] \text{ is } 6.$$

[1]Arthur Cayley (1821-1895) showed his mathematical talent early and graduated from Cambridge in 1842 as senior wrangler. With no employment in mathematics in view, he took legal training and worked as a lawyer while continuing to do mathematics, publishing nearly 300 papers in fourteen years. Finally, in 1863, he accepted the Sadlerian professorship in Cambridge and remained there for the rest of his life, valued for his administrative and teaching skills as well as for his scholarship. His mathematical achievements were of the first rank. In addition to originating matrix theory and the theory of determinants, he did fundamental work in group theory, in higher-dimensional geometry, and in the theory of invariants. He was one of the most prolific mathematicians of all time and produced 966 papers.

A special notation is commonly used for the entries of a matrix. If A is an $m \times n$ matrix, and if the $(i,\ j)$-entry of A is denoted as a_{ij}, then A is displayed as follows:

$$A = \left[\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array} \right]$$

This is usually denoted simply as $A = \left[a_{ij} \right]$. Thus a_{ij} is the entry in row i and column j of A. For example, a 3×4 matrix in this notation is written

$$A = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array} \right]$$

It is worth pointing out a convention regarding rows and columns: *Rows are mentioned before columns*. For example:

- *If a matrix has size $m \times n$, it has m rows and n columns.*
- *If we speak of the $(i,\ j)$-entry of a matrix, it lies in row i and column j.*
- *If an entry is denoted a_{ij}, the first subscript i refers to the row and the second subscript j to the column in which a_{ij} lies.*

Two points $(x_1,\ y_1)$ and $(x_2,\ y_2)$ in the plane are equal if and only if[2] they have the same coordinates, that is $x_1 = x_2$ and $y_1 = y_2$. Similarly, two matrices A and B are called **equal** (written $A = B$) if and only if:

1. *They have the same size.*
2. *Corresponding entries are equal.*

If the entries of A and B are written in the form $A = \left[a_{ij} \right]$, $B = \left[b_{ij} \right]$, described earlier, then the second condition takes the following form:

$$A = \left[a_{ij} \right] = \left[b_{ij} \right] \text{ means } a_{ij} = b_{ij} \text{ for all } i \text{ and } j$$

Example 2.1.1

Given $A = \left[\begin{array}{cc} a & b \\ c & d \end{array} \right]$, $B = \left[\begin{array}{rrr} 1 & 2 & -1 \\ 3 & 0 & 1 \end{array} \right]$ and $C = \left[\begin{array}{rr} 1 & 0 \\ -1 & 2 \end{array} \right]$ discuss the possibility that $A = B$, $B = C$, $A = C$.

Solution. $A = B$ is impossible because A and B are of different sizes: A is 2×2 whereas B is 2×3. Similarly, $B = C$ is impossible. But $A = C$ is possible provided that corresponding entries are

[2] If p and q are statements, we say that p implies q if q is true whenever p is true. Then "p if and only if q" means that both p implies q and q implies p. See Appendix B for more on this.

equal: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$ means $a = 1$, $b = 0$, $c = -1$, and $d = 2$.

Matrix Addition

Definition 2.1 Matrix Addition

If A and B are matrices of the same size, their **sum** *$A + B$ is the matrix formed by adding corresponding entries.*

If $A = [a_{ij}]$ and $B = [b_{ij}]$, this takes the form

$$A + B = [a_{ij} + b_{ij}]$$

Note that addition is *not* defined for matrices of different sizes.

Example 2.1.2

If $A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 6 \end{bmatrix}$, compute $A + B$.

Solution.

$$A + B = \begin{bmatrix} 2+1 & 1+1 & 3-1 \\ -1+2 & 2+0 & 0+6 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 2 & 6 \end{bmatrix}$$

Example 2.1.3

Find a, b, and c if $\begin{bmatrix} a & b & c \end{bmatrix} + \begin{bmatrix} c & a & b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$.

Solution. Add the matrices on the left side to obtain

$$\begin{bmatrix} a+c & b+a & c+b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$$

Because corresponding entries must be equal, this gives three equations: $a + c = 3$, $b + a = 2$, and $c + b = -1$. Solving these yields $a = 3$, $b = -1$, $c = 0$.

If A, B, and C are any matrices *of the same size*, then

$$A + B = B + A \qquad \text{(commutative law)}$$
$$A + (B + C) = (A + B) + C \qquad \text{(associative law)}$$

In fact, if $A = [a_{ij}]$ and $B = [b_{ij}]$, then the $(i,\ j)$-entries of $A + B$ and $B + A$ are, respectively, $a_{ij} + b_{ij}$ and $b_{ij} + a_{ij}$. Since these are equal for all i and j, we get

$$A + B = \begin{bmatrix} a_{ij} + b_{ij} \end{bmatrix} = \begin{bmatrix} b_{ij} + a_{ij} \end{bmatrix} = B + A$$

The associative law is verified similarly.

The $m \times n$ matrix in which every entry is zero is called the $m \times n$ **zero matrix** and is denoted as 0 (or 0_{mn} if it is important to emphasize the size). Hence,

$$0 + X = X$$

holds for all $m \times n$ matrices X. The **negative** of an $m \times n$ matrix A (written $-A$) is defined to be the $m \times n$ matrix obtained by multiplying each entry of A by -1. If $A = \left[a_{ij} \right]$, this becomes $-A = \left[-a_{ij} \right]$. Hence,

$$A + (-A) = 0$$

holds for all matrices A where, of course, 0 is the zero matrix of the same size as A.

A closely related notion is that of subtracting matrices. If A and B are two $m \times n$ matrices, their **difference** $A - B$ is defined by

$$A - B = A + (-B)$$

Note that if $A = \left[a_{ij} \right]$ and $B = \left[b_{ij} \right]$, then

$$A - B = \left[a_{ij} \right] + \left[-b_{ij} \right] = \left[a_{ij} - b_{ij} \right]$$

is the $m \times n$ matrix formed by *subtracting* corresponding entries.

Example 2.1.4

Let $A = \left[\begin{array}{rrr} 3 & -1 & 0 \\ 1 & 2 & -4 \end{array} \right]$, $B = \left[\begin{array}{rrr} 1 & -1 & 1 \\ -2 & 0 & 6 \end{array} \right]$, $C = \left[\begin{array}{rrr} 1 & 0 & -2 \\ 3 & 1 & 1 \end{array} \right]$. Compute $-A$, $A - B$, and $A + B - C$.

Solution.

$$\begin{aligned} -A &= \left[\begin{array}{rrr} -3 & 1 & 0 \\ -1 & -2 & 4 \end{array} \right] \\ A - B &= \left[\begin{array}{ccc} 3-1 & -1-(-1) & 0-1 \\ 1-(-2) & 2-0 & -4-6 \end{array} \right] = \left[\begin{array}{rrr} 2 & 0 & -1 \\ 3 & 2 & -10 \end{array} \right] \\ A + B - C &= \left[\begin{array}{ccc} 3+1-1 & -1-1-0 & 0+1-(-2) \\ 1-2-3 & 2+0-1 & -4+6-1 \end{array} \right] = \left[\begin{array}{rrr} 3 & -2 & 3 \\ -4 & 1 & 1 \end{array} \right] \end{aligned}$$

Example 2.1.5

Solve $\left[\begin{array}{rr} 3 & 2 \\ -1 & 1 \end{array} \right] + X = \left[\begin{array}{rr} 1 & 0 \\ -1 & 2 \end{array} \right]$ where X is a matrix.

Solution. We solve a numerical equation $a + x = b$ by subtracting the number a from both sides to obtain $x = b - a$. This also works for matrices. To solve $\left[\begin{array}{rr} 3 & 2 \\ -1 & 1 \end{array} \right] + X = \left[\begin{array}{rr} 1 & 0 \\ -1 & 2 \end{array} \right]$ simply subtract the matrix $\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ from both sides to get

$$X = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1-3 & 0-2 \\ -1-(-1) & 2-1 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 1 \end{bmatrix}$$

The reader should verify that this matrix X does indeed satisfy the original equation.

The solution in Example 2.1.5 solves the single matrix equation $A+X=B$ directly via matrix subtraction: $X=B-A$. This ability to work with matrices as entities lies at the heart of matrix algebra.

It is important to note that the sizes of matrices involved in some calculations are often determined by the context. For example, if

$$A+C = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

then A and C must be the same size (so that $A+C$ makes sense), and that size must be 2×3 (so that the sum is 2×3). For simplicity we shall often omit reference to such facts when they are clear from the context.

Scalar Multiplication

In gaussian elimination, multiplying a row of a matrix by a number k means multiplying *every* entry of that row by k.

Definition 2.2 Matrix Scalar Multiplication

*More generally, if A is any matrix and k is any number, the **scalar multiple** kA is the matrix obtained from A by multiplying each entry of A by k.*

If $A = \left[a_{ij}\right]$, this is

$$kA = \left[ka_{ij}\right]$$

Thus $1A = A$ and $(-1)A = -A$ for any matrix A.

The term *scalar* arises here because the set of numbers from which the entries are drawn is usually referred to as the set of scalars. We have been using real numbers as scalars, but we could equally well have been using complex numbers.

Example 2.1.6

If $A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 0 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$ compute $5A$, $\frac{1}{2}B$, and $3A-2B$.

Solution.

$$5A = \begin{bmatrix} 15 & -5 & 20 \\ 10 & 0 & 30 \end{bmatrix}, \quad \tfrac{1}{2}B = \begin{bmatrix} \frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & \frac{3}{2} & 1 \end{bmatrix}$$

$$3A-2B = \begin{bmatrix} 9 & -3 & 12 \\ 6 & 0 & 18 \end{bmatrix} - \begin{bmatrix} 2 & 4 & -2 \\ 0 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 7 & -7 & 14 \\ 6 & -6 & 14 \end{bmatrix}$$

If A is any matrix, note that kA is the same size as A for all scalars k. We also have

$$0A = 0 \quad \text{and} \quad k0 = 0$$

because the zero matrix has every entry zero. In other words, $kA = 0$ if either $k = 0$ or $A = 0$. The converse of this statement is also true, as Example 2.1.7 shows.

Example 2.1.7

If $kA = 0$, show that either $k = 0$ or $A = 0$.

Solution. Write $A = \begin{bmatrix} a_{ij} \end{bmatrix}$ so that $kA = 0$ means $ka_{ij} = 0$ for all i and j. If $k = 0$, there is nothing to do. If $k \neq 0$, then $ka_{ij} = 0$ implies that $a_{ij} = 0$ for all i and j; that is, $A = 0$.

For future reference, the basic properties of matrix addition and scalar multiplication are listed in Theorem 2.1.1.

Theorem 2.1.1

Let A, B, and C denote arbitrary $m \times n$ matrices where m and n are fixed. Let k and p denote arbitrary real numbers. Then

1. $A + B = B + A$.
2. $A + (B + C) = (A + B) + C$.
3. *There is an $m \times n$ matrix 0, such that $0 + A = A$ for each A.*
4. *For each A there is an $m \times n$ matrix, $-A$, such that $A + (-A) = 0$.*
5. $k(A + B) = kA + kB$.
6. $(k + p)A = kA + pA$.
7. $(kp)A = k(pA)$.
8. $1A = A$.

Proof. Properties 1–4 were given previously. To check Property 5, let $A = \begin{bmatrix} a_{ij} \end{bmatrix}$ and $B = \begin{bmatrix} b_{ij} \end{bmatrix}$ denote matrices of the same size. Then $A + B = \begin{bmatrix} a_{ij} + b_{ij} \end{bmatrix}$, as before, so the $(i,\ j)$-entry of $k(A + B)$ is

$$k(a_{ij} + b_{ij}) = ka_{ij} + kb_{ij}$$

But this is just the $(i,\ j)$-entry of $kA + kB$, and it follows that $k(A + B) = kA + kB$. The other Properties can be similarly verified; the details are left to the reader. □

The Properties in Theorem 2.1.1 enable us to do calculations with matrices in much the same way that numerical calculations are carried out. To begin, Property 2 implies that the sum

$$(A + B) + C = A + (B + C)$$

is the same no matter how it is formed and so is written as $A+B+C$. Similarly, the sum

$$A+B+C+D$$

is independent of how it is formed; for example, it equals both $(A+B)+(C+D)$ and $A+[B+(C+D)]$. Furthermore, property 1 ensures that, for example,

$$B+D+A+C=A+B+C+D$$

In other words, the *order* in which the matrices are added does not matter. A similar remark applies to sums of five (or more) matrices.

Properties 5 and 6 in Theorem 2.1.1 are called **distributive laws** for scalar multiplication, and they extend to sums of more than two terms. For example,

$$k(A+B-C)=kA+kB-kC$$

$$(k+p-m)A=kA+pA-mA$$

Similar observations hold for more than three summands. These facts, together with properties 7 and 8, enable us to simplify expressions by collecting like terms, expanding, and taking common factors in exactly the same way that algebraic expressions involving variables and real numbers are manipulated. The following example illustrates these techniques.

Example 2.1.8

Simplify $2(A+3C)-3(2C-B)-3\left[2(2A+B-4C)-4(A-2C)\right]$ where A, B, and C are all matrices of the same size.

Solution. The reduction proceeds as though A, B, and C were variables.

$$\begin{aligned} &2(A+3C)-3(2C-B)-3\left[2(2A+B-4C)-4(A-2C)\right] \\ &\quad = 2A+6C-6C+3B-3\left[4A+2B-8C-4A+8C\right] \\ &\quad = 2A+3B-3\left[2B\right] \\ &\quad = 2A-3B \end{aligned}$$

Transpose of a Matrix

Many results about a matrix A involve the *rows* of A, and the corresponding result for columns is derived in an analogous way, essentially by replacing the word *row* by the word *column* throughout. The following definition is made with such applications in mind.

Definition 2.3 Transpose of a Matrix

*If A is an $m\times n$ matrix, the **transpose** of A, written A^T, is the $n\times m$ matrix whose rows are just the columns of A in the same order.*

In other words, the first row of A^T is the first column of A (that is it consists of the entries of column 1 in order). Similarly the second row of A^T is the second column of A, and so on.

Example 2.1.9

Write down the transpose of each of the following matrices.

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 2 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad D = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$$

Solution.

$$A^T = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}, \; B^T = \begin{bmatrix} 5 \\ 2 \\ 6 \end{bmatrix}, \; C^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}, \text{ and } D^T = D.$$

If $A = \begin{bmatrix} a_{ij} \end{bmatrix}$ is a matrix, write $A^T = \begin{bmatrix} b_{ij} \end{bmatrix}$. Then b_{ij} is the jth element of the ith row of A^T and so is the jth element of the ith *column* of A. This means $b_{ij} = a_{ji}$, so the definition of A^T can be stated as follows:

$$\text{If } A = \begin{bmatrix} a_{ij} \end{bmatrix}, \text{ then } A^T = \begin{bmatrix} a_{ji} \end{bmatrix}. \tag{2.1}$$

This is useful in verifying the following properties of transposition.

Theorem 2.1.2

Let A and B denote matrices of the same size, and let k denote a scalar.

1. *If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix.*
2. $(A^T)^T = A$.
3. $(kA)^T = kA^T$.
4. $(A+B)^T = A^T + B^T$.

Proof. Property 1 is part of the definition of A^T, and Property 2 follows from (2.1). As to Property 3: If $A = \begin{bmatrix} a_{ij} \end{bmatrix}$, then $kA = \begin{bmatrix} ka_{ij} \end{bmatrix}$, so (2.1) gives

$$(kA)^T = \begin{bmatrix} ka_{ji} \end{bmatrix} = k\begin{bmatrix} a_{ji} \end{bmatrix} = kA^T$$

Finally, if $B = \begin{bmatrix} b_{ij} \end{bmatrix}$, then $A+B = \begin{bmatrix} c_{ij} \end{bmatrix}$ where $c_{ij} = a_{ij} + b_{ij}$ Then (2.1) gives Property 4:

$$(A+B)^T = \begin{bmatrix} c_{ij} \end{bmatrix}^T = \begin{bmatrix} c_{ji} \end{bmatrix} = \begin{bmatrix} a_{ji} + b_{ji} \end{bmatrix} = \begin{bmatrix} a_{ji} \end{bmatrix} + \begin{bmatrix} b_{ji} \end{bmatrix} = A^T + B^T$$

□

There is another useful way to think of transposition. If $A = \left[a_{ij}\right]$ is an $m \times n$ matrix, the elements $a_{11}, a_{22}, a_{33}, \dots$ are called the **main diagonal** of A. Hence the main diagonal extends down and to the right from the upper left corner of the matrix A; it is outlined in the following examples:

$$\left[\begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{array}\right] \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{array}\right] \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array}\right] \left[\begin{array}{c} a_{11} \\ a_{21} \end{array}\right]$$

Thus forming the transpose of a matrix A can be viewed as "flipping" A about its main diagonal, or as "rotating" A through 180° about the line containing the main diagonal. This makes Property 2 in Theorem 2.1.2 transparent.

Example 2.1.10

Solve for A if $\left(2A^T - 3\left[\begin{array}{rr} 1 & 2 \\ -1 & 1 \end{array}\right]\right)^T = \left[\begin{array}{rr} 2 & 3 \\ -1 & 2 \end{array}\right]$.

Solution. Using Theorem 2.1.2, the left side of the equation is

$$\left(2A^T - 3\left[\begin{array}{rr} 1 & 2 \\ -1 & 1 \end{array}\right]\right)^T = 2\left(A^T\right)^T - 3\left[\begin{array}{rr} 1 & 2 \\ -1 & 1 \end{array}\right]^T = 2A - 3\left[\begin{array}{rr} 1 & -1 \\ 2 & 1 \end{array}\right]$$

Hence the equation becomes

$$2A - 3\left[\begin{array}{rr} 1 & -1 \\ 2 & 1 \end{array}\right] = \left[\begin{array}{rr} 2 & 3 \\ -1 & 2 \end{array}\right]$$

Thus $2A = \left[\begin{array}{rr} 2 & 3 \\ -1 & 2 \end{array}\right] + 3\left[\begin{array}{rr} 1 & -1 \\ 2 & 1 \end{array}\right] = \left[\begin{array}{rr} 5 & 0 \\ 5 & 5 \end{array}\right]$, so finally $A = \frac{1}{2}\left[\begin{array}{rr} 5 & 0 \\ 5 & 5 \end{array}\right] = \frac{5}{2}\left[\begin{array}{rr} 1 & 0 \\ 1 & 1 \end{array}\right]$.

Note that Example 2.1.10 can also be solved by first transposing both sides, then solving for A^T, and so obtaining $A = (A^T)^T$. The reader should do this.

The matrix $D = \left[\begin{array}{rr} 1 & 2 \\ 2 & 5 \end{array}\right]$ in Example 2.1.9 has the property that $D = D^T$. Such matrices are important; a matrix A is called **symmetric** if $A = A^T$. A symmetric matrix A is necessarily square (if A is $m \times n$, then A^T is $n \times m$, so $A = A^T$ forces $n = m$). The name comes from the fact that these matrices exhibit a symmetry about the main diagonal. That is, entries that are directly across the main diagonal from each other are equal.

For example, $\left[\begin{array}{ccc} a & b & c \\ b' & d & e \\ c' & e' & f \end{array}\right]$ is symmetric when $b = b'$, $c = c'$, and $e = e'$.

Example 2.1.11

If A and B are symmetric $n \times n$ matrices, show that $A+B$ is symmetric.

Solution. We have $A^T = A$ and $B^T = B$, so, by Theorem 2.1.2, we have $(A+B)^T = A^T + B^T = A+B$. Hence $A+B$ is symmetric.

Example 2.1.12

Suppose a square matrix A satisfies $A = 2A^T$. Show that necessarily $A = 0$.

Solution. If we iterate the given equation, Theorem 2.1.2 gives

$$A = 2A^T = 2\left[2A^T\right]^T = 2\left[2(A^T)^T\right] = 4A$$

Subtracting A from both sides gives $3A = 0$, so $A = \frac{1}{3}(0) = 0$.

Exercises for 2.1

Exercise 2.1.1 Find a, b, c, and d if

a. $\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = \left[\begin{array}{cc} c-3d & -d \\ 2a+d & a+b \end{array}\right]$

b. $\left[\begin{array}{cc} a-b & b-c \\ c-d & d-a \end{array}\right] = 2\left[\begin{array}{rr} 1 & 1 \\ -3 & 1 \end{array}\right]$

c. $3\left[\begin{array}{c} a \\ b \end{array}\right] + 2\left[\begin{array}{c} b \\ a \end{array}\right] = \left[\begin{array}{c} 1 \\ 2 \end{array}\right]$

d. $\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = \left[\begin{array}{cc} b & c \\ d & a \end{array}\right]$

Exercise 2.1.2 Compute the following:

a. $\left[\begin{array}{rrr} 3 & 2 & 1 \\ 5 & 1 & 0 \end{array}\right] - 5\left[\begin{array}{rrr} 3 & 0 & -2 \\ 1 & -1 & 2 \end{array}\right]$

b. $3\left[\begin{array}{r} 3 \\ -1 \end{array}\right] - 5\left[\begin{array}{r} 6 \\ 2 \end{array}\right] + 7\left[\begin{array}{r} 1 \\ -1 \end{array}\right]$

c. $\left[\begin{array}{rr} -2 & 1 \\ 3 & 2 \end{array}\right] - 4\left[\begin{array}{rr} 1 & -2 \\ 0 & -1 \end{array}\right] + 3\left[\begin{array}{rr} 2 & -3 \\ -1 & -2 \end{array}\right]$

d. $\left[\begin{array}{rrr} 3 & -1 & 2 \end{array}\right] - 2\left[\begin{array}{rrr} 9 & 3 & 4 \end{array}\right] + \left[\begin{array}{rrr} 3 & 11 & -6 \end{array}\right]$

e. $\left[\begin{array}{rrrr} 1 & -5 & 4 & 0 \\ 2 & 1 & 0 & 6 \end{array}\right]^T$

f. $\left[\begin{array}{rrr} 0 & -1 & 2 \\ 1 & 0 & -4 \\ -2 & 4 & 0 \end{array}\right]^T$

g. $\left[\begin{array}{rr} 3 & -1 \\ 2 & 1 \end{array}\right] - 2\left[\begin{array}{rr} 1 & -2 \\ 1 & 1 \end{array}\right]^T$

h. $3\left[\begin{array}{rr} 2 & 1 \\ -1 & 0 \end{array}\right]^T - 2\left[\begin{array}{rr} 1 & -1 \\ 2 & 3 \end{array}\right]$

Exercise 2.1.3 Let $A = \left[\begin{array}{rr} 2 & 1 \\ 0 & -1 \end{array}\right]$,
$B = \left[\begin{array}{rrr} 3 & -1 & 2 \\ 0 & 1 & 4 \end{array}\right]$, $C = \left[\begin{array}{rr} 3 & -1 \\ 2 & 0 \end{array}\right]$,
$D = \left[\begin{array}{rr} 1 & 3 \\ -1 & 0 \\ 1 & 4 \end{array}\right]$, and $E = \left[\begin{array}{rrr} 1 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$.

Compute the following (where possible).

a. $3A-2B$
b. $5C$
c. $3E^T$
d. $B+D$
e. $4A^T-3C$
f. $(A+C)^T$
g. $2B-3E$
h. $A-D$
i. $(B-2E)^T$

Exercise 2.1.4 Find A if:

a. $5A-\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}=3A-\begin{bmatrix} 5 & 2 \\ 6 & 1 \end{bmatrix}$

b. $3A-\begin{bmatrix} 2 \\ 1 \end{bmatrix}=5A-2\begin{bmatrix} 3 \\ 0 \end{bmatrix}$

Exercise 2.1.5 Find A in terms of B if:

a. $A+B=3A+2B$
b. $2A-B=5(A+2B)$

Exercise 2.1.6 If X, Y, A, and B are matrices of the same size, solve the following systems of equations to obtain X and Y in terms of A and B.

a. $\begin{aligned} 5X+3Y&=A \\ 2X+Y&=B \end{aligned}$
b. $\begin{aligned} 4X+3Y&=A \\ 5X+4Y&=B \end{aligned}$

Exercise 2.1.7 Find all matrices X and Y such that:

a. $3X-2Y=\begin{bmatrix} 3 & -1 \end{bmatrix}$
b. $2X-5Y=\begin{bmatrix} 1 & 2 \end{bmatrix}$

Exercise 2.1.8 Simplify the following expressions where A, B, and C are matrices.

a. $2[9(A-B)+7(2B-A)]$
$-2[3(2B+A)-2(A+3B)-5(A+B)]$

b. $5[3(A-B+2C)-2(3C-B)-A]$
$+2[3(3A-B+C)+2(B-2A)-2C]$

Exercise 2.1.9 If A is any 2×2 matrix, show that:

a. $A = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}+b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}+c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}+$
$d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ for some numbers a, b, c, and d.

b. $A = p\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}+q\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}+r\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}+$
$s\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ for some numbers p, q, r, and s.

Exercise 2.1.10 Let $A=\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$, $B=\begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$, and $C=\begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$. If $rA+sB+tC=0$ for some scalars r, s, and t, show that necessarily $r=s=t=0$.

Exercise 2.1.11

a. If $Q+A=A$ holds for every $m\times n$ matrix A, show that $Q=0_{mn}$.

b. If A is an $m\times n$ matrix and $A+A'=0_{mn}$, show that $A'=-A$.

Exercise 2.1.12 If A denotes an $m\times n$ matrix, show that $A=-A$ if and only if $A=0$.

Exercise 2.1.13 A square matrix is called a **diagonal** matrix if all the entries off the main diagonal are zero. If A and B are diagonal matrices, show that the following matrices are also diagonal.

a. $A+B$
b. $A-B$
c. kA for any number k

Exercise 2.1.14 In each case determine all s and t such that the given matrix is symmetric:

a. $\begin{bmatrix} 1 & s \\ -2 & t \end{bmatrix}$
b. $\begin{bmatrix} s & t \\ st & 1 \end{bmatrix}$
c. $\begin{bmatrix} s & 2s & st \\ t & -1 & s \\ t & s^2 & s \end{bmatrix}$
d. $\begin{bmatrix} 2 & s & t \\ 2s & 0 & s+t \\ 3 & 3 & t \end{bmatrix}$

Exercise 2.1.15 In each case find the matrix A.

a. $\left(A+3\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix}\right)^T=\begin{bmatrix} 2 & 1 \\ 0 & 5 \\ 3 & 8 \end{bmatrix}$

b. $\left(3A^T + 2\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right)^T = \begin{bmatrix} 8 & 0 \\ 3 & 1 \end{bmatrix}$

c. $\left(2A - 3\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}\right)^T = 3A^T + \begin{bmatrix} 2 & 1 & -1 \end{bmatrix}^T$

d. $\left(2A^T - 5\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}\right)^T = 4A - 9\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

Exercise 2.1.16 Let A and B be symmetric (of the same size). Show that each of the following is symmetric.

a. $(A - B)$

b. kA for any scalar k

Exercise 2.1.17 Show that $A + A^T$ is symmetric for *any* square matrix A.

Exercise 2.1.18 If A is a square matrix and $A = kA^T$ where $k \neq \pm 1$, show that $A = 0$.

Exercise 2.1.19 In each case either show that the statement is true or give an example showing it is false.

a. If $A + B = A + C$, then B and C have the same size.

b. If $A + B = 0$, then $B = 0$.

c. If the $(3,\ 1)$-entry of A is 5, then the $(1,\ 3)$-entry of A^T is -5.

d. A and A^T have the same main diagonal for every matrix A.

e. If B is symmetric and $A^T = 3B$, then $A = 3B$.

f. If A and B are symmetric, then $kA + mB$ is symmetric for any scalars k and m.

Exercise 2.1.20 A square matrix W is called **skew-symmetric** if $W^T = -W$. Let A be any square matrix.

a. Show that $A - A^T$ is skew-symmetric.

b. Find a symmetric matrix S and a skew-symmetric matrix W such that $A = S + W$.

c. Show that S and W in part (b) are uniquely determined by A.

Exercise 2.1.21 If W is skew-symmetric (Exercise 2.1.20), show that the entries on the main diagonal are zero.

Exercise 2.1.22 Prove the following parts of Theorem 2.1.1.

a. $(k + p)A = kA + pA$

b. $(kp)A = k(pA)$

Exercise 2.1.23 Let A, A_1, A_2, $\dots$, A_n denote matrices of the same size. Use induction on n to verify the following extensions of properties 5 and 6 of Theorem 2.1.1.

a. $k(A_1 + A_2 + \cdots + A_n) = kA_1 + kA_2 + \cdots + kA_n$ for any number k

b. $(k_1 + k_2 + \cdots + k_n)A = k_1A + k_2A + \cdots + k_nA$ for any numbers k_1, k_2, $\dots$, k_n

Exercise 2.1.24 Let A be a square matrix. If $A = pB^T$ and $B = qA^T$ for some matrix B and numbers p and q, show that either $A = 0 = B$ or $pq = 1$.
[*Hint*: Example 2.1.7.]

2.2 Matrix-Vector Multiplication

Up to now we have used matrices to solve systems of linear equations by manipulating the rows of the augmented matrix. In this section we introduce a different way of describing linear systems that makes more use of the coefficient matrix of the system and leads to a useful way of "multiplying" matrices.

Vectors

It is a well-known fact in analytic geometry that two points in the plane with coordinates (a_1, a_2) and (b_1, b_2) are equal if and only if $a_1 = b_1$ and $a_2 = b_2$. Moreover, a similar condition applies to points (a_1, a_2, a_3) in space. We extend this idea as follows.

An ordered sequence $(a_1, a_2, \dots, a_n)$ of real numbers is called an **ordered *n*-tuple**. The word "ordered" here reflects our insistence that two ordered n-tuples are equal if and only if corresponding entries are the same. In other words,

$$(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n) \quad \text{if and only if} \quad a_1 = b_1,\ a_2 = b_2,\ \dots,\ \text{and } a_n = b_n.$$

Thus the ordered 2-tuples and 3-tuples are just the ordered pairs and triples familiar from geometry.

Definition 2.4 The set $\mathbb{R}^n$ of ordered n-tuples of real numbers

Let $\mathbb{R}$ denote the set of all real numbers. The set of all *ordered n-tuples from $\mathbb{R}$ has a special notation:*

$$\mathbb{R}^n \text{ denotes the set of all ordered } n\text{-tuples of real numbers.}$$

There are two commonly used ways to denote the n-tuples in $\mathbb{R}^n$: As rows $(r_1, r_2, \dots, r_n)$ or columns $\left[\begin{array}{c} r_1 \\ r_2 \\ \vdots \\ r_n \end{array} \right]$; the notation we use depends on the context. In any event they are called **vectors** or n-**vectors** and will be denoted using bold type such as $\mathbf{x}$ or $\mathbf{v}$. For example, an $m \times n$ matrix A will be written as a row of columns:

$$A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array} \right] \text{ where } \mathbf{a}_j \text{ denotes column } j \text{ of } A \text{ for each } j.$$

If $\mathbf{x}$ and $\mathbf{y}$ are two n-vectors in $\mathbb{R}^n$, it is clear that their matrix sum $\mathbf{x} + \mathbf{y}$ is also in $\mathbb{R}^n$ as is the scalar multiple $k\mathbf{x}$ for any real number k. We express this observation by saying that $\mathbb{R}^n$ is **closed** under addition and scalar multiplication. In particular, all the basic properties in Theorem 2.1.1 are true of these n-vectors. These properties are fundamental and will be used frequently below without comment. As for matrices in general, the $n \times 1$ zero matrix is called the **zero *n*-vector** in $\mathbb{R}^n$ and, if $\mathbf{x}$ is an n-vector, the n-vector $-\mathbf{x}$ is called the **negative $\mathbf{x}$**.

Of course, we have already encountered these n-vectors in Section 1.3 as the solutions to systems of linear equations with n variables. In particular we defined the notion of a linear combination of vectors and showed that a linear combination of solutions to a homogeneous system is again a solution. Clearly, a linear combination of n-vectors in $\mathbb{R}^n$ is again in $\mathbb{R}^n$, a fact that we will be using.

Matrix-Vector Multiplication

Given a system of linear equations, the left sides of the equations depend only on the coefficient matrix A and the column $\mathbf{x}$ of variables, and not on the constants. This observation leads to a fundamental idea in linear algebra: We view the left sides of the equations as the "product" $A\mathbf{x}$ of the matrix A and the vector $\mathbf{x}$. This simple change of perspective leads to a completely new way of viewing linear systems—one that is very useful and will occupy our attention throughout this book.

To motivate the definition of the "product" $A\mathbf{x}$, consider first the following system of two equations in three variables:

$$\begin{aligned} ax_1 + \ bx_2 + \ cx_3 &= b_1 \\ a'x_1 + b'x_2 + c'x_3 &= b_1 \end{aligned} \tag{2.2}$$

and let $A = \left[\begin{array}{ccc} a & b & c \\ a' & b' & c' \end{array} \right]$, $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right]$, $\mathbf{b} = \left[\begin{array}{c} b_1 \\ b_2 \end{array} \right]$ denote the coefficient matrix, the variable matrix, and the constant matrix, respectively. The system (2.2) can be expressed as a single vector equation

$$\left[\begin{array}{c} ax_1 + \ bx_2 + \ cx_3 \\ a'x_1 + b'x_2 + c'x_3 \end{array} \right] = \left[\begin{array}{c} b_1 \\ b_2 \end{array} \right]$$

which in turn can be written as follows:

$$x_1 \left[\begin{array}{c} a \\ a' \end{array} \right] + x_2 \left[\begin{array}{c} b \\ b' \end{array} \right] + x_3 \left[\begin{array}{c} c \\ c' \end{array} \right] = \left[\begin{array}{c} b_1 \\ b_2 \end{array} \right]$$

Now observe that the vectors appearing on the left side are just the columns

$$\mathbf{a}_1 = \left[\begin{array}{c} a \\ a' \end{array} \right], \ \mathbf{a}_2 = \left[\begin{array}{c} b \\ b' \end{array} \right], \ \text{and } \mathbf{a}_3 = \left[\begin{array}{c} c \\ c' \end{array} \right]$$

of the coefficient matrix A. Hence the system (2.2) takes the form

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b} \tag{2.3}$$

This shows that the system (2.2) has a solution if and only if the constant matrix $\mathbf{b}$ is a linear combination[3] of the columns of A, and that in this case the entries of the solution are the coefficients x_1, x_2, and x_3 in this linear combination.

Moreover, this holds in general. If A is any $m \times n$ matrix, it is often convenient to view A as a row of columns. That is, if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of A, we write

$$A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array} \right]$$

and say that $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array} \right]$ is *given in terms of its columns*.

Now consider any system of linear equations with $m \times n$ coefficient matrix A. If $\mathbf{b}$ is the constant matrix of the system, and if $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right]$ is the matrix of variables then, exactly as above, the system can be written as a single vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b} \tag{2.4}$$

[3]Linear combinations were introduced in Section 1.3 to describe the solutions of homogeneous systems of linear equations. They will be used extensively in what follows.

Example 2.2.1

Write the system $\left\{\begin{array}{rl} 3x_1 + 2x_2 - 4x_3 = & 0 \\ x_1 - 3x_2 + \ x_3 = & 3 \\ x_2 - 5x_3 = & -1 \end{array}\right.$ in the form given in (2.4).

Solution.

$$x_1 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 1 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}$$

As mentioned above, we view the left side of (2.4) as the *product* of the matrix A and the vector $\mathbf{x}$. This basic idea is formalized in the following definition:

Definition 2.5 Matrix-Vector Multiplication

Let $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right]$ be an $m \times n$ matrix, written in terms of its columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is any n-vector, the ***product*** *$A\mathbf{x}$ is defined to be the m-vector given by:*

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

In other words, if A is $m \times n$ and $\mathbf{x}$ is an n-vector, the product $A\mathbf{x}$ is the linear combination of the columns of A where the coefficients are the entries of $\mathbf{x}$ (in order).

Note that if A is an $m \times n$ matrix, the product $A\mathbf{x}$ is only defined if $\mathbf{x}$ is an n-vector and then the vector $A\mathbf{x}$ is an m-vector because this is true of each column $\mathbf{a}_j$ of A. But in this case the *system* of linear equations with coefficient matrix A and constant vector $\mathbf{b}$ takes the form of a *single* matrix equation

$$A\mathbf{x} = \mathbf{b}$$

The following theorem combines Definition 2.5 and equation (2.4) and summarizes the above discussion. Recall that a system of linear equations is said to be *consistent* if it has at least one solution.

Theorem 2.2.1

1. *Every system of linear equations has the form $A\mathbf{x} = \mathbf{b}$ where A is the coefficient matrix, $\mathbf{b}$ is the constant matrix, and $\mathbf{x}$ is the matrix of variables.*

2. *The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is a linear combination of the columns of A.*

3. *If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of A and if $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, then $\mathbf{x}$ is a solution to the linear*

system $A\mathbf{x} = \mathbf{b}$ if and only if $x_1, x_2, \dots, x_n$ are a solution of the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

A system of linear equations in the form $A\mathbf{x} = \mathbf{b}$ as in (1) of Theorem 2.2.1 is said to be written in **matrix form**. This is a useful way to view linear systems as we shall see.

Theorem 2.2.1 transforms the problem of solving the linear system $A\mathbf{x} = \mathbf{b}$ into the problem of expressing the constant matrix B as a linear combination of the columns of the coefficient matrix A. Such a change in perspective is very useful because one approach or the other may be better in a particular situation; the importance of the theorem is that there is a choice.

Example 2.2.2

If $A = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ -2 \end{bmatrix}$, compute $A\mathbf{x}$.

Solution. By Definition 2.5: $A\mathbf{x} = 2\begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} + 1\begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + 0\begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -7 \\ 0 \\ -6 \end{bmatrix}$.

Example 2.2.3

Given columns $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ in $\mathbb{R}^3$, write $2\mathbf{a}_1 - 3\mathbf{a}_2 + 5\mathbf{a}_3 + \mathbf{a}_4$ in the form $A\mathbf{x}$ where A is a matrix and $\mathbf{x}$ is a vector.

Solution. Here the column of coefficients is $\mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 5 \\ 1 \end{bmatrix}$. Hence Definition 2.5 gives

$$A\mathbf{x} = 2\mathbf{a}_1 - 3\mathbf{a}_2 + 5\mathbf{a}_3 + \mathbf{a}_4$$

where $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix}$ is the matrix with $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ as its columns.

Example 2.2.4

Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix}$ be the 3×4 matrix given in terms of its columns $\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$,

$\mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 3 \\ -1 \\ -3 \end{bmatrix}$, and $\mathbf{a}_4 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$. In each case below, either express $\mathbf{b}$ as a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$, or show that it is not such a linear combination. Explain what your answer means for the corresponding system $A\mathbf{x} = \mathbf{b}$ of linear equations.

a. $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

b. $\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

Solution. By Theorem 2.2.1, $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ if and only if the system $A\mathbf{x} = \mathbf{b}$ is consistent (that is, it has a solution). So in each case we carry the augmented matrix $[A|\mathbf{b}]$ of the system $A\mathbf{x} = \mathbf{b}$ to reduced form.

a. Here $\left[\begin{array}{rrrr|r} 2 & 1 & 3 & 3 & 1 \\ 0 & 1 & -1 & 1 & 2 \\ -1 & 1 & -3 & 0 & 3 \end{array}\right] \to \left[\begin{array}{rrrr|r} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$, so the system $A\mathbf{x} = \mathbf{b}$ has no solution in this case. Hence $\mathbf{b}$ is *not* a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$.

b. Now $\left[\begin{array}{rrrr|r} 2 & 1 & 3 & 3 & 4 \\ 0 & 1 & -1 & 1 & 2 \\ -1 & 1 & -3 & 0 & 1 \end{array}\right] \to \left[\begin{array}{rrrr|r} 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$, so the system $A\mathbf{x} = \mathbf{b}$ is consistent.

Thus $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ in this case. In fact the general solution is $x_1 = 1 - 2s - t$, $x_2 = 2 + s - t$, $x_3 = s$, and $x_4 = t$ where s and t are arbitrary parameters. Hence $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 + x_4\mathbf{a}_4 = \mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ for *any* choice of s and t. If we take $s = 0$ and $t = 0$, this becomes $\mathbf{a}_1 + 2\mathbf{a}_2 = \mathbf{b}$, whereas taking $s = 1 = t$ gives $-2\mathbf{a}_1 + 2\mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4 = \mathbf{b}$.

Example 2.2.5

Taking A to be the zero matrix, we have $0\mathbf{x} = \mathbf{0}$ for all vectors $\mathbf{x}$ by Definition 2.5 because every column of the zero matrix is zero. Similarly, $A\mathbf{0} = \mathbf{0}$ for all matrices A because every entry of the zero vector is zero.

Example 2.2.6

If $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $I\mathbf{x} = \mathbf{x}$ for any vector $\mathbf{x}$ in $\mathbb{R}^3$.

Solution. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ then Definition 2.5 gives

$$I\mathbf{x} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{x}$$

The matrix I in Example 2.2.6 is called the 3×3 **identity matrix**, and we will encounter such matrices again in Example 2.2.11 below. Before proceeding, we develop some algebraic properties of matrix-vector multiplication that are used extensively throughout linear algebra.

Theorem 2.2.2

Let A and B be $m \times n$ matrices, and let $\mathbf{x}$ and $\mathbf{y}$ be n-vectors in $\mathbb{R}^n$. Then:

1. $A(\mathbf{x}+\mathbf{y}) = A\mathbf{x} + A\mathbf{y}$.
2. $A(a\mathbf{x}) = a(A\mathbf{x}) = (aA)\mathbf{x}$ *for all scalars* a.
3. $(A+B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Proof. We prove (3); the other verifications are similar and are left as exercises. Let $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array} \right]$ and $B = \left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{array} \right]$ be given in terms of their columns. Since adding two matrices is the same as adding their columns, we have

$$A + B = \left[\begin{array}{cccc} \mathbf{a}_1 + \mathbf{b}_1 & \mathbf{a}_2 + \mathbf{b}_2 & \cdots & \mathbf{a}_n + \mathbf{b}_n \end{array} \right]$$

If we write $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right]$ Definition 2.5 gives

$$\begin{aligned} (A+B)\mathbf{x} &= x_1(\mathbf{a}_1 + \mathbf{b}_1) + x_2(\mathbf{a}_2 + \mathbf{b}_2) + \cdots + x_n(\mathbf{a}_n + \mathbf{b}_n) \\ &= (x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n) + (x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \cdots + x_n\mathbf{b}_n) \\ &= A\mathbf{x} + B\mathbf{x} \end{aligned}$$

□

Theorem 2.2.2 allows matrix-vector computations to be carried out much as in ordinary arithmetic. For example, for any $m \times n$ matrices A and B and any n-vectors $\mathbf{x}$ and $\mathbf{y}$, we have:

$$A(2\mathbf{x} - 5\mathbf{y}) = 2A\mathbf{x} - 5A\mathbf{y} \quad \text{and} \quad (3A - 7B)\mathbf{x} = 3A\mathbf{x} - 7B\mathbf{x}$$

We will use such manipulations throughout the book, often without mention.

Linear Equations

Theorem 2.2.2 also gives a useful way to describe the solutions to a system

$$A\mathbf{x} = \mathbf{b}$$

of linear equations. There is a related system

$$A\mathbf{x} = \mathbf{0}$$

called the **associated homogeneous system**, obtained from the original system $A\mathbf{x} = \mathbf{b}$ by replacing all the constants by zeros. Suppose $\mathbf{x}_1$ is a solution to $A\mathbf{x} = \mathbf{b}$ and $\mathbf{x}_0$ is a solution to $A\mathbf{x} = \mathbf{0}$ (that is $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_0 = \mathbf{0}$). Then $\mathbf{x}_1 + \mathbf{x}_0$ is another solution to $A\mathbf{x} = \mathbf{b}$. Indeed, Theorem 2.2.2 gives

$$A(\mathbf{x}_1 + \mathbf{x}_0) = A\mathbf{x}_1 + A\mathbf{x}_0 = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

This observation has a useful converse.

Theorem 2.2.3

Suppose $\mathbf{x}_1$ is any particular solution to the system $A\mathbf{x} = \mathbf{b}$ of linear equations. Then every solution $\mathbf{x}_2$ to $A\mathbf{x} = \mathbf{b}$ has the form

$$\mathbf{x}_2 = \mathbf{x}_0 + \mathbf{x}_1$$

for some solution $\mathbf{x}_0$ of the associated homogeneous system $A\mathbf{x} = \mathbf{0}$.

Proof. Suppose $\mathbf{x}_2$ is also a solution to $A\mathbf{x} = \mathbf{b}$, so that $A\mathbf{x}_2 = \mathbf{b}$. Write $\mathbf{x}_0 = \mathbf{x}_2 - \mathbf{x}_1$. Then $\mathbf{x}_2 = \mathbf{x}_0 + \mathbf{x}_1$ and, using Theorem 2.2.2, we compute

$$A\mathbf{x}_0 = A(\mathbf{x}_2 - \mathbf{x}_1) = A\mathbf{x}_2 - A\mathbf{x}_1 = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

Hence $\mathbf{x}_0$ is a solution to the associated homogeneous system $A\mathbf{x} = \mathbf{0}$. □

Note that gaussian elimination provides one such representation.

Example 2.2.7

Express every solution to the following system as the sum of a specific solution plus a solution to the associated homogeneous system.

$$\begin{aligned} x_1 - x_2 - x_3 + 3x_4 &= 2 \\ 2x_1 - x_2 - 3x_3 + 4x_4 &= 6 \\ x_1 \phantom{{}- x_2} - 2x_3 + x_4 &= 4 \end{aligned}$$

Solution. Gaussian elimination gives $x_1 = 4 + 2s - t$, $x_2 = 2 + s + 2t$, $x_3 = s$, and $x_4 = t$ where s and t are arbitrary parameters. Hence the general solution can be written

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 + 2s - t \\ 2 + s + 2t \\ s \\ t \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 0 \\ 0 \end{bmatrix} + \left(s \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right)$$

Thus $\mathbf{x}_1 = \begin{bmatrix} 4 \\ 2 \\ 0 \\ 0 \end{bmatrix}$ is a particular solution (where $s = 0 = t$), and $\mathbf{x}_0 = s \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$ gives *all* solutions to the associated homogeneous system. (To see why this is so, carry out the gaussian elimination again but with all the constants set equal to zero.)

The following useful result is included with no proof.

Theorem 2.2.4

Let $A\mathbf{x} = \mathbf{b}$ be a system of equations with augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$. Write $\operatorname{rank} A = r$.

1. $\operatorname{rank} \left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$ *is either r or $r+1$.*
2. *The system is consistent if and only if* $\operatorname{rank} \left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right] = r$.
3. *The system is inconsistent if and only if* $\operatorname{rank} \left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right] = r+1$.

The Dot Product

Definition 2.5 is not always the easiest way to compute a matrix-vector product $A\mathbf{x}$ because it requires that the columns of A be explicitly identified. There is another way to find such a product which uses the matrix A as a whole with no reference to its columns, and hence is useful in practice. The method depends on the following notion.

Definition 2.6 Dot Product in $\mathbb{R}^n$

*If $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$ are two ordered n-tuples, their **dot product** is defined to be the number*

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

obtained by multiplying corresponding entries and adding the results.

To see how this relates to matrix products, let A denote a 3×4 matrix and let $\mathbf{x}$ be a 4-vector. Writing

$$\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array}\right] \quad \text{and} \quad A = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right]$$

in the notation of Section 2.1, we compute

$$\begin{aligned} A\mathbf{x} = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array}\right] &= x_1 \left[\begin{array}{c} a_{11} \\ a_{21} \\ a_{31} \end{array}\right] + x_2 \left[\begin{array}{c} a_{12} \\ a_{22} \\ a_{32} \end{array}\right] + x_3 \left[\begin{array}{c} a_{13} \\ a_{23} \\ a_{33} \end{array}\right] + x_4 \left[\begin{array}{c} a_{14} \\ a_{24} \\ a_{34} \end{array}\right] \\ &= \left[\begin{array}{c} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 \end{array}\right] \end{aligned}$$

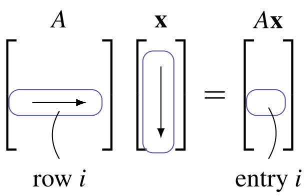
From this we see that each entry of $A\mathbf{x}$ is the dot product of the corresponding row of A with $\mathbf{x}$. This computation goes through in general, and we record the result in Theorem 2.2.5.

Theorem 2.2.5: Dot Product Rule

Let A be an $m \times n$ matrix and let $\mathbf{x}$ be an n-vector. Then each entry of the vector $A\mathbf{x}$ is the dot product of the corresponding row of A with $\mathbf{x}$.

This result is used extensively throughout linear algebra.

If A is $m \times n$ and $\mathbf{x}$ is an n-vector, the computation of $A\mathbf{x}$ by the dot product rule is simpler than using Definition 2.5 because the computation can be carried out directly with no explicit reference to the columns of A (as in Definition 2.5). The first entry of $A\mathbf{x}$ is the dot product of row 1 of A with $\mathbf{x}$. In hand calculations this is computed by going *across* row one of A, going *down* the column $\mathbf{x}$, multiplying corresponding entries, and adding the results. The other entries of $A\mathbf{x}$ are computed in the same way using the other rows of A with the column $\mathbf{x}$.



In general, compute entry i of $A\mathbf{x}$ as follows (see the diagram):

> Go *across* row i of A and *down* column $\mathbf{x}$, multiply corresponding entries, and add the results.

As an illustration, we rework Example 2.2.2 using the dot product rule instead of Definition 2.5.

Example 2.2.8

If $A = \left[\begin{array}{rrrr} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{array} \right]$ and $\mathbf{x} = \left[\begin{array}{r} 2 \\ 1 \\ 0 \\ -2 \end{array} \right]$, compute $A\mathbf{x}$.

Solution. The entries of $A\mathbf{x}$ are the dot products of the rows of A with $\mathbf{x}$:

$$A\mathbf{x} = \left[\begin{array}{rrrr} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{array} \right] \left[\begin{array}{r} 2 \\ 1 \\ 0 \\ -2 \end{array} \right] = \left[\begin{array}{rcrcrcr} 2 \cdot 2 & + & (-1)1 & + & 3 \cdot 0 & + & 5(-2) \\ 0 \cdot 2 & + & 2 \cdot 1 & + & (-3)0 & + & 1(-2) \\ (-3)2 & + & 4 \cdot 1 & + & 1 \cdot 0 & + & 2(-2) \end{array} \right] = \left[\begin{array}{r} -7 \\ 0 \\ -6 \end{array} \right]$$

Of course, this agrees with the outcome in Example 2.2.2.

Example 2.2.9

Write the following system of linear equations in the form $A\mathbf{x} = \mathbf{b}$.

$$\begin{array}{rcr} 5x_1 - x_2 + 2x_3 + x_4 - 3x_5 & = & 8 \\ x_1 + x_2 + 3x_3 - 5x_4 + 2x_5 & = & -2 \\ -x_1 + x_2 - 2x_3 + - 3x_5 & = & 0 \end{array}$$

Solution. Write $A = \begin{bmatrix} 5 & -1 & 2 & 1 & -3 \\ 1 & 1 & 3 & -5 & 2 \\ -1 & 1 & -2 & 0 & -3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$. Then the dot product rule gives $A\mathbf{x} = \begin{bmatrix} 5x_1 - x_2 + 2x_3 + x_4 - 3x_5 \\ x_1 + x_2 + 3x_3 - 5x_4 + 2x_5 \\ -x_1 + x_2 - 2x_3 \qquad - 3x_5 \end{bmatrix}$, so the entries of $A\mathbf{x}$ are the left sides of the equations in the linear system. Hence the system becomes $A\mathbf{x} = \mathbf{b}$ because matrices are equal if and only corresponding entries are equal.

Example 2.2.10

If A is the zero $m \times n$ matrix, then $A\mathbf{x} = \mathbf{0}$ for each n-vector $\mathbf{x}$.

Solution. For each k, entry k of $A\mathbf{x}$ is the dot product of row k of A with $\mathbf{x}$, and this is zero because row k of A consists of zeros.

Definition 2.7 The Identity Matrix

*For each $n > 2$, the **identity matrix** I_n is the $n \times n$ matrix with 1s on the main diagonal (upper left to lower right), and zeros elsewhere.*

The first few identity matrices are

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \dots$$

In Example 2.2.6 we showed that $I_3\mathbf{x} = \mathbf{x}$ for each 3-vector $\mathbf{x}$ using Definition 2.5. The following result shows that this holds in general, and is the reason for the name.

Example 2.2.11

For each $n \geq 2$ we have $I_n\mathbf{x} = \mathbf{x}$ for each n-vector $\mathbf{x}$ in $\mathbb{R}^n$.

Solution. We verify the case $n = 4$. Given the 4-vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ the dot product rule gives

$$I_4\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + 0 + 0 + 0 \\ 0 + x_2 + 0 + 0 \\ 0 + 0 + x_3 + 0 \\ 0 + 0 + 0 + x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \mathbf{x}$$

In general, $I_n\mathbf{x} = \mathbf{x}$ because entry k of $I_n\mathbf{x}$ is the dot product of row k of I_n with $\mathbf{x}$, and row k of I_n has 1 in position k and zeros elsewhere.

Example 2.2.12

Let $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right]$ be any $m \times n$ matrix with columns $\mathbf{a}_1,\ \mathbf{a}_2,\ \dots,\ \mathbf{a}_n$. If $\mathbf{e}_j$ denotes column j of the $n \times n$ identity matrix I_n, then $A\mathbf{e}_j = \mathbf{a}_j$ for each $j = 1,\ 2,\ \dots,\ n$.

Solution. Write $\mathbf{e}_j = \left[\begin{array}{c} t_1 \\ t_2 \\ \vdots \\ t_n \end{array}\right]$ where $t_j = 1$, but $t_i = 0$ for all $i \neq j$. Then Theorem 2.2.5 gives

$$A\mathbf{e}_j = t_1\mathbf{a}_1 + \cdots + t_j\mathbf{a}_j + \cdots + t_n\mathbf{a}_n = 0 + \cdots + \mathbf{a}_j + \cdots + 0 = \mathbf{a}_j$$

Example 2.2.12 will be referred to later; for now we use it to prove:

Theorem 2.2.6

Let A and B be $m \times n$ matrices. If $A\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$, then $A = B$.

Proof. Write $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right]$ and $B = \left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{array}\right]$ and in terms of their columns. It is enough to show that $\mathbf{a}_k = \mathbf{b}_k$ holds for all k. But we are assuming that $A\mathbf{e}_k = B\mathbf{e}_k$, which gives $\mathbf{a}_k = \mathbf{b}_k$ by Example 2.2.12. □

We have introduced matrix-vector multiplication as a new way to think about systems of linear equations. But it has several other uses as well. It turns out that many geometric operations can be described using matrix multiplication, and we now investigate how this happens. As a bonus, this description provides a geometric "picture" of a matrix by revealing the effect on a vector when it is multiplied by A. This "geometric view" of matrices is a fundamental tool in understanding them.

Transformations

The set $\mathbb{R}^2$ has a geometrical interpretation as the euclidean plane where a vector $\left[\begin{array}{c} a_1 \\ a_2 \end{array}\right]$ in $\mathbb{R}^2$ represents the point $(a_1,\ a_2)$ in the plane (see Figure 2.2.1). In this way we regard $\mathbb{R}^2$ as the set of all points in the plane. Accordingly, we will refer to vectors in $\mathbb{R}^2$ as points, and denote their coordinates as a column rather than a row. To enhance this geometrical interpretation of the vector $\left[\begin{array}{c} a_1 \\ a_2 \end{array}\right]$, it is denoted graphically by an arrow from the origin $\left[\begin{array}{c} 0 \\ 0 \end{array}\right]$ to the vector as in Figure 2.2.1.

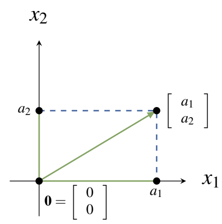


Figure 2.2.1

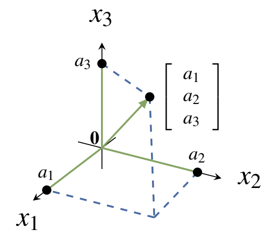


Figure 2.2.2

Similarly we identify $\mathbb{R}^3$ with 3-dimensional space by writing a point (a_1, a_2, a_3) as the vector $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ in $\mathbb{R}^3$, again represented by an arrow[4] from the origin to the point as in Figure 2.2.2. In this way the terms "point" and "vector" mean the same thing in the plane or in space.

We begin by describing a particular geometrical transformation of the plane $\mathbb{R}^2$.

Example 2.2.13

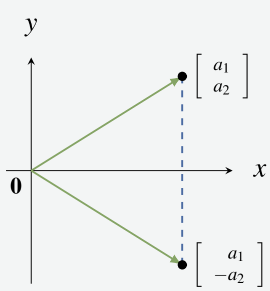


Figure 2.2.3

Consider the transformation of $\mathbb{R}^2$ given by *reflection* in the x axis. This operation carries the vector $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ to its reflection $\begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}$ as in Figure 2.2.3. Now observe that

$$\begin{bmatrix} a_1 \\ -a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

so reflecting $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ in the x axis can be achieved by multiplying by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

If we write $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, Example 2.2.13 shows that reflection in the x axis carries each vector $\mathbf{x}$ in $\mathbb{R}^2$ to the vector $A\mathbf{x}$ in $\mathbb{R}^2$. It is thus an example of a function

$$T : \mathbb{R}^2 \to \mathbb{R}^2 \quad \text{where} \quad T(\mathbf{x}) = A\mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^2$$

As such it is a generalization of the familiar functions $f : \mathbb{R} \to \mathbb{R}$ that carry a *number* x to another real *number* $f(x)$.

[4]This "arrow" representation of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ will be used extensively in Chapter 4.

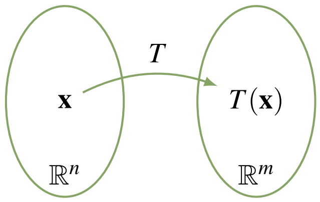


Figure 2.2.4

More generally, functions $T : \mathbb{R}^n \to \mathbb{R}^m$ are called **transformations** from $\mathbb{R}^n$ to $\mathbb{R}^m$. Such a transformation T is a rule that assigns to every vector $\mathbf{x}$ in $\mathbb{R}^n$ a uniquely determined vector $T(\mathbf{x})$ in $\mathbb{R}^m$ called the **image** of $\mathbf{x}$ under T. We denote this state of affairs by writing

$$T : \mathbb{R}^n \to \mathbb{R}^m \quad \text{or} \quad \mathbb{R}^n \xrightarrow{T} \mathbb{R}^m$$

The transformation T can be visualized as in Figure 2.2.4.

To describe a transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ we must specify the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ for every $\mathbf{x}$ in $\mathbb{R}^n$. This is referred to as **defining** T, or as specifying the **action** of T. Saying that the action *defines* the transformation means that we regard two transformations $S : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^n \to \mathbb{R}^m$ as **equal** if they have the **same action**; more formally

$$S = T \quad \text{if and only if} \quad S(\mathbf{x}) = T(\mathbf{x}) \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n.$$

Again, this what we mean by $f = g$ where $f,\ g : \mathbb{R} \to \mathbb{R}$ are ordinary functions.

Functions $f : \mathbb{R} \to \mathbb{R}$ are often described by a formula, examples being $f(x) = x^2 + 1$ and $f(x) = \sin x$. The same is true of transformations; here is an example.

Example 2.2.14

The formula $T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \\ x_3 + x_4 \end{bmatrix}$ defines a transformation $\mathbb{R}^4 \to \mathbb{R}^3$.

Example 2.2.13 suggests that matrix multiplication is an important way of defining transformations $\mathbb{R}^n \to \mathbb{R}^m$. If A is any $m \times n$ matrix, multiplication by A gives a transformation

$$T_A : \mathbb{R}^n \to \mathbb{R}^m \quad \text{defined by} \quad T_A(\mathbf{x}) = A\mathbf{x} \text{ for every } \mathbf{x} \text{ in } \mathbb{R}^n$$

Definition 2.8 Matrix Transformation T_A

T_A is called the ***matrix transformation induced*** *by A.*

Thus Example 2.2.13 shows that reflection in the x axis is the matrix transformation $\mathbb{R}^2 \to \mathbb{R}^2$ induced by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Also, the transformation $R : \mathbb{R}^4 \to \mathbb{R}^3$ in Example 2.2.13 is the matrix transformation induced by the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \text{because} \quad \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \\ x_3 + x_4 \end{bmatrix}$$

Example 2.2.15

Let $R_{\frac{\pi}{2}} : \mathbb{R}^2 \to \mathbb{R}^2$ denote counterclockwise rotation about the origin through $\frac{\pi}{2}$ radians (that is, $90°$)[5]. Show that $R_{\frac{\pi}{2}}$ is induced by the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Solution.

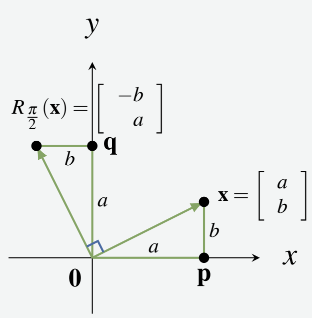


Figure 2.2.5

The effect of $R_{\frac{\pi}{2}}$ is to rotate the vector $\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix}$ counterclockwise through $\frac{\pi}{2}$ to produce the vector $R_{\frac{\pi}{2}}(\mathbf{x})$ shown in Figure 2.2.5. Since triangles $\mathbf{0px}$ and $\mathbf{0q}R_{\frac{\pi}{2}}(\mathbf{x})$ are identical, we obtain $R_{\frac{\pi}{2}}(\mathbf{x}) = \begin{bmatrix} -b \\ a \end{bmatrix}$. But $\begin{bmatrix} -b \\ a \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$, so we obtain $R_{\frac{\pi}{2}}(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^2$ where $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. In other words, $R_{\frac{\pi}{2}}$ is the matrix transformation induced by A.

If A is the $m \times n$ zero matrix, then A induces the transformation

$$T : \mathbb{R}^n \to \mathbb{R}^m \quad \text{given by} \quad T(\mathbf{x}) = A\mathbf{x} = \mathbf{0} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

This is called the **zero transformation**, and is denoted $T = 0$.

Another important example is the **identity transformation**

$$1_{\mathbb{R}^n} : \mathbb{R}^n \to \mathbb{R}^n \quad \text{given by} \quad 1_{\mathbb{R}^n}(\mathbf{x}) = \mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

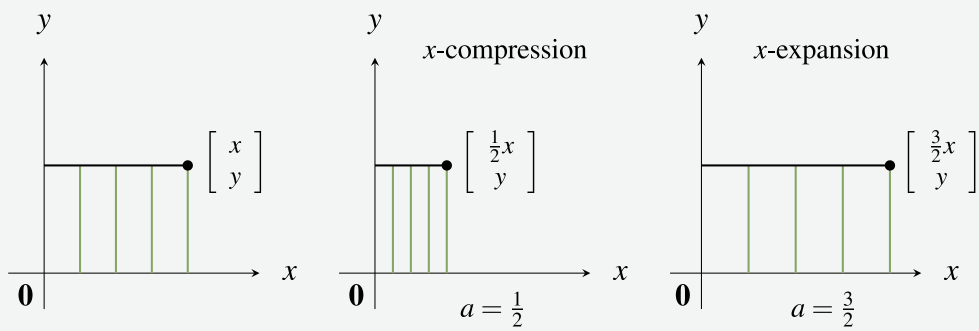
That is, the action of $1_{\mathbb{R}^n}$ on $\mathbf{x}$ is to do nothing to it. If I_n denotes the $n \times n$ identity matrix, we showed in Example 2.2.11 that $I_n\mathbf{x} = \mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Hence $1_{\mathbb{R}^n}(\mathbf{x}) = I_n\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$; that is, the identity matrix I_n induces the identity transformation.

Here are two more examples of matrix transformations with a clear geometric description.

Example 2.2.16

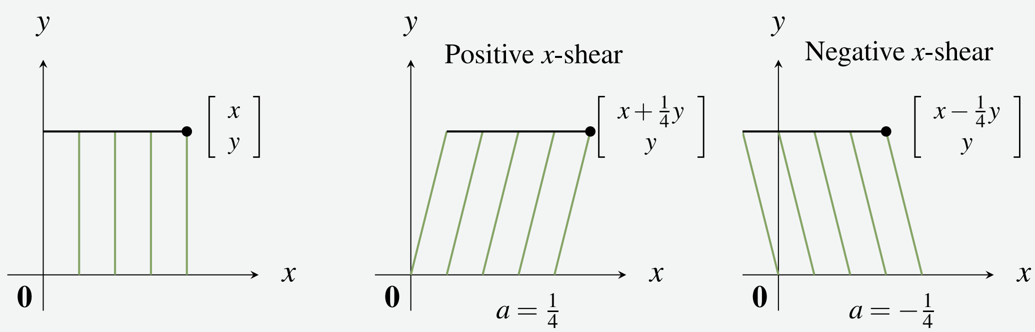
If $a > 0$, the matrix transformation $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ y \end{bmatrix}$ induced by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$ is called an ***x*-expansion** of $\mathbb{R}^2$ if $a > 1$, and an ***x*-compression** if $0 < a < 1$. The reason for the names is clear in the diagram below. Similarly, if $b > 0$ the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix}$ gives rise to ***y*-expansions** and ***y*-compressions**.

[5] Radian measure for angles is based on the fact that $360°$ equals 2π radians. Hence π radians $= 180°$ and $\frac{\pi}{2}$ radians $= 90°$.



Example 2.2.17

If a is a number, the matrix transformation $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+ay \\ y \end{bmatrix}$ induced by the matrix $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ is called an ***x*-shear** of $\mathbb{R}^2$ (**positive** if $a > 0$ and **negative** if $a < 0$). Its effect is illustrated below when $a = \frac{1}{4}$ and $a = -\frac{1}{4}$.



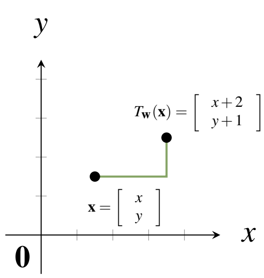


Figure 2.2.6

We hasten to note that there are important geometric transformations that are *not* matrix transformations. For example, if $\mathbf{w}$ is a fixed column in $\mathbb{R}^n$, define the transformation $T_{\mathbf{w}} : \mathbb{R}^n \to \mathbb{R}^n$ by

$$T_{\mathbf{w}}(\mathbf{x}) = \mathbf{x} + \mathbf{w} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

Then $T_{\mathbf{w}}$ is called **translation** by $\mathbf{w}$. In particular, if $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ in $\mathbb{R}^2$, the effect of $T_{\mathbf{w}}$ on $\begin{bmatrix} x \\ y \end{bmatrix}$ is to translate it two units to the right and one unit up (see Figure 2.2.6).

The translation $T_{\mathbf{w}}$ is not a matrix transformation unless $\mathbf{w} = \mathbf{0}$. Indeed, if $T_{\mathbf{w}}$ were induced by a matrix A, then $A\mathbf{x} = T_{\mathbf{w}}(\mathbf{x}) = \mathbf{x} + \mathbf{w}$ would hold for every $\mathbf{x}$ in $\mathbb{R}^n$. In particular, taking $\mathbf{x} = \mathbf{0}$ gives $\mathbf{w} = A\mathbf{0} = \mathbf{0}$.

Exercises for 2.2

Exercise 2.2.1 In each case find a system of equations that is equivalent to the given vector equation. (Do not solve the system.)

a. $x_1 \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -3 \end{bmatrix}$

b. $x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 8 \\ 2 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 2 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 2 \\ 0 \end{bmatrix}$

Exercise 2.2.2 In each case find a vector equation that is equivalent to the given system of equations. (Do not solve the equation.)

a. $$\begin{aligned} x_1 - x_2 + 3x_3 &= 5 \\ -3x_1 + x_2 + x_3 &= -6 \\ 5x_1 - 8x_2 &= 9 \end{aligned}$$

b. $$\begin{aligned} x_1 - 2x_2 - x_3 + x_4 &= 5 \\ -x_1 + x_3 - 2x_4 &= -3 \\ 2x_1 - 2x_2 + 7x_3 &= 8 \\ 3x_1 - 4x_2 + 9x_3 - 2x_4 &= 12 \end{aligned}$$

Exercise 2.2.3 In each case compute $A\mathbf{x}$ using: (i) Definition 2.5. (ii) Theorem 2.2.5.

a. $A = \begin{bmatrix} 3 & -2 & 0 \\ 5 & -4 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

b. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

c. $A = \begin{bmatrix} -2 & 0 & 5 & 4 \\ 1 & 2 & 0 & 3 \\ -5 & 6 & -7 & 8 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$.

d. $A = \begin{bmatrix} 3 & -4 & 1 & 6 \\ 0 & 2 & 1 & 5 \\ -8 & 7 & -3 & 0 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$.

Exercise 2.2.4 Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix}$ be the 3×4 matrix given in terms of its columns $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, and $\mathbf{a}_4 = \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix}$. In each case either express $\mathbf{b}$ as a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$, or show that it is not such a linear combination. Explain what your answer means for the corresponding system $A\mathbf{x} = \mathbf{b}$ of linear equations.

a. $\mathbf{b} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$

b. $\mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$

Exercise 2.2.5 In each case, express every solution of the system as a sum of a specific solution plus a solution of the associated homogeneous system.

a. $$\begin{aligned} x + y + z &= 2 \\ 2x + y &= 3 \\ x - y - 3z &= 0 \end{aligned}$$

b. $$\begin{aligned} x - y - 4z &= -4 \\ x + 2y + 5z &= 2 \\ x + y + 2z &= 0 \end{aligned}$$

c. $$\begin{aligned} x_1 + x_2 - x_3 - 5x_5 &= 2 \\ x_2 + x_3 - 4x_5 &= -1 \\ x_2 + x_3 + x_4 - x_5 &= -1 \\ 2x_1 - 4x_3 + x_4 + x_5 &= 6 \end{aligned}$$

d. $$\begin{aligned} 2x_1 + x_2 - x_3 - x_4 &= -1 \\ 3x_1 + x_2 + x_3 - 2x_4 &= -2 \\ -x_1 - x_2 + 2x_3 + x_4 &= 2 \\ -2x_1 - x_2 + 2x_4 &= 3 \end{aligned}$$

Exercise 2.2.6 If $\mathbf{x}_0$ and $\mathbf{x}_1$ are solutions to the homogeneous system of equations $A\mathbf{x} = \mathbf{0}$, use Theorem 2.2.2 to show that $s\mathbf{x}_0 + t\mathbf{x}_1$ is also a solution for any scalars s and t (called a **linear combination** of $\mathbf{x}_0$ and $\mathbf{x}_1$).

Exercise 2.2.7 Assume that $A \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \mathbf{0} = A \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$. Show that $\mathbf{x}_0 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ is a solution to $A\mathbf{x} = \mathbf{b}$. Find a two-parameter family of solutions to $A\mathbf{x} = \mathbf{b}$.

Exercise 2.2.8 In each case write the system in the form $A\mathbf{x} = \mathbf{b}$, use the gaussian algorithm to solve the system, and express the solution as a particular solution plus a linear combination of basic solutions to the associated homogeneous system $A\mathbf{x} = \mathbf{0}$.

a. $$\begin{aligned} x_1 - 2x_2 + x_3 + 4x_4 - x_5 &= 8 \\ -2x_1 + 4x_2 + x_3 - 2x_4 - 4x_5 &= -1 \\ 3x_1 - 6x_2 + 8x_3 + 4x_4 - 13x_5 &= 1 \\ 8x_1 - 16x_2 + 7x_3 + 12x_4 - 6x_5 &= 11 \end{aligned}$$

b. $$\begin{aligned} x_1 - 2x_2 + x_3 + 2x_4 + 3x_5 &= -4 \\ -3x_1 + 6x_2 - 2x_3 - 3x_4 - 11x_5 &= 11 \\ -2x_1 + 4x_2 - x_3 + x_4 - 8x_5 &= 7 \\ -x_1 + 2x_2 \qquad + 3x_4 - 5x_5 &= 3 \end{aligned}$$

Exercise 2.2.9 Given vectors $\mathbf{a}_1 = \left[\begin{array}{r} 1 \\ 0 \\ 1 \end{array}\right]$, $\mathbf{a}_2 = \left[\begin{array}{r} 1 \\ 1 \\ 0 \end{array}\right]$, and $\mathbf{a}_3 = \left[\begin{array}{r} 0 \\ -1 \\ 1 \end{array}\right]$, find a vector $\mathbf{b}$ that is *not* a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$. Justify your answer. [*Hint*: Part (2) of Theorem 2.2.1.]

Exercise 2.2.10 In each case either show that the statement is true, or give an example showing that it is false.

a. $\left[\begin{array}{r} 3 \\ 2 \end{array}\right]$ is a linear combination of $\left[\begin{array}{r} 1 \\ 0 \end{array}\right]$ and $\left[\begin{array}{r} 0 \\ 1 \end{array}\right]$.

b. If $A\mathbf{x}$ has a zero entry, then A has a row of zeros.

c. If $A\mathbf{x} = \mathbf{0}$ where $\mathbf{x} \neq \mathbf{0}$, then $A = 0$.

d. Every linear combination of vectors in $\mathbb{R}^n$ can be written in the form $A\mathbf{x}$.

e. If $A = \left[\begin{array}{ccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{array}\right]$ in terms of its columns, and if $\mathbf{b} = 3\mathbf{a}_1 - 2\mathbf{a}_2$, then the system $A\mathbf{x} = \mathbf{b}$ has a solution.

f. If $A = \left[\begin{array}{ccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{array}\right]$ in terms of its columns, and if the system $A\mathbf{x} = \mathbf{b}$ has a solution, then $\mathbf{b} = s\mathbf{a}_1 + t\mathbf{a}_2$ for some s, t.

g. If A is $m \times n$ and $m < n$, then $A\mathbf{x} = \mathbf{b}$ has a solution for every column $\mathbf{b}$.

h. If $A\mathbf{x} = \mathbf{b}$ has a solution for some column $\mathbf{b}$, then it has a solution for every column $\mathbf{b}$.

i. If $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions to $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.

j. Let $A = \left[\begin{array}{ccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{array}\right]$ in terms of its columns. If $\mathbf{a}_3 = s\mathbf{a}_1 + t\mathbf{a}_2$, then $A\mathbf{x} = \mathbf{0}$, where $\mathbf{x} = \left[\begin{array}{r} s \\ t \\ -1 \end{array}\right]$.

Exercise 2.2.11 Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a transformation. In each case show that T is induced by a matrix and find the matrix.

a. T is a reflection in the y axis.

b. T is a reflection in the line $y = x$.

c. T is a reflection in the line $y = -x$.

d. T is a clockwise rotation through $\frac{\pi}{2}$.

Exercise 2.2.12 The **projection** $P : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by $P\left[\begin{array}{c} x \\ y \\ z \end{array}\right] = \left[\begin{array}{c} x \\ y \end{array}\right]$ for all $\left[\begin{array}{c} x \\ y \\ z \end{array}\right]$ in $\mathbb{R}^3$. Show that P is induced by a matrix and find the matrix.

Exercise 2.2.13 Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a transformation. In each case show that T is induced by a matrix and find the matrix.

a. T is a reflection in the $x - y$ plane.

b. T is a reflection in the $y - z$ plane.

Exercise 2.2.14 Fix $a > 0$ in $\mathbb{R}$, and define $T_a : \mathbb{R}^4 \to \mathbb{R}^4$ by $T_a(\mathbf{x}) = a\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^4$. Show that T is induced by a matrix and find the matrix. [T is called a **dilation** if $a > 1$ and a **contraction** if $a < 1$.]

Exercise 2.2.15 Let A be $m \times n$ and let $\mathbf{x}$ be in $\mathbb{R}^n$. If A has a row of zeros, show that $A\mathbf{x}$ has a zero entry.

Exercise 2.2.16 If a vector $\mathbf{b}$ is a linear combination of the columns of A, show that the system $A\mathbf{x} = \mathbf{b}$ is consistent (that is, it has at least one solution.)

Exercise 2.2.17 If a system $A\mathbf{x} = \mathbf{b}$ is inconsistent (no solution), show that $\mathbf{b}$ is not a linear combination of the columns of A.

Exercise 2.2.18 Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$.

a. Show that $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.

b. Show that $t\mathbf{x}_1$ is a solution to $A\mathbf{x} = \mathbf{0}$ for any scalar t.

Exercise 2.2.19 Suppose $\mathbf{x}_1$ is a solution to the system $A\mathbf{x} = \mathbf{b}$. If $\mathbf{x}_0$ is any nontrivial solution to the associated homogeneous system $A\mathbf{x} = \mathbf{0}$, show that $\mathbf{x}_1 + t\mathbf{x}_0$, t a scalar, is an infinite one parameter family of solutions to $A\mathbf{x} = \mathbf{b}$. [*Hint*: Example 2.1.7 Section 2.1.]

Exercise 2.2.20 Let A and B be matrices of the same size. If $\mathbf{x}$ is a solution to both the system $A\mathbf{x} = \mathbf{0}$ and the system $B\mathbf{x} = \mathbf{0}$, show that $\mathbf{x}$ is a solution to the system $(A+B)\mathbf{x} = \mathbf{0}$.

Exercise 2.2.21 If A is $m \times n$ and $A\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$ in $\mathbb{R}^n$, show that $A = 0$ is the zero matrix. [*Hint*: Consider $A\mathbf{e}_j$ where $\mathbf{e}_j$ is the jth column of I_n; that is, $\mathbf{e}_j$ is the vector in $\mathbb{R}^n$ with 1 as entry j and every other entry 0.]

Exercise 2.2.22 Prove part (1) of Theorem 2.2.2.

Exercise 2.2.23 Prove part (2) of Theorem 2.2.2.

2.3 Matrix Multiplication

In Section 2.2 matrix-vector products were introduced. If A is an $m \times n$ matrix, the product $A\mathbf{x}$ was defined for any n-column $\mathbf{x}$ in $\mathbb{R}^n$ as follows: If $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ where the $\mathbf{a}_j$ are the columns of A, and if $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, Definition 2.5 reads

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n \tag{2.5}$$

This was motivated as a way of describing systems of linear equations with coefficient matrix A. Indeed every such system has the form $A\mathbf{x} = \mathbf{b}$ where $\mathbf{b}$ is the column of constants.

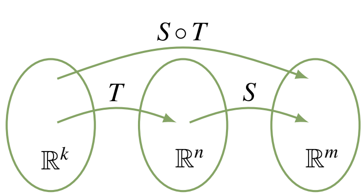
In this section we extend this matrix-vector multiplication to a way of multiplying matrices in general, and then investigate matrix algebra for its own sake. While it shares several properties of ordinary arithmetic, it will soon become clear that matrix arithmetic is different in a number of ways.

Matrix multiplication is closely related to composition of transformations.

Composition and Matrix Multiplication

Sometimes two transformations "link" together as follows:

$$\mathbb{R}^k \xrightarrow{T} \mathbb{R}^n \xrightarrow{S} \mathbb{R}^m$$



In this case we can apply T first and then apply S, and the result is a new transformation

$$S \circ T : \mathbb{R}^k \to \mathbb{R}^m$$

called the **composite** of S and T, defined by

$$(S \circ T)(\mathbf{x}) = S[T(\mathbf{x})] \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^k$$

The action of $S \circ T$ can be described as "first T then S" (note the order!)[6]. This new transformation is described in the diagram. The reader will have encountered composition of ordinary functions: For example, consider $\mathbb{R} \xrightarrow{g} \mathbb{R} \xrightarrow{f} \mathbb{R}$ where $f(x) = x^2$ and $g(x) = x+1$ for all x in $\mathbb{R}$. Then

$$(f \circ g)(x) = f[g(x)] = f(x+1) = (x+1)^2$$

[6]When reading the notation $S \circ T$, we read S first and then T even though the action is "first T then S". This annoying state of affairs results because we write $T(\mathbf{x})$ for the effect of the transformation T on $\mathbf{x}$, with T on the left. If we wrote this instead as $(\mathbf{x})T$, the confusion would not occur. However the notation $T(\mathbf{x})$ is well established.

$$(g \circ f)(x) = g[f(x)] = g(x^2) = x^2 + 1$$

for all x in $\mathbb{R}$.

Our concern here is with matrix transformations. Suppose that A is an $m \times n$ matrix and B is an $n \times k$ matrix, and let $\mathbb{R}^k \xrightarrow{T_B} \mathbb{R}^n \xrightarrow{T_A} \mathbb{R}^m$ be the matrix transformations induced by B and A respectively, that is:

$$T_B(\mathbf{x}) = B\mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^k \quad \text{and} \quad T_A(\mathbf{y}) = A\mathbf{y} \text{ for all } \mathbf{y} \text{ in } \mathbb{R}^n$$

Write $B = \left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_k \end{array} \right]$ where $\mathbf{b}_j$ denotes column j of B for each j. Hence each $\mathbf{b}_j$ is an n-vector (B is $n \times k$) so we can form the matrix-vector product $A\mathbf{b}_j$. In particular, we obtain an $m \times k$ matrix

$$\left[\begin{array}{cccc} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_k \end{array} \right]$$

with columns $A\mathbf{b}_1, A\mathbf{b}_2, \cdots, A\mathbf{b}_k$. Now compute $(T_A \circ T_B)(\mathbf{x})$ for any $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_k \end{array} \right]$ in $\mathbb{R}^k$:

$$\begin{array}{rcll}
(T_A \circ T_B)(\mathbf{x}) & = & T_A[T_B(\mathbf{x})] & \text{Definition of } T_A \circ T_B \\
& = & A(B\mathbf{x}) & A \text{ and } B \text{ induce } T_A \text{ and } T_B \\
& = & A(x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \cdots + x_k\mathbf{b}_k) & \text{Equation 2.5 above} \\
& = & A(x_1\mathbf{b}_1) + A(x_2\mathbf{b}_2) + \cdots + A(x_k\mathbf{b}_k) & \text{Theorem 2.2.2} \\
& = & x_1(A\mathbf{b}_1) + x_2(A\mathbf{b}_2) + \cdots + x_k(A\mathbf{b}_k) & \text{Theorem 2.2.2} \\
& = & \left[\begin{array}{cccc} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_k \end{array} \right] \mathbf{x} & \text{Equation 2.5 above}
\end{array}$$

Because $\mathbf{x}$ was an arbitrary vector in $\mathbb{R}^n$, this shows that $T_A \circ T_B$ is the matrix transformation induced by the matrix $\left[\begin{array}{cccc} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \end{array} \right]$. This motivates the following definition.

Definition 2.9 Matrix Multiplication

Let A be an $m \times n$ matrix, let B be an $n \times k$ matrix, and write $B = \left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_k \end{array} \right]$ where $\mathbf{b}_j$ is column j of B for each j. The product matrix AB is the $m \times k$ matrix defined as follows:

$$AB = A\left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_k \end{array} \right] = \left[\begin{array}{cccc} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_k \end{array} \right]$$

Thus the product matrix AB is given in terms of its columns $A\mathbf{b}_1, A\mathbf{b}_2, \ldots, A\mathbf{b}_n$: Column j of AB is the matrix-vector product $A\mathbf{b}_j$ of A and the corresponding column $\mathbf{b}_j$ of B. Note that each such product $A\mathbf{b}_j$ makes sense by Definition 2.5 because A is $m \times n$ and each $\mathbf{b}_j$ is in $\mathbb{R}^n$ (since B has n rows). Note also that if B is a column matrix, this definition reduces to Definition 2.5 for matrix-vector multiplication.

Given matrices A and B, Definition 2.9 and the above computation give

$$A(B\mathbf{x}) = \left[\begin{array}{cccc} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \end{array} \right] \mathbf{x} = (AB)\mathbf{x}$$

for all $\mathbf{x}$ in $\mathbb{R}^k$. We record this for reference.

Theorem 2.3.1

Let A be an $m \times n$ matrix and let B be an $n \times k$ matrix. Then the product matrix AB is $m \times k$ and satisfies

$$A(B\mathbf{x}) = (AB)\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^k$$

Here is an example of how to compute the product AB of two matrices using Definition 2.9.

Example 2.3.1

Compute AB if $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$.

Solution. The columns of B are $\mathbf{b}_1 = \begin{bmatrix} 8 \\ 7 \\ 6 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 9 \\ 2 \\ 1 \end{bmatrix}$, so Definition 2.5 gives

$$A\mathbf{b}_1 = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 8 \\ 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 67 \\ 78 \\ 55 \end{bmatrix} \text{ and } A\mathbf{b}_2 = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 9 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 29 \\ 24 \\ 10 \end{bmatrix}$$

Hence Definition 2.9 above gives $AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 \end{bmatrix} = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$.

Example 2.3.2

If A is $m \times n$ and B is $n \times k$, Theorem 2.3.1 gives a simple formula for the composite of the matrix transformations T_A and T_B:

$$T_A \circ T_B = T_{AB}$$

Solution. Given any $\mathbf{x}$ in $\mathbb{R}^k$,

$$\begin{aligned} (T_A \circ T_B)(\mathbf{x}) &= T_A[T_B(\mathbf{x})] \\ &= A[B\mathbf{x}] \\ &= (AB)\mathbf{x} \\ &= T_{AB}(\mathbf{x}) \end{aligned}$$

While Definition 2.9 is important, there is another way to compute the matrix product AB that gives a way to calculate each individual entry. In Section 2.2 we defined the dot product of two n-tuples to be the sum of the products of corresponding entries. We went on to show (Theorem 2.2.5) that if A is an $m \times n$ matrix and $\mathbf{x}$ is an n-vector, then entry j of the product $A\mathbf{x}$ is the dot product of row j of A with $\mathbf{x}$. This observation was called the "dot product rule" for matrix-vector multiplication, and the next theorem shows that it extends to matrix multiplication in general.

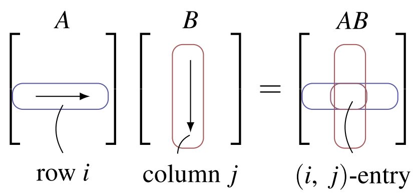
Theorem 2.3.2: Dot Product Rule

Let A and B be matrices of sizes $m \times n$ and $n \times k$, respectively. Then the $(i,\ j)$-entry of AB is the dot product of row i of A with column j of B.

Proof. Write $B = \left[\begin{array}{cccc} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{array} \right]$ in terms of its columns. Then $A\mathbf{b}_j$ is column j of AB for each j. Hence the $(i,\ j)$-entry of AB is entry i of $A\mathbf{b}_j$, which is the dot product of row i of A with $\mathbf{b}_j$. This proves the theorem. □

Thus to compute the $(i,\ j)$-entry of AB, proceed as follows (see the diagram):

Go *across* row i of A, and *down* column j of B, multiply corresponding entries, and add the results.



Note that this requires that the rows of A must be the same length as the columns of B. The following rule is useful for remembering this and for deciding the size of the product matrix AB.

Compatibility Rule

$$\begin{array}{cc} A & B \\ m \times n & n' \times k \end{array}$$

Let A and B denote matrices. If A is $m \times n$ and B is $n' \times k$, the product AB can be formed if and only if $n = n'$. In this case the size of the product matrix AB is $m \times k$, and we say that AB is **defined**, or that A and B are **compatible** for multiplication.

The diagram provides a useful mnemonic for remembering this. We adopt the following convention:

Convention

Whenever a product of matrices is written, it is tacitly assumed that the sizes of the factors are such that the product is defined.

To illustrate the dot product rule, we recompute the matrix product in Example 2.3.1.

Example 2.3.3

Compute AB if $A = \left[\begin{array}{ccc} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{array} \right]$ and $B = \left[\begin{array}{cc} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{array} \right]$.

Solution. Here A is 3×3 and B is 3×2, so the product matrix AB is defined and will be of size 3×2. Theorem 2.3.2 gives each entry of AB as the dot product of the corresponding row of A with the corresponding column of B_j that is,

$$AB = \left[\begin{array}{ccc} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{array} \right] \left[\begin{array}{cc} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{array} \right] = \left[\begin{array}{cc} 2 \cdot 8 + 3 \cdot 7 + 5 \cdot 6 & 2 \cdot 9 + 3 \cdot 2 + 5 \cdot 1 \\ 1 \cdot 8 + 4 \cdot 7 + 7 \cdot 6 & 1 \cdot 9 + 4 \cdot 2 + 7 \cdot 1 \\ 0 \cdot 8 + 1 \cdot 7 + 8 \cdot 6 & 0 \cdot 9 + 1 \cdot 2 + 8 \cdot 1 \end{array} \right] = \left[\begin{array}{cc} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{array} \right]$$

Of course, this agrees with Example 2.3.1.

Example 2.3.4

Compute the $(1,\ 3)$- and $(2,\ 4)$-entries of AB where

$$A = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 & 6 & 0 \\ 0 & 2 & 3 & 4 \\ -1 & 0 & 5 & 8 \end{bmatrix}.$$

Then compute AB.

Solution. The $(1,\ 3)$-entry of AB is the dot product of row 1 of A and column 3 of B (highlighted in the following display), computed by multiplying corresponding entries and adding the results.

$$\begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 & 6 & 0 \\ 0 & 2 & 3 & 4 \\ -1 & 0 & 5 & 8 \end{bmatrix} \quad (1,\ 3)\text{-entry} = 3 \cdot 6 + (-1) \cdot 3 + 2 \cdot 5 = 25$$

Similarly, the $(2,\ 4)$-entry of AB involves row 2 of A and column 4 of B.

$$\begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 & 6 & 0 \\ 0 & 2 & 3 & 4 \\ -1 & 0 & 5 & 8 \end{bmatrix} \quad (2,\ 4)\text{-entry} = 0 \cdot 0 + 1 \cdot 4 + 4 \cdot 8 = 36$$

Since A is 2×3 and B is 3×4, the product is 2×4.

$$AB = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 & 6 & 0 \\ 0 & 2 & 3 & 4 \\ -1 & 0 & 5 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 1 & 25 & 12 \\ -4 & 2 & 23 & 36 \end{bmatrix}$$

Example 2.3.5

If $A = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 6 \\ 4 \end{bmatrix}$, compute A^2, AB, BA, and B^2 when they are defined.[7]

Solution. Here, A is a 1×3 matrix and B is a 3×1 matrix, so A^2 and B^2 are not defined. However, the compatibility rule reads

$$\begin{array}{cc} A & B \\ 1 \times 3 & 3 \times 1 \end{array} \quad \text{and} \quad \begin{array}{cc} B & A \\ 3 \times 1 & 1 \times 3 \end{array}$$

so both AB and BA can be formed and these are 1×1 and 3×3 matrices, respectively.

[7] As for numbers, we write $A^2 = A \cdot A$, $A^3 = A \cdot A \cdot A$, etc. Note that A^2 is defined if and only if A is of size $n \times n$ for some n.

$$AB = \left[\begin{array}{ccc} 1 & 3 & 2 \end{array}\right] \left[\begin{array}{c} 5 \\ 6 \\ 4 \end{array}\right] = \left[\begin{array}{c} 1\cdot 5 + 3\cdot 6 + 2\cdot 4 \end{array}\right] = \left[31\right]$$

$$BA = \left[\begin{array}{c} 5 \\ 6 \\ 4 \end{array}\right] \left[\begin{array}{ccc} 1 & 3 & 2 \end{array}\right] = \left[\begin{array}{ccc} 5\cdot 1 & 5\cdot 3 & 5\cdot 2 \\ 6\cdot 1 & 6\cdot 3 & 6\cdot 2 \\ 4\cdot 1 & 4\cdot 3 & 4\cdot 2 \end{array}\right] = \left[\begin{array}{rrr} 5 & 15 & 10 \\ 6 & 18 & 12 \\ 4 & 12 & 8 \end{array}\right]$$

Unlike numerical multiplication, matrix products AB and BA *need not be equal*. In fact they need not even be the same size, as Example 2.3.5 shows. It turns out to be rare that $AB = BA$ (although it is by no means impossible), and A and B are said to **commute** when this happens.

Example 2.3.6

Let $A = \left[\begin{array}{rr} 6 & 9 \\ -4 & -6 \end{array}\right]$ and $B = \left[\begin{array}{rr} 1 & 2 \\ -1 & 0 \end{array}\right]$. Compute A^2, AB, BA.

Solution. $A^2 = \left[\begin{array}{rr} 6 & 9 \\ -4 & -6 \end{array}\right] \left[\begin{array}{rr} 6 & 9 \\ -4 & -6 \end{array}\right] = \left[\begin{array}{rr} 0 & 0 \\ 0 & 0 \end{array}\right]$, so $A^2 = 0$ can occur even if $A \neq 0$. Next,

$$AB = \left[\begin{array}{rr} 6 & 9 \\ -4 & -6 \end{array}\right] \left[\begin{array}{rr} 1 & 2 \\ -1 & 0 \end{array}\right] = \left[\begin{array}{rr} -3 & 12 \\ 2 & -8 \end{array}\right]$$
$$BA = \left[\begin{array}{rr} 1 & 2 \\ -1 & 0 \end{array}\right] \left[\begin{array}{rr} 6 & 9 \\ -4 & -6 \end{array}\right] = \left[\begin{array}{rr} -2 & -3 \\ -6 & -9 \end{array}\right]$$

Hence $AB \neq BA$, even though AB and BA are the same size.

Example 2.3.7

If A is any matrix, then $IA = A$ and $AI = A$, and where I denotes an identity matrix of a size so that the multiplications are defined.

Solution. These both follow from the dot product rule as the reader should verify. For a more formal proof, write $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right]$ where $\mathbf{a}_j$ is column j of A. Then Definition 2.9 and Example 2.2.11 give

$$IA = \left[\begin{array}{cccc} I\mathbf{a}_1 & I\mathbf{a}_2 & \cdots & I\mathbf{a}_n \end{array}\right] = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right] = A$$

If $\mathbf{e}_j$ denotes column j of I, then $A\mathbf{e}_j = \mathbf{a}_j$ for each j by Example 2.2.12. Hence Definition 2.9 gives:

$$AI = A\left[\begin{array}{cccc} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{array}\right] = \left[\begin{array}{cccc} A\mathbf{e}_1 & A\mathbf{e}_2 & \cdots & A\mathbf{e}_n \end{array}\right] = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array}\right] = A$$

The following theorem collects several results about matrix multiplication that are used everywhere in linear algebra.

Theorem 2.3.3

Assume that a is any scalar, and that A, B, and C are matrices of sizes such that the indicated matrix products are defined. Then:

1. $IA = A$ and $AI = A$ where I denotes an identity matrix.
2. $A(BC) = (AB)C$.
3. $A(B+C) = AB + AC$.
4. $(B+C)A = BA + CA$.
5. $a(AB) = (aA)B = A(aB)$.
6. $(AB)^T = B^T A^T$.

Proof. Condition (1) is Example 2.3.7; we prove (2), (4), and (6) and leave (3) and (5) as exercises.

2. If $C = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_k \end{bmatrix}$ in terms of its columns, then $BC = \begin{bmatrix} B\mathbf{c}_1 & B\mathbf{c}_2 & \cdots & B\mathbf{c}_k \end{bmatrix}$ by Definition 2.9, so

$$\begin{aligned} A(BC) &= \begin{bmatrix} A(B\mathbf{c}_1) & A(B\mathbf{c}_2) & \cdots & A(B\mathbf{c}_k) \end{bmatrix} && \text{Definition 2.9} \\ &= \begin{bmatrix} (AB)\mathbf{c}_1 & (AB)\mathbf{c}_2 & \cdots & (AB)\mathbf{c}_k \end{bmatrix} && \text{Theorem 2.3.1} \\ &= (AB)C && \text{Definition 2.9} \end{aligned}$$

4. We know (Theorem 2.2.2) that $(B+C)\mathbf{x} = B\mathbf{x} + C\mathbf{x}$ holds for every column $\mathbf{x}$. If we write $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ in terms of its columns, we get

$$\begin{aligned} (B+C)A &= \begin{bmatrix} (B+C)\mathbf{a}_1 & (B+C)\mathbf{a}_2 & \cdots & (B+C)\mathbf{a}_n \end{bmatrix} && \text{Definition 2.9} \\ &= \begin{bmatrix} B\mathbf{a}_1 + C\mathbf{a}_1 & B\mathbf{a}_2 + C\mathbf{a}_2 & \cdots & B\mathbf{a}_n + C\mathbf{a}_n \end{bmatrix} && \text{Theorem 2.2.2} \\ &= \begin{bmatrix} B\mathbf{a}_1 & B\mathbf{a}_2 & \cdots & B\mathbf{a}_n \end{bmatrix} + \begin{bmatrix} C\mathbf{a}_1 & C\mathbf{a}_2 & \cdots & C\mathbf{a}_n \end{bmatrix} && \text{Adding Columns} \\ &= BA + CA && \text{Definition 2.9} \end{aligned}$$

6. As in Section 2.1, write $A = [a_{ij}]$ and $B = [b_{ij}]$, so that $A^T = [a'_{ij}]$ and $B^T = [b'_{ij}]$ where $a'_{ij} = a_{ji}$ and $b'_{ji} = b_{ij}$ for all i and j. If c_{ij} denotes the $(i,\ j)$-entry of $B^T A^T$, then c_{ij} is the dot product of row i of B^T with column j of A^T. Hence

$$\begin{aligned} c_{ij} = b'_{i1}a'_{1j} + b'_{i2}a'_{2j} + \cdots + b'_{im}a'_{mj} &= b_{1i}a_{j1} + b_{2i}a_{j2} + \cdots + b_{mi}a_{jm} \\ &= a_{j1}b_{1i} + a_{j2}b_{2i} + \cdots + a_{jm}b_{mi} \end{aligned}$$

But this is the dot product of row j of A with column i of B; that is, the $(j,\ i)$-entry of AB; that is, the $(i,\ j)$-entry of $(AB)^T$. This proves (6). □

Property 2 in Theorem 2.3.3 is called the **associative law** of matrix multiplication. It asserts that the equation $A(BC) = (AB)C$ holds for all matrices (if the products are defined). Hence this product is the same no matter how it is formed, and so is written simply as ABC. This extends: The product $ABCD$ of four matrices can be formed several ways—for example, $(AB)(CD)$, $[A(BC)]D$, and $A[B(CD)]$—but the associative law implies that they are all equal and so are written as $ABCD$. A similar remark applies in general: Matrix products can be written unambiguously with no parentheses.

However, a note of caution about matrix multiplication must be taken: The fact that AB and BA need *not* be equal means that the *order* of the factors is important in a product of matrices. For example $ABCD$ and $ADCB$ may *not* be equal.

Warning

> *If the order of the factors in a product of matrices is changed, the product matrix may change (or may not be defined). Ignoring this warning is a source of many errors by students of linear algebra!*

Properties 3 and 4 in Theorem 2.3.3 are called **distributive laws**. They assert that $A(B+C) = AB+AC$ and $(B+C)A = BA+CA$ hold whenever the sums and products are defined. These rules extend to more than two terms and, together with Property 5, ensure that many manipulations familiar from ordinary algebra extend to matrices. For example

$$\begin{aligned} A(2B-3C+D-5E) &= 2AB-3AC+AD-5AE \\ (A+3C-2D)B &= AB+3CB-2DB \end{aligned}$$

Note again that the warning is in effect: For example $A(B-C)$ need *not* equal $AB-CA$. These rules make possible a lot of simplification of matrix expressions.

Example 2.3.8

Simplify the expression $A(BC-CD)+A(C-B)D-AB(C-D)$.

Solution.

$$\begin{aligned} A(BC-CD)+A(C-B)D-AB(C-D) &= A(BC)-A(CD)+(AC-AB)D-(AB)C+(AB)D \\ &= ABC-ACD+ACD-ABD-ABC+ABD \\ &= 0 \end{aligned}$$

Example 2.3.9 and Example 2.3.10 below show how we can use the properties in Theorem 2.3.2 to deduce other facts about matrix multiplication. Matrices A and B are said to **commute** if $AB = BA$.

Example 2.3.9

Suppose that A, B, and C are $n \times n$ matrices and that both A and B commute with C; that is, $AC = CA$ and $BC = CB$. Show that AB commutes with C.

Solution. Showing that AB commutes with C means verifying that $(AB)C = C(AB)$. The computation uses the associative law several times, as well as the given facts that $AC = CA$ and $BC = CB$.

$$(AB)C = A(BC) = A(CB) = (AC)B = (CA)B = C(AB)$$

Example 2.3.10

Show that $AB = BA$ if and only if $(A-B)(A+B) = A^2 - B^2$.

Solution. The following *always* holds:

$$(A-B)(A+B) = A(A+B) - B(A+B) = A^2 + AB - BA - B^2 \tag{2.6}$$

Hence if $AB = BA$, then $(A-B)(A+B) = A^2 - B^2$ follows. Conversely, if this last equation holds, then equation (2.6) becomes

$$A^2 - B^2 = A^2 + AB - BA - B^2$$

This gives $0 = AB - BA$, and $AB = BA$ follows.

In Section 2.2 we saw (in Theorem 2.2.1) that every system of linear equations has the form

$$A\mathbf{x} = \mathbf{b}$$

where A is the coefficient matrix, $\mathbf{x}$ is the column of variables, and $\mathbf{b}$ is the constant matrix. Thus the *system* of linear equations becomes a single matrix equation. Matrix multiplication can yield information about such a system.

Example 2.3.11

Consider a system $A\mathbf{x} = \mathbf{b}$ of linear equations where A is an $m \times n$ matrix. Assume that a matrix C exists such that $CA = I_n$. If the system $A\mathbf{x} = \mathbf{b}$ *has* a solution, show that this solution must be $C\mathbf{b}$. Give a condition guaranteeing that $C\mathbf{b}$ *is in fact* a solution.

Solution. Suppose that $\mathbf{x}$ is any solution to the system, so that $A\mathbf{x} = \mathbf{b}$. Multiply both sides of this matrix equation by C to obtain, successively,

$$C(A\mathbf{x}) = C\mathbf{b}, \quad (CA)\mathbf{x} = C\mathbf{b}, \quad I_n\mathbf{x} = C\mathbf{b}, \quad \mathbf{x} = C\mathbf{b}$$

This shows that *if* the system has a solution $\mathbf{x}$, then that solution must be $\mathbf{x} = C\mathbf{b}$, as required. But it does *not* guarantee that the system *has* a solution. However, if we write $\mathbf{x}_1 = C\mathbf{b}$, then

$$A\mathbf{x}_1 = A(C\mathbf{b}) = (AC)\mathbf{b}$$

Thus $\mathbf{x}_1 = C\mathbf{b}$ will be a solution if the condition $AC = I_m$ is satisfied.

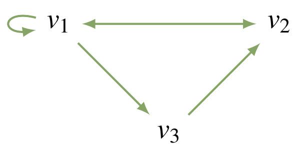
The ideas in Example 2.3.11 lead to important information about matrices; this will be pursued in the next section.

Directed Graphs

The study of directed graphs illustrates how matrix multiplication arises in ways other than the study of linear equations or matrix transformations.

A **directed graph** consists of a set of points (called **vertices**) connected by arrows (called **edges**). For example, the vertices could represent cities and the edges available flights. If the graph has n vertices v_1, $v_2, \dots, v_n$, the **adjacency** matrix $A = \left[a_{ij}\right]$ is the $n \times n$ matrix whose (i, j)-entry a_{ij} is 1 if there is an

edge from v_j to v_i (note the order), and zero otherwise. For example, the adjacency matrix of the directed graph shown is $A = \left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{array}\right]$.



A **path of length** r (or an r-**path**) from vertex j to vertex i is a sequence of r edges leading from v_j to v_i. Thus $v_1 \to v_2 \to v_1 \to v_1 \to v_3$ is a 4-path from v_1 to v_3 in the given graph. The edges are just the paths of length 1, so the $(i,\ j)$-entry a_{ij} of the adjacency matrix A is the number of 1-paths from v_j to v_i. This observation has an important extension:

Theorem 2.3.6

If A is the adjacency matrix of a directed graph with n vertices, then the $(i,\ j)$-entry of A^r is the number of r-paths $v_j \to v_i$.

As an illustration, consider the adjacency matrix A in the graph shown. Then

$$A = \left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right], \quad A^2 = \left[\begin{array}{ccc} 2 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right], \quad \text{and} \quad A^3 = \left[\begin{array}{ccc} 4 & 2 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{array} \right]$$

Hence, since the $(2,\ 1)$-entry of A^2 is 2, there are two 2-paths $v_1 \to v_2$ (in fact they are $v_1 \to v_1 \to v_2$ and $v_1 \to v_3 \to v_2$). Similarly, the $(2,\ 3)$-entry of A^2 is zero, so there are *no* 2-paths $v_3 \to v_2$, as the reader can verify. The fact that no entry of A^3 is zero shows that it is possible to go from any vertex to any other vertex in exactly three steps.

To see why Theorem 2.3.6 is true, observe that it asserts that

$$\text{the } (i,\ j)\text{-entry of } A^r \text{ equals the number of } r\text{-paths } v_j \to v_i \tag{2.7}$$

holds for each $r \geq 1$. We proceed by induction on r (see Appendix C). The case $r = 1$ is the definition of the adjacency matrix. So assume inductively that (2.7) is true for some $r \geq 1$; we must prove that (2.7) also holds for $r+1$. But every $(r+1)$-path $v_j \to v_i$ is the result of an r-path $v_j \to v_k$ for some k, followed by a 1-path $v_k \to v_i$. Writing $A = \left[a_{ij} \right]$ and $A^r = \left[b_{ij} \right]$, there are b_{kj} paths of the former type (by induction) and a_{ik} of the latter type, and so there are $a_{ik}b_{kj}$ such paths in all. Summing over k, this shows that there are

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \quad (r+1)\text{-paths } v_j \to v_i$$

But this sum is the dot product of the ith row $\left[\begin{array}{cccc} a_{i1} & a_{i2} & \cdots & a_{in} \end{array} \right]$ of A with the jth column $\left[\begin{array}{cccc} b_{1j} & b_{2j} & \cdots & b_{nj} \end{array} \right]^T$ of A^r. As such, it is the $(i,\ j)$-entry of the matrix product $A^rA = A^{r+1}$. This shows that (2.7) holds for $r+1$, as required.

Exercises for 2.3

Exercise 2.3.1 Compute the following matrix products.

a. $\left[\begin{array}{rr} 1 & 3 \\ 0 & -2 \end{array} \right] \left[\begin{array}{rr} 2 & -1 \\ 0 & 1 \end{array} \right]$

b. $\left[\begin{array}{rrr} 1 & -1 & 2 \\ 2 & 0 & 4 \end{array} \right] \left[\begin{array}{rrr} 2 & 3 & 1 \\ 1 & 9 & 7 \\ -1 & 0 & 2 \end{array} \right]$

c. $\left[\begin{array}{rrr} 5 & 0 & -7 \\ 1 & 5 & 9 \end{array} \right] \left[\begin{array}{r} 3 \\ 1 \\ -1 \end{array} \right]$

d. $\left[\begin{array}{rrr} 1 & 3 & -3 \end{array} \right] \left[\begin{array}{rr} 3 & 0 \\ -2 & 1 \\ 0 & 6 \end{array} \right]$

e. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 5 & -7 \\ 9 & 7 \end{bmatrix}$

f. $\begin{bmatrix} 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -8 \end{bmatrix}$

g. $\begin{bmatrix} 2 \\ 1 \\ -7 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \end{bmatrix}$

h. $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$

i. $\begin{bmatrix} 2 & 3 & 1 \\ 5 & 7 & 4 \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

j. $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} a' & 0 & 0 \\ 0 & b' & 0 \\ 0 & 0 & c' \end{bmatrix}$

Exercise 2.3.2 In each of the following cases, find all possible products A^2, AB, AC, and so on.

a. $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ \frac{1}{2} & 3 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 0 \\ 2 & 5 \\ 0 & 3 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 6 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 1 & 2 \end{bmatrix}$

Exercise 2.3.3 Find a, b, a_1, and b_1 if:

a. $\begin{bmatrix} a & b \\ a_1 & b_1 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} a & b \\ a_1 & b_1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ -1 & 4 \end{bmatrix}$

Exercise 2.3.4 Verify that $A^2 - A - 6I = 0$ if:

a. $\begin{bmatrix} 3 & -1 \\ 0 & -2 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$

Exercise 2.3.5

Given $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 5 & 8 \end{bmatrix}$, and $D = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 5 \end{bmatrix}$, verify the following facts from Theorem 2.3.1.

a. $A(B-D) = AB - AD$

b. $A(BC) = (AB)C$

c. $(CD)^T = D^T C^T$

Exercise 2.3.6 Let A be a 2×2 matrix.

a. If A commutes with $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, show that $A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ for some a and b.

b. If A commutes with $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, show that $A = \begin{bmatrix} a & 0 \\ c & a \end{bmatrix}$ for some a and c.

c. Show that A commutes with *every* 2×2 matrix if and only if $A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ for some a.

Exercise 2.3.7

a. If A^2 can be formed, what can be said about the size of A?

b. If AB and BA can both be formed, describe the sizes of A and B.

c. If ABC can be formed, A is 3×3, and C is 5×5, what size is B?

Exercise 2.3.8

a. Find two 2×2 matrices A such that $A^2 = 0$.

b. Find three 2×2 matrices A such that (i) $A^2 = I$; (ii) $A^2 = A$.

c. Find 2×2 matrices A and B such that $AB = 0$ but $BA \neq 0$.

Exercise 2.3.9 Write $P = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right]$, and let A be $3 \times n$ and B be $m \times 3$.

a. Describe PA in terms of the rows of A.

b. Describe BP in terms of the columns of B.

Exercise 2.3.10 Let A, B, and C be as in Exercise 2.3.5. Find the (3, 1)-entry of CAB using exactly six numerical multiplications.

Exercise 2.3.14 Let A denote an $m \times n$ matrix.

a. If $AX = 0$ for every $n \times 1$ matrix X, show that $A = 0$.

b. If $YA = 0$ for every $1 \times m$ matrix Y, show that $A = 0$.

Exercise 2.3.15

a. If $U = \left[\begin{array}{cc} 1 & 2 \\ 0 & -1 \end{array} \right]$, and $AU = 0$, show that $A = 0$.

b. Let U be such that $AU = 0$ implies that $A = 0$. If $PU = QU$, show that $P = Q$.

Exercise 2.3.16 Simplify the following expressions where A, B, and C represent matrices.

a. $A(3B - C) + (A - 2B)C + 2B(C + 2A)$

b. $A(B + C - D) + B(C - A + D) - (A + B)C$
$+ (A - B)D$

c. $AB(BC - CB) + (CA - AB)BC + CA(A - B)C$

d. $(A - B)(C - A) + (C - B)(A - C) + (C - A)^2$

Exercise 2.3.17 If $A = \left[\begin{array}{cc} a & b \\ c & d \end{array} \right]$ where $a \neq 0$, show that A factors in the form $A = \left[\begin{array}{cc} 1 & 0 \\ x & 1 \end{array} \right] \left[\begin{array}{cc} y & z \\ 0 & w \end{array} \right]$.

Exercise 2.3.18 If A and B commute with C, show that the same is true of:

a. $A + B$

b. kA, k any scalar

Exercise 2.3.19 If A is any matrix, show that both AA^T and A^TA are symmetric.

Exercise 2.3.20 If A and B are symmetric, show that AB is symmetric if and only if $AB = BA$.

Exercise 2.3.21 If A is a 2×2 matrix, show that $A^TA = AA^T$ if and only if A is symmetric or $A = \left[\begin{array}{cc} a & b \\ -b & a \end{array} \right]$ for some a and b.

Exercise 2.3.22

a. Find all symmetric 2×2 matrices A such that $A^2 = 0$.

b. Repeat (a) if A is 3×3.

c. Repeat (a) if A is $n \times n$.

Exercise 2.3.23 Show that there exist no 2×2 matrices A and B such that $AB - BA = I$. [*Hint*: Examine the (1, 1)- and (2, 2)-entries.]

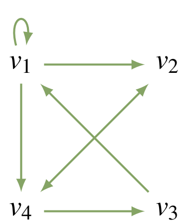
Exercise 2.3.24 Let B be an $n \times n$ matrix. Suppose $AB = 0$ for some nonzero $m \times n$ matrix A. Show that no $n \times n$ matrix C exists such that $BC = I$.

Exercise 2.3.25 An autoparts manufacturer makes fenders, doors, and hoods. Each requires assembly and packaging carried out at factories: Plant 1, Plant 2, and Plant 3. Matrix A below gives the number of hours for assembly and packaging, and matrix B gives the hourly rates at the three plants. Explain the meaning of the (3, 2)-entry in the matrix AB. Which plant is the most economical to operate? Give reasons.

$$\begin{array}{l} \\ \text{Fenders} \\ \text{Doors} \\ \text{Hoods} \end{array} \begin{array}{c} \begin{array}{cc} \text{Assembly} & \text{Packaging} \end{array} \\ \left[\begin{array}{cc} 12 & 2 \\ 21 & 3 \\ 10 & 2 \end{array}\right] \end{array} = A$$

$$\begin{array}{l} \\ \text{Assembly} \\ \text{Packaging} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Plant 1} & \text{Plant 2} & \text{Plant 3} \end{array} \\ \left[\begin{array}{ccc} 21 & 18 & 20 \\ 14 & 10 & 13 \end{array}\right] \end{array} = B$$

Exercise 2.3.26 For the directed graph below, find the adjacency matrix A, compute A^3, and determine the number of paths of length 3 from v_1 to v_4 and from v_2 to v_3.



Exercise 2.3.27 In each case either show the statement is true, or give an example showing that it is false.

a. If $A^2 = I$, then $A = I$.

b. If $AJ = A$, then $J = I$.

c. If A is square, then $(A^T)^3 = (A^3)^T$.

d. If A is symmetric, then $I + A$ is symmetric.

e. If $AB = AC$ and $A \neq 0$, then $B = C$.

f. If $A \neq 0$, then $A^2 \neq 0$.

g. If A has a row of zeros, so also does BA for all B.

h. If A commutes with $A + B$, then A commutes with B.

i. If B has a column of zeros, so also does AB.

j. If AB has a column of zeros, so also does B.

k. If A has a row of zeros, so also does AB.

l. If AB has a row of zeros, so also does A.

Exercise 2.3.28

a. If A and B are 2×2 matrices whose rows sum to 1, show that the rows of AB also sum to 1.

b. Repeat part (a) for the case where A and B are $n \times n$.

Exercise 2.3.29 Let A and B be $n \times n$ matrices for which the systems of equations $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ each have only the trivial solution $\mathbf{x} = \mathbf{0}$. Show that the system $(AB)\mathbf{x} = \mathbf{0}$ has only the trivial solution.

Exercise 2.3.30 The **trace** of a square matrix A, denoted $\text{tr}\, A$, is the sum of the elements on the main diagonal of A. Show that, if A and B are $n \times n$ matrices:

a. $\text{tr}\,(A + B) = \text{tr}\, A + \text{tr}\, B$.

b. $\text{tr}\,(kA) = k\, \text{tr}\,(A)$ for any number k.

c. $\text{tr}\,(A^T) = \text{tr}\,(A)$.

d. $\text{tr}\,(AB) = \text{tr}\,(BA)$.

e. $\text{tr}\,(AA^T)$ is the sum of the squares of all entries of A.

Exercise 2.3.31 Show that $AB - BA = I$ is impossible. [*Hint*: See the preceding exercise.]

Exercise 2.3.32 A square matrix P is called an **idempotent** if $P^2 = P$. Show that:

a. 0 and I are idempotents.

b. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, and $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, are idempotents.

c. If P is an idempotent, so is $I - P$. Show further that $P(I - P) = 0$.

d. If P is an idempotent, so is P^T.

e. If P is an idempotent, so is $Q = P + AP - PAP$ for any square matrix A (of the same size as P).

f. If A is $n \times m$ and B is $m \times n$, and if $AB = I_n$, then BA is an idempotent.

Exercise 2.3.33 Let A and B be $n \times n$ **diagonal matrices** (all entries off the main diagonal are zero).

a. Show that AB is diagonal and $AB = BA$.

b. Formulate a rule for calculating XA if X is $m \times n$.

c. Formulate a rule for calculating AY if Y is $n \times k$.

Exercise 2.3.34 If A and B are $n \times n$ matrices, show that:

a. $AB = BA$ if and only if

$$(A+B)^2 = A^2 + 2AB + B^2$$

b. $AB = BA$ if and only if

$$(A+B)(A-B) = (A-B)(A+B)$$

Exercise 2.3.35 In Theorem 2.3.3, prove

a. part 3;

b. part 5.

2.4 Matrix Inverses

Three basic operations on matrices, addition, multiplication, and subtraction, are analogs for matrices of the same operations for numbers. In this section we introduce the matrix analog of numerical division.

To begin, consider how a numerical equation $ax = b$ is solved when a and b are known numbers. If $a = 0$, there is no solution (unless $b = 0$). But if $a \neq 0$, we can multiply both sides by the inverse $a^{-1} = \frac{1}{a}$ to obtain the solution $x = a^{-1}b$. Of course multiplying by a^{-1} is just dividing by a, and the property of a^{-1} that makes this work is that $a^{-1}a = 1$. Moreover, we saw in Section 2.2 that the role that 1 plays in arithmetic is played in matrix algebra by the identity matrix I. This suggests the following definition.

Definition 2.11 Matrix Inverses

*If A is a square matrix, a matrix B is called an **inverse** of A if and only if*

$$AB = I \quad \text{and} \quad BA = I$$

*A matrix A that has an inverse is called an **invertible matrix**.*[8]

Example 2.4.1

Show that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ is an inverse of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

[8]Only square matrices have inverses. Even though it is plausible that nonsquare matrices A and B could exist such that $AB = I_m$ and $BA = I_n$, where A is $m \times n$ and B is $n \times m$, we claim that this forces $n = m$. Indeed, if $m < n$ there exists a nonzero column $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$ (by Theorem 1.3.1), so $\mathbf{x} = I_n\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B(\mathbf{0}) = \mathbf{0}$, a contradiction. Hence $m \geq n$. Similarly, the condition $AB = I_m$ implies that $n \geq m$. Hence $m = n$ so A is square.

Solution. Compute AB and BA.

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad BA = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence $AB = I = BA$, so B is indeed an inverse of A.

Example 2.4.2

Show that $A = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix}$ has no inverse.

Solution. Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ denote an arbitrary 2×2 matrix. Then

$$AB = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ a+3c & b+3d \end{bmatrix}$$

so AB has a row of zeros. Hence AB cannot equal I for any B.

The argument in Example 2.4.2 shows that no zero matrix has an inverse. But Example 2.4.2 also shows that, unlike arithmetic, *it is possible for a nonzero matrix to have no inverse*. However, if a matrix *does* have an inverse, it has only one.

Theorem 2.4.1

If B and C are both inverses of A, then $B = C$.

Proof. Since B and C are both inverses of A, we have $CA = I = AB$. Hence

$$B = IB = (CA)B = C(AB) = CI = C$$

□

If A is an invertible matrix, the (unique) inverse of A is denoted A^{-1}. Hence A^{-1} (when it exists) is a square matrix of the same size as A with the property that

$$AA^{-1} = I \quad \text{and} \quad A^{-1}A = I$$

These equations characterize A^{-1} in the following sense:

> **Inverse Criterion:** *If somehow a matrix B can be found such that $AB = I$ and $BA = I$, then A is invertible and B is the inverse of A; in symbols, $B = A^{-1}$.*

This is a way to verify that the inverse of a matrix exists. Example 2.4.3 and Example 2.4.4 offer illustrations.

Example 2.4.3

If $A = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$, show that $A^3 = I$ and so find A^{-1}.

Solution. We have $A^2 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$, and so

$$A^3 = A^2A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence $A^3 = I$, as asserted. This can be written as $A^2A = I = AA^2$, so it shows that A^2 is the inverse of A. That is, $A^{-1} = A^2 = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$.

The next example presents a useful formula for the inverse of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ when it exists. To state it, we define the **determinant** $\det A$ and the **adjugate** $\operatorname{adj} A$ of the matrix A as follows:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc, \quad \text{and} \quad \operatorname{adj} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 2.4.4

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, show that A has an inverse if and only if $\det A \neq 0$, and in this case

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

Solution. For convenience, write $e = \det A = ad - bc$ and $B = \operatorname{adj} A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Then $AB = eI = BA$ as the reader can verify. So if $e \neq 0$, scalar multiplication by $\frac{1}{e}$ gives

$$A(\tfrac{1}{e}B) = I = (\tfrac{1}{e}B)A$$

Hence A is invertible and $A^{-1} = \frac{1}{e}B$. Thus it remains only to show that if A^{-1} exists, then $e \neq 0$. We prove this by showing that assuming $e = 0$ leads to a contradiction. In fact, if $e = 0$, then $AB = eI = 0$, so left multiplication by A^{-1} gives $A^{-1}AB = A^{-1}0$; that is, $IB = 0$, so $B = 0$. But this implies that a, b, c, and d are *all* zero, so $A = 0$, contrary to the assumption that A^{-1} exists.

As an illustration, if $A = \begin{bmatrix} 2 & 4 \\ -3 & 8 \end{bmatrix}$ then $\det A = 2 \cdot 8 - 4 \cdot (-3) = 28 \neq 0$. Hence A is invertible and $A^{-1} = \frac{1}{\det A} \operatorname{adj} A = \frac{1}{28}\begin{bmatrix} 8 & -4 \\ 3 & 2 \end{bmatrix}$, as the reader is invited to verify.

The determinant and adjugate will be defined in Chapter 3 for any square matrix, and the conclusions in Example 2.4.4 will be proved in full generality.

Inverses and Linear Systems

Matrix inverses can be used to solve certain systems of linear equations. Recall that a *system* of linear equations can be written as a *single* matrix equation

$$A\mathbf{x} = \mathbf{b}$$

where A and $\mathbf{b}$ are known and $\mathbf{x}$ is to be determined. If A is invertible, we multiply each side of the equation on the left by A^{-1} to get

$$\begin{aligned} A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b} \\ I\mathbf{x} &= A^{-1}\mathbf{b} \\ \mathbf{x} &= A^{-1}\mathbf{b} \end{aligned}$$

This gives the solution to the system of equations (the reader should verify that $\mathbf{x} = A^{-1}\mathbf{b}$ really does satisfy $A\mathbf{x} = \mathbf{b}$). Furthermore, the argument shows that if $\mathbf{x}$ is *any* solution, then necessarily $\mathbf{x} = A^{-1}\mathbf{b}$, so the solution is unique. Of course the technique works only when the coefficient matrix A has an inverse. This proves Theorem 2.4.2.

Theorem 2.4.2

Suppose a system of n equations in n variables is written in matrix form as

$$A\mathbf{x} = \mathbf{b}$$

If the $n \times n$ coefficient matrix A is invertible, the system has the unique solution

$$\mathbf{x} = A^{-1}\mathbf{b}$$

Example 2.4.5

Use Example 2.4.4 to solve the system $\left\{ \begin{array}{rl} 5x_1 - 3x_2 = & -4 \\ 7x_1 + 4x_2 = & 8 \end{array} \right.$.

Solution. In matrix form this is $A\mathbf{x} = \mathbf{b}$ where $A = \left[\begin{array}{rr} 5 & -3 \\ 7 & 4 \end{array} \right]$, $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right]$, and $\mathbf{b} = \left[\begin{array}{r} -4 \\ 8 \end{array} \right]$. Then $\det A = 5 \cdot 4 - (-3) \cdot 7 = 41$, so A is invertible and $A^{-1} = \frac{1}{41} \left[\begin{array}{rr} 4 & 3 \\ -7 & 5 \end{array} \right]$ by Example 2.4.4. Thus Theorem 2.4.2 gives

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{41} \left[\begin{array}{rr} 4 & 3 \\ -7 & 5 \end{array} \right] \left[\begin{array}{r} -4 \\ 8 \end{array} \right] = \frac{1}{41} \left[\begin{array}{r} 8 \\ 68 \end{array} \right]$$

so the solution is $x_1 = \frac{8}{41}$ and $x_2 = \frac{68}{41}$.

An Inversion Method

If a matrix A is $n \times n$ and invertible, it is desirable to have an efficient technique for finding the inverse. The following procedure will be justified in Section 2.5.

Matrix Inversion Algorithm

If A is an invertible (square) matrix, there exists a sequence of elementary row operations that carry A to the identity matrix I of the same size, written $A \to I$. This same series of row operations carries I to A^{-1}; that is, $I \to A^{-1}$. The algorithm can be summarized as follows:

$$\left[\begin{array}{cc} A & I \end{array}\right] \to \left[\begin{array}{cc} I & A^{-1} \end{array}\right]$$

where the row operations on A and I are carried out simultaneously.

Example 2.4.6

Use the inversion algorithm to find the inverse of the matrix

$$A = \left[\begin{array}{rrr} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{array}\right]$$

Solution. Apply elementary row operations to the double matrix

$$\left[\begin{array}{cc} A & I \end{array}\right] = \left[\begin{array}{rrr|rrr} 2 & 7 & 1 & 1 & 0 & 0 \\ 1 & 4 & -1 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 \end{array}\right]$$

so as to carry A to I. First interchange rows 1 and 2.

$$\left[\begin{array}{rrr|rrr} 1 & 4 & -1 & 0 & 1 & 0 \\ 2 & 7 & 1 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 \end{array}\right]$$

Next subtract 2 times row 1 from row 2, and subtract row 1 from row 3.

$$\left[\begin{array}{rrr|rrr} 1 & 4 & -1 & 0 & 1 & 0 \\ 0 & -1 & 3 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right]$$

Continue to reduced row-echelon form.

$$\left[\begin{array}{rrr|rrr} 1 & 0 & 11 & 4 & -7 & 0 \\ 0 & 1 & -3 & -1 & 2 & 0 \\ 0 & 0 & -2 & -1 & 1 & 1 \end{array}\right]$$

$$\left[\begin{array}{rrr|rrr} 1 & 0 & 0 & \frac{-3}{2} & \frac{-3}{2} & \frac{11}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{-3}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{-1}{2} & \frac{-1}{2} \end{array}\right]$$

Hence $A^{-1} = \frac{1}{2}\begin{bmatrix} -3 & -3 & 11 \\ 1 & 1 & -3 \\ 1 & -1 & -1 \end{bmatrix}$, as is readily verified.

Given any $n \times n$ matrix A, Theorem 1.2.1 shows that A can be carried by elementary row operations to a matrix R in reduced row-echelon form. If $R = I$, the matrix A is invertible (this will be proved in the next section), so the algorithm produces A^{-1}. If $R \neq I$, then R has a row of zeros (it is square), so no system of linear equations $A\mathbf{x} = \mathbf{b}$ can have a unique solution. But then A is not invertible by Theorem 2.4.2. Hence, the algorithm is effective in the sense conveyed in Theorem 2.4.3.

Theorem 2.4.3

If A is an $n \times n$ matrix, either A can be reduced to I by elementary row operations or it cannot. In the first case, the algorithm produces A^{-1}; in the second case, A^{-1} does not exist.

Properties of Inverses

The following properties of an invertible matrix are used everywhere.

Example 2.4.7: Cancellation Laws

Let A be an invertible matrix. Show that:

1. If $AB = AC$, then $B = C$.
2. If $BA = CA$, then $B = C$.

Solution. Given the equation $AB = AC$, left multiply both sides by A^{-1} to obtain $A^{-1}AB = A^{-1}AC$. Thus $IB = IC$, that is $B = C$. This proves (1) and the proof of (2) is left to the reader.

Properties (1) and (2) in Example 2.4.7 are described by saying that an invertible matrix can be "left cancelled" and "right cancelled", respectively. Note however that "mixed" cancellation does not hold in general: If A is invertible and $AB = CA$, then B and C may *not* be equal, even if both are 2×2. Here is a specific example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Sometimes the inverse of a matrix is given by a formula. Example 2.4.4 is one illustration; Example 2.4.8 and Example 2.4.9 provide two more. The idea is the *Inverse Criterion*: If a matrix B can be found such that $AB = I = BA$, then A is invertible and $A^{-1} = B$.

Example 2.4.8

If A is an invertible matrix, show that the transpose A^T is also invertible. Show further that the inverse of A^T is just the transpose of A^{-1}; in symbols, $(A^T)^{-1} = (A^{-1})^T$.

Solution. A^{-1} exists (by assumption). Its transpose $(A^{-1})^T$ is the candidate proposed for the inverse of A^T. Using the inverse criterion, we test it as follows:

$$\begin{aligned} A^T(A^{-1})^T &= (A^{-1}A)^T = I^T = I \\ (A^{-1})^T A^T &= (AA^{-1})^T = I^T = I \end{aligned}$$

Hence $(A^{-1})^T$ is indeed the inverse of A^T; that is, $(A^T)^{-1} = (A^{-1})^T$.

Example 2.4.9

If A and B are invertible $n \times n$ matrices, show that their product AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Solution. We are given a candidate for the inverse of AB, namely $B^{-1}A^{-1}$. We test it as follows:

$$\begin{aligned} (B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \\ (AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \end{aligned}$$

Hence $B^{-1}A^{-1}$ is the inverse of AB; in symbols, $(AB)^{-1} = B^{-1}A^{-1}$.

We now collect several basic properties of matrix inverses for reference.

Theorem 2.4.4

All the following matrices are square matrices of the same size.

1. *I is invertible and $I^{-1} = I$.*
2. *If A is invertible, so is A^{-1}, and $(A^{-1})^{-1} = A$.*
3. *If A and B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.*
4. *If $A_1, A_2, \dots, A_k$ are all invertible, so is their product $A_1A_2\cdots A_k$, and*

$$(A_1A_2\cdots A_k)^{-1} = A_k^{-1}\cdots A_2^{-1}A_1^{-1}.$$

5. *If A is invertible, so is A^k for any $k \geq 1$, and $(A^k)^{-1} = (A^{-1})^k$.*
6. *If A is invertible and $a \neq 0$ is a number, then aA is invertible and $(aA)^{-1} = \frac{1}{a}A^{-1}$.*
7. *If A is invertible, so is its transpose A^T, and $(A^T)^{-1} = (A^{-1})^T$.*

Proof.

1. This is an immediate consequence of the fact that $I^2 = I$.
2. The equations $AA^{-1} = I = A^{-1}A$ show that A is the inverse of A^{-1}; in symbols, $(A^{-1})^{-1} = A$.

3. This is Example 2.4.9.

4. Use induction on k. If $k = 1$, there is nothing to prove, and if $k = 2$, the result is property 3. If $k > 2$, assume inductively that $(A_1A_2\cdots A_{k-1})^{-1} = A_{k-1}^{-1}\cdots A_2^{-1}A_1^{-1}$. We apply this fact together with property 3 as follows:

$$\begin{aligned}
[A_1A_2\cdots A_{k-1}A_k]^{-1} &= [(A_1A_2\cdots A_{k-1})A_k]^{-1} \\
&= A_k^{-1}(A_1A_2\cdots A_{k-1})^{-1} \\
&= A_k^{-1}\left(A_{k-1}^{-1}\cdots A_2^{-1}A_1^{-1}\right)
\end{aligned}$$

   So the proof by induction is complete.

5. This is property 4 with $A_1 = A_2 = \cdots = A_k = A$.

6. This is left as Exercise 2.4.29.

7. This is Example 2.4.8. □

The reversal of the order of the inverses in properties 3 and 4 of Theorem 2.4.4 is a consequence of the fact that matrix multiplication is not commutative. Another manifestation of this comes when matrix equations are dealt with. If a matrix equation $B = C$ is given, it can be *left-multiplied* by a matrix A to yield $AB = AC$. Similarly, *right-multiplication* gives $BA = CA$. However, we cannot mix the two: If $B = C$, it need *not* be the case that $AB = CA$ even if A is invertible, for example, $A = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$, $B = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array}\right] = C$.

Part 7 of Theorem 2.4.4 together with the fact that $(A^T)^T = A$ gives

Corollary 2.4.1

A square matrix A is invertible if and only if A^T is invertible.

Example 2.4.10

Find A if $(A^T - 2I)^{-1} = \left[\begin{array}{rr} 2 & 1 \\ -1 & 0 \end{array}\right]$.

Solution. By Theorem 2.4.4(2) and Example 2.4.4, we have

$$(A^T - 2I) = \left[\left(A^T - 2I\right)^{-1}\right]^{-1} = \left[\begin{array}{rr} 2 & 1 \\ -1 & 0 \end{array}\right]^{-1} = \left[\begin{array}{rr} 0 & -1 \\ 1 & 2 \end{array}\right]$$

Hence $A^T = 2I + \left[\begin{array}{rr} 0 & -1 \\ 1 & 2 \end{array}\right] = \left[\begin{array}{rr} 2 & -1 \\ 1 & 4 \end{array}\right]$, so $A = \left[\begin{array}{rr} 2 & 1 \\ -1 & 4 \end{array}\right]$ by Theorem 2.4.4(7).

The following important theorem collects a number of conditions all equivalent[9] to invertibility. It will be referred to frequently below.

[9] If p and q are statements, we say that p **implies** q (written $p \Rightarrow q$) if q is true whenever p is true. The statements are called **equivalent** if both $p \Rightarrow q$ and $q \Rightarrow p$ (written $p \Leftrightarrow q$, spoken "p if and only if q"). See Appendix B.

Theorem 2.4.5: Inverse Theorem

The following conditions are equivalent for an $n \times n$ matrix A:

1. *A is invertible.*
2. *The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.*
3. *A can be carried to the identity matrix I_n by elementary row operations.*
4. *The system $A\mathbf{x} = \mathbf{b}$ has at least one solution $\mathbf{x}$ for every choice of column $\mathbf{b}$.*
5. *There exists an $n \times n$ matrix C such that $AC = I_n$.*

Proof. We show that each of these conditions implies the next, and that (5) implies (1).

(1) $\Rightarrow$ (2). If A^{-1} exists, then $A\mathbf{x} = \mathbf{0}$ gives $\mathbf{x} = I_n\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$.

(2) $\Rightarrow$ (3). Assume that (2) is true. Certainly $A \to R$ by row operations where R is a reduced, row-echelon matrix. It suffices to show that $R = I_n$. Suppose that this is not the case. Then R has a row of zeros (being square). Now consider the augmented matrix $\left[\begin{array}{c|c} A & \mathbf{0} \end{array}\right]$ of the system $A\mathbf{x} = \mathbf{0}$. Then $\left[\begin{array}{c|c} A & \mathbf{0} \end{array}\right] \to \left[\begin{array}{c|c} R & \mathbf{0} \end{array}\right]$ is the reduced form, and $\left[\begin{array}{c|c} R & \mathbf{0} \end{array}\right]$ also has a row of zeros. Since R is square there must be at least one nonleading variable, and hence at least one parameter. Hence the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions, contrary to (2). So $R = I_n$ after all.

(3) $\Rightarrow$ (4). Consider the augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$ of the system $A\mathbf{x} = \mathbf{b}$. Using (3), let $A \to I_n$ by a sequence of row operations. Then these same operations carry $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right] \to \left[\begin{array}{c|c} I_n & \mathbf{c} \end{array}\right]$ for some column $\mathbf{c}$. Hence the system $A\mathbf{x} = \mathbf{b}$ has a solution (in fact unique) by gaussian elimination. This proves (4).

(4) $\Rightarrow$ (5). Write $I_n = \left[\begin{array}{cccc} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{array}\right]$ where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the columns of I_n. For each $j = 1, 2, \dots, n$, the system $A\mathbf{x} = \mathbf{e}_j$ has a solution $\mathbf{c}_j$ by (4), so $A\mathbf{c}_j = \mathbf{e}_j$. Now let $C = \left[\begin{array}{cccc} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{array}\right]$ be the $n \times n$ matrix with these matrices $\mathbf{c}_j$ as its columns. Then Definition 2.9 gives (5):

$$AC = A\left[\begin{array}{cccc} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{array}\right] = \left[\begin{array}{cccc} A\mathbf{c}_1 & A\mathbf{c}_2 & \cdots & A\mathbf{c}_n \end{array}\right] = \left[\begin{array}{cccc} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{array}\right] = I_n$$

(5) $\Rightarrow$ (1). Assume that (5) is true so that $AC = I_n$ for some matrix C. Then $C\mathbf{x} = 0$ implies $\mathbf{x} = \mathbf{0}$ (because $\mathbf{x} = I_n\mathbf{x} = AC\mathbf{x} = A\mathbf{0} = \mathbf{0}$). Thus condition (2) holds for the matrix C rather than A. Hence the argument above that (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) (with A replaced by C) shows that a matrix C' exists such that $CC' = I_n$. But then

$$A = AI_n = A(CC') = (AC)C' = I_nC' = C'$$

Thus $CA = CC' = I_n$ which, together with $AC = I_n$, shows that C is the inverse of A. This proves (1). □

The proof of (5) $\Rightarrow$ (1) in Theorem 2.4.5 shows that if $AC = I$ for square matrices, then necessarily $CA = I$, and hence that C and A are inverses of each other. We record this important fact for reference.

Corollary 2.4.1

If A and C are square matrices such that $AC = I$, then also $CA = I$. In particular, both A and C are invertible, $C = A^{-1}$, and $A = C^{-1}$.

Here is a quick way to remember Corollary 2.4.1. If A is a square matrix, then

1. If $AC = I$ then $C = A^{-1}$.
2. If $CA = I$ then $C = A^{-1}$.

Observe that Corollary 2.4.1 is false if A and C are not square matrices. For example, we have

$$\left[\begin{array}{rrr} 1 & 2 & 1 \\ 1 & 1 & 1 \end{array}\right] \left[\begin{array}{rr} -1 & 1 \\ 1 & -1 \\ 0 & 1 \end{array}\right] = I_2 \quad \text{but} \quad \left[\begin{array}{rr} -1 & 1 \\ 1 & -1 \\ 0 & 1 \end{array}\right] \left[\begin{array}{rrr} 1 & 2 & 1 \\ 1 & 1 & 1 \end{array}\right] \neq I_3$$

In fact, it is verified in the footnote on page 79 that if $AB = I_m$ and $BA = I_n$, where A is $m \times n$ and B is $n \times m$, then $m = n$ and A and B are (square) inverses of each other.

An $n \times n$ matrix A has rank n if and only if (3) of Theorem 2.4.5 holds. Hence

Corollary 2.4.2

An $n \times n$ matrix A is invertible if and only if $\text{rank } A = n$.

Exercises for 2.4

Exercise 2.4.1 In each case, show that the matrices are inverses of each other.

a. $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$

b. $\begin{bmatrix} 3 & 0 \\ 1 & -4 \end{bmatrix}, \frac{1}{2}\begin{bmatrix} 4 & 0 \\ 1 & -3 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 1 & 3 & 1 \end{bmatrix}, \begin{bmatrix} 7 & 2 & -6 \\ -3 & -1 & 3 \\ 2 & 1 & -2 \end{bmatrix}$

d. $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}, \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$

Exercise 2.4.2 Find the inverse of each of the following matrices.

a. $\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$

b. $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \\ -1 & -1 & 0 \end{bmatrix}$

d. $\begin{bmatrix} 1 & -1 & 2 \\ -5 & 7 & -11 \\ -2 & 3 & -5 \end{bmatrix}$

e. $\begin{bmatrix} 3 & 5 & 0 \\ 3 & 7 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ f. $\begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & 5 & -1 \end{bmatrix}$

g. $\begin{bmatrix} 2 & 4 & 1 \\ 3 & 3 & 2 \\ 4 & 1 & 4 \end{bmatrix}$ h. $\begin{bmatrix} 3 & 1 & -1 \\ 5 & 2 & 0 \\ 1 & 1 & -1 \end{bmatrix}$

i. $\begin{bmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$ j. $\begin{bmatrix} -1 & 4 & 5 & 2 \\ 0 & 0 & 0 & -1 \\ 1 & -2 & -2 & 0 \\ 0 & -1 & -1 & 0 \end{bmatrix}$

k. $\begin{bmatrix} 1 & 0 & 7 & 5 \\ 0 & 1 & 3 & 6 \\ 1 & -1 & 5 & 2 \\ 1 & -1 & 5 & 1 \end{bmatrix}$ l. $\begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Exercise 2.4.3 In each case, solve the systems of equations by finding the inverse of the coefficient matrix.

a. $\begin{aligned} 3x - y &= 5 \\ 2x + 2y &= 1 \end{aligned}$ b. $\begin{aligned} 2x - 3y &= 0 \\ x - 4y &= 1 \end{aligned}$

c. $\begin{aligned} x + y + 2z &= 5 \\ x + y + z &= 0 \\ x + 2y + 4z &= -2 \end{aligned}$ d. $\begin{aligned} x + 4y + 2z &= 1 \\ 2x + 3y + 3z &= -1 \\ 4x + y + 4z &= 0 \end{aligned}$

Exercise 2.4.4 Given $A^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$:

a. Solve the system of equations $A\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$.

b. Find a matrix B such that $AB = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

c. Find a matrix C such that $CA = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \end{bmatrix}$.

Exercise 2.4.5 Find A when

a. $(3A)^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ b. $(2A)^T = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}^{-1}$

c. $(I + 3A)^{-1} = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$

d. $(I - 2A^T)^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

e. $\left(A \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\right)^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$

f. $\left(\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A\right)^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

g. $\left(A^T - 2I\right)^{-1} = 2\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

h. $\left(A^{-1} - 2I\right)^T = -2\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Exercise 2.4.6 Find A when:

a. $A^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 1 & 1 \\ 0 & 2 & -2 \end{bmatrix}$ b. $A^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

Exercise 2.4.7 Given $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ and $\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 0 \\ -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, express the variables x_1, x_2, and x_3 in terms of z_1, z_2, and z_3.

Exercise 2.4.8

a. In the system $\begin{aligned} 3x + 4y &= 7 \\ 4x + 5y &= 1 \end{aligned}$, substitute the new variables x' and y' given by $\begin{aligned} x &= -5x' + 4y' \\ y &= 4x' - 3y' \end{aligned}$. Then find x and y.

b. Explain part (a) by writing the equations as $A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} x \\ y \end{bmatrix} = B\begin{bmatrix} x' \\ y' \end{bmatrix}$. What is the relationship between A and B?

Exercise 2.4.9 In each case either prove the assertion or give an example showing that it is false.

a. If $A \neq 0$ is a square matrix, then A is invertible.

b. If A and B are both invertible, then $A + B$ is invertible.

c. If A and B are both invertible, then $(A^{-1}B)^T$ is invertible.

d. If $A^4 = 3I$, then A is invertible.

e. If $A^2 = A$ and $A \neq 0$, then A is invertible.

f. If $AB = B$ for some $B \neq 0$, then A is invertible.

g. If A is invertible and skew symmetric ($A^T = -A$), the same is true of A^{-1}.

h. If A^2 is invertible, then A is invertible.

i. If $AB = I$, then A and B commute.

Exercise 2.4.10

a. If A, B, and C are square matrices and $AB = I$, $I = CA$, show that A is invertible and $B = C = A^{-1}$.

b. If $C^{-1} = A$, find the inverse of C^T in terms of A.

Exercise 2.4.11 Suppose $CA = I_m$, where C is $m \times n$ and A is $n \times m$. Consider the system $A\mathbf{x} = \mathbf{b}$ of n equations in m variables.

a. Show that this system has a unique solution CB if it is consistent.

b. If $C = \left[\begin{array}{rrr} 0 & -5 & 1 \\ 3 & 0 & -1 \end{array}\right]$ and $A = \left[\begin{array}{rr} 2 & -3 \\ 1 & -2 \\ 6 & -10 \end{array}\right]$, find $\mathbf{x}$ (if it exists) when (i) $\mathbf{b} = \left[\begin{array}{r} 1 \\ 0 \\ 3 \end{array}\right]$; and (ii) $\mathbf{b} = \left[\begin{array}{r} 7 \\ 4 \\ 22 \end{array}\right]$.

Exercise 2.4.12 Verify that $A = \left[\begin{array}{rr} 1 & -1 \\ 0 & 2 \end{array}\right]$ satisfies $A^2 - 3A + 2I = 0$, and use this fact to show that $A^{-1} = \frac{1}{2}(3I - A)$.

Exercise 2.4.13 Let $Q = \left[\begin{array}{rrrr} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & -c & b & a \end{array}\right]$. Compute QQ^T and so find Q^{-1} if $Q \neq 0$.

Exercise 2.4.14 Let $U = \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right]$. Show that each of U, $-U$, and $-I_2$ is its own inverse and that the product of any two of these is the third.

Exercise 2.4.15 Consider $A = \left[\begin{array}{rr} 1 & 1 \\ -1 & 0 \end{array}\right]$, $B = \left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array}\right]$, $C = \left[\begin{array}{rrr} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & 0 & 0 \end{array}\right]$. Find the inverses by computing (a) A^6; (b) B^4; and (c) C^3.

Exercise 2.4.16 Find the inverse of $\left[\begin{array}{ccc} 1 & 0 & 1 \\ c & 1 & c \\ 3 & c & 2 \end{array}\right]$ in terms of c.

Exercise 2.4.17 If $c \neq 0$, find the inverse of $\left[\begin{array}{rrr} 1 & -1 & 1 \\ 2 & -1 & 2 \\ 0 & 2 & c \end{array}\right]$ in terms of c.

Exercise 2.4.18 Show that A has no inverse when:

a. A has a row of zeros.

b. A has a column of zeros.

c. each row of A sums to 0.
[*Hint*: Theorem 2.4.5(2).]

d. each column of A sums to 0.
[*Hint*: Corollary 2.4.1, Theorem 2.4.4.]

Exercise 2.4.19 Let A denote a square matrix.

a. Let $YA = 0$ for some matrix $Y \neq 0$. Show that A has no inverse. [*Hint*: Corollary 2.4.1, Theorem 2.4.4.]

b. Use part (a) to show that (i) $\left[\begin{array}{rrr} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{array}\right]$; and (ii) $\left[\begin{array}{rrr} 2 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{array}\right]$ have no inverse.
[*Hint*: For part (ii) compare row 3 with the difference between row 1 and row 2.]

Exercise 2.4.20 If A is invertible, show that

a. $A^2 \neq 0$.

b. $A^k \neq 0$ for all $k = 1, 2, \dots$.

Exercise 2.4.21 Suppose $AB = 0$, where A and B are square matrices. Show that:

a. If one of A and B has an inverse, the other is zero.

b. It is impossible for both A and B to have inverses.

c. $(BA)^2 = 0$.

Exercise 2.4.22 Find the inverse of the x-expansion in Example 2.2.16 and describe it geometrically.

Exercise 2.4.23 Find the inverse of the shear transformation in Example 2.2.17 and describe it geometrically.

Exercise 2.4.24 In each case assume that A is a square matrix that satisfies the given condition. Show that A is invertible and find a formula for A^{-1} in terms of A.

a. $A^3 - 3A + 2I = 0$.

b. $A^4 + 2A^3 - A - 4I = 0$.

Exercise 2.4.25 Let A and B denote $n \times n$ matrices.

a. If A and AB are invertible, show that B is invertible using only (2) and (3) of Theorem 2.4.4.

b. If AB is invertible, show that both A and B are invertible using Theorem 2.4.5.

Exercise 2.4.26 In each case find the inverse of the matrix A using Example 2.4.11.

a. $A = \left[\begin{array}{rrr} -1 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & -1 \end{array}\right]$ b. $A = \left[\begin{array}{rrr} 3 & 1 & 0 \\ 5 & 2 & 0 \\ 1 & 3 & -1 \end{array}\right]$

c. $A = \left[\begin{array}{rrrr} 3 & 4 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 1 & -1 & 1 & 3 \\ 3 & 1 & 1 & 4 \end{array}\right]$

d. $A = \left[\begin{array}{rrrr} 2 & 1 & 5 & 2 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -2 \end{array}\right]$

Exercise 2.4.27 If A and B are invertible symmetric matrices such that $AB = BA$, show that A^{-1}, AB, AB^{-1}, and $A^{-1}B^{-1}$ are also invertible and symmetric.

Exercise 2.4.28 Let A be an $n \times n$ matrix and let I be the $n \times n$ identity matrix.

a. If $A^2 = 0$, verify that $(I - A)^{-1} = I + A$.

b. If $A^3 = 0$, verify that $(I - A)^{-1} = I + A + A^2$.

c. Find the inverse of $\left[\begin{array}{rrr} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right]$.

d. If $A^n = 0$, find the formula for $(I - A)^{-1}$.

Exercise 2.4.29 Prove property 6 of Theorem 2.4.4: If A is invertible and $a \neq 0$, then aA is invertible and $(aA)^{-1} = \frac{1}{a}A^{-1}$

Exercise 2.4.30 Let A, B, and C denote $n \times n$ matrices. Using only Theorem 2.4.4, show that:

a. If A, C, and ABC are all invertible, B is invertible.

b. If AB and BA are both invertible, A and B are both invertible.

Exercise 2.4.31 Let A and B denote invertible $n \times n$ matrices.

a. If $A^{-1} = B^{-1}$, does it mean that $A = B$? Explain.

b. Show that $A = B$ if and only if $A^{-1}B = I$.

Exercise 2.4.32 Let A, B, and C be $n \times n$ matrices, with A and B invertible. Show that

a. If A commutes with C, then A^{-1} commutes with C.

b. If A commutes with B, then A^{-1} commutes with B^{-1}.

Exercise 2.4.33 Let A and B be square matrices of the same size.

a. Show that $(AB)^2 = A^2B^2$ if $AB = BA$.

b. If A and B are invertible and $(AB)^2 = A^2B^2$, show that $AB = BA$.

c. If $A = \left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array}\right]$ and $B = \left[\begin{array}{rr} 1 & 1 \\ 0 & 0 \end{array}\right]$, show that $(AB)^2 = A^2B^2$ but $AB \neq BA$.

Exercise 2.4.34 Let A and B be $n \times n$ matrices for which AB is invertible. Show that A and B are both invertible.

Exercise 2.4.35 Consider $A = \left[\begin{array}{rrr} 1 & 3 & -1 \\ 2 & 1 & 5 \\ 1 & -7 & 13 \end{array}\right]$,

$B = \left[\begin{array}{rrr} 1 & 1 & 2 \\ 3 & 0 & -3 \\ -2 & 5 & 17 \end{array}\right]$.

a. Show that A is not invertible by finding a nonzero 1×3 matrix Y such that $YA = 0$.

[*Hint*: Row 3 of A equals 2(row 2) $-$ 3(row 1).]

b. Show that B is not invertible.

[*Hint*: Column 3 = 3(column 2) $-$ column 1.]

Exercise 2.4.36 Show that a square matrix A is invertible if and only if it can be left-cancelled: $AB = AC$ implies $B = C$.

Exercise 2.4.37 If $U^2 = I$, show that $I + U$ is not invertible unless $U = I$.

Exercise 2.4.38

a. If J is the 4×4 matrix with every entry 1, show that $I - \frac{1}{2}J$ is self-inverse and symmetric.

b. If X is $n \times m$ and satisfies $X^T X = I_m$, show that $I_n - 2XX^T$ is self-inverse and symmetric.

Exercise 2.4.39 An $n \times n$ matrix P is called an idempotent if $P^2 = P$. Show that:

a. I is the only invertible idempotent.

b. P is an idempotent if and only if $I - 2P$ is self-inverse.

c. U is self-inverse if and only if $U = I - 2P$ for some idempotent P.

d. $I - aP$ is invertible for any $a \neq 1$, and that $(I - aP)^{-1} = I + \left(\frac{a}{1-a}\right)^P$.

Exercise 2.4.40 If $A^2 = kA$, where $k \neq 0$, show that A is invertible if and only if $A = kI$.

Exercise 2.4.41 Let A and B denote $n \times n$ invertible matrices.

a. Show that $A^{-1} + B^{-1} = A^{-1}(A + B)B^{-1}$.

b. If $A + B$ is also invertible, show that $A^{-1} + B^{-1}$ is invertible and find a formula for $(A^{-1} + B^{-1})^{-1}$.

Exercise 2.4.42 Let A and B be $n \times n$ matrices, and let I be the $n \times n$ identity matrix.

a. Verify that $A(I + BA) = (I + AB)A$ and that $(I + BA)B = B(I + AB)$.

b. If $I + AB$ is invertible, verify that $I + BA$ is also invertible and that $(I + BA)^{-1} = I - B(I + AB)^{-1}A$.

2.5 Elementary Matrices

It is now clear that elementary row operations are important in linear algebra: They are essential in solving linear systems (using the gaussian algorithm) and in inverting a matrix (using the matrix inversion algorithm). It turns out that they can be performed by left multiplying by certain invertible matrices. These matrices are the subject of this section.

Definition 2.12 Elementary Matrices

An $n \times n$ matrix E is called an ***elementary matrix*** *if it can be obtained from the identity matrix I_n by a single elementary row operation (called the operation* ***corresponding*** *to E). We say that E is of type I, II, or III if the operation is of that type (see Definition 1.2).*

Hence

$$E_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right], \quad E_2 = \left[\begin{array}{cc} 1 & 0 \\ 0 & 9 \end{array}\right], \quad \text{and} \quad E_3 = \left[\begin{array}{cc} 1 & 5 \\ 0 & 1 \end{array}\right]$$

are elementary of types I, II, and III, respectively, obtained from the 2×2 identity matrix by interchanging rows 1 and 2, multiplying row 2 by 9, and adding 5 times row 2 to row 1.

Suppose now that the matrix $A = \left[\begin{array}{ccc} a & b & c \\ p & q & r \end{array}\right]$ is left multiplied by the above elementary matrices E_1,

a. Show that A is not invertible by finding a nonzero 1×3 matrix Y such that $YA = 0$.

[*Hint*: Row 3 of A equals 2(row 2) $-$3(row 1).]

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Prelude to Section 2.5

In Section 2.2, we learned how to calculate the product $A\mathbf{x}$ where A is $m \times n$ and $\mathbf{x}$ is $n \times k$, thinking of this product as a linear combination of the columns of A. To review, assuming $\mathbf{a}_i$ is the i^{th} row of A,

$$A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [\ \mathbf{a}_1\ \mathbf{a}_2 \cdots \mathbf{a}_n\ ] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots x_n\mathbf{a}_n.$$

To summarize,

> If A is $m \times n$ and $\mathbf{x}$ is $n \times k$, then $A\mathbf{x}$ is a linear combination of the columns of A where the entries of $\mathbf{x}$ are the scalars (or weights) in the linear combination.

In Section 2.3, we learned how to multiply two compatible matrices together via two strategies: Definition 2.9, the so-called "matrix-column" rule for matrix multiplication, and Theorem 2.3.2, the dot product rule for matrix multiplication. Each of these views of matrix multiplication is important and useful in different settings. In particular, we made the following observation.

> **Rule for the Columns of a Product** If A is $m \times n$ and B is $n \times k$, then the i^{th} column of AB is A times the i^{th} column of B. This means that the i^{th} column of AB is the linear combination of the columns of A, found by using the entries in the i^{th} column of B as the weights.

This rule tells us how to find the i^{th} column of AB. Let's figure out how to find the i^{th} row of AB:

- The i^{th} row of AB is the same as the i^{th} column of B^TA^T because $(AB)^T = B^TA^T$.
- By the rule for the columns of a product, the i^{th} column of B^TA^T is the linear combination of the columns of B^T found by using the entries in the i^{th} column of A^T as the weights.
- The columns of B^T are rows in B and the entries in the i^{th} column of A^T are entries in the i^{th} row of A.

Thus, we make the following conclusion.

> **Rule for the Rows of a Product** If A is $m \times n$ and B is $n \times k$, the the i^{th} row of AB is the linear combination of the rows of B, found by using the entries in the i^{th} row of A as the weights.

Here is an example using this rule.

$$\overbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}^{\text{on the left}} \overbrace{\begin{bmatrix} 7 & 8 \\ 9 & 10 \\ -1 & -2 \end{bmatrix}}^{\text{on the right}} = \begin{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ -1 & -2 \end{bmatrix} \\ \\ \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ -1 & -2 \end{bmatrix} \end{bmatrix} \tag{1}$$

$$= \begin{bmatrix} 1\begin{bmatrix} 7 & 8 \end{bmatrix} + 2\begin{bmatrix} 9 & 10 \end{bmatrix} + 3\begin{bmatrix} -1 & -2 \end{bmatrix} \\ \\ 4\begin{bmatrix} 7 & 8 \end{bmatrix} + 5\begin{bmatrix} 9 & 10 \end{bmatrix} + 6\begin{bmatrix} -1 & -2 \end{bmatrix} \end{bmatrix} \tag{2}$$

$$= \begin{bmatrix} \begin{bmatrix} 7 & 8 \end{bmatrix} + \begin{bmatrix} 18 & 20 \end{bmatrix} + \begin{bmatrix} -3 & -6 \end{bmatrix} \\ \\ \begin{bmatrix} 23 & 32 \end{bmatrix} + \begin{bmatrix} 45 & 50 \end{bmatrix} + \begin{bmatrix} -6 & -12 \end{bmatrix} \end{bmatrix} \tag{3}$$

$$= \begin{bmatrix} 22 & 22 \\ 62 & 70 \end{bmatrix} \tag{4}$$

Notice what is happening in step (2) above. Each row of the product is a linear combination of the rows of the matrix on the right. The weights in this linear combination come from a row in the matrix on the left. This is important.

Perhaps you are wondering, "Why we need yet another rule for matrix multiplication?" When we perform row operations, we are finding linear combinations of rows of a matrix. Thus, if we multiply a matrix on the left by the appropriate matrix, a desired row operation will be performed. This is the subject of Section 2.5.

For completeness, we end with a third view of matrix multiplication.

The Row-Matrix Rule for Matrix Multiplication Let A be an $m \times n$ matrix and B an $n \times k$ matrix where

$$A = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$$

so that $\mathbf{a}_i$ denotes the i^{th} row of A. Then

$$AB = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} B = \begin{bmatrix} \mathbf{a}_1 B \\ \mathbf{a}_2 B \\ \vdots \\ \mathbf{a}_m B \end{bmatrix}.$$

That is, row i of AB is the product of the i^{th} row of A with B.

Proof. From Definition 2.9, we know that

$$B^T A^T = B^T \left[\mathbf{a}_1^T \ \mathbf{a}_2^T \cdots \mathbf{a}_m^T\right] = \left[B^T \mathbf{a}_1^T \ B^T \mathbf{a}_2^T \cdots B^T \mathbf{a}_m^T\right].$$

This means that

$$\begin{aligned} AB &= (B^T A^T)^T \\ &= \left(\left[B^T \mathbf{a}_1^T \ B^T \mathbf{a}_2^T \cdots B^T \mathbf{a}_m^T\right]\right)^T \\ &= \begin{bmatrix} (B^T \mathbf{a}_1^T)^T \\ (B^T \mathbf{a}_2^T)^T \\ \vdots \\ (B^T \mathbf{a}_m^T)^T \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{a}_1 B \\ \mathbf{a}_2 B \\ \vdots \\ \\ \mathbf{a}_m B \end{bmatrix}. \end{aligned}$$

$\square$

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$$B^T A^T = B^T \left[\mathbf{a}_1^T \ \mathbf{a}_2^T \cdots \mathbf{a}_m^T\right] = \left[B^T\mathbf{a}_1^T \ B^T\mathbf{a}_2^T \cdots B^T\mathbf{a}_m^T\right].$$

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□

2.5 Elementary Matrices

It is now clear that elementary row operations are important in linear algebra: They are essential in solving linear systems (using the gaussian algorithm) and in inverting a matrix (using the matrix inversion algorithm). It turns out that they can be performed by left multiplying by certain invertible matrices. These matrices are the subject of this section.

Definition 2.12 Elementary Matrices

An $n \times n$ matrix E is called an ***elementary matrix*** *if it can be obtained from the identity matrix I_n by a single elementary row operation (called the operation* ***corresponding*** *to E). We say that E is of type I, II, or III if the operation is of that type (see Definition 1.2).*

Hence

$$E_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], \quad E_2 = \left[\begin{array}{cc} 1 & 0 \\ 0 & 9 \end{array} \right], \quad \text{and} \quad E_3 = \left[\begin{array}{cc} 1 & 5 \\ 0 & 1 \end{array} \right]$$

are elementary of types I, II, and III, respectively, obtained from the 2×2 identity matrix by interchanging rows 1 and 2, multiplying row 2 by 9, and adding 5 times row 2 to row 1.

Suppose now that the matrix $A = \left[\begin{array}{ccc} a & b & c \\ p & q & r \end{array} \right]$ is left multiplied by the above elementary matrices E_1,

E_2, and E_3. The results are:

$$E_1A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix} = \begin{bmatrix} p & q & r \\ a & b & c \end{bmatrix}$$
$$E_2A = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix} = \begin{bmatrix} a & b & c \\ 9p & 9q & 9r \end{bmatrix}$$
$$E_3A = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix} = \begin{bmatrix} a+5p & b+5q & c+5r \\ p & q & r \end{bmatrix}$$

In each case, left multiplying A by the elementary matrix has the *same* effect as doing the corresponding row operation to A. This works in general.

Lemma 2.5.1: [10]

If an elementary row operation is performed on an $m \times n$ matrix A, the result is EA where E is the elementary matrix obtained by performing the same operation on the $m \times m$ identity matrix.

Proof. We prove it for operations of type III; the proofs for types I and II are left as exercises. Let E be the elementary matrix corresponding to the operation that adds k times row p to row $q \neq p$. The proof depends on the fact that each row of EA is equal to the corresponding row of E times A. Let $K_1, K_2, \dots, K_m$ denote the rows of I_m. Then row i of E is K_i if $i \neq q$, while row q of E is $K_q + kK_p$. Hence:

$$\begin{aligned} \text{If } i \neq q \text{ then row } i \text{ of } EA = {} & K_iA = (\text{row } i \text{ of } A). \\ \text{Row } q \text{ of } EA = (K_q + kK_p)A = {} & K_qA + k(K_pA) \\ = {} & (\text{row } q \text{ of } A) \text{ plus } k\ (\text{row } p \text{ of } A). \end{aligned}$$

Thus EA is the result of adding k times row p of A to row q, as required. □

The effect of an elementary row operation can be reversed by another such operation (called its inverse) which is also elementary of the same type (see the discussion following (Example 1.1.3). It follows that each elementary matrix E is invertible. In fact, if a row operation on I produces E, then the inverse operation carries E back to I. If F is the elementary matrix corresponding to the inverse operation, this means $FE = I$ (by Lemma 2.5.1). Thus $F = E^{-1}$ and we have proved

Lemma 2.5.2

Every elementary matrix E is invertible, and E^{-1} is also a elementary matrix (of the same type). Moreover, E^{-1} corresponds to the inverse of the row operation that produces E.

The following table gives the inverse of each type of elementary row operation:

Type	Operation	Inverse Operation
I	Interchange rows p and q	Interchange rows p and q
II	Multiply row p by $k \neq 0$	Multiply row p by $1/k$, $k \neq 0$
III	Add k times row p to row $q \neq p$	Subtract k times row p from row q, $q \neq p$

Note that elementary matrices of type I are self-inverse.

[10] A lemma is an auxiliary theorem used in the proof of other theorems.

Example 2.5.1

Find the inverse of each of the elementary matrices

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{bmatrix}, \quad \text{and} \quad E_3 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution. E_1, E_2, and E_3 are of type I, II, and III respectively, so the table gives

$$E_1^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_1, \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{9} \end{bmatrix}, \quad \text{and} \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Inverses and Elementary Matrices

Suppose that an $m \times n$ matrix A is carried to a matrix B (written $A \to B$) by a series of k elementary row operations. Let E_1, E_2, $\dots$, E_k denote the corresponding elementary matrices. By Lemma 2.5.1, the reduction becomes

$$A \to E_1 A \to E_2 E_1 A \to E_3 E_2 E_1 A \to \cdots \to E_k E_{k-1} \cdots E_2 E_1 A = B$$

In other words,

$$A \to UA = B \quad \text{where } U = E_k E_{k-1} \cdots E_2 E_1$$

The matrix $U = E_k E_{k-1} \cdots E_2 E_1$ is invertible, being a product of invertible matrices by Lemma 2.5.2. Moreover, U can be computed without finding the E_i as follows: If the above series of operations carrying $A \to B$ is performed on I_m in place of A, the result is $I_m \to UI_m = U$. Hence this series of operations carries the block matrix $\left[\begin{array}{cc} A & I_m \end{array} \right] \to \left[\begin{array}{cc} B & U \end{array} \right]$. This, together with the above discussion, proves

Theorem 2.5.1

Suppose A is $m \times n$ and $A \to B$ by elementary row operations.

1. *$B = UA$ where U is an $m \times m$ invertible matrix.*
2. *U can be computed by $\left[\begin{array}{cc} A & I_m \end{array} \right] \to \left[\begin{array}{cc} B & U \end{array} \right]$ using the operations carrying $A \to B$.*
3. *$U = E_k E_{k-1} \cdots E_2 E_1$ where E_1, E_2, $\dots$, E_k are the elementary matrices corresponding (in order) to the elementary row operations carrying A to B.*

Example 2.5.2

If $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 1 \end{bmatrix}$, express the reduced row-echelon form R of A as $R = UA$ where U is invertible.

Solution. Reduce the double matrix $\left[\begin{array}{cc} A & I \end{array}\right] \to \left[\begin{array}{cc} R & U \end{array}\right]$ as follows:

$$\left[\begin{array}{cc} A & I \end{array}\right] = \left[\begin{array}{rrr|rr} 2 & 3 & 1 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{rrr|rr} 1 & 2 & 1 & 0 & 1 \\ 2 & 3 & 1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{rrr|rr} 1 & 2 & 1 & 0 & 1 \\ 0 & -1 & -1 & 1 & -2 \end{array}\right]$$
$$\to \left[\begin{array}{rrr|rr} 1 & 0 & -1 & 2 & -3 \\ 0 & 1 & 1 & -1 & 2 \end{array}\right]$$

Hence $R = \left[\begin{array}{rrr} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array}\right]$ and $U = \left[\begin{array}{rr} 2 & -3 \\ -1 & 2 \end{array}\right]$.

Now suppose that A is invertible. We know that $A \to I$ by Theorem 2.4.5, so taking $B = I$ in Theorem 2.5.1 gives $\left[\begin{array}{cc} A & I \end{array}\right] \to \left[\begin{array}{cc} I & U \end{array}\right]$ where $I = UA$. Thus $U = A^{-1}$, so we have $\left[\begin{array}{cc} A & I \end{array}\right] \to \left[\begin{array}{cc} I & A^{-1} \end{array}\right]$. This is the matrix inversion algorithm in Section 2.4. However, more is true: Theorem 2.5.1 gives $A^{-1} = U = E_k E_{k-1} \cdots E_2 E_1$ where $E_1, E_2, \dots, E_k$ are the elementary matrices corresponding (in order) to the row operations carrying $A \to I$. Hence

$$A = \left(A^{-1}\right)^{-1} = (E_k E_{k-1} \cdots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_{k-1}^{-1} E_k^{-1} \tag{2.10}$$

By Lemma 2.5.2, this shows that every invertible matrix A is a product of elementary matrices. Since elementary matrices are invertible (again by Lemma 2.5.2), this proves the following important characterization of invertible matrices.

Theorem 2.5.2

A square matrix is invertible if and only if it is a product of elementary matrices.

It follows from Theorem 2.5.1 that $A \to B$ by row operations if and only if $B = UA$ for some invertible matrix U. In this case we say that A and B are **row-equivalent**. (See Exercise 2.5.17.)

Example 2.5.3

Express $A = \left[\begin{array}{rr} -2 & 3 \\ 1 & 0 \end{array}\right]$ as a product of elementary matrices.

Solution. Using Lemma 2.5.1, the reduction of $A \to I$ is as follows:

$$A = \left[\begin{array}{rr} -2 & 3 \\ 1 & 0 \end{array}\right] \to E_1 A = \left[\begin{array}{rr} 1 & 0 \\ -2 & 3 \end{array}\right] \to E_2 E_1 A = \left[\begin{array}{rr} 1 & 0 \\ 0 & 3 \end{array}\right] \to E_3 E_2 E_1 A = \left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array}\right]$$

where the corresponding elementary matrices are

$$E_1 = \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right], \quad E_2 = \left[\begin{array}{rr} 1 & 0 \\ 2 & 1 \end{array}\right], \quad E_3 = \left[\begin{array}{rr} 1 & 0 \\ 0 & \frac{1}{3} \end{array}\right]$$

Hence $(E_3\ E_2\ E_1)A = I$, so:

$$A = (E_3 E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} E_3^{-1} = \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right] \left[\begin{array}{rr} 1 & 0 \\ -2 & 1 \end{array}\right] \left[\begin{array}{rr} 1 & 0 \\ 0 & 3 \end{array}\right]$$

Exercises for 2.5

Exercise 2.5.1 For each of the following elementary matrices, describe the corresponding elementary row operation and write the inverse.

a. $E = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b. $E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

c. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

d. $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

e. $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

f. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Exercise 2.5.2 In each case find an elementary matrix E such that $B = EA$.

a. $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$

b. $A = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix}$

d. $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$

e. $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$

f. $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}$

Exercise 2.5.3 Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$.

a. Find elementary matrices E_1 and E_2 such that $C = E_2E_1A$.

b. Show that there is *no* elementary matrix E such that $C = EA$.

Exercise 2.5.4 If E is elementary, show that A and EA differ in at most two rows.

Exercise 2.5.5

a. Is I an elementary matrix? Explain.

b. Is 0 an elementary matrix? Explain.

Exercise 2.5.6 In each case find an invertible matrix U such that $UA = R$ is in reduced row-echelon form, and express U as a product of elementary matrices.

a. $A = \left[\begin{array}{rrr} 1 & -1 & 2 \\ -2 & 1 & 0 \end{array} \right]$ b. $A = \left[\begin{array}{rrr} 1 & 2 & 1 \\ 5 & 12 & -1 \end{array} \right]$

c. $A = \left[\begin{array}{rrrr} 1 & 2 & -1 & 0 \\ 3 & 1 & 1 & 2 \\ 1 & -3 & 3 & 2 \end{array} \right]$

d. $A = \left[\begin{array}{rrrr} 2 & 1 & -1 & 0 \\ 3 & -1 & 2 & 1 \\ 1 & -2 & 3 & 1 \end{array} \right]$

Exercise 2.5.7 In each case find an invertible matrix U such that $UA = B$, and express U as a product of elementary matrices.

a. $A = \left[\begin{array}{rrr} 2 & 1 & 3 \\ -1 & 1 & 2 \end{array} \right]$, $B = \left[\begin{array}{rrr} 1 & -1 & -2 \\ 3 & 0 & 1 \end{array} \right]$

b. $A = \left[\begin{array}{rrr} 2 & -1 & 0 \\ 1 & 1 & 1 \end{array} \right]$, $B = \left[\begin{array}{rrr} 3 & 0 & 1 \\ 2 & -1 & 0 \end{array} \right]$

Exercise 2.5.8 In each case factor A as a product of elementary matrices.

a. $A = \left[\begin{array}{rr} 1 & 1 \\ 2 & 1 \end{array} \right]$ b. $A = \left[\begin{array}{rr} 2 & 3 \\ 1 & 2 \end{array} \right]$

c. $A = \left[\begin{array}{rrr} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 2 & 1 & 6 \end{array} \right]$ d. $A = \left[\begin{array}{rrr} 1 & 0 & -3 \\ 0 & 1 & 4 \\ -2 & 2 & 15 \end{array} \right]$

Exercise 2.5.9 Let E be an elementary matrix.

a. Show that E^T is also elementary of the same type.

b. Show that $E^T = E$ if E is of type I or II.

Exercise 2.5.10 Show that every matrix A can be factored as $A = UR$ where U is invertible and R is in reduced row-echelon form.

Exercise 2.5.11 If $A = \left[\begin{array}{rr} 1 & 2 \\ 1 & -3 \end{array} \right]$ and $B = \left[\begin{array}{rr} 5 & 2 \\ -5 & -3 \end{array} \right]$ find an elementary matrix F such that $AF = B$.

[*Hint*: See Exercise 2.5.9.]

Exercise 2.5.13 Prove Lemma 2.5.1 for elementary matrices of:

a. type I; b. type II.

Exercise 2.5.14 While trying to invert A, $\left[\begin{array}{cc} A & I \end{array} \right]$ is carried to $\left[\begin{array}{cc} P & Q \end{array} \right]$ by row operations. Show that $P = QA$.

Exercise 2.5.15 If A and B are $n \times n$ matrices and AB is a product of elementary matrices, show that the same is true of A.

Exercise 2.5.16 If U is invertible, show that the reduced row-echelon form of a matrix $\left[\begin{array}{cc} U & A \end{array}\right]$ is $\left[\begin{array}{cc} I & U^{-1}A \end{array}\right]$.

Exercise 2.5.17 Two matrices A and B are called **row-equivalent** (written $A \overset{r}{\sim} B$) if there is a sequence of elementary row operations carrying A to B.

a. Show that $A \overset{r}{\sim} B$ if and only if $A = UB$ for some invertible matrix U.

b. Show that:

   i. $A \overset{r}{\sim} A$ for all matrices A.

   ii. If $A \overset{r}{\sim} B$, then $B \overset{r}{\sim} A$

   iii. If $A \overset{r}{\sim} B$ and $B \overset{r}{\sim} C$, then $A \overset{r}{\sim} C$.

c. Show that, if A and B are both row-equivalent to some third matrix, then $A \overset{r}{\sim} B$.

d. Show that $\left[\begin{array}{rrrr} 1 & -1 & 3 & 2 \\ 0 & 1 & 4 & 1 \\ 1 & 0 & 8 & 6 \end{array}\right]$ and $\left[\begin{array}{rrrr} 1 & -1 & 4 & 5 \\ -2 & 1 & -11 & -8 \\ -1 & 2 & 2 & 2 \end{array}\right]$ are row-equivalent. [*Hint*: Consider (c) and Theorem 1.2.1.]

Exercise 2.5.18 If U and V are invertible $n \times n$ matrices, show that $U \overset{r}{\sim} V$. (See Exercise 2.5.17.)

Exercise 2.5.19 (See Exercise 2.5.17.) Find all matrices that are row-equivalent to:

a. $\left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$

b. $\left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right]$

c. $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$

d. $\left[\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 0 & 1 \end{array}\right]$

Exercise 2.5.20 Let A and B be $m \times n$ and $n \times m$ matrices, respectively. If $m > n$, show that AB is not invertible. [*Hint*: Use Theorem 1.3.1 to find $\mathbf{x} \neq \mathbf{0}$ with $B\mathbf{x} = \mathbf{0}$.]

Exercise 2.5.21 Define an *elementary column operation* on a matrix to be one of the following: (I) Interchange two columns. (II) Multiply a column by a nonzero scalar. (III) Add a multiple of a column to another column. Show that:

a. If an elementary column operation is done to an $m \times n$ matrix A, the result is AF, where F is an $n \times n$ elementary matrix.

b. Given any $m \times n$ matrix A, there exist $m \times m$ elementary matrices $E_1, \dots, E_k$ and $n \times n$ elementary matrices $F_1, \dots, F_p$ such that, in block form,

$$E_k \cdots E_1 A F_1 \cdots F_p = \left[\begin{array}{cc} I_r & 0 \\ 0 & 0 \end{array}\right]$$

Exercise 2.5.22 Suppose B is obtained from A by:

a. interchanging rows i and j;

b. multiplying row i by $k \neq 0$;

c. adding k times row i to row j $(i \neq j)$.

In each case describe how to obtain B^{-1} from A^{-1}. [*Hint*: See part (a) of the preceding exercise.]

Exercise 2.5.23 Two $m \times n$ matrices A and B are called **equivalent** (written $A \overset{e}{\sim} B$) if there exist invertible matrices U and V (sizes $m \times m$ and $n \times n$) such that $A = UBV$.

a. Prove the following the properties of equivalence.

   i. $A \overset{e}{\sim} A$ for all $m \times n$ matrices A.

   ii. If $A \overset{e}{\sim} B$, then $B \overset{e}{\sim} A$.

   iii. If $A \overset{e}{\sim} B$ and $B \overset{e}{\sim} C$, then $A \overset{e}{\sim} C$.

b. Prove that two $m \times n$ matrices are equivalent if they have the same rank. [*Hint*: Use part (a) and Theorem 2.5.3.]

2.6 Linear Transformations

If A is an $m \times n$ matrix, recall that the transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

is called the *matrix transformation induced* by A. In Section 2.2, we saw that many important geometric transformations were in fact matrix transformations. These transformations can be characterized in a different way. The new idea is that of a linear transformation, one of the basic notions in linear algebra. We define these transformations in this section, and show that they are really just the matrix transformations looked at in another way. Having these two ways to view them turns out to be useful because, in a given situation, one perspective or the other may be preferable.

Linear Transformations

Definition 2.13 Linear Transformations $\mathbb{R}^n \to \mathbb{R}^m$

A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is called a ***linear transformation*** *if it satisfies the following two conditions for all vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ and all scalars a:*

T1 $\quad T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$

T2 $\quad T(a\mathbf{x}) = aT(\mathbf{x})$

Of course, $\mathbf{x}+\mathbf{y}$ and $a\mathbf{x}$ here are computed in $\mathbb{R}^n$, while $T(\mathbf{x})+T(\mathbf{y})$ and $aT(\mathbf{x})$ are in $\mathbb{R}^m$. We say that T *preserves addition* if T1 holds, and that T *preserves scalar multiplication* if T2 holds. Moreover, taking $a = 0$ and $a = -1$ in T2 gives

$$T(\mathbf{0}) = \mathbf{0} \quad \text{and} \quad T(-\mathbf{x}) = -T(\mathbf{x}) \quad \text{for all } \mathbf{x}$$

Hence T preserves the zero vector and the negative of a vector. Even more is true.

Recall that a vector $\mathbf{y}$ in $\mathbb{R}^n$ is called a **linear combination** of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ if $\mathbf{y}$ has the form

$$\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k$$

for some scalars $a_1, a_2, \dots, a_k$. Conditions T1 and T2 combine to show that every linear transformation *T preserves linear combinations* in the sense of the following theorem. This result is used repeatedly in linear algebra.

Theorem 2.6.1: Linearity Theorem

If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then for each $k = 1, 2, \dots$

$$T(a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k) = a_1T(\mathbf{x}_1) + a_2T(\mathbf{x}_2) + \cdots + a_kT(\mathbf{x}_k)$$

for all scalars a_i and all vectors $\mathbf{x}_i$ in $\mathbb{R}^n$.

Proof. If $k = 1$, it reads $T(a_1\mathbf{x}_1) = a_1T(\mathbf{x}_1)$ which is Condition T1. If $k = 2$, we have

$$\begin{aligned} T(a_1\mathbf{x}_1 + a_2\mathbf{x}_2) &= T(a_1\mathbf{x}_1) + T(a_2\mathbf{x}_2) && \text{by Condition T1} \\ &= a_1T(\mathbf{x}_1) + a_2T(\mathbf{x}_2) && \text{by Condition T2} \end{aligned}$$

If $k=3$, we use the case $k=2$ to obtain

$$\begin{aligned} T(a_1\mathbf{x}_1+a_2\mathbf{x}_2+a_3\mathbf{x}_3) &= T[(a_1\mathbf{x}_1+a_2\mathbf{x}_2)+a_3\mathbf{x}_3] && \text{collect terms} \\ &= T(a_1\mathbf{x}_1+a_2\mathbf{x}_2)+T(a_3\mathbf{x}_3) && \text{by Condition T1} \\ &= [a_1T(\mathbf{x}_1)+a_2T(\mathbf{x}_2)]+T(a_3\mathbf{x}_3) && \text{by the case } k=2 \\ &= [a_1T(\mathbf{x}_1)+a_2T(\mathbf{x}_2)]+a_3T(\mathbf{x}_3) && \text{by Condition T2} \end{aligned}$$

The proof for any k is similar, using the previous case $k-1$ and Conditions T1 and T2. □

The method of proof in Theorem 2.6.1 is called *mathematical induction* (Appendix C).

Theorem 2.6.1 shows that if T is a linear transformation and $T(\mathbf{x}_1)$, $T(\mathbf{x}_2)$, $\dots$, $T(\mathbf{x}_k)$ are all known, then $T(\mathbf{y})$ can be easily computed for any linear combination $\mathbf{y}$ of $\mathbf{x}_1$, $\mathbf{x}_2$, $\dots$, $\mathbf{x}_k$. This is a very useful property of linear transformations, and is illustrated in the next example.

Example 2.6.1

If $T:\mathbb{R}^2\to\mathbb{R}^2$ is a linear transformation, $T\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}2\\-3\end{bmatrix}$ and $T\begin{bmatrix}1\\-2\end{bmatrix}=\begin{bmatrix}5\\1\end{bmatrix}$, find $T\begin{bmatrix}4\\3\end{bmatrix}$.

Solution. Write $\mathbf{z}=\begin{bmatrix}4\\3\end{bmatrix}$, $\mathbf{x}=\begin{bmatrix}1\\1\end{bmatrix}$, and $\mathbf{y}=\begin{bmatrix}1\\-2\end{bmatrix}$ for convenience. Then we know $T(\mathbf{x})$ and $T(\mathbf{y})$ and we want $T(\mathbf{z})$, so it is enough by Theorem 2.6.1 to express $\mathbf{z}$ as a linear combination of $\mathbf{x}$ and $\mathbf{y}$. That is, we want to find numbers a and b such that $\mathbf{z}=a\mathbf{x}+b\mathbf{y}$. Equating entries gives two equations $4=a+b$ and $3=a-2b$. The solution is, $a=\frac{11}{3}$ and $b=\frac{1}{3}$, so $\mathbf{z}=\frac{11}{3}\mathbf{x}+\frac{1}{3}\mathbf{y}$. Thus Theorem 2.6.1 gives

$$T(\mathbf{z})=\tfrac{11}{3}T(\mathbf{x})+\tfrac{1}{3}T(\mathbf{y})=\tfrac{11}{3}\begin{bmatrix}2\\-3\end{bmatrix}+\tfrac{1}{3}\begin{bmatrix}5\\1\end{bmatrix}=\tfrac{1}{3}\begin{bmatrix}27\\-32\end{bmatrix}$$

This is what we wanted.

Example 2.6.2

If A is $m\times n$, the matrix transformation $T_A:\mathbb{R}^n\to\mathbb{R}^m$, is a linear transformation.

Solution. We have $T_A(\mathbf{x})=A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$, so Theorem 2.2.2 gives

$$T_A(\mathbf{x}+\mathbf{y})=A(\mathbf{x}+\mathbf{y})=A\mathbf{x}+A\mathbf{y}=T_A(\mathbf{x})+T_A(\mathbf{y})$$

and

$$T_A(a\mathbf{x})=A(a\mathbf{x})=a(A\mathbf{x})=aT_A(\mathbf{x})$$

hold for all $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ and all scalars a. Hence T_A satisfies T1 and T2, and so is linear.

The remarkable thing is that the *converse* of Example 2.6.2 is true: Every linear transformation $T:\mathbb{R}^n\to\mathbb{R}^m$ is actually a matrix transformation. To see why, we define the **standard basis** of $\mathbb{R}^n$ to be the set of columns

$$\{\mathbf{e}_1,\,\mathbf{e}_2,\,\dots,\,\mathbf{e}_n\}$$

of the identity matrix I_n. Then each $\mathbf{e}_i$ is in $\mathbb{R}^n$ and every vector $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right]$ in $\mathbb{R}^n$ is a linear combination of the $\mathbf{e}_i$. In fact:

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$$

as the reader can verify. Hence Theorem 2.6.1 shows that

$$T(\mathbf{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n)$$

Now observe that each $T(\mathbf{e}_i)$ is a column in $\mathbb{R}^m$, so

$$A = \left[\begin{array}{cccc} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{array} \right]$$

is an $m \times n$ matrix. Hence we can apply Definition 2.5 to get

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n) = \left[\begin{array}{cccc} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right] = A\mathbf{x}$$

Since this holds for every $\mathbf{x}$ in $\mathbb{R}^n$, it shows that T is the matrix transformation induced by A, and so proves most of the following theorem.

Theorem 2.6.2

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a transformation.

1. *T is linear if and only if it is a matrix transformation.*
2. *In this case $T = T_A$ is the matrix transformation induced by a unique $m \times n$ matrix A, given in terms of its columns by*

$$A = \left[\begin{array}{cccc} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{array} \right]$$

where $\{\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$.

Proof. It remains to verify that the matrix A is unique. Suppose that T is induced by another matrix B. Then $T(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$. But $T(\mathbf{x}) = A\mathbf{x}$ for each $\mathbf{x}$, so $B\mathbf{x} = A\mathbf{x}$ for every $\mathbf{x}$. Hence $A = B$ by Theorem 2.2.6. □

Hence we can speak of *the* matrix of a linear transformation. Because of Theorem 2.6.2 we may (and shall) use the phrases “linear transformation” and “matrix transformation” interchangeably.

Example 2.6.3

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ for all $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ in $\mathbb{R}^3$. Show that T is a linear transformation and use Theorem 2.6.2 to find its matrix.

Solution. Write $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, so that $\mathbf{x}+\mathbf{y} = \begin{bmatrix} x_1+y_1 \\ x_2+y_2 \\ x_3+y_3 \end{bmatrix}$. Hence

$$T(\mathbf{x}+\mathbf{y}) = \begin{bmatrix} x_1+y_1 \\ x_2+y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = T(\mathbf{x}) + T(\mathbf{y})$$

Similarly, the reader can verify that $T(a\mathbf{x}) = aT(\mathbf{x})$ for all a in $\mathbb{R}$, so T is a linear transformation. Now the standard basis of $\mathbb{R}^3$ is

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

so, by Theorem 2.6.2, the matrix of T is

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Of course, the fact that $T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ shows directly that T is a matrix transformation (hence linear) and reveals the matrix.

To illustrate how Theorem 2.6.2 is used, we rederive the matrices of the transformations in Examples 2.2.13 and 2.2.15.

Example 2.6.4

Let $Q_0 : \mathbb{R}^2 \to \mathbb{R}^2$ denote reflection in the x axis (as in Example 2.2.13) and let $R_{\frac{\pi}{2}} : \mathbb{R}^2 \to \mathbb{R}^2$ denote counterclockwise rotation through $\frac{\pi}{2}$ about the origin (as in Example 2.2.15). Use Theorem 2.6.2 to find the matrices of Q_0 and $R_{\frac{\pi}{2}}$.

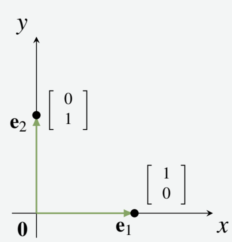


Figure 2.6.1

Solution. Observe that Q_0 and $R_{\frac{\pi}{2}}$ are linear by Example 2.6.2 (they are matrix transformations), so Theorem 2.6.2 applies to them. The standard basis of $\mathbb{R}^2$ is $\{\mathbf{e}_1, \mathbf{e}_2\}$ where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ points along the positive x axis, and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ points along the positive y axis (see Figure 2.6.1).

The reflection of $\mathbf{e}_1$ in the x axis is $\mathbf{e}_1$ itself because $\mathbf{e}_1$ points along the x axis, and the reflection of $\mathbf{e}_2$ in the x axis is $-\mathbf{e}_2$ because $\mathbf{e}_2$ is perpendicular to the x axis. In other words, $Q_0(\mathbf{e}_1) = \mathbf{e}_1$ and $Q_0(\mathbf{e}_2) = -\mathbf{e}_2$. Hence Theorem 2.6.2 shows that the matrix of Q_0 is

$$\left[\begin{array}{cc} Q_0(\mathbf{e}_1) & Q_0(\mathbf{e}_2) \end{array}\right] = \left[\begin{array}{cc} \mathbf{e}_1 & -\mathbf{e}_2 \end{array}\right] = \left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array}\right]$$

which agrees with Example 2.2.13.

Similarly, rotating $\mathbf{e}_1$ through $\frac{\pi}{2}$ counterclockwise about the origin produces $\mathbf{e}_2$, and rotating $\mathbf{e}_2$ through $\frac{\pi}{2}$ counterclockwise about the origin gives $-\mathbf{e}_1$. That is, $R_{\frac{\pi}{2}}(\mathbf{e}_1) = \mathbf{e}_2$ and $R_{\frac{\pi}{2}}(\mathbf{e}_2) = -\mathbf{e}_2$. Hence, again by Theorem 2.6.2, the matrix of $R_{\frac{\pi}{2}}$ is

$$\left[\begin{array}{cc} R_{\frac{\pi}{2}}(\mathbf{e}_1) & R_{\frac{\pi}{2}}(\mathbf{e}_2) \end{array}\right] = \left[\begin{array}{cc} \mathbf{e}_2 & -\mathbf{e}_1 \end{array}\right] = \left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array}\right]$$

agreeing with Example 2.2.15.

Example 2.6.5

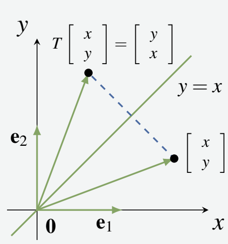


Figure 2.6.2

Let $Q_1 : \mathbb{R}^2 \to \mathbb{R}^2$ denote reflection in the line $y = x$. Show that Q_1 is a matrix transformation, find its matrix, and use it to illustrate Theorem 2.6.2.

Solution. Figure 2.6.2 shows that $Q_1 \left[\begin{array}{c} x \\ y \end{array}\right] = \left[\begin{array}{c} y \\ x \end{array}\right]$. Hence $Q_1 \left[\begin{array}{c} x \\ y \end{array}\right] = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right] \left[\begin{array}{c} y \\ x \end{array}\right]$, so Q_1 is the matrix transformation induced by the matrix $A = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$. Hence Q_1 is linear (by Example 2.6.2) and so Theorem 2.6.2 applies. If $\mathbf{e}_1 = \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ and $\mathbf{e}_2 = \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$ are the standard basis of $\mathbb{R}^2$, then it is clear geometrically that $Q_1(\mathbf{e}_1) = \mathbf{e}_2$ and $Q_1(\mathbf{e}_2) = \mathbf{e}_1$. Thus (by Theorem 2.6.2) the matrix of Q_1 is $\left[\begin{array}{cc} Q_1(\mathbf{e}_1) & Q_1(\mathbf{e}_2) \end{array}\right] = \left[\begin{array}{cc} \mathbf{e}_2 & \mathbf{e}_1 \end{array}\right] = A$ as before.

Recall that, given two "linked" transformations

$$\mathbb{R}^k \xrightarrow{T} \mathbb{R}^n \xrightarrow{S} \mathbb{R}^m$$

we can apply T first and then apply S, and so obtain a new transformation

$$S \circ T : \mathbb{R}^k \to \mathbb{R}^m$$

called the **composite** of S and T, defined by

$$(S \circ T)(\mathbf{x}) = S[T(\mathbf{x})] \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^k$$

If S and T are linear, the action of $S \circ T$ can be computed by multiplying their matrices.

Theorem 2.6.3

Let $\mathbb{R}^k \xrightarrow{T} \mathbb{R}^n \xrightarrow{S} \mathbb{R}^m$ be linear transformations, and let A and B be the matrices of S and T respectively. Then $S \circ T$ is linear with matrix AB.

Proof. $(S \circ T)(\mathbf{x}) = S[T(\mathbf{x})] = A[B\mathbf{x}] = (AB)\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^k$. □

Theorem 2.6.3 shows that the action of the composite $S \circ T$ is determined by the matrices of S and T. But it also provides a very useful interpretation of matrix multiplication. If A and B are matrices, the product matrix AB induces the transformation resulting from first applying B and then applying A. Thus the study of matrices can cast light on geometrical transformations and vice-versa. Here is an example.

Example 2.6.6

Show that reflection in the x axis followed by rotation through $\frac{\pi}{2}$ is reflection in the line $y = x$.

Solution. The composite in question is $R_{\frac{\pi}{2}} \circ Q_0$ where Q_0 is reflection in the x axis and $R_{\frac{\pi}{2}}$ is rotation through $\frac{\pi}{2}$. By Example 2.6.4, $R_{\frac{\pi}{2}}$ has matrix $A = \left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array} \right]$ and Q_0 has matrix $B = \left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array} \right]$. Hence Theorem 2.6.3 shows that the matrix of $R_{\frac{\pi}{2}} \circ Q_0$ is $AB = \left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array} \right] \left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array} \right] = \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array} \right]$, which is the matrix of reflection in the line $y = x$ by Example 2.6.3.

This conclusion can also be seen geometrically. Let $\mathbf{x}$ be a typical point in $\mathbb{R}^2$, and assume that $\mathbf{x}$ makes an angle α with the positive x axis. The effect of first applying Q_0 and then applying $R_{\frac{\pi}{2}}$ is shown in Figure 2.6.3. The fact that $R_{\frac{\pi}{2}}[Q_0(\mathbf{x})]$ makes the angle α with the positive y axis shows that $R_{\frac{\pi}{2}}[Q_0(\mathbf{x})]$ is the reflection of $\mathbf{x}$ in the line $y = x$.

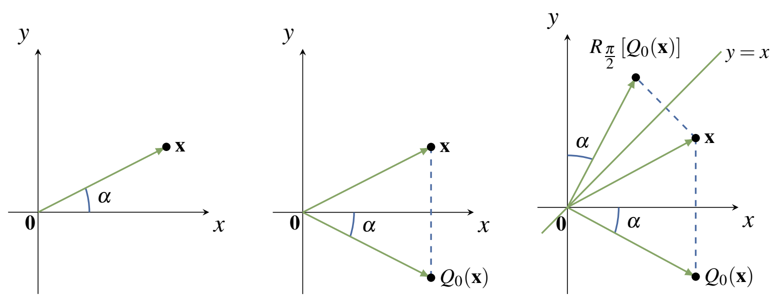


Figure 2.6.3

In Theorem 2.6.3, we saw that the matrix of the composite of two linear transformations is the product of their matrices (in fact, matrix products were defined so that this is the case). We are going to apply this fact to rotations, reflections, and projections in the plane. Before proceeding, we pause to present useful geometrical descriptions of vector addition and scalar multiplication in the plane, and to give a short review of angles and the trigonometric functions.

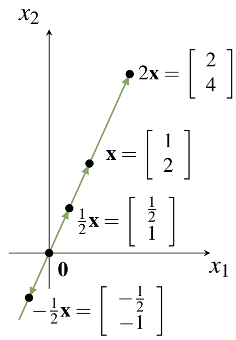


Figure 2.6.4

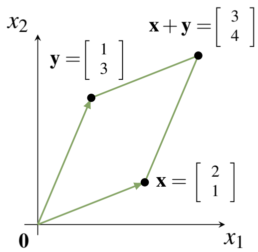


Figure 2.6.5

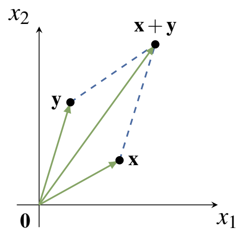


Figure 2.6.6

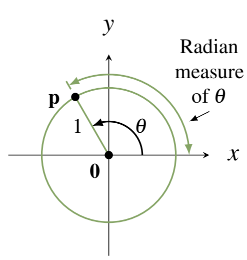


Figure 2.6.7

Some Geometry

As we have seen, it is convenient to view a vector $\mathbf{x}$ in $\mathbb{R}^2$ as an arrow from the origin to the point $\mathbf{x}$ (see Section 2.2). This enables us to visualize what sums and scalar multiples mean geometrically. For example consider $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Then $2\mathbf{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\frac{1}{2}\mathbf{x} = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$ and $-\frac{1}{2}\mathbf{x} = \begin{bmatrix} -\frac{1}{2} \\ -1 \end{bmatrix}$, and these are shown as arrows in Figure 2.6.4.

Observe that the arrow for $2\mathbf{x}$ is twice as long as the arrow for $\mathbf{x}$ and in the same direction, and that the arrows for $\frac{1}{2}\mathbf{x}$ is also in the same direction as the arrow for $\mathbf{x}$, but only half as long. On the other hand, the arrow for $-\frac{1}{2}\mathbf{x}$ is half as long as the arrow for $\mathbf{x}$, but in the *opposite* direction. More generally, we have the following geometrical description of scalar multiplication in $\mathbb{R}^2$:

Scalar Multiple Law

Let $\mathbf{x}$ be a vector in $\mathbb{R}^2$. The arrow for $k\mathbf{x}$ is $|k|$ times[12] *as long as the arrow for $\mathbf{x}$, and is in the same direction as the arrow for $\mathbf{x}$ if $k > 0$, and in the opposite direction if $k < 0$.*

Now consider two vectors $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ in $\mathbb{R}^2$. They are plotted in Figure 2.6.5 along with their sum $\mathbf{x}+\mathbf{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. It is a routine matter to verify that the four points $\mathbf{0}$, $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{x}+\mathbf{y}$ form the vertices of a **parallelogram**–that is opposite sides are parallel and of the same length. (The reader should verify that the side from $\mathbf{0}$ to $\mathbf{x}$ has slope of $\frac{1}{2}$, as does the side from $\mathbf{y}$ to $\mathbf{x}+\mathbf{y}$, so these sides are parallel.) We state this as follows:

Parallelogram Law

Consider vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^2$. If the arrows for $\mathbf{x}$ and $\mathbf{y}$ are drawn (see Figure 2.6.6), the arrow for $\mathbf{x}+\mathbf{y}$ corresponds to the fourth vertex of the parallelogram determined by the points $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{0}$.

We will have more to say about this in Chapter 4.

Before proceeding we turn to a brief review of angles and the trigonometric functions. Recall that an angle θ is said to be in **standard position** if it is measured counterclockwise from the positive x axis (as in Figure 2.6.7). Then θ uniquely determines a point $\mathbf{p}$ on the **unit circle** (radius 1, centre at the origin). The **radian** measure of θ is the length of the arc on the unit circle from the positive x axis to $\mathbf{p}$. Thus $360° = 2\pi$ radians, $180° = \pi$, $90° = \frac{\pi}{2}$, and so on.

[12]If k is a real number, $|k|$ denotes the **absolute value** of k; that is, $|k| = k$ if $k \geq 0$ and $|k| = -k$ if $k < 0$.

The point $\mathbf{p}$ in Figure 2.6.7 is also closely linked to the trigonometric functions **cosine** and **sine**, written $\cos\theta$ and $\sin\theta$ respectively. In fact these functions are *defined* to be the x and y coordinates of $\mathbf{p}$; that is $\mathbf{p} = \left[\begin{array}{c} \cos\theta \\ \sin\theta \end{array}\right]$. This defines $\cos\theta$ and $\sin\theta$ for the arbitrary angle θ (possibly negative), and agrees with the usual values when θ is an acute angle $\left(0 \leq \theta \leq \frac{\pi}{2}\right)$ as the reader should verify. For more discussion of this, see Appendix A.

Rotations

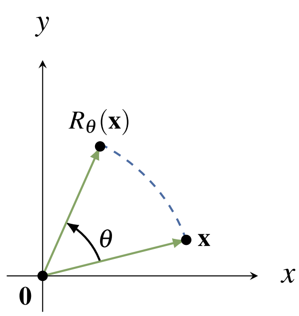


Figure 2.6.8

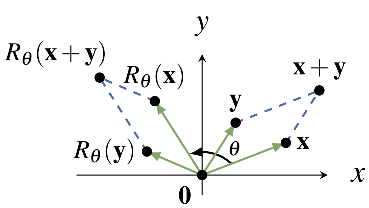


Figure 2.6.9

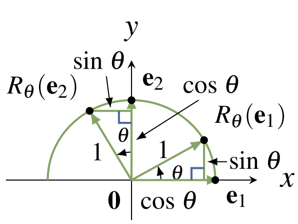


Figure 2.6.10

We can now describe rotations in the plane. Given an angle θ, let

$$R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$$

denote counterclockwise rotation of $\mathbb{R}^2$ about the origin through the angle θ. The action of R_θ is depicted in Figure 2.6.8. We have already looked at $R_{\frac{\pi}{2}}$ (in Example 2.2.15) and found it to be a matrix transformation. It turns out that R_θ is a matrix transformation for *every* angle θ (with a simple formula for the matrix), but it is not clear how to find the matrix. Our approach is to first establish the (somewhat surprising) fact that R_θ is *linear*, and then obtain the matrix from Theorem 2.6.2.

Let $\mathbf{x}$ and $\mathbf{y}$ be two vectors in $\mathbb{R}^2$. Then $\mathbf{x}+\mathbf{y}$ is the diagonal of the parallelogram determined by $\mathbf{x}$ and $\mathbf{y}$ as in Figure 2.6.9.

The effect of R_θ is to rotate the *entire* parallelogram to obtain the new parallelogram determined by $R_\theta(\mathbf{x})$ and $R_\theta(\mathbf{y})$, with diagonal $R_\theta(\mathbf{x}+\mathbf{y})$. But this diagonal is $R_\theta(\mathbf{x}) + R_\theta(\mathbf{y})$ by the parallelogram law (applied to the new parallelogram). It follows that

$$R_\theta(\mathbf{x}+\mathbf{y}) = R_\theta(\mathbf{x}) + R_\theta(\mathbf{y})$$

A similar argument shows that $R_\theta(a\mathbf{x}) = aR_\theta(\mathbf{x})$ for any scalar a, so $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ is indeed a linear transformation.

With linearity established we can find the matrix of R_θ. Let $\mathbf{e}_1 = \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ and $\mathbf{e}_2 = \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$ denote the standard basis of $\mathbb{R}^2$. By Figure 2.6.10 we see that

$$R_\theta(\mathbf{e}_1) = \left[\begin{array}{c} \cos\theta \\ \sin\theta \end{array}\right] \quad \text{and} \quad R_\theta(\mathbf{e}_2) = \left[\begin{array}{c} -\sin\theta \\ \cos\theta \end{array}\right]$$

Hence Theorem 2.6.2 shows that R_θ is induced by the matrix

$$\left[\begin{array}{cc} R_\theta(\mathbf{e}_1) & R_\theta(\mathbf{e}_2) \end{array}\right] = \left[\begin{array}{cc} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{array}\right]$$

We record this as

Theorem 2.6.4

The rotation $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ is the linear transformation with matrix $\left[\begin{array}{rr} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{array} \right]$.

For example, $R_{\frac{\pi}{2}}$ and R_π have matrices $\left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array} \right]$ and $\left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array} \right]$, respectively, by Theorem 2.6.4. The first of these confirms the result in Example 2.2.15. The second shows that rotating a vector $\mathbf{x} = \left[\begin{array}{c} x \\ y \end{array} \right]$ through the angle π results in $R_\pi(\mathbf{x}) = \left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{c} x \\ y \end{array} \right] = \left[\begin{array}{c} -x \\ -y \end{array} \right] = -\mathbf{x}$. Thus applying R_π is the same as negating $\mathbf{x}$, a fact that is evident without Theorem 2.6.4.

Example 2.6.8

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation through $-\frac{\pi}{2}$ followed by reflection in the y axis. Show that T is a reflection in a line through the origin and find the line.

Solution. The matrix of $R_{-\frac{\pi}{2}}$ is $\left[\begin{array}{cc} \cos(-\frac{\pi}{2}) & -\sin(-\frac{\pi}{2}) \\ \sin(-\frac{\pi}{2}) & \cos(-\frac{\pi}{2}) \end{array}\right] = \left[\begin{array}{rr} 0 & 1 \\ -1 & 0 \end{array}\right]$ and the matrix of reflection in the y axis is $\left[\begin{array}{rr} -1 & 0 \\ 0 & 1 \end{array}\right]$. Hence the matrix of T is $\left[\begin{array}{rr} -1 & 0 \\ 0 & 1 \end{array}\right]\left[\begin{array}{rr} 0 & 1 \\ -1 & 0 \end{array}\right] = \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array}\right]$ and this is reflection in the line $y = -x$ (take $m = -1$ in Theorem 2.6.5).

Exercises for 2.6

Exercise 2.6.1 Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation.

a. Find $T\begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix}$ if $T\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $T\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

b. Find $T\begin{bmatrix} 5 \\ 6 \\ -13 \end{bmatrix}$ if $T\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and $T\begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

Exercise 2.6.2 Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation.

a. Find $T\begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix}$ if $T\begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ and $T\begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$.

b. Find $T\begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix}$ if $T\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix}$ and $T\begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$.

Exercise 2.6.3 In each case assume that the transformation T is linear, and use Theorem 2.6.2 to obtain the matrix A of T.

a. $T : \mathbb{R}^2 \to \mathbb{R}^2$ is reflection in the line $y = -x$.

b. $T : \mathbb{R}^2 \to \mathbb{R}^2$ is given by $T(\mathbf{x}) = -\mathbf{x}$ for each $\mathbf{x}$ in $\mathbb{R}^2$.

c. $T : \mathbb{R}^2 \to \mathbb{R}^2$ is clockwise rotation through $\frac{\pi}{4}$.

d. $T : \mathbb{R}^2 \to \mathbb{R}^2$ is counterclockwise rotation through $\frac{\pi}{4}$.

Exercise 2.6.4 In each case use Theorem 2.6.2 to obtain the matrix A of the transformation T. You may assume that T is linear in each case.

a. $T : \mathbb{R}^3 \to \mathbb{R}^3$ is reflection in the $x-z$ plane.

b. $T : \mathbb{R}^3 \to \mathbb{R}^3$ is reflection in the $y-z$ plane.

Exercise 2.6.5 Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.

a. If $\mathbf{x}$ is in $\mathbb{R}^n$, we say that $\mathbf{x}$ is in the *kernel* of T if $T(\mathbf{x}) = \mathbf{0}$. If $\mathbf{x}_1$ and $\mathbf{x}_2$ are both in the kernel of T, show that $a\mathbf{x}_1 + b\mathbf{x}_2$ is also in the kernel of T for all scalars a and b.

b. If $\mathbf{y}$ is in $\mathbb{R}^n$, we say that $\mathbf{y}$ is in the *image* of T if $\mathbf{y} = T(\mathbf{x})$ for some $\mathbf{x}$ in $\mathbb{R}^n$. If $\mathbf{y}_1$ and $\mathbf{y}_2$ are both in the image of T, show that $a\mathbf{y}_1 + b\mathbf{y}_2$ is also in the image of T for all scalars a and b.

Exercise 2.6.6 Use Theorem 2.6.2 to find the matrix of the **identity transformation** $1_{\mathbb{R}^n} : \mathbb{R}^n \to \mathbb{R}^n$ defined by $1_{\mathbb{R}^n}(\mathbf{x}) = \mathbf{x}$ for each $\mathbf{x}$ in $\mathbb{R}^n$.

Exercise 2.6.7 In each case show that $T : \mathbb{R}^2 \to \mathbb{R}^2$ is not a linear transformation.

a. $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} xy \\ 0 \end{bmatrix}$

b. $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ y^2 \end{bmatrix}$

Exercise 2.6.9 Express reflection in the line $y = -x$ as the composition of a rotation followed by reflection in the line $y = x$.

Exercise 2.6.10 Find the matrix of $T : \mathbb{R}^3 \to \mathbb{R}^3$ in each case:

a. T is rotation through θ about the x axis (from the y axis to the z axis).

b. T is rotation through θ about the y axis (from the x axis to the z axis).

Exercise 2.6.11 Let $T_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ denote reflection in the line making an angle θ with the positive x axis.

a. Show that the matrix of T_θ is $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$ for all θ.

b. Show that $T_\theta \circ R_{2\phi} = T_{\theta-\phi}$ for all θ and ϕ.

Exercise 2.6.12 In each case find a rotation or reflection that equals the given transformation.

a. Reflection in the y axis followed by rotation through $\frac{\pi}{2}$.

b. Rotation through π followed by reflection in the x axis.

c. Rotation through $\frac{\pi}{2}$ followed by reflection in the line $y = x$.

d. Reflection in the x axis followed by rotation through $\frac{\pi}{2}$.

e. Reflection in the line $y = x$ followed by reflection in the x axis.

f. Reflection in the x axis followed by reflection in the line $y = x$.

Exercise 2.6.13 Let R and S be matrix transformations $\mathbb{R}^n \to \mathbb{R}^m$ induced by matrices A and B respectively. In each case, show that T is a matrix transformation and describe its matrix in terms of A and B.

a. $T(\mathbf{x}) = R(\mathbf{x}) + S(\mathbf{x})$ for all $\mathbf{x}$ in $\mathbb{R}^n$.

b. $T(\mathbf{x}) = aR(\mathbf{x})$ for all $\mathbf{x}$ in $\mathbb{R}^n$ (where a is a fixed real number).

Exercise 2.6.14 Show that the following hold for all linear transformations $T : \mathbb{R}^n \to \mathbb{R}^m$:

a. $T(\mathbf{0}) = \mathbf{0}$

b. $T(-\mathbf{x}) = -T(\mathbf{x})$ for all $\mathbf{x}$ in $\mathbb{R}^n$

Exercise 2.6.15 The transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x}$ in $\mathbb{R}^n$ is called the **zero transformation**.

a. Show that the zero transformation is linear and find its matrix.

b. Let $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ denote the columns of the $n \times n$ identity matrix. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear and $T(\mathbf{e}_i) = \mathbf{0}$ for each i, show that T is the zero transformation. [*Hint*: Theorem 2.6.1.]

Exercise 2.6.16 Write the elements of $\mathbb{R}^n$ and $\mathbb{R}^m$ as rows. If A is an $m \times n$ matrix, define $T : \mathbb{R}^m \to \mathbb{R}^n$ by $T(\mathbf{y}) = \mathbf{y}A$ for all rows $\mathbf{y}$ in $\mathbb{R}^m$. Show that:

a. T is a linear transformation.

b. the rows of A are $T(\mathbf{f}_1), T(\mathbf{f}_2), \dots, T(\mathbf{f}_m)$ where $\mathbf{f}_i$ denotes row i of I_m. [*Hint*: Show that $\mathbf{f}_i A$ is row i of A.]

Exercise 2.6.17 Let $S : \mathbb{R}^n \to \mathbb{R}^n$ and $T : \mathbb{R}^n \to \mathbb{R}^n$ be linear transformations with matrices A and B respectively.

a. Show that $B^2 = B$ if and only if $T^2 = T$ (where T^2 means $T \circ T$).

b. Show that $B^2 = I$ if and only if $T^2 = 1_{\mathbb{R}^n}$.

c. Show that $AB = BA$ if and only if $S \circ T = T \circ S$.

[*Hint*: Theorem 2.6.3.]

Exercise 2.6.18 Let $Q_0 : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the x axis, let $Q_1 : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the line $y = x$, let $Q_{-1} : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the line $y = -x$, and let $R_{\frac{\pi}{2}} : \mathbb{R}^2 \to \mathbb{R}^2$ be counterclockwise rotation through $\frac{\pi}{2}$.

a. Show that $Q_1 \circ R_{\frac{\pi}{2}} = Q_0$.

b. Show that $Q_1 \circ Q_0 = R_{\frac{\pi}{2}}$.

c. Show that $R_{\frac{\pi}{2}} \circ Q_0 = Q_1$.

d. Show that $Q_0 \circ R_{\frac{\pi}{2}} = Q_{-1}$.

Exercise 2.6.19 For any slope m, show that:

a. $Q_m \circ P_m = P_m$

b. $P_m \circ Q_m = P_m$

Exercise 2.6.20 Define $T : \mathbb{R}^n \to \mathbb{R}$ by $T(x_1, x_2, \dots, x_n) = x_1 + x_2 + \cdots + x_n$. Show that T is a linear transformation and find its matrix.

Exercise 2.6.21 Given c in $\mathbb{R}$, define $T_c : \mathbb{R}^n \to \mathbb{R}$ by $T_c(\mathbf{x}) = c\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Show that T_c is a linear transformation and find its matrix.

Exercise 2.6.22 Given vectors $\mathbf{w}$ and $\mathbf{x}$ in $\mathbb{R}^n$, denote their dot product by $\mathbf{w} \cdot \mathbf{x}$.

a. Given $\mathbf{w}$ in $\mathbb{R}^n$, define $T_{\mathbf{w}} : \mathbb{R}^n \to \mathbb{R}$ by $T_{\mathbf{w}}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Show that $T_{\mathbf{w}}$ is a linear transformation.

b. Show that *every* linear transformation $T : \mathbb{R}^n \to \mathbb{R}$ is given as in (a); that is $T = T_{\mathbf{w}}$ for some $\mathbf{w}$ in $\mathbb{R}^n$.

Exercise 2.6.23 If $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{y}$ are vectors in $\mathbb{R}^n$, show that there is a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{y}$. [*Hint*: By Definition 2.5, find a matrix A such that $A\mathbf{x} = \mathbf{y}$.]

Exercise 2.6.24 Let $\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m \xrightarrow{S} \mathbb{R}^k$ be two linear transformations. Show directly that $S \circ T$ is linear. That is:

a. Show that $(S \circ T)(\mathbf{x} + \mathbf{y}) = (S \circ T)\mathbf{x} + (S \circ T)\mathbf{y}$ for all $\mathbf{x}$, $\mathbf{y}$ in $\mathbb{R}^n$.

b. Show that $(S \circ T)(a\mathbf{x}) = a[(S \circ T)\mathbf{x}]$ for all $\mathbf{x}$ in $\mathbb{R}^n$ and all a in $\mathbb{R}$.

Exercise 2.6.25 Let $\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m \xrightarrow{S} \mathbb{R}^k \xrightarrow{R} \mathbb{R}^k$ be linear. Show that $R \circ (S \circ T) = (R \circ S) \circ T$ by showing directly that $[R \circ (S \circ T)](\mathbf{x}) = [(R \circ S) \circ T)](\mathbf{x})$ holds for each vector $\mathbf{x}$ in $\mathbb{R}^n$.

2.9 An Application to Markov Chains

Many natural phenomena progress through various stages and can be in a variety of states at each stage. For example, the weather in a given city progresses day by day and, on any given day, may be sunny or rainy. Here the states are "sun" and "rain," and the weather progresses from one state to another in daily stages. Another example might be a football team: The stages of its evolution are the games it plays, and the possible states are "win," "draw," and "loss."

The general setup is as follows: A real conceptual "system" is run generating a sequence of outcomes. The system evolves through a series of "stages," and at any stage it can be in any one of a finite number of "states." At any given stage, the state to which it will go at the next stage depends on the past and present history of the system—that is, on the sequence of states it has occupied to date.

Definition 2.15 Markov Chain

A ***Markov chain*** *is such an evolving system wherein the state to which it will go next depends only on its present state and does not depend on the earlier history of the system.*[19]

Even in the case of a Markov chain, the state the system will occupy at any stage is determined only in terms of probabilities. In other words, chance plays a role. For example, if a football team wins a particular game, we do not know whether it will win, draw, or lose the next game. On the other hand, we may know that the team tends to persist in winning streaks; for example, if it wins one game it may win the next game $\frac{1}{2}$ of the time, lose $\frac{4}{10}$ of the time, and draw $\frac{1}{10}$ of the time. These fractions are called the **probabilities** of these various possibilities. Similarly, if the team loses, it may lose the next game with probability $\frac{1}{2}$ (that is, half the time), win with probability $\frac{1}{4}$, and draw with probability $\frac{1}{4}$. The probabilities of the various outcomes after a drawn game will also be known.

We shall treat probabilities informally here: *The probability that a given event will occur is the long-run proportion of the time that the event does indeed occur*. Hence, all probabilities are numbers between 0 and 1. A probability of 0 means the event is impossible and never occurs; events with probability 1 are certain to occur.

If a Markov chain is in a particular state, the probabilities that it goes to the various states at the next stage of its evolution are called the **transition probabilities** for the chain, and they are assumed to be known quantities. To motivate the general conditions that follow, consider the following simple example. Here the system is a man, the stages are his successive lunches, and the states are the two restaurants he chooses.

Example 2.9.1

A man always eats lunch at one of two restaurants, A and B. He never eats at A twice in a row. However, if he eats at B, he is three times as likely to eat at B next time as at A. Initially, he is equally likely to eat at either restaurant.

a. What is the probability that he eats at A on the third day after the initial one?

b. What proportion of his lunches does he eat at A?

[19]The name honours Andrei Andreyevich Markov (1856–1922) who was a professor at the university in St. Petersburg, Russia.

Solution. The table of transition probabilities follows. The A column indicates that if he eats at A on one day, he never eats there again on the next day and so is certain to go to B.

		Present Lunch	
		A	B
Next	A	0	0.25
Lunch	B	1	0.75

The B column shows that, if he eats at B on one day, he will eat there on the next day $\frac{3}{4}$ of the time and switches to A only $\frac{1}{4}$ of the time.
The restaurant he visits on a given day is not determined. The most that we can expect is to know the probability that he will visit A or B on that day.

Let $\mathbf{s}_m = \left[\begin{array}{c} s_1^{(m)} \\ s_2^{(m)} \end{array}\right]$ denote the *state vector* for day m. Here $s_1^{(m)}$ denotes the probability that he eats at A on day m, and $s_2^{(m)}$ is the probability that he eats at B on day m. It is convenient to let $\mathbf{s}_0$ correspond to the initial day. Because he is equally likely to eat at A or B on that initial day, $s_1^{(0)} = 0.5$ and $s_2^{(0)} = 0.5$, so $\mathbf{s}_0 = \left[\begin{array}{c} 0.5 \\ 0.5 \end{array}\right]$. Now let

$$P = \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]$$

denote the *transition matrix*. We claim that the relationship

$$\mathbf{s}_{m+1} = P\mathbf{s}_m$$

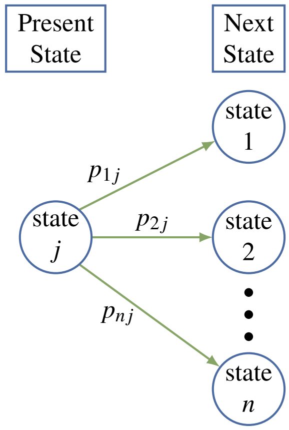
holds for all integers $m \geq 0$. This will be derived later; for now, we use it as follows to successively compute $\mathbf{s}_1,\ \mathbf{s}_2,\ \mathbf{s}_3,\ \dots$.

$$\begin{aligned}
\mathbf{s}_1 = P\mathbf{s}_0 &= \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]\left[\begin{array}{c} 0.5 \\ 0.5 \end{array}\right] = \left[\begin{array}{c} 0.125 \\ 0.875 \end{array}\right] \\
\mathbf{s}_2 = P\mathbf{s}_1 &= \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]\left[\begin{array}{c} 0.125 \\ 0.875 \end{array}\right] = \left[\begin{array}{c} 0.21875 \\ 0.78125 \end{array}\right] \\
\mathbf{s}_3 = P\mathbf{s}_2 &= \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]\left[\begin{array}{c} 0.21875 \\ 0.78125 \end{array}\right] = \left[\begin{array}{c} 0.1953125 \\ 0.8046875 \end{array}\right]
\end{aligned}$$

Hence, the probability that his third lunch (after the initial one) is at A is approximately 0.195, whereas the probability that it is at B is 0.805. If we carry these calculations on, the next state vectors are (to five figures):

$$\begin{array}{cc}
\mathbf{s}_4 = \left[\begin{array}{c} 0.20117 \\ 0.79883 \end{array}\right] & \mathbf{s}_5 = \left[\begin{array}{c} 0.19971 \\ 0.80029 \end{array}\right] \\
\mathbf{s}_6 = \left[\begin{array}{c} 0.20007 \\ 0.79993 \end{array}\right] & \mathbf{s}_7 = \left[\begin{array}{c} 0.19998 \\ 0.80002 \end{array}\right]
\end{array}$$

Moreover, as m increases the entries of $\mathbf{s}_m$ get closer and closer to the corresponding entries of $\left[\begin{array}{c} 0.2 \\ 0.8 \end{array}\right]$. Hence, in the long run, he eats 20% of his lunches at A and 80% at B.



Example 2.9.1 incorporates most of the essential features of all Markov chains. The general model is as follows: The system evolves through various stages and at each stage can be in exactly one of n distinct states. It progresses through a sequence of states as time goes on. If a Markov chain is in state j at a particular stage of its development, the probability p_{ij} that it goes to state i at the next stage is called the **transition probability**. The $n \times n$ matrix $P = \left[p_{ij}\right]$ is called the **transition matrix** for the Markov chain. The situation is depicted graphically in the diagram.

We make one important assumption about the transition matrix $P = \left[p_{ij}\right]$: It does *not* depend on which stage the process is in. This assumption means that the transition probabilities are *independent of time*—that is, they do not change as time goes on. It is this assumption that distinguishes Markov chains in the literature of this subject.

Example 2.9.2

Suppose the transition matrix of a three-state Markov chain is

$$P = \left[\begin{array}{ccc} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{array}\right] = \begin{array}{c} \text{Present state} \\ \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \left[\begin{array}{ccc} 0.3 & 0.1 & 0.6 \\ 0.5 & 0.9 & 0.2 \\ 0.2 & 0.0 & 0.2 \end{array}\right] \end{array} \begin{array}{l} \\ \\ 1 \\ 2 \quad \text{Next state} \\ 3 \end{array}$$

If, for example, the system is in state 2, then column 2 lists the probabilities of where it goes next. Thus, the probability is $p_{12} = 0.1$ that it goes from state 2 to state 1, and the probability is $p_{22} = 0.9$ that it goes from state 2 to state 2. The fact that $p_{32} = 0$ means that it is impossible for it to go from state 2 to state 3 at the next stage.

Consider the jth column of the transition matrix P.

$$\left[\begin{array}{c} p_{1j} \\ p_{2j} \\ \vdots \\ p_{nj} \end{array}\right]$$

If the system is in state j at some stage of its evolution, the transition probabilities $p_{1j}, p_{2j}, \dots, p_{nj}$ represent the fraction of the time that the system will move to state 1, state 2, ..., state n, respectively, at the next stage. We assume that it has to go to *some* state at each transition, so the sum of these probabilities is 1:

$$p_{1j} + p_{2j} + \cdots + p_{nj} = 1 \quad \text{for each } j$$

Thus, the columns of P all sum to 1 and the entries of P lie between 0 and 1. Hence P is called a **stochastic matrix**.

As in Example 2.9.1, we introduce the following notation: Let $s_i^{(m)}$ denote the probability that the system is in state i after m transitions. The $n \times 1$ matrices

$$\mathbf{s}_m = \begin{bmatrix} s_1^{(m)} \\ s_2^{(m)} \\ \vdots \\ s_n^{(m)} \end{bmatrix} \quad m = 0,\ 1,\ 2,\ \dots$$

are called the **state vectors** for the Markov chain. Note that the sum of the entries of $\mathbf{s}_m$ must equal 1 because the system must be in *some* state after m transitions. The matrix $\mathbf{s}_0$ is called the **initial state vector** for the Markov chain and is given as part of the data of the particular chain. For example, if the chain has only two states, then an initial vector $\mathbf{s}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ means that it started in state 1. If it started in state 2, the initial vector would be $\mathbf{s}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. If $\mathbf{s}_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$, it is equally likely that the system started in state 1 or in state 2.

Theorem 2.9.1

Let P be the transition matrix for an n-state Markov chain. If $\mathbf{s}_m$ is the state vector at stage m, then

$$\mathbf{s}_{m+1} = P\mathbf{s}_m$$

for each $m = 0,\ 1,\ 2,\ \dots$.

Heuristic Proof. Suppose that the Markov chain has been run N times, each time starting with the same initial state vector. Recall that p_{ij} is the proportion of the time the system goes from state j at some stage to state i at the next stage, whereas $s_i^{(m)}$ is the proportion of the time it is in state i at stage m. Hence

$$s_i^{m+1}N$$

is (approximately) the number of times the system is in state i at stage $m+1$. We are going to calculate this number another way. The system got to state i at stage $m+1$ through *some* other state (say state j) at stage m. The number of times it was *in* state j at that stage is (approximately) $s_j^{(m)}N$, so the number of times it got to state i via state j is $p_{ij}(s_j^{(m)}N)$. Summing over j gives the number of times the system is in state i (at stage $m+1$). This is the number we calculated before, so

$$s_i^{(m+1)}N = p_{i1}s_1^{(m)}N + p_{i2}s_2^{(m)}N + \cdots + p_{in}s_n^{(m)}N$$

Dividing by N gives $s_i^{(m+1)} = p_{i1}s_1^{(m)} + p_{i2}s_2^{(m)} + \cdots + p_{in}s_n^{(m)}$ for each i, and this can be expressed as the matrix equation $\mathbf{s}_{m+1} = P\mathbf{s}_m$. □

If the initial probability vector $\mathbf{s}_0$ and the transition matrix P are given, Theorem 2.9.1 gives $\mathbf{s}_1,\ \mathbf{s}_2,\ \mathbf{s}_3,\ \dots,$ one after the other, as follows:

$$\begin{aligned} \mathbf{s}_1 &= P\mathbf{s}_0 \\ \mathbf{s}_2 &= P\mathbf{s}_1 \\ \mathbf{s}_3 &= P\mathbf{s}_2 \\ &\vdots \end{aligned}$$

Hence, the state vector $\mathbf{s}_m$ is completely determined for each $m = 0, 1, 2, \dots$ by P and $\mathbf{s}_0$.

Example 2.9.3

A wolf pack always hunts in one of three regions R_1, R_2, and R_3. Its hunting habits are as follows:

1. If it hunts in some region one day, it is as likely as not to hunt there again the next day.
2. If it hunts in R_1, it never hunts in R_2 the next day.
3. If it hunts in R_2 or R_3, it is equally likely to hunt in each of the other regions the next day.

If the pack hunts in R_1 on Monday, find the probability that it hunts there on Thursday.

Solution. The stages of this process are the successive days; the states are the three regions. The transition matrix P is determined as follows (see the table): The first habit asserts that $p_{11} = p_{22} = p_{33} = \frac{1}{2}$. Now column 1 displays what happens when the pack starts in R_1: It never goes to state 2, so $p_{21} = 0$ and, because the column must sum to 1, $p_{31} = \frac{1}{2}$. Column 2 describes what happens if it starts in R_2: $p_{22} = \frac{1}{2}$ and p_{12} and p_{32} are equal (by habit 3), so $p_{12} = p_{32} = \frac{1}{2}$ because the column sum must equal 1. Column 3 is filled in a similar way.

	R_1	R_2	R_3
R_1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
R_2	0	$\frac{1}{2}$	$\frac{1}{4}$
R_3	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$

Now let Monday be the initial stage. Then $\mathbf{s}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ because the pack hunts in R_1 on that day. Then $\mathbf{s}_1$, $\mathbf{s}_2$, and $\mathbf{s}_3$ describe Tuesday, Wednesday, and Thursday, respectively, and we compute them using Theorem 2.9.1.

$$\mathbf{s}_1 = P\mathbf{s}_0 = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix} \quad \mathbf{s}_2 = P\mathbf{s}_1 = \begin{bmatrix} \frac{3}{8} \\ \frac{1}{8} \\ \frac{4}{8} \end{bmatrix} \quad \mathbf{s}_3 = P\mathbf{s}_2 = \begin{bmatrix} \frac{11}{32} \\ \frac{6}{32} \\ \frac{15}{32} \end{bmatrix}$$

Hence, the probability that the pack hunts in Region R_1 on Thursday is $\frac{11}{32}$.

Steady State Vector

Another phenomenon that was observed in Example 2.9.1 can be expressed in general terms. The state vectors $\mathbf{s}_0, \mathbf{s}_1, \mathbf{s}_2, \dots$ were calculated in that example and were found to "approach" $\mathbf{s} = \left[\begin{array}{c} 0.2 \\ 0.8 \end{array}\right]$. This means that the first component of $\mathbf{s}_m$ becomes and remains very close to 0.2 as m becomes large, whereas the second component gets close to 0.8 as m increases. When this is the case, we say that $\mathbf{s}_m$ **converges** to $\mathbf{s}$. For large m, then, there is very little error in taking $\mathbf{s}_m = \mathbf{s}$, so the long-term probability that the system is in state 1 is 0.2, whereas the probability that it is in state 2 is 0.8. In Example 2.9.1, enough state vectors were computed for the limiting vector $\mathbf{s}$ to be apparent. However, there is a better way to do this that works in most cases.

Suppose P is the transition matrix of a Markov chain, and assume that the state vectors $\mathbf{s}_m$ converge to a limiting vector $\mathbf{s}$. Then $\mathbf{s}_m$ is very close to $\mathbf{s}$ for sufficiently large m, so $\mathbf{s}_{m+1}$ is also very close to $\mathbf{s}$. Thus, the equation $\mathbf{s}_{m+1} = P\mathbf{s}_m$ from Theorem 2.9.1 is closely approximated by

$$\mathbf{s} = P\mathbf{s}$$

so it is not surprising that $\mathbf{s}$ should be a solution to this matrix equation. Moreover, it is easily solved because it can be written as a system of homogeneous linear equations

$$(I - P)\mathbf{s} = \mathbf{0}$$

with the entries of $\mathbf{s}$ as variables.

In Example 2.9.1, where $P = \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]$, the general solution to $(I - P)\mathbf{s} = \mathbf{0}$ is $\mathbf{s} = \left[\begin{array}{c} t \\ 4t \end{array}\right]$, where t is a parameter. But if we insist that the entries of S sum to 1 (as must be true of all state vectors), we find $t = 0.2$ and so $\mathbf{s} = \left[\begin{array}{c} 0.2 \\ 0.8 \end{array}\right]$ as before.

All this is predicated on the existence of a limiting vector for the sequence of state vectors of the Markov chain, and such a vector may not always exist. However, it does exist in one commonly occurring situation. A stochastic matrix P is called **regular** if some power P^m of P has every entry greater than zero. The matrix $P = \left[\begin{array}{cc} 0 & 0.25 \\ 1 & 0.75 \end{array}\right]$ of Example 2.9.1 is regular (in this case, each entry of P^2 is positive), and the general theorem is as follows:

Theorem 2.9.2

Let P be the transition matrix of a Markov chain and assume that P is regular. Then there is a unique column matrix $\mathbf{s}$ satisfying the following conditions:

1. *$P\mathbf{s} = \mathbf{s}$.*
2. *The entries of $\mathbf{s}$ are positive and sum to 1.*

Moreover, condition 1 can be written as

$$(I - P)\mathbf{s} = \mathbf{0}$$

and so gives a homogeneous system of linear equations for $\mathbf{s}$. Finally, the sequence of state vectors $\mathbf{s}_0, \mathbf{s}_1, \mathbf{s}_2, \dots$ converges to $\mathbf{s}$ in the sense that if m is large enough, each entry of $\mathbf{s}_m$ is closely approximated by the corresponding entry of $\mathbf{s}$.

This theorem will not be proved here.[20]

If P is the regular transition matrix of a Markov chain, the column $\mathbf{s}$ satisfying conditions 1 and 2 of Theorem 2.9.2 is called the **steady-state vector** for the Markov chain. The entries of $\mathbf{s}$ are the long-term probabilities that the chain will be in each of the various states.

Example 2.9.4

A man eats one of three soups—beef, chicken, and vegetable—each day. He never eats the same soup two days in a row. If he eats beef soup on a certain day, he is equally likely to eat each of the others the next day; if he does not eat beef soup, he is twice as likely to eat it the next day as the alternative.

a. If he has beef soup one day, what is the probability that he has it again two days later?

b. What are the long-run probabilities that he eats each of the three soups?

Solution. The states here are B, C, and V, the three soups. The transition matrix P is given in the table. (Recall that, for each state, the corresponding column lists the probabilities for the next state.)

	B	C	V
B	0	$\frac{2}{3}$	$\frac{2}{3}$
C	$\frac{1}{2}$	0	$\frac{1}{3}$
V	$\frac{1}{2}$	$\frac{1}{3}$	0

If he has beef soup initially, then the initial state vector is

$$\mathbf{s}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Then two days later the state vector is $\mathbf{s}_2$. If P is the transition matrix, then

$$\mathbf{s}_1 = P\mathbf{s}_0 = \frac{1}{2}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{s}_2 = P\mathbf{s}_1 = \frac{1}{6}\begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

so he eats beef soup two days later with probability $\frac{2}{3}$. This answers (a.) and also shows that he eats chicken and vegetable soup each with probability $\frac{1}{6}$.

To find the long-run probabilities, we must find the steady-state vector $\mathbf{s}$. Theorem 2.9.2 applies because P is regular (P^2 has positive entries), so $\mathbf{s}$ satisfies $P\mathbf{s} = \mathbf{s}$. That is, $(I - P)\mathbf{s} = \mathbf{0}$ where

$$I - P = \frac{1}{6}\begin{bmatrix} 6 & -4 & -4 \\ -3 & 6 & -2 \\ -3 & -2 & 6 \end{bmatrix}$$

[20]The interested reader can find an elementary proof in J. Kemeny, H. Mirkil, J. Snell, and G. Thompson, Finite Mathematical Structures (Englewood Cliffs, N.J.: Prentice-Hall, 1958).

The solution is $\mathbf{s} = \begin{bmatrix} 4t \\ 3t \\ 3t \end{bmatrix}$, where t is a parameter, and we use $\mathbf{s} = \begin{bmatrix} 0.4 \\ 0.3 \\ 0.3 \end{bmatrix}$ because the entries of $\mathbf{s}$ must sum to 1. Hence, in the long run, he eats beef soup 40% of the time and eats chicken soup and vegetable soup each 30% of the time.

Exercises for 2.9

Exercise 2.9.1 Which of the following stochastic matrices is regular?

a. $\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$ b. $\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{4} & 1 & \frac{1}{3} \\ \frac{1}{4} & 0 & \frac{1}{3} \end{bmatrix}$

Exercise 2.9.2 In each case find the steady-state vector and, assuming that it starts in state 1, find the probability that it is in state 2 after 3 transitions.

a. $\begin{bmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{bmatrix}$ b. $\begin{bmatrix} \frac{1}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix}$

c. $\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{4} \\ 1 & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ d. $\begin{bmatrix} 0.4 & 0.1 & 0.5 \\ 0.2 & 0.6 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$

e. $\begin{bmatrix} 0.8 & 0.0 & 0.2 \\ 0.1 & 0.6 & 0.1 \\ 0.1 & 0.4 & 0.7 \end{bmatrix}$ f. $\begin{bmatrix} 0.1 & 0.3 & 0.3 \\ 0.3 & 0.1 & 0.6 \\ 0.6 & 0.6 & 0.1 \end{bmatrix}$

Exercise 2.9.3 A fox hunts in three territories A, B, and C. He never hunts in the same territory on two successive days. If he hunts in A, then he hunts in C the next day. If he hunts in B or C, he is twice as likely to hunt in A the next day as in the other territory.

a. What proportion of his time does he spend in A, in B, and in C?

b. If he hunts in A on Monday (C on Monday), what is the probability that he will hunt in B on Thursday?

Exercise 2.9.4 Assume that there are three social classes—upper, middle, and lower—and that social mobility behaves as follows:

1. Of the children of upper-class parents, 70% remain upper-class, whereas 10% become middle-class and 20% become lower-class.
2. Of the children of middle-class parents, 80% remain middle-class, whereas the others are evenly split between the upper class and the lower class.
3. For the children of lower-class parents, 60% remain lower-class, whereas 30% become middle-class and 10% upper-class.

a. Find the probability that the grandchild of lower-class parents becomes upper-class.

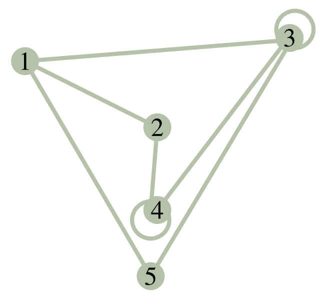
b. Find the long-term breakdown of society into classes.

Exercise 2.9.5 The prime minister says she will call an election. This gossip is passed from person to person with a probability $p \neq 0$ that the information is passed incorrectly at any stage. Assume that when a person hears the gossip he or she passes it to one person who does not know. Find the long-term probability that a person will hear that there is going to be an election.

Exercise 2.9.6 John makes it to work on time one Monday out of four. On other work days his behaviour is as follows: If he is late one day, he is twice as likely to come to work on time the next day as to be late. If he is on time one day, he is as likely to be late as not the next day. Find the probability of his being late and that of his being on time Wednesdays.

Exercise 2.9.7 Suppose you have 1¢ and match coins with a friend. At each match you either win or lose 1¢ with equal probability. If you go broke or ever get 4¢, you quit. Assume your friend never quits. If the states are 0, 1, 2, 3, and 4 representing your wealth, show that the corresponding transition matrix P is not regular. Find the probability that you will go broke after 3 matches.

Exercise 2.9.8 A mouse is put into a maze of compartments, as in the diagram. Assume that he always leaves any compartment he enters and that he is equally likely to take any tunnel entry.



a. If he starts in compartment 1, find the probability that he is in compartment 1 again after 3 moves.

b. Find the compartment in which he spends most of his time if he is left for a long time.

Exercise 2.9.9 If a stochastic matrix has a 1 on its main diagonal, show that it cannot be regular. Assume it is not 1×1.

Exercise 2.9.10 If $\mathbf{s}_m$ is the stage-m state vector for a Markov chain, show that $\mathbf{s}_{m+k} = P^k \mathbf{s}_m$ holds for all $m \geq 1$ and $k \geq 1$ (where P is the transition matrix).

Exercise 2.9.11 A stochastic matrix is **doubly stochastic** if all the row sums also equal 1. Find the steady-state vector for a doubly stochastic matrix.

Exercise 2.9.12 Consider the 2×2 stochastic matrix

$$P = \left[\begin{array}{cc} 1-p & q \\ p & 1-q \end{array} \right],$$

where $0 < p < 1$ and $0 < q < 1$.

a. Show that $\frac{1}{p+q} \left[\begin{array}{c} q \\ p \end{array} \right]$ is the steady-state vector for P.

b. Show that P^m converges to the matrix $\frac{1}{p+q} \left[\begin{array}{cc} q & q \\ p & p \end{array} \right]$ by first verifying inductively that $P^m = \frac{1}{p+q} \left[\begin{array}{cc} q & q \\ p & p \end{array} \right] + \frac{(1-p-q)^m}{p+q} \left[\begin{array}{cc} p & -q \\ -p & q \end{array} \right]$ for $m = 1, 2, \dots$. (It can be shown that the sequence of powers P, P^2, P^3, $\dots$ of any regular transition matrix converges to the matrix each of whose columns equals the steady-state vector for P.)

Supplementary Exercises for Chapter 2

Exercise 2.1 Solve for the matrix X if:

a. $PXQ = R$;

b. $XP = S$;

where $P = \left[\begin{array}{rr} 1 & 0 \\ 2 & -1 \\ 0 & 3 \end{array} \right]$, $Q = \left[\begin{array}{rrr} 1 & 1 & -1 \\ 2 & 0 & 3 \end{array} \right]$,
$R = \left[\begin{array}{rrr} -1 & 1 & -4 \\ -4 & 0 & -6 \\ 6 & 6 & -6 \end{array} \right]$, $S = \left[\begin{array}{rr} 1 & 6 \\ 3 & 1 \end{array} \right]$

Exercise 2.2 Consider

$$p(X) = X^3 - 5X^2 + 11X - 4I.$$

a. If $p(U) = \left[\begin{array}{rr} 1 & 3 \\ -1 & 0 \end{array} \right]$ compute $p(U^T)$.

b. If $p(U) = 0$ where U is $n \times n$, find U^{-1} in terms of U.

Exercise 2.3 Show that, if a (possibly nonhomogeneous) system of equations is consistent and has more variables than equations, then it must have infinitely many solutions. [*Hint*: Use Theorem 2.2.2 and Theorem 1.3.1.]

Exercise 2.4 Assume that a system $A\mathbf{x} = \mathbf{b}$ of linear equations has at least two distinct solutions $\mathbf{y}$ and $\mathbf{z}$.

a. Show that $\mathbf{x}_k = \mathbf{y} + k(\mathbf{y} - \mathbf{z})$ is a solution for every k.

b. Show that $\mathbf{x}_k = \mathbf{x}_m$ implies $k = m$. [*Hint*: See Example 2.1.7.]

c. Deduce that $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.

Exercise 2.5

a. Let A be a 3×3 matrix with all entries on and below the main diagonal zero. Show that $A^3 = 0$.

b. Generalize to the $n \times n$ case and prove your answer.

Exercise 2.6 Let I_{pq} denote the $n \times n$ matrix with (p, q)-entry equal to 1 and all other entries 0. Show that:

a. $I_n = I_{11} + I_{22} + \cdots + I_{nn}$.

b. $I_{pq}I_{rs} = \begin{cases} I_{ps} & \text{if } q = r \\ 0 & \text{if } q \neq r \end{cases}$.

c. If $A = [a_{ij}]$ is $n \times n$, then $A = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}I_{ij}$.

d. If $A = [a_{ij}]$, then $I_{pq}AI_{rs} = a_{qr}I_{ps}$ for all p, q, r, and s.

Exercise 2.7 A matrix of the form aI_n, where a is a number, is called an $n \times n$ **scalar matrix**.

a. Show that each $n \times n$ scalar matrix commutes with every $n \times n$ matrix.

b. Show that A is a scalar matrix if it commutes with every $n \times n$ matrix. [*Hint*: See part (d.) of Exercise 2.6.]

Exercise 2.9 If A is 2×2, show that $A^{-1} = A^T$ if and only if $A = \left[\begin{array}{rr} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{array} \right]$ for some θ or $A = \left[\begin{array}{rr} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{array} \right]$ for some θ.

[*Hint*: If $a^2 + b^2 = 1$, then $a = \cos\theta$, $b = \sin\theta$ for some θ. Use

$$\cos(\theta - \phi) = \cos\theta\cos\phi + \sin\theta\sin\phi.]$$

Exercise 2.10

a. If $A = \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array} \right]$, show that $A^2 = I$.

b. What is wrong with the following argument? If $A^2 = I$, then $A^2 - I = 0$, so $(A - I)(A + I) = 0$, whence $A = I$ or $A = -I$.

Exercise 2.11 Let E and F be elementary matrices obtained from the identity matrix by adding multiples of row k to rows p and q. If $k \neq p$ and $k \neq q$, show that $EF = FE$.

Exercise 2.12 If A is a 2×2 real matrix, $A^2 = A$ and $A^T = A$, show that either A is one of $\left[\begin{array}{rr} 0 & 0 \\ 0 & 0 \end{array} \right]$, $\left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array} \right]$, $\left[\begin{array}{rr} 0 & 0 \\ 0 & 1 \end{array} \right]$, $\left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array} \right]$, or $A = \left[\begin{array}{cc} a & b \\ b & 1-a \end{array} \right]$ where $a^2 + b^2 = a$, $-\frac{1}{2} \leq b \leq \frac{1}{2}$ and $b \neq 0$.

Exercise 2.13 Show that the following are equivalent for matrices P, Q:

1. P, Q, and $P + Q$ are all invertible and

$$(P + Q)^{-1} = P^{-1} + Q^{-1}$$

2. P is invertible and $Q = PG$ where $G^2 + G + I = 0$.

Chapter 3

Determinants and Diagonalization

With each square matrix we can calculate a number, called the determinant of the matrix, which tells us whether or not the matrix is invertible. In fact, determinants can be used to give a formula for the inverse of a matrix. They also arise in calculating certain numbers (called eigenvalues) associated with the matrix. These eigenvalues are essential to a technique called diagonalization that is used in many applications where it is desired to predict the future behaviour of a system. For example, we use it to predict whether a species will become extinct.

Determinants were first studied by Leibnitz in 1696, and the term "determinant" was first used in 1801 by Gauss is his *Disquisitiones Arithmeticae*. Determinants are much older than matrices (which were introduced by Cayley in 1878) and were used extensively in the eighteenth and nineteenth centuries, primarily because of their significance in geometry (see Section 4.4). Although they are somewhat less important today, determinants still play a role in the theory and application of matrix algebra.

3.1 The Cofactor Expansion

In Section 2.4 we defined the determinant of a 2×2 matrix $A = \left[\begin{array}{cc} a & b \\ c & d \end{array} \right]$ as follows:[1]

$$\det A = \left| \begin{array}{cc} a & b \\ c & d \end{array} \right| = ad - bc$$

and showed (in Example 2.4.4) that A has an inverse if and only if $\det A \neq 0$. One objective of this chapter is to do this for *any* square matrix A. There is no difficulty for 1×1 matrices: If $A = [a]$, we define $\det A = \det\, [a] = a$ and note that A is invertible if and only if $a \neq 0$.

If A is 3×3 and invertible, we look for a suitable definition of $\det A$ by trying to carry A to the identity matrix by row operations. The first column is not zero (A is invertible); suppose the (1, 1)-entry a is not zero. Then row operations give

$$A = \left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array} \right] \rightarrow \left[\begin{array}{ccc} a & b & c \\ ad & ae & af \\ ag & ah & ai \end{array} \right] \rightarrow \left[\begin{array}{ccc} a & b & c \\ 0 & ae-bd & af-cd \\ 0 & ah-bg & ai-cg \end{array} \right] = \left[\begin{array}{ccc} a & b & c \\ 0 & u & af-cd \\ 0 & v & ai-cg \end{array} \right]$$

where $u = ae - bd$ and $v = ah - bg$. Since A is invertible, one of u and v is nonzero (by Example 2.4.11); suppose that $u \neq 0$. Then the reduction proceeds

$$A \rightarrow \left[\begin{array}{ccc} a & b & c \\ 0 & u & af-cd \\ 0 & v & ai-cg \end{array} \right] \rightarrow \left[\begin{array}{ccc} a & b & c \\ 0 & u & af-cd \\ 0 & uv & u(ai-cg) \end{array} \right] \rightarrow \left[\begin{array}{ccc} a & b & c \\ 0 & u & af-cd \\ 0 & 0 & w \end{array} \right]$$

where $w = u(ai - cg) - v(af - cd) = a(aei + bfg + cdh - ceg - afh - bdi)$. We define

$$\det A = aei + bfg + cdh - ceg - afh - bdi \tag{3.1}$$

[1]Determinants are commonly written $|A| = \det A$ using vertical bars. We will use both notations.

and observe that $\det A \neq 0$ because $a \det A = w \neq 0$ (is invertible).

To motivate the definition below, collect the terms in Equation 3.1 involving the entries a, b, and c in row 1 of A:

$$\begin{aligned}
\det A = \left|\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right| &= aei + bfg + cdh - ceg - afh - bdi \\
&= a(ei - fh) - b(di - fg) + c(dh - eg) \\
&= a\left|\begin{array}{cc} e & f \\ h & i \end{array}\right| - b\left|\begin{array}{cc} d & f \\ g & i \end{array}\right| + c\left|\begin{array}{cc} d & e \\ g & h \end{array}\right|
\end{aligned}$$

This last expression can be described as follows: To compute the determinant of a 3×3 matrix A, multiply each entry in row 1 by a sign times the determinant of the 2×2 matrix obtained by deleting the row and column of that entry, and add the results. The signs alternate down row 1, starting with $+$. It is this observation that we generalize below.

Example 3.1.1

$$\begin{aligned}
\det \left[\begin{array}{rrr} 2 & 3 & 7 \\ -4 & 0 & 6 \\ 1 & 5 & 0 \end{array}\right] &= 2\left|\begin{array}{cc} 0 & 6 \\ 5 & 0 \end{array}\right| - 3\left|\begin{array}{rr} -4 & 6 \\ 1 & 0 \end{array}\right| + 7\left|\begin{array}{rr} -4 & 0 \\ 1 & 5 \end{array}\right| \\
&= 2(-30) - 3(-6) + 7(-20) \\
&= -182
\end{aligned}$$

This suggests an inductive method of defining the determinant of any square matrix in terms of determinants of matrices one size smaller. The idea is to define determinants of 3×3 matrices in terms of determinants of 2×2 matrices, then we do 4×4 matrices in terms of 3×3 matrices, and so on.

To describe this, we need some terminology.

Definition 3.1 Cofactors of a Matrix

Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. Given the $n \times n$ matrix A, let

A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j.

*Then the $(i,\ j)$-**cofactor** $c_{ij}(A)$ is the scalar defined by*

$$c_{ij}(A) = (-1)^{i+j} \det (A_{ij})$$

*Here $(-1)^{i+j}$ is called the **sign** of the $(i,\ j)$-position.*

The sign of a position is clearly 1 or -1, and the following diagram is useful for remembering it:

$$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{bmatrix}$$

Note that the signs alternate along each row and column with $+$ in the upper left corner.

Example 3.1.2

Find the cofactors of positions $(1, 2)$, $(3, 1)$, and $(2, 3)$ in the following matrix.

$$A = \begin{bmatrix} 3 & -1 & 6 \\ 5 & 2 & 7 \\ 8 & 9 & 4 \end{bmatrix}$$

Solution. Here A_{12} is the matrix $\begin{bmatrix} 5 & 7 \\ 8 & 4 \end{bmatrix}$ that remains when row 1 and column 2 are deleted. The sign of position $(1, 2)$ is $(-1)^{1+2} = -1$ (this is also the $(1, 2)$-entry in the sign diagram), so the $(1, 2)$-cofactor is

$$c_{12}(A) = (-1)^{1+2} \begin{vmatrix} 5 & 7 \\ 8 & 4 \end{vmatrix} = (-1)(5 \cdot 4 - 7 \cdot 8) = (-1)(-36) = 36$$

Turning to position $(3, 1)$, we find

$$c_{31}(A) = (-1)^{3+1} A_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 6 \\ 2 & 7 \end{vmatrix} = (+1)(-7-12) = -19$$

Finally, the $(2, 3)$-cofactor is

$$c_{23}(A) = (-1)^{2+3} A_{23} = (-1)^{2+3} \begin{vmatrix} 3 & -1 \\ 8 & 9 \end{vmatrix} = (-1)(27+8) = -35$$

Clearly other cofactors can be found—there are nine in all, one for each position in the matrix.

We can now define $\det A$ for any square matrix A

Definition 3.2 Cofactor expansion of a Matrix

Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. If $A = \left[a_{ij}\right]$ is $n \times n$ define

$$\det A = a_{11}c_{11}(A) + a_{12}c_{12}(A) + \cdots + a_{1n}c_{1n}(A)$$

This is called the ***cofactor expansion*** *of $\det A$ along row 1.*

It asserts that $\det A$ can be computed by multiplying the entries of row 1 by the corresponding cofactors, and adding the results. The astonishing thing is that $\det A$ can be computed by taking the cofactor expansion along *any row or column*: Simply multiply each entry of that row or column by the corresponding cofactor and add.

Theorem 3.1.1: Cofactor Expansion Theorem[2]

The determinant of an $n \times n$ matrix A can be computed by using the cofactor expansion along any row or column of A. That is $\det A$ can be computed by multiplying each entry of the row or column by the corresponding cofactor and adding the results.

The proof will be given in Section 3.6.

Example 3.1.3

Compute the determinant of $A = \left[\begin{array}{rrr} 3 & 4 & 5 \\ 1 & 7 & 2 \\ 9 & 8 & -6 \end{array} \right]$.

Solution. The cofactor expansion along the first row is as follows:

$$\begin{aligned} \det A &= 3c_{11}(A) + 4c_{12}(A) + 5c_{13}(A) \\ &= 3 \left| \begin{array}{rr} 7 & 2 \\ 8 & -6 \end{array} \right| - 4 \left| \begin{array}{rr} 1 & 2 \\ 9 & -6 \end{array} \right| + 5 \left| \begin{array}{rr} 1 & 7 \\ 9 & 8 \end{array} \right| \\ &= 3(-58) - 4(-24) + 5(-55) \\ &= -353 \end{aligned}$$

Note that the signs alternate along the row (indeed along *any* row or column). Now we compute $\det A$ by expanding along the first column.

$$\begin{aligned} \det A &= 3c_{11}(A) + 1c_{21}(A) + 9c_{31}(A) \\ &= 3 \left| \begin{array}{rr} 7 & 2 \\ 8 & -6 \end{array} \right| - \left| \begin{array}{rr} 4 & 5 \\ 8 & -6 \end{array} \right| + 9 \left| \begin{array}{rr} 4 & 5 \\ 7 & 2 \end{array} \right| \\ &= 3(-58) - (-64) + 9(-27) \\ &= -353 \end{aligned}$$

The reader is invited to verify that $\det A$ can be computed by expanding along any other row or column.

The fact that the cofactor expansion along *any row or column* of a matrix A always gives the same result (the determinant of A) is remarkable, to say the least. The choice of a particular row or column can simplify the calculation.

[2]The cofactor expansion is due to Pierre Simon de Laplace (1749–1827), who discovered it in 1772 as part of a study of linear differential equations. Laplace is primarily remembered for his work in astronomy and applied mathematics.

Example 3.1.4

Compute $\det A$ where $A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 5 & 1 & 2 & 0 \\ 2 & 6 & 0 & -1 \\ -6 & 3 & 1 & 0 \end{bmatrix}$.

Solution. The first choice we must make is which row or column to use in the cofactor expansion. The expansion involves multiplying entries by cofactors, so the work is minimized when the row or column contains as many zero entries as possible. Row 1 is a best choice in this matrix (column 4 would do as well), and the expansion is

$$\begin{aligned}\det A &= 3c_{11}(A) + 0c_{12}(A) + 0c_{13}(A) + 0c_{14}(A) \\ &= 3\begin{vmatrix} 1 & 2 & 0 \\ 6 & 0 & -1 \\ 3 & 1 & 0 \end{vmatrix}\end{aligned}$$

This is the first stage of the calculation, and we have succeeded in expressing the determinant of the 4×4 matrix A in terms of the determinant of a 3×3 matrix. The next stage involves this 3×3 matrix. Again, we can use any row or column for the cofactor expansion. The third column is preferred (with two zeros), so

$$\begin{aligned}\det A &= 3\left(0\begin{vmatrix} 6 & 0 \\ 3 & 1 \end{vmatrix} - (-1)\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 2 \\ 6 & 0 \end{vmatrix}\right) \\ &= 3[0 + 1(-5) + 0] \\ &= -15\end{aligned}$$

This completes the calculation.

Computing the determinant of a matrix A can be tedious. For example, if A is a 4×4 matrix, the cofactor expansion along any row or column involves calculating four cofactors, each of which involves the determinant of a 3×3 matrix. And if A is 5×5, the expansion involves five determinants of 4×4 matrices! There is a clear need for some techniques to cut down the work.[3]

The motivation for the method is the observation (see Example 3.1.4) that calculating a determinant is simplified a great deal when a row or column consists mostly of zeros. (In fact, when a row or column consists *entirely* of zeros, the determinant is zero—simply expand along that row or column.)

Recall next that one method of *creating* zeros in a matrix is to apply elementary row operations to it. Hence, a natural question to ask is what effect such a row operation has on the determinant of the matrix. It turns out that the effect is easy to determine and that elementary *column* operations can be used in the same way. These observations lead to a technique for evaluating determinants that greatly reduces the

[3] If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ we can calculate $\det A$ by considering $\begin{bmatrix} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{bmatrix}$ obtained from A by adjoining columns 1 and 2 on the right. Then $\det A = aei + bfg + cdh - ceg - afh - bdi$, where the positive terms aei, bfg, and cdh are the products down and to the right starting at a, b, and c, and the negative terms ceg, afh, and bdi are the products down and to the left starting at c, a, and b. **Warning**: This rule does **not** apply to $n \times n$ matrices where $n > 3$ or $n = 2$.

labour involved. The necessary information is given in Theorem 3.1.2.

Theorem 3.1.2

Let A denote an $n \times n$ matrix.

1. *If A has a row or column of zeros,* $\det A = 0$.
2. *If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is* $-\det A$.
3. *If a row (or column) of A is multiplied by a constant u, the determinant of the resulting matrix is* $u(\det A)$.
4. *If two distinct rows (or columns) of A are identical,* $\det A = 0$.
5. *If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is* $\det A$.

Proof. We prove properties 2, 4, and 5 and leave the rest as exercises.

Property 2. If A is $n \times n$, this follows by induction on n. If $n = 2$, the verification is left to the reader. If $n > 2$ and two rows are interchanged, let B denote the resulting matrix. Expand $\det A$ and $\det B$ along a row *other than* the two that were interchanged. The entries in this row are the same for both A and B, but the cofactors in B are the negatives of those in A (by induction) because the corresponding $(n-1) \times (n-1)$ matrices have two rows interchanged. Hence, $\det B = -\det A$, as required. A similar argument works if two columns are interchanged.

Property 4. If two rows of A are equal, let B be the matrix obtained by interchanging them. Then $B = A$, so $\det B = detA$. But $\det B = -\det A$ by property 2, so $\det A = \det B = 0$. Again, the same argument works for columns.

Property 5. Let B be obtained from $A = \left[a_{ij}\right]$ by adding u times row p to row q. Then row q of B is

$$(a_{q1} + ua_{p1},\ a_{q2} + ua_{p2},\ \dots,\ a_{qn} + ua_{pn})$$

The cofactors of these elements in B are the same as in A (they do not involve row q): in symbols, $c_{qj}(B) = c_{qj}(A)$ for each j. Hence, expanding B along row q gives

$$\begin{aligned} \det B &= (a_{q1} + ua_{p1})c_{q1}(A) + (a_{q2} + ua_{p2})c_{q2}(A) + \cdots + (a_{qn} + ua_{pn})c_{qn}(A) \\ &= [a_{q1}c_{q1}(A) + a_{q2}c_{q2}(A) + \cdots + a_{qn}c_{qn}(A)] + u[a_{p1}c_{q1}(A) + a_{p2}c_{q2}(A) + \cdots + a_{pn}c_{qn}(A)] \\ &= \det A + u \det C \end{aligned}$$

where C is the matrix obtained from A by replacing row q by row p (and both expansions are along row q). Because rows p and q of C are equal, $\det C = 0$ by property 4. Hence, $\det B = \det A$, as required. As before, a similar proof holds for columns. □

To illustrate Theorem 3.1.2, consider the following determinants.

$$\begin{vmatrix} 3 & -1 & 2 \\ 2 & 5 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$ (because the last row consists of zeros)

$$\begin{vmatrix} 3 & -1 & 5 \\ 2 & 8 & 7 \\ 1 & 2 & -1 \end{vmatrix} = -\begin{vmatrix} 5 & -1 & 3 \\ 7 & 8 & 2 \\ -1 & 2 & 1 \end{vmatrix}$$ (because two columns are interchanged)

$$\begin{vmatrix} 8 & 1 & 2 \\ 3 & 0 & 9 \\ 1 & 2 & -1 \end{vmatrix} = 3\begin{vmatrix} 8 & 1 & 2 \\ 1 & 0 & 3 \\ 1 & 2 & -1 \end{vmatrix}$$ (because the second row of the matrix on the left is 3 times the second row of the matrix on the right)

$$\begin{vmatrix} 2 & 1 & 2 \\ 4 & 0 & 4 \\ 1 & 3 & 1 \end{vmatrix} = 0$$ (because two columns are identical)

$$\begin{vmatrix} 2 & 5 & 2 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 9 & 20 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix}$$ (because twice the second row of the matrix on the left was added to the first row)

The following four examples illustrate how Theorem 3.1.2 is used to evaluate determinants.

Example 3.1.5

Evaluate $\det A$ when $A = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 0 & -1 \\ 2 & 1 & 6 \end{bmatrix}$.

Solution. The matrix does have zero entries, so expansion along (say) the second row would involve somewhat less work. However, a column operation can be used to get a zero in position $(2, 3)$—namely, add column 1 to column 3. Because this does not change the value of the determinant, we obtain

$$\det A = \begin{vmatrix} 1 & -1 & 3 \\ 1 & 0 & -1 \\ 2 & 1 & 6 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 4 \\ 1 & 0 & 0 \\ 2 & 1 & 8 \end{vmatrix} = -\begin{vmatrix} -1 & 4 \\ 1 & 8 \end{vmatrix} = 12$$

where we expanded the second 3×3 matrix along row 2.

Example 3.1.6

If $\det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} = 6$, evaluate $\det A$ where $A = \begin{bmatrix} a+x & b+y & c+z \\ 3x & 3y & 3z \\ -p & -q & -r \end{bmatrix}$.

Solution. First take common factors out of rows 2 and 3.

$$\det A = 3(-1) \det \begin{bmatrix} a+x & b+y & c+z \\ x & y & z \\ p & q & r \end{bmatrix}$$

Now subtract the second row from the first and interchange the last two rows.

$$\det A = -3 \det \begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix} = 3 \det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} = 3 \cdot 6 = 18$$

The determinant of a matrix is a sum of products of its entries. In particular, if these entries are polynomials in x, then the determinant itself is a polynomial in x. It is often of interest to determine which values of x make the determinant zero, so it is very useful if the determinant is given in factored form. Theorem 3.1.2 can help.

Example 3.1.7

Find the values of x for which $\det A = 0$, where $A = \begin{bmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{bmatrix}$.

Solution. To evaluate $\det A$, first subtract x times row 1 from rows 2 and 3.

$$\det A = \begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix} = \begin{vmatrix} 1 & x & x \\ 0 & 1-x^2 & x-x^2 \\ 0 & x-x^2 & 1-x^2 \end{vmatrix} = \begin{vmatrix} 1-x^2 & x-x^2 \\ x-x^2 & 1-x^2 \end{vmatrix}$$

At this stage we could simply evaluate the determinant (the result is $2x^3 - 3x^2 + 1$). But then we would have to factor this polynomial to find the values of x that make it zero. However, this factorization can be obtained directly by first factoring each entry in the determinant and taking a common factor of $(1-x)$ from each row.

$$\begin{aligned} \det A = \begin{vmatrix} (1-x)(1+x) & x(1-x) \\ x(1-x) & (1-x)(1+x) \end{vmatrix} &= (1-x)^2 \begin{vmatrix} 1+x & x \\ x & 1+x \end{vmatrix} \\ &= (1-x)^2(2x+1) \end{aligned}$$

Hence, $\det A = 0$ means $(1-x)^2(2x+1) = 0$, that is $x = 1$ or $x = -\frac{1}{2}$.

Example 3.1.8

If a_1, a_2, and a_3 are given show that

$$\det \left[\begin{array}{ccc} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{array} \right] = (a_3 - a_1)(a_3 - a_2)(a_2 - a_1)$$

Solution. Begin by subtracting row 1 from rows 2 and 3, and then expand along column 1:

$$\det \left[\begin{array}{ccc} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{array} \right] = \det \left[\begin{array}{ccc} 1 & a_1 & a_1^2 \\ 0 & a_2 - a_1 & a_2^2 - a_1^2 \\ 0 & a_3 - a_1 & a_3^2 - a_1^2 \end{array} \right] = \det \left[\begin{array}{cc} a_2 - a_1 & a_2^2 - a_1^2 \\ a_3 - a_1 & a_3^2 - a_1^2 \end{array} \right]$$

Now $(a_2 - a_1)$ and $(a_3 - a_1)$ are common factors in rows 1 and 2, respectively, so

$$\begin{aligned} \det \left[\begin{array}{ccc} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{array} \right] &= (a_2 - a_1)(a_3 - a_1) \det \left[\begin{array}{cc} 1 & a_2 + a_1 \\ 1 & a_3 + a_1 \end{array} \right] \\ &= (a_2 - a_1)(a_3 - a_1)(a_3 - a_2) \end{aligned}$$

The matrix in Example 3.1.8 is called a Vandermonde matrix, and the formula for its determinant can be generalized to the $n \times n$ case (see Theorem 3.2.7).

If A is an $n \times n$ matrix, forming uA means multiplying *every* row of A by u. Applying property 3 of Theorem 3.1.2, we can take the common factor u out of each row and so obtain the following useful result.

Theorem 3.1.3

If A is an $n \times n$ matrix, then $\det(uA) = u^n \det A$ for any number u.

The next example displays a type of matrix whose determinant is easy to compute.

Example 3.1.9

Evaluate $\det A$ if $A = \left[\begin{array}{cccc} a & 0 & 0 & 0 \\ u & b & 0 & 0 \\ v & w & c & 0 \\ x & y & z & d \end{array} \right]$.

Solution. Expand along row 1 to get $\det A = a \left| \begin{array}{ccc} b & 0 & 0 \\ w & c & 0 \\ y & z & d \end{array} \right|$. Now expand this along the top row to get $\det A = ab \left| \begin{array}{cc} c & 0 \\ z & d \end{array} \right| = abcd$, the product of the main diagonal entries.

A square matrix is called a **lower triangular matrix** if all entries above the main diagonal are zero (as in Example 3.1.9). Similarly, an **upper triangular matrix** is one for which all entries below the main diagonal are zero. A **triangular matrix** is one that is either upper or lower triangular. Theorem 3.1.4 gives an easy rule for calculating the determinant of any triangular matrix. The proof is like the solution to Example 3.1.9.

Theorem 3.1.4

If A is a square triangular matrix, then $\det A$ is the product of the entries on the main diagonal.

Theorem 3.1.4 is useful in computer calculations because it is a routine matter to carry a matrix to triangular form using row operations.

Exercises for 3.1

Exercise 3.1.1 Compute the determinants of the following matrices.

a. $\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

b. $\begin{bmatrix} 6 & 9 \\ 8 & 12 \end{bmatrix}$

c. $\begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$

d. $\begin{bmatrix} a+1 & a \\ a & a-1 \end{bmatrix}$

e. $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

f. $\begin{bmatrix} 2 & 0 & -3 \\ 1 & 2 & 5 \\ 0 & 3 & 0 \end{bmatrix}$

g. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

h. $\begin{bmatrix} 0 & a & 0 \\ b & c & d \\ 0 & e & 0 \end{bmatrix}$

i. $\begin{bmatrix} 1 & b & c \\ b & c & 1 \\ c & 1 & b \end{bmatrix}$

j. $\begin{bmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{bmatrix}$

k. $\begin{bmatrix} 0 & 1 & -1 & 0 \\ 3 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 5 & 0 & 0 & 7 \end{bmatrix}$

l. $\begin{bmatrix} 1 & 0 & 3 & 1 \\ 2 & 2 & 6 & 0 \\ -1 & 0 & -3 & 1 \\ 4 & 1 & 12 & 0 \end{bmatrix}$

m. $\begin{bmatrix} 3 & 1 & -5 & 2 \\ 1 & 3 & 0 & 1 \\ 1 & 0 & 5 & 2 \\ 1 & 1 & 2 & -1 \end{bmatrix}$

n. $\begin{bmatrix} 4 & -1 & 3 & -1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 1 & 2 & -1 & 1 \end{bmatrix}$

o. $\begin{bmatrix} 1 & -1 & 5 & 5 \\ 3 & 1 & 2 & 4 \\ -1 & -3 & 8 & 0 \\ 1 & 1 & 2 & -1 \end{bmatrix}$

p. $\begin{bmatrix} 0 & 0 & 0 & a \\ 0 & 0 & b & p \\ 0 & c & q & k \\ d & s & t & u \end{bmatrix}$

Exercise 3.1.2 Show that $\det A = 0$ if A has a row or column consisting of zeros.

Exercise 3.1.3 Show that the sign of the position in the last row and the last column of A is always $+1$.

Exercise 3.1.4 Show that $\det I = 1$ for any identity matrix I.

Exercise 3.1.5 Evaluate the determinant of each matrix by reducing it to upper triangular form.

a. $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 1 \\ 2 & -1 & 3 \end{bmatrix}$

b. $\begin{bmatrix} -1 & 3 & 1 \\ 2 & 5 & 3 \\ 1 & -2 & 1 \end{bmatrix}$

c. $\begin{bmatrix} -1 & -1 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & -1 & 2 \end{bmatrix}$

d. $\begin{bmatrix} 2 & 3 & 1 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 5 & 1 & 1 \\ 1 & 1 & 2 & 5 \end{bmatrix}$

Exercise 3.1.6 Evaluate by cursory inspection:

a. $\det \begin{bmatrix} a & b & c \\ a+1 & b+1 & c+1 \\ a-1 & b-1 & c-1 \end{bmatrix}$

b. $\det \begin{bmatrix} a & b & c \\ a+b & 2b & c+b \\ 2 & 2 & 2 \end{bmatrix}$

Exercise 3.1.7 If $\det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} = -1$ compute:

a. $\det \begin{bmatrix} -x & -y & -z \\ 3p+a & 3q+b & 3r+c \\ 2p & 2q & 2r \end{bmatrix}$

b. $\det \begin{bmatrix} -2a & -2b & -2c \\ 2p+x & 2q+y & 2r+z \\ 3x & 3y & 3z \end{bmatrix}$

Exercise 3.1.8 Show that:

a. $\det \begin{bmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{bmatrix} = 2 \det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$

b. $\det \begin{bmatrix} 2a+p & 2b+q & 2c+r \\ 2p+x & 2q+y & 2r+z \\ 2x+a & 2y+b & 2z+c \end{bmatrix} = 9 \det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$

Exercise 3.1.9 In each case either prove the statement or give an example showing that it is false:

a. $\det (A+B) = \det A + \det B$.

b. If $\det A = 0$, then A has two equal rows.

c. If A is 2×2, then $\det (A^T) = \det A$.

d. If R is the reduced row-echelon form of A, then $\det A = \det R$.

e. If A is 2×2, then $\det (7A) = 49 \det A$.

f. $\det (A^T) = -\det A$.

g. $\det (-A) = -\det A$.

h. If $\det A = \det B$ where A and B are the same size, then $A = B$.

Exercise 3.1.12 If A has three columns with only the top two entries nonzero, show that $\det A = 0$.

Exercise 3.1.13

a. Find $\det A$ if A is 3×3 and $\det (2A) = 6$.

b. Under what conditions is $\det (-A) = \det A$?

Exercise 3.1.14 Evaluate by first adding all other rows to the first row.

a. $\det \begin{bmatrix} x-1 & 2 & 3 \\ 2 & -3 & x-2 \\ -2 & x & -2 \end{bmatrix}$

b. $\det \begin{bmatrix} x-1 & -3 & 1 \\ 2 & -1 & x-1 \\ -3 & x+2 & -2 \end{bmatrix}$

Exercise 3.1.15

a. Find b if $\det \begin{bmatrix} 5 & -1 & x \\ 2 & 6 & y \\ -5 & 4 & z \end{bmatrix} = ax + by + cz$.

b. Find c if $\det \begin{bmatrix} 2 & x & -1 \\ 1 & y & 3 \\ -3 & z & 4 \end{bmatrix} = ax+by+cz$.

Exercise 3.1.16 Find the real numbers x and y such that $\det A = 0$ if:

a. $A = \begin{bmatrix} 0 & x & y \\ y & 0 & x \\ x & y & 0 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & x & x \\ -x & -2 & x \\ -x & -x & -3 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & x & x^2 & x^3 \\ x & x^2 & x^3 & 1 \\ x^2 & x^3 & 1 & x \\ x^3 & 1 & x & x^2 \end{bmatrix}$

d. $A = \begin{bmatrix} x & y & 0 & 0 \\ 0 & x & y & 0 \\ 0 & 0 & x & y \\ y & 0 & 0 & x \end{bmatrix}$

Exercise 3.1.17 Show that

$$\det \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & x & x \\ 1 & x & 0 & x \\ 1 & x & x & 0 \end{bmatrix} = -3x^2$$

Exercise 3.1.18 Show that

$$\det \begin{bmatrix} 1 & x & x^2 & x^3 \\ a & 1 & x & x^2 \\ p & b & 1 & x \\ q & r & c & 1 \end{bmatrix} = (1-ax)(1-bx)(1-cx).$$

Exercise 3.1.19

Given the polynomial $p(x) = a+bx+cx^2+dx^3+x^4$, the matrix $C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a & -b & -c & -d \end{bmatrix}$ is called the **companion matrix** of $p(x)$. Show that $\det(xI-C) = p(x)$.

Exercise 3.1.20 Show that

$$\det \begin{bmatrix} a+x & b+x & c+x \\ b+x & c+x & a+x \\ c+x & a+x & b+x \end{bmatrix}$$
$$= (a+b+c+3x)[(ab+ac+bc)-(a^2+b^2+c^2)]$$

Exercise 3.1.21 . Prove Theorem 3.1.6. [*Hint*: Expand the determinant along column j.]

Exercise 3.1.22 Show that

$$\det \begin{bmatrix} 0 & 0 & \cdots & 0 & a_1 \\ 0 & 0 & \cdots & a_2 & * \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n-1} & \cdots & * & * \\ a_n & * & \cdots & * & * \end{bmatrix} = (-1)^k a_1 a_2 \cdots a_n$$

where either $n = 2k$ or $n = 2k+1$, and $*$-entries are arbitrary.

Exercise 3.1.23 By expanding along the first column, show that:

$$\det \begin{bmatrix} 1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} = 1+(-1)^{n+1}$$

if the matrix is $n \times n$, $n \geq 2$.

Exercise 3.1.24 Form matrix B from a matrix A by writing the columns of A in reverse order. Express $\det B$ in terms of $\det A$.

Exercise 3.1.25 Prove property 3 of Theorem 3.1.2 by expanding along the row (or column) in question.

Exercise 3.1.26 Show that the line through two distinct points (x_1, y_1) and (x_2, y_2) in the plane has equation

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0$$

Exercise 3.1.27 Let A be an $n \times n$ matrix. Given a polynomial $p(x) = a_0 + a_1 x + \cdots + a_m x^m$, we write $p(A) = a_0 I + a_1 A + \cdots + a_m A^m$.

For example, if $p(x) = 2-3x+5x^2$, then $p(A) = 2I - 3A + 5A^2$. The *characteristic polynomial* of A is defined to be $c_A(x) = \det[xI-A]$, and the Cayley-Hamilton theorem asserts that $c_A(A) = 0$ for any matrix A.

a. Verify the theorem for

i. $A = \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$

ii. $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 8 & 2 & 2 \end{bmatrix}$

b. Prove the theorem for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

3.2 Determinants and Matrix Inverses

In this section, several theorems about determinants are derived. One consequence of these theorems is that a square matrix A is invertible if and only if $\det A \neq 0$. Moreover, determinants are used to give a formula for A^{-1} which, in turn, yields a formula (called Cramer's rule) for the solution of any system of linear equations with an invertible coefficient matrix.

We begin with a remarkable theorem (due to Cauchy in 1812) about the determinant of a product of matrices. The proof is given at the end of this section.

Theorem 3.2.1: Product Theorem

If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B$.

The complexity of matrix multiplication makes the product theorem quite unexpected. Here is an example where it reveals an important numerical identity.

Example 3.2.1

If $A = \left[\begin{array}{rr} a & b \\ -b & a \end{array}\right]$ and $B = \left[\begin{array}{rr} c & d \\ -d & c \end{array}\right]$ then $AB = \left[\begin{array}{cc} ac-bd & ad+bc \\ -(ad+bc) & ac-bd \end{array}\right]$.

Hence $\det A \det B = \det(AB)$ gives the identity

$$(a^2+b^2)(c^2+d^2) = (ac-bd)^2 + (ad+bc)^2$$

Theorem 3.2.1 extends easily to $\det(ABC) = \det A \det B \det C$. In fact, induction gives

$$\det(A_1 A_2 \cdots A_{k-1} A_k) = \det A_1 \det A_2 \cdots \det A_{k-1} \det A_k$$

for any square matrices $A_1, \ldots, A_k$ of the same size. In particular, if each $A_i = A$, we obtain

$$\det(A^k) = (detA)^k, \text{ for any } k \geq 1$$

We can now give the invertibility condition.

Theorem 3.2.2

An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When this is the case, $\det(A^{-1}) = \frac{1}{\det A}$

Proof. If A is invertible, then $AA^{-1} = I$; so the product theorem gives

$$1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$$

Hence, $\det A \neq 0$ and also $\det A^{-1} = \frac{1}{\det A}$.

Conversely, if $\det A \neq 0$, we show that A can be carried to I by elementary row operations (and invoke Theorem 2.4.5). Certainly, A can be carried to its reduced row-echelon form R, so $R = E_k \cdots E_2 E_1 A$ where the E_i are elementary matrices (Theorem 2.5.1). Hence the product theorem gives

$$\det R = \det E_k \cdots \det E_2 \det E_1 \det A$$

Since $\det E \neq 0$ for all elementary matrices E, this shows $\det R \neq 0$. In particular, R has no row of zeros, so $R = I$ because R is square and reduced row-echelon. This is what we wanted. □

Example 3.2.2

For which values of c does $A = \left[\begin{array}{rrr} 1 & 0 & -c \\ -1 & 3 & 1 \\ 0 & 2c & -4 \end{array} \right]$ have an inverse?

Solution. Compute $\det A$ by first adding c times column 1 to column 3 and then expanding along row 1.

$$\det A = \det \left[\begin{array}{rrr} 1 & 0 & -c \\ -1 & 3 & 1 \\ 0 & 2c & -4 \end{array} \right] = \det \left[\begin{array}{rrc} 1 & 0 & 0 \\ -1 & 3 & 1-c \\ 0 & 2c & -4 \end{array} \right] = 2(c+2)(c-3)$$

Hence, $\det A = 0$ if $c = -2$ or $c = 3$, and A has an inverse if $c \neq -2$ and $c \neq 3$.

Example 3.2.3

If a product $A_1A_2 \cdots A_k$ of square matrices is invertible, show that each A_i is invertible.

Solution. We have $\det A_1 \det A_2 \cdots \det A_k = \det (A_1A_2 \cdots A_k)$ by the product theorem, and $\det (A_1A_2 \cdots A_k) \neq 0$ by Theorem 3.2.2 because $A_1A_2 \cdots A_k$ is invertible. Hence

$$\det A_1 \det A_2 \cdots \det A_k \neq 0$$

so $\det A_i \neq 0$ for each i. This shows that each A_i is invertible, again by Theorem 3.2.2.

Theorem 3.2.3

If A is any square matrix, $\det A^T = \det A$.

Proof. Consider first the case of an elementary matrix E. If E is of type I or II, then $E^T = E$; so certainly $\det E^T = \det E$. If E is of type III, then E^T is also of type III; so $\det E^T = 1 = \det E$ by Theorem 3.1.2. Hence, $\det E^T = \det E$ for every elementary matrix E.

Now let A be any square matrix. If A is not invertible, then neither is A^T; so $\det A^T = 0 = \det A$ by Theorem 3.2.2. On the other hand, if A is invertible, then $A = E_k \cdots E_2E_1$, where the E_i are elementary matrices (Theorem 2.5.2). Hence, $A^T = E_1^T E_2^T \cdots E_k^T$ so the product theorem gives

$$\begin{aligned}\det A^T = \det E_1^T \det E_2^T \cdots \det E_k^T &= \det E_1 \det E_2 \cdots \det E_k \\ &= \det E_k \cdots \det E_2 \det E_1 \\ &= \det A\end{aligned}$$

This completes the proof. □

Example 3.2.4

If $\det A = 2$ and $\det B = 5$, calculate $\det (A^3 B^{-1} A^T B^2)$.

Solution. We use several of the facts just derived.

$$\begin{aligned}\det (A^3 B^{-1} A^T B^2) &= \det (A^3) \det (B^{-1}) \det (A^T) \det (B^2) \\ &= (\det A)^3 \frac{1}{\det B} \det A (\det B)^2 \\ &= 2^3 \cdot \frac{1}{5} \cdot 2 \cdot 5^2 \\ &= 80\end{aligned}$$

Example 3.2.5

A square matrix is called **orthogonal** if $A^{-1} = A^T$. What are the possible values of $\det A$ if A is orthogonal?

Solution. If A is orthogonal, we have $I = AA^T$. Take determinants to obtain

$$1 = \det I = \det (AA^T) = \det A \det A^T = (\det A)^2$$

Since $\det A$ is a number, this means $\det A = \pm 1$.

Hence Theorems 2.6.4 and 2.6.5 imply that rotation about the origin and reflection about a line through the origin in $\mathbb{R}^2$ have orthogonal matrices with determinants 1 and -1 respectively. In fact they are the *only* such transformations of $\mathbb{R}^2$. We have more to say about this in Section 8.2.

□

Proof of Theorem 3.2.1. If A and B are $n \times n$ matrices we must show that

$$\det(AB) = \det A \det B \tag{3.5}$$

Recall that if E is an elementary matrix obtained by doing one row operation to I_n, then doing that operation to a matrix C (Lemma 2.5.1) results in EC. By looking at the three types of elementary matrices separately, Theorem 3.1.2 shows that

$$\det(EC) = \det E \det C \quad \text{for any matrix } C \tag{3.6}$$

Thus if $E_1, E_2, \ldots, E_k$ are all elementary matrices, it follows by induction that

$$\det(E_k \cdots E_2 E_1 C) = \det E_k \cdots \det E_2 \det E_1 \det C \text{ for any matrix } C \tag{3.7}$$

Lemma. If A has no inverse, then $\det A = 0$.

Proof. Let $A \to R$ where R is reduced row-echelon, say $E_n \cdots E_2 E_1 A = R$. Then R has a row of zeros by Part (4) of Theorem 2.4.5, and hence $\det R = 0$. But then Equation 3.7 gives $\det A = 0$ because $\det E \neq 0$ for any elementary matrix E. This proves the Lemma.

Now we can prove Equation 3.5 by considering two cases.

Case 1. A has no inverse. Then AB also has no inverse (otherwise $A[B(AB)^{-1}] = I$ so A is invertible by Corollary 2.4.1 to Theorem 2.4.5). Hence the above Lemma (twice) gives

$$\det(AB) = 0 = 0 \det B = \det A \det B$$

proving Equation 3.5 in this case.

Case 2. A has an inverse. Then A is a product of elementary matrices by Theorem 2.5.2, say $A = E_1 E_2 \cdots E_k$. Then Equation 3.7 with $C = I$ gives

$$\det A = \det(E_1 E_2 \cdots E_k) = \det E_1 \det E_2 \cdots \det E_k$$

But then Equation 3.7 with $C = B$ gives

$$\det(AB) = \det[(E_1 E_2 \cdots E_k)B] = \det E_1 \det E_2 \cdots \det E_k \det B = \det A \det B$$

and Equation 3.5 holds in this case too. □

Exercises for 3.2

Exercise 3.2.2 Use determinants to find which real values of c make each of the following matrices invertible.

a. $\begin{bmatrix} 1 & 0 & 3 \\ 3 & -4 & c \\ 2 & 5 & 8 \end{bmatrix}$ b. $\begin{bmatrix} 0 & c & -c \\ -1 & 2 & 1 \\ c & -c & c \end{bmatrix}$

c. $\begin{bmatrix} c & 1 & 0 \\ 0 & 2 & c \\ -1 & c & 5 \end{bmatrix}$ d. $\begin{bmatrix} 4 & c & 3 \\ c & 2 & c \\ 5 & c & 4 \end{bmatrix}$

e. $\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & c \\ 2 & c & 1 \end{bmatrix}$ f. $\begin{bmatrix} 1 & c & -1 \\ c & 1 & 1 \\ 0 & 1 & c \end{bmatrix}$

Exercise 3.2.3 Let A, B, and C denote $n \times n$ matrices and assume that $\det A = -1$, $\det B = 2$, and $\det C = 3$. Evaluate:

a. $\det(A^3BC^TB^{-1})$ b. $\det(B^2C^{-1}AB^{-1}C^T)$

Exercise 3.2.4 Let A and B be invertible $n \times n$ matrices. Evaluate:

a. $\det(B^{-1}AB)$ b. $\det(A^{-1}B^{-1}AB)$

Exercise 3.2.5 If A is 3×3 and $\det(2A^{-1}) = -4$ and $\det(A^3(B^{-1})^T) = -4$, find $\det A$ and $\det B$.

Exercise 3.2.6 Let $A = \begin{bmatrix} a & b & c \\ p & q & r \\ u & v & w \end{bmatrix}$ and assume that $\det A = 3$. Compute:

a. $\det(2B^{-1})$ where $B = \begin{bmatrix} 4u & 2a & -p \\ 4v & 2b & -q \\ 4w & 2c & -r \end{bmatrix}$

b. $\det(2C^{-1})$ where $C = \begin{bmatrix} 2p & -a+u & 3u \\ 2q & -b+v & 3v \\ 2r & -c+w & 3w \end{bmatrix}$

Exercise 3.2.7 If $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = -2$ calculate:

a. $\det \begin{bmatrix} 2 & -2 & 0 \\ c+1 & -1 & 2a \\ d-2 & 2 & 2b \end{bmatrix}$

b. $\det \begin{bmatrix} 2b & 0 & 4d \\ 1 & 2 & -2 \\ a+1 & 2 & 2(c-1) \end{bmatrix}$

c. $\det(3A^{-1})$ where $A = \begin{bmatrix} 3c & a+c \\ 3d & b+d \end{bmatrix}$

Exercise 3.2.9 Use Theorem 3.2.4 to find the (2, 3)-entry of A^{-1} if:

a. $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$ b. $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \\ 0 & 4 & 7 \end{bmatrix}$

Exercise 3.2.10 Explain what can be said about $\det A$ if:

a. $A^2 = A$

b. $A^2 = I$

c. $A^3 = A$

d. $PA = P$ and P is invertible

e. $A^2 = uA$ and A is $n \times n$

f. $A = -A^T$ and A is $n \times n$

g. $A^2 + I = 0$ and A is $n \times n$

Exercise 3.2.11 Let A be $n \times n$. Show that $uA = (uI)A$, and use this with Theorem 3.2.1 to deduce the result in Theorem 3.1.3: $\det(uA) = u^n \det A$.

Exercise 3.2.12 If A and B are $n \times n$ matrices, if $AB = -BA$, and if n is odd, show that either A or B has no inverse.

Exercise 3.2.13 Show that $\det AB = \det BA$ holds for any two $n \times n$ matrices A and B.

Exercise 3.2.14 If $A^k = 0$ for some $k \geq 1$, show that A is not invertible.

Exercise 3.2.15 If $A^{-1} = A^T$, describe the cofactor matrix of A in terms of A.

Exercise 3.2.16 Show that no 3×3 matrix A exists such that $A^2 + I = 0$. Find a 2×2 matrix A with this property.

Exercise 3.2.17 Show that $\det(A + B^T) = \det(A^T + B)$ for any $n \times n$ matrices A and B.

Exercise 3.2.18 Let A and B be invertible $n \times n$ matrices. Show that $\det A = \det B$ if and only if $A = UB$ where U is a matrix with $\det U = 1$.

Exercise 3.2.19 For each of the matrices in Exercise 2, find the inverse for those values of c for which it exists.

Exercise 3.2.20 In each case either prove the statement or give an example showing that it is false:

a. $\det(AB) = \det(B^T A)$.

b. If $\det A \neq 0$ and $AB = AC$, then $B = C$.

c. If $A^T = -A$, then $\det A = -1$.

d. If A is invertible, then $\operatorname{adj} A$ is invertible.

e. If A has a row of zeros, so also does $\operatorname{adj} A$.

f. $\det(A^T A) > 0$ for all square matrices A.

g. $\det(I + A) = 1 + \det A$.

h. If AB is invertible, then A and B are invertible.

Exercise 3.2.21 If A is 2×2 and $\det A = 0$, show that one column of A is a scalar multiple of the other. [*Hint*: Definition 2.5 and Part (2) of Theorem 2.4.5.]

Exercise 3.2.22 Find a polynomial $p(x)$ of degree 2 such that:

a. $p(0) = 2$, $p(1) = 3$, $p(3) = 8$

b. $p(0) = 5$, $p(1) = 3$, $p(2) = 5$

Exercise 3.2.23 Find a polynomial $p(x)$ of degree 3 such that:

a. $p(0) = p(1) = 1$, $p(-1) = 4$, $p(2) = -5$

b. $p(0) = p(1) = 1$, $p(-1) = 2$, $p(-2) = -3$

Exercise 3.2.24 Given the following data pairs, find the interpolating polynomial of degree at most 3 and estimate the value of y corresponding to $x = 1.5$.

a. $(0,\ 1)$, $(1,\ 2)$, $(2,\ 5)$, $(3,\ 10)$

b. $(0,\ 1)$, $(1,\ 1.49)$, $(2,\ -0.42)$, $(3,\ -11.33)$

c. $(0,\ 2)$, $(1,\ 2.03)$, $(2,\ -0.40)$, $(-1,\ 0.89)$

Exercise 3.2.25 If $A = \begin{bmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{bmatrix}$ show that $\det A = 1 + a^2 + b^2 + c^2$. Hence, find A^{-1} for any a, b, and c.

Exercise 3.2.26

a. Show that $A = \begin{bmatrix} a & p & q \\ 0 & b & r \\ 0 & 0 & c \end{bmatrix}$ has an inverse if and only if $abc \neq 0$, and find A^{-1} in that case.

b. Show that if an upper triangular matrix is invertible, the inverse is also upper triangular.

Exercise 3.2.27 Let A be a matrix each of whose entries are integers. Show that each of the following conditions implies the other.

1. A is invertible and A^{-1} has integer entries.

2. $\det A = 1$ or -1.

3.3 Diagonalization and Eigenvalues

The world is filled with examples of systems that evolve in time—the weather in a region, the economy of a nation, the diversity of an ecosystem, etc. Describing such systems is difficult in general and various methods have been developed in special cases. In this section we describe one such method, called *diagonalization,* which is one of the most important techniques in linear algebra. A very fertile example of this procedure is in modelling the growth of the population of an animal species. This has attracted more attention in recent years with the ever increasing awareness that many species are endangered. To motivate the technique, we begin by setting up a simple model of a bird population in which we make assumptions about survival and reproduction rates.

Example 3.3.1

Consider the evolution of the population of a species of birds. Because the number of males and females are nearly equal, we count only females. We assume that each female remains a juvenile for one year and then becomes an adult, and that only adults have offspring. We make three assumptions about reproduction and survival rates:

1. The number of juvenile females hatched in any year is twice the number of adult females alive the year before (we say the **reproduction rate** is 2).
2. Half of the adult females in any year survive to the next year (the **adult survival rate** is $\frac{1}{2}$).
3. One quarter of the juvenile females in any year survive into adulthood (the **juvenile survival rate** is $\frac{1}{4}$).

If there were 100 adult females and 40 juvenile females alive initially, compute the population of females k years later.

Solution. Let a_k and j_k denote, respectively, the number of adult and juvenile females after k years, so that the total female population is the sum $a_k + j_k$. Assumption 1 shows that $j_{k+1} = 2a_k$, while assumptions 2 and 3 show that $a_{k+1} = \frac{1}{2}a_k + \frac{1}{4}j_k$. Hence the numbers a_k and j_k in successive years are related by the following equations:

$$\begin{aligned} a_{k+1} &= \tfrac{1}{2}a_k + \tfrac{1}{4}j_k \\ j_{k+1} &= 2a_k \end{aligned}$$

If we write $\mathbf{v}_k = \left[\begin{array}{c} a_k \\ j_k \end{array} \right]$ and $A = \left[\begin{array}{cc} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{array} \right]$ these equations take the matrix form

$$\mathbf{v}_{k+1} = A\mathbf{v}_k, \quad \text{for each } k = 0,\ 1,\ 2,\ \dots$$

Taking $k = 0$ gives $\mathbf{v}_1 = A\mathbf{v}_0$, then taking $k = 1$ gives $\mathbf{v}_2 = A\mathbf{v}_1 = A^2\mathbf{v}_0$, and taking $k = 2$ gives $\mathbf{v}_3 = A\mathbf{v}_2 = A^3\mathbf{v}_0$. Continuing in this way, we get

$$\mathbf{v}_k = A^k\mathbf{v}_0, \quad \text{for each } k = 0,\ 1,\ 2,\ \dots$$

Since $\mathbf{v}_0 = \begin{bmatrix} a_0 \\ j_0 \end{bmatrix} = \begin{bmatrix} 100 \\ 40 \end{bmatrix}$ is known, finding the population profile $\mathbf{v}_k$ amounts to computing A^k for all $k \geq 0$. We will complete this calculation in Example 3.3.12 after some new techniques have been developed.

Let A be a fixed $n \times n$ matrix. A sequence $\mathbf{v}_0,\ \mathbf{v}_1,\ \mathbf{v}_2,\ \dots$ of column vectors in $\mathbb{R}^n$ is called a **linear dynamical system**[8] if $\mathbf{v}_0$ is known and the other $\mathbf{v}_k$ are determined (as in Example 3.3.1) by the conditions

$$\mathbf{v}_{k+1} = A\mathbf{v}_k \text{ for each } k = 0,\ 1,\ 2,\ \dots$$

These conditions are called a **matrix recurrence** for the vectors $\mathbf{v}_k$. As in Example 3.3.1, they imply that

$$\mathbf{v}_k = A^k\mathbf{v}_0 \text{ for all } k \geq 0$$

so finding the columns $\mathbf{v}_k$ amounts to calculating A^k for $k \geq 0$.

Direct computation of the powers A^k of a square matrix A can be time-consuming, so we adopt an indirect method that is commonly used. The idea is to first **diagonalize** the matrix A, that is, to find an invertible matrix P such that

$$P^{-1}AP = D \text{ is a diagonal matrix} \tag{3.8}$$

This works because the powers D^k of the diagonal matrix D are easy to compute, and Equation 3.8 enables us to compute powers A^k of the matrix A in terms of powers D^k of D. Indeed, we can solve Equation 3.8 for A to get $A = PDP^{-1}$. Squaring this gives

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$$

Using this we can compute A^3 as follows:

$$A^3 = AA^2 = (PDP^{-1})(PD^2P^{-1}) = PD^3P^{-1}$$

Continuing in this way we obtain Theorem 3.3.1 (even if D is not diagonal).

Theorem 3.3.1

If $A = PDP^{-1}$ then $A^k = PD^kP^{-1}$ for each $k = 1,\ 2,\ \dots$.

Hence computing A^k comes down to finding an invertible matrix P as in equation Equation 3.8. To do this it is necessary to first compute certain numbers (called eigenvalues) associated with the matrix A.

[8]More precisely, this is a linear discrete dynamical system. Many models regard $\mathbf{v}_t$ as a continuous function of the time t, and replace our condition between $\mathbf{b}_{k+1}$ and $A\mathbf{v}_k$ with a differential relationship viewed as functions of time.

Eigenvalues and Eigenvectors

Definition 3.4 Eigenvalues and Eigenvectors of a Matrix

If A is an $n \times n$ matrix, a number λ is called an ***eigenvalue*** *of A if*

$$A\mathbf{x} = \lambda \mathbf{x} \text{ for some column } \mathbf{x} \neq \mathbf{0} \text{ in } \mathbb{R}^n$$

In this case, $\mathbf{x}$ is called an ***eigenvector*** *of A corresponding to the eigenvalue λ, or a λ-**eigenvector** for short.*

Example 3.3.2

If $A = \left[\begin{array}{rr} 3 & 5 \\ 1 & -1 \end{array} \right]$ and $\mathbf{x} = \left[\begin{array}{r} 5 \\ 1 \end{array} \right]$ then $A\mathbf{x} = 4\mathbf{x}$ so $\lambda = 4$ is an eigenvalue of A with corresponding eigenvector $\mathbf{x}$.

The matrix A in Example 3.3.2 has another eigenvalue in addition to $\lambda = 4$. To find it, we develop a general procedure for *any* $n \times n$ matrix A.

By definition a number λ is an eigenvalue of the $n \times n$ matrix A if and only if $A\mathbf{x} = \lambda\mathbf{x}$ for some column $\mathbf{x} \neq \mathbf{0}$. This is equivalent to asking that the homogeneous system

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

of linear equations has a nontrivial solution $\mathbf{x} \neq \mathbf{0}$. By Theorem 2.4.5 this happens if and only if the matrix $\lambda I - A$ is not invertible and this, in turn, holds if and only if the determinant of the coefficient matrix is zero:

$$\det(\lambda I - A) = 0$$

This last condition prompts the following definition:

Definition 3.5 Characteristic Polynomial of a Matrix

If A is an $n \times n$ matrix, the ***characteristic polynomial*** *$c_A(x)$ of A is defined by*

$$c_A(x) = \det(xI - A)$$

Note that $c_A(x)$ is indeed a polynomial in the variable x, and it has degree n when A is an $n \times n$ matrix (this is illustrated in the examples below). The above discussion shows that a number λ is an eigenvalue of A if and only if $c_A(\lambda) = 0$, that is if and only if λ is a **root** of the characteristic polynomial $c_A(x)$. We record these observations in

Theorem 3.3.2

Let A be an $n \times n$ matrix.

1. *The eigenvalues λ of A are the roots of the characteristic polynomial $c_A(x)$ of A.*

2. *The λ-eigenvectors $\mathbf{x}$ are the nonzero solutions to the homogeneous system*

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

of linear equations with $\lambda I - A$ as coefficient matrix.

In practice, solving the equations in part 2 of Theorem 3.3.2 is a routine application of gaussian elimination, but finding the eigenvalues can be difficult, often requiring computers (see Section 8.5). For now, the examples and exercises will be constructed so that the roots of the characteristic polynomials are relatively easy to find (usually integers). However, the reader should not be misled by this into thinking that eigenvalues are so easily obtained for the matrices that occur in practical applications!

Example 3.3.3

Find the characteristic polynomial of the matrix $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ discussed in Example 3.3.2, and then find all the eigenvalues and their eigenvectors.

Solution. Since $xI - A = \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} x-3 & -5 \\ -1 & x+1 \end{bmatrix}$ we get

$$c_A(x) = \det \begin{bmatrix} x-3 & -5 \\ -1 & x+1 \end{bmatrix} = x^2 - 2x - 8 = (x-4)(x+2)$$

Hence, the roots of $c_A(x)$ are $\lambda_1 = 4$ and $\lambda_2 = -2$, so these are the eigenvalues of A. Note that $\lambda_1 = 4$ was the eigenvalue mentioned in Example 3.3.2, but we have found a new one: $\lambda_2 = -2$. To find the eigenvectors corresponding to $\lambda_2 = -2$, observe that in this case

$$(\lambda_2 I - A)\mathbf{x} = \begin{bmatrix} \lambda_2 - 3 & -5 \\ -1 & \lambda_2 + 1 \end{bmatrix} = \begin{bmatrix} -5 & -5 \\ -1 & -1 \end{bmatrix}$$

so the general solution to $(\lambda_2 I - A)\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ where t is an arbitrary real number.

Hence, the eigenvectors $\mathbf{x}$ corresponding to λ_2 are $\mathbf{x} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ where $t \neq 0$ is arbitrary. Similarly, $\lambda_1 = 4$ gives rise to the eigenvectors $\mathbf{x} = t \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, $t \neq 0$ which includes the observation in Example 3.3.2.

Note that a square matrix A has *many* eigenvectors associated with any given eigenvalue λ. In fact *every* nonzero solution $\mathbf{x}$ of $(\lambda I - A)\mathbf{x} = \mathbf{0}$ is an eigenvector. Recall that these solutions are all linear combinations of certain basic solutions determined by the gaussian algorithm (see Theorem 1.3.2). Observe that any nonzero multiple of an eigenvector is again an eigenvector,[9] and such multiples are often more convenient.[10] Any set of nonzero multiples of the basic solutions of $(\lambda I - A)\mathbf{x} = \mathbf{0}$ will be called a set of

[9] In fact, any nonzero linear combination of λ-eigenvectors is again a λ-eigenvector.

[10] Allowing nonzero multiples helps eliminate round-off error when the eigenvectors involve fractions.

basic eigenvectors corresponding to λ.

Example 3.3.4

Find the characteristic polynomial, eigenvalues, and basic eigenvectors for

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$$

Solution. Here the characteristic polynomial is given by

$$c_A(x) = \det \begin{bmatrix} x-2 & 0 & 0 \\ -1 & x-2 & 1 \\ -1 & -3 & x+2 \end{bmatrix} = (x-2)(x-1)(x+1)$$

so the eigenvalues are $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = -1$. To find all eigenvectors for $\lambda_1 = 2$, compute

$$\lambda_1 I - A = \begin{bmatrix} \lambda_1 - 2 & 0 & 0 \\ -1 & \lambda_1 - 2 & 1 \\ -1 & -3 & \lambda_1 + 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & -3 & 4 \end{bmatrix}$$

We want the (nonzero) solutions to $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$. The augmented matrix becomes

$$\left[\begin{array}{rrr|r} 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & -3 & 4 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrr|r} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

using row operations. Hence, the general solution $\mathbf{x}$ to $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ where t is arbitrary, so we can use $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as the basic eigenvector corresponding to $\lambda_1 = 2$. As the reader can verify, the gaussian algorithm gives basic eigenvectors $\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{x}_3 = \begin{bmatrix} 0 \\ \frac{1}{3} \\ 1 \end{bmatrix}$ corresponding to $\lambda_2 = 1$ and $\lambda_3 = -1$, respectively. Note that to eliminate fractions, we could instead use $3\mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ as the basic λ_3-eigenvector.

Example 3.3.5

If A is a square matrix, show that A and A^T have the same characteristic polynomial, and hence the same eigenvalues.

Solution. We use the fact that $xI - A^T = (xI - A)^T$. Then

$$c_{A^T}(x) = \det\left(xI - A^T\right) = \det\left[(xI - A)^T\right] = \det(xI - A) = c_A(x)$$

by Theorem 3.2.3. Hence $c_{A^T}(x)$ and $c_A(x)$ have the same roots, and so A^T and A have the same eigenvalues (by Theorem 3.3.2).

The eigenvalues of a matrix need not be distinct. For example, if $A = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$ the characteristic polynomial is $(x-1)^2$ so the eigenvalue 1 occurs twice. Furthermore, eigenvalues are usually not computed as the roots of the characteristic polynomial. There are iterative, numerical methods (for example the QR-algorithm in Section 8.5) that are much more efficient for large matrices.

Diagonalization

An $n \times n$ matrix D is called a **diagonal matrix** if all its entries off the main diagonal are zero, that is if D has the form

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are numbers. Calculations with diagonal matrices are very easy. Indeed, if $D = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ and $E = \operatorname{diag}(\mu_1, \mu_2, \dots, \mu_n)$ are two diagonal matrices, their product DE and sum $D+E$ are again diagonal, and are obtained by doing the same operations to corresponding diagonal elements:

$$\begin{aligned} DE &= \operatorname{diag}(\lambda_1\mu_1, \lambda_2\mu_2, \dots, \lambda_n\mu_n) \\ D+E &= \operatorname{diag}(\lambda_1+\mu_1, \lambda_2+\mu_2, \dots, \lambda_n+\mu_n) \end{aligned}$$

Because of the simplicity of these formulas, and with an eye on Theorem 3.3.1 and the discussion preceding it, we make another definition:

Definition 3.6 Diagonalizable Matrices

*An $n \times n$ matrix A is called **diagonalizable** if*

$$P^{-1}AP \text{ is diagonal for some invertible } n \times n \text{ matrix } P$$

*Here the invertible matrix P is called a **diagonalizing matrix** for A.*

To discover when such a matrix P exists, we let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ denote the columns of P and look for ways to determine when such $\mathbf{x}_i$ exist and how to compute them. To this end, write P in terms of its columns as follows:

$$P = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n]$$

Observe that $P^{-1}AP = D$ for some diagonal matrix D holds if and only if

$$AP = PD$$

If we write $D = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, where the λ_i are numbers to be determined, the equation $AP = PD$ becomes

$$A[\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n] = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

By the definition of matrix multiplication, each side simplifies as follows

$$\begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 & \cdots & A\mathbf{x}_n \end{bmatrix} = \begin{bmatrix} \lambda_1\mathbf{x}_1 & \lambda_2\mathbf{x}_2 & \cdots & \lambda_n\mathbf{x}_n \end{bmatrix}$$

Comparing columns shows that $A\mathbf{x}_i = \lambda_i\mathbf{x}_i$ for each i, so

$$P^{-1}AP = D \quad \text{if and only if } A\mathbf{x}_i = \lambda_i\mathbf{x}_i \text{ for each } i$$

In other words, $P^{-1}AP = D$ holds if and only if the diagonal entries of D are eigenvalues of A and the columns of P are corresponding eigenvectors. This proves the following fundamental result.

Theorem 3.3.4

Let A be an $n \times n$ matrix.

1. *A is diagonalizable if and only if it has eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ such that the matrix $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_n \end{bmatrix}$ is invertible.*
2. *When this is the case, $P^{-1}AP = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ where, for each i, λ_i is the eigenvalue of A corresponding to $\mathbf{x}_i$.*

Example 3.3.8

Diagonalize the matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$ in Example 3.3.4.

Solution. By Example 3.3.4, the eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = -1$, with corresponding basic eigenvectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ respectively. Since the matrix $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$ is invertible, Theorem 3.3.4 guarantees that

$$P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = D$$

The reader can verify this directly—easier to check $AP = PD$.

In Example 3.3.8, suppose we let $Q = \begin{bmatrix} \mathbf{x}_2 & \mathbf{x}_1 & \mathbf{x}_3 \end{bmatrix}$ be the matrix formed from the eigenvectors $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ of A, but in a *different order* than that used to form P. Then $Q^{-1}AQ = \operatorname{diag}(\lambda_2, \lambda_1, \lambda_3)$ is diagonal by Theorem 3.3.4, but the eigenvalues are in the *new* order. Hence we can choose the diagonalizing matrix P so that the eigenvalues λ_i appear in any order we want along the main diagonal of D.

In every example above each eigenvalue has had only one basic eigenvector. Here is a diagonalizable matrix where this is not the case.

Example 3.3.9

Diagonalize the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Solution. To compute the characteristic polynomial of A first add rows 2 and 3 of $xI - A$ to row 1:

$$c_A(x) = \det \begin{bmatrix} x & -1 & -1 \\ -1 & x & -1 \\ -1 & -1 & x \end{bmatrix} = \det \begin{bmatrix} x-2 & x-2 & x-2 \\ -1 & x & -1 \\ -1 & -1 & x \end{bmatrix}$$

$$= \det \begin{bmatrix} x-2 & 0 & 0 \\ -1 & x+1 & 0 \\ -1 & 0 & x+1 \end{bmatrix} = (x-2)(x+1)^2$$

Hence the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$, with λ_2 repeated twice (we say that λ_2 has *multiplicity* two). However, A *is* diagonalizable. For $\lambda_1 = 2$, the system of equations $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$ has general solution $\mathbf{x} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as the reader can verify, so a basic λ_1-eigenvector is $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Turning to the repeated eigenvalue $\lambda_2 = -1$, we must solve $(\lambda_2 I - A)\mathbf{x} = \mathbf{0}$. By gaussian elimination, the general solution is $\mathbf{x} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ where s and t are arbitrary. Hence the gaussian algorithm produces *two* basic λ_2-eigenvectors $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ If we take $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{y}_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ we find that P is invertible. Hence $P^{-1}AP = \text{diag}\,(2,\ -1,\ -1)$ by Theorem 3.3.4.

Example 3.3.9 typifies every diagonalizable matrix. To describe the general case, we need some terminology.

Definition 3.7 Multiplicity of an Eigenvalue

An eigenvalue λ of a square matrix A is said to have ***multiplicity*** *m if it occurs m times as a root of the characteristic polynomial $c_A(x)$.*

For example, the eigenvalue $\lambda_2 = -1$ in Example 3.3.9 has multiplicity 2. In that example the gaussian algorithm yields two basic λ_2-eigenvectors, the same number as the multiplicity. This works in general.

Theorem 3.3.5

A square matrix A is diagonalizable if and only if every eigenvalue λ of multiplicity m yields exactly m basic eigenvectors; that is, if and only if the general solution of the system $(\lambda I - A)\mathbf{x} = \mathbf{0}$ has exactly m parameters.

One case of Theorem 3.3.5 deserves mention.

Theorem 3.3.6

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

The proofs of Theorem 3.3.5 and Theorem 3.3.6 require more advanced techniques and are given in Chapter 5. The following procedure summarizes the method.

Diagonalization Algorithm

To diagonalize an $n \times n$ matrix A:

Step 1. Find the distinct eigenvalues λ of A.

Step 2. Compute a set of basic eigenvectors corresponding to each of these eigenvalues λ as basic solutions of the homogeneous system $(\lambda I - A)\mathbf{x} = \mathbf{0}$.

Step 3. The matrix A is diagonalizable if and only if there are n basic eigenvectors in all.

Step 4. If A is diagonalizable, the $n \times n$ matrix P with these basic eigenvectors as its columns is a diagonalizing matrix for A, that is, P is invertible and $P^{-1}AP$ is diagonal.

The diagonalization algorithm is valid even if the eigenvalues are nonreal complex numbers. In this case the eigenvectors will also have complex entries, but we will not pursue this here.

Example 3.3.10

Show that $A = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array} \right]$ is not diagonalizable.

Solution 1. The characteristic polynomial is $c_A(x) = (x-1)^2$, so A has only one eigenvalue $\lambda_1 = 1$ of multiplicity 2. But the system of equations $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$ has general solution $t \left[\begin{array}{c} 1 \\ 0 \end{array} \right]$, so there is only one parameter, and so only one basic eigenvector $\left[\begin{array}{c} 1 \\ 2 \end{array} \right]$. Hence A is not diagonalizable.

Solution 2. We have $c_A(x) = (x-1)^2$ so the only eigenvalue of A is $\lambda = 1$. Hence, if A were diagonalizable, Theorem 3.3.4 would give $P^{-1}AP = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] = I$ for some invertible matrix P. But then $A = PIP^{-1} = I$, which is not the case. So A cannot be diagonalizable.

Diagonalizable matrices share many properties of their eigenvalues. The following example illustrates why.

Example 3.3.11

If $\lambda^3 = 5\lambda$ for every eigenvalue of the diagonalizable matrix A, show that $A^3 = 5A$.

Solution. Let $P^{-1}AP = D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Because $\lambda_i^3 = 5\lambda_i$ for each i, we obtain

$$D^3 = \text{diag}(\lambda_1^3, \dots, \lambda_n^3) = \text{diag}(5\lambda_1, \dots, 5\lambda_n) = 5D$$

Hence $A^3 = (PDP^{-1})^3 = PD^3P^{-1} = P(5D)P^{-1} = 5(PDP^{-1}) = 5A$ using Theorem 3.3.1. This is what we wanted.

If $p(x)$ is any polynomial and $p(\lambda) = 0$ for every eigenvalue of the diagonalizable matrix A, an argument similar to that in Example 3.3.11 shows that $p(A) = 0$. Thus Example 3.3.11 deals with the case $p(x) = x^3 - 5x$. In general, $p(A)$ is called the *evaluation* of the polynomial $p(x)$ at the matrix A. For example, if $p(x) = 2x^3 - 3x + 5$, then $p(A) = 2A^3 - 3A + 5I$—note the use of the identity matrix.

In particular, if $c_A(x)$ denotes the characteristic polynomial of A, we certainly have $c_A(\lambda) = 0$ for each eigenvalue λ of A (Theorem 3.3.2). Hence $c_A(A) = 0$ for every diagonalizable matrix A. This is, in fact, true for *any* square matrix, diagonalizable or not, and the general result is called the Cayley-Hamilton theorem. It is proved in Section 8.7 and again in Section 11.1.

Linear Dynamical Systems

We began Section 3.3 with an example from ecology which models the evolution of the population of a species of birds as time goes on. As promised, we now complete the example—Example 3.3.12 below.

The bird population was described by computing the female population profile $\mathbf{v}_k = \left[\begin{array}{c} a_k \\ j_k \end{array}\right]$ of the species, where a_k and j_k represent the number of adult and juvenile females present k years after the initial values a_0 and j_0 were observed. The model assumes that these numbers are related by the following equations:

$$\begin{aligned} a_{k+1} &= \tfrac{1}{2}a_k + \tfrac{1}{4}j_k \\ j_{k+1} &= 2a_k \end{aligned}$$

If we write $A = \left[\begin{array}{cc} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{array}\right]$ the columns $\mathbf{v}_k$ satisfy $\mathbf{v}_{k+1} = A\mathbf{v}_k$ for each $k = 0, 1, 2, \dots$.

Hence $\mathbf{v}_k = A^k\mathbf{v}_0$ for each $k = 1, 2, \dots$. We can now use our diagonalization techniques to determine the population profile $\mathbf{v}_k$ for all values of k in terms of the initial values.

Example 3.3.12

Assuming that the initial values were $a_0 = 100$ adult females and $j_0 = 40$ juvenile females, compute a_k and j_k for $k = 1, 2, \dots$.

Solution. The characteristic polynomial of the matrix $A = \left[\begin{array}{cc} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{array}\right]$ is
$c_A(x) = x^2 - \frac{1}{2}x - \frac{1}{2} = (x-1)(x+\frac{1}{2})$, so the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = -\frac{1}{2}$ and gaussian

elimination gives corresponding basic eigenvectors $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$. For convenience, we can use multiples $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ respectively. Hence a diagonalizing matrix is $P = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$ and we obtain

$$P^{-1}AP = D \text{ where } D = \begin{bmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

This gives $A = PDP^{-1}$ so, for each $k \geq 0$, we can compute A^k explicitly:

$$\begin{aligned} A^k = PD^kP^{-1} &= \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (-\frac{1}{2})^k \end{bmatrix} \frac{1}{6} \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 4+2(-\frac{1}{2})^k & 1-(-\frac{1}{2})^k \\ 8-8(-\frac{1}{2})^k & 2+4(-\frac{1}{2})^k \end{bmatrix} \end{aligned}$$

Hence we obtain

$$\begin{aligned} \begin{bmatrix} a_k \\ j_k \end{bmatrix} = \mathbf{v}_k = A^k\mathbf{v}_0 &= \frac{1}{6} \begin{bmatrix} 4+2(-\frac{1}{2})^k & 1-(-\frac{1}{2})^k \\ 8-8(-\frac{1}{2})^k & 2+4(-\frac{1}{2})^k \end{bmatrix} \begin{bmatrix} 100 \\ 40 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 440+160(-\frac{1}{2})^k \\ 880-640(-\frac{1}{2})^k \end{bmatrix} \end{aligned}$$

Equating top and bottom entries, we obtain exact formulas for a_k and j_k:

$$a_k = \frac{220}{3} + \frac{80}{3}\left(-\frac{1}{2}\right)^k \text{ and } j_k = \frac{440}{3} + \frac{320}{3}\left(-\frac{1}{2}\right)^k \text{ for } k = 1,\ 2,\ \cdots$$

In practice, the exact values of a_k and j_k are not usually required. What is needed is a measure of how these numbers behave for large values of k. This is easy to obtain here. Since $(-\frac{1}{2})^k$ is nearly zero for large k, we have the following approximate values

$$a_k \approx \frac{220}{3} \text{ and } j_k \approx \frac{440}{3} \text{ if } k \text{ is large}$$

Hence, in the long term, the female population stabilizes with approximately twice as many juveniles as adults.

Definition 3.8 Linear Dynamical System

If A is an $n \times n$ matrix, a sequence $\mathbf{v}_0,\ \mathbf{v}_1,\ \mathbf{v}_2,\ \ldots$ of columns in $\mathbb{R}^n$ is called a ***linear dynamical system*** *if $\mathbf{v}_0$ is specified and $\mathbf{v}_1,\ \mathbf{v}_2,\ \ldots$ are given by the matrix recurrence $\mathbf{v}_{k+1} = A\mathbf{v}_k$ for each $k \geq 0$. We call A the* ***migration*** *matrix of the system.*

We have $\mathbf{v}_1 = A\mathbf{v}_0$, then $\mathbf{v}_2 = A\mathbf{v}_1 = A^2\mathbf{v}_0$, and continuing we find

$$\mathbf{v}_k = A^k\mathbf{v}_0 \text{ for each } k = 1,\ 2,\ \cdots \tag{3.9}$$

Hence the columns $\mathbf{v}_k$ are determined by the powers A^k of the matrix A and, as we have seen, these powers can be efficiently computed if A is diagonalizable. In fact Equation 3.9 can be used to give a nice "formula" for the columns $\mathbf{v}_k$ in this case.

Assume that A is diagonalizable with eigenvalues $\lambda_1,\ \lambda_2,\ \dots,\ \lambda_n$ and corresponding basic eigenvectors $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_n$. If $P = \left[\begin{array}{cccc} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_n \end{array}\right]$ is a diagonalizing matrix with the $\mathbf{x}_i$ as columns, then P is invertible and

$$P^{-1}AP = D = \text{diag}\,(\lambda_1,\ \lambda_2,\ \cdots,\ \lambda_n)$$

by Theorem 3.3.4. Hence $A = PDP^{-1}$ so Equation 3.9 and Theorem 3.3.1 give

$$\mathbf{v}_k = A^k\mathbf{v}_0 = (PDP^{-1})^k\mathbf{v}_0 = (PD^kP^{-1})\mathbf{v}_0 = PD^k(P^{-1}\mathbf{v}_0)$$

for each $k = 1,\ 2,\ \dots$. For convenience, we denote the column $P^{-1}\mathbf{v}_0$ arising here as follows:

$$\mathbf{b} = P^{-1}\mathbf{v}_0 = \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_n \end{array}\right]$$

Then matrix multiplication gives

$$\begin{aligned}
\mathbf{v}_k &= PD^k(P^{-1}\mathbf{v}_0) \\
&= \left[\begin{array}{cccc} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \end{array}\right] \left[\begin{array}{cccc} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{array}\right] \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_n \end{array}\right] \\
&= \left[\begin{array}{cccc} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \end{array}\right] \left[\begin{array}{c} b_1\lambda_1^k \\ b_2\lambda_2^k \\ \vdots \\ b_3\lambda_n^k \end{array}\right] \\
&= b_1\lambda_1^k\mathbf{x}_1 + b_2\lambda_2^k\mathbf{x}_2 + \cdots + b_n\lambda_n^k\mathbf{x}_n
\end{aligned} \tag{3.10}$$

for each $k \geq 0$. This is a useful **exact formula** for the columns $\mathbf{v}_k$. Note that, in particular,

$$\mathbf{v}_0 = b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \cdots + b_n\mathbf{x}_n$$

However, such an exact formula for $\mathbf{v}_k$ is often not required in practice; all that is needed is to *estimate* $\mathbf{v}_k$ for large values of k (as was done in Example 3.3.12). This can be easily done if A has a largest eigenvalue. An eigenvalue λ of a matrix A is called a **dominant eigenvalue** of A if it has multiplicity 1 and

$$|\lambda| > |\mu| \text{ for all eigenvalues } \mu \neq \lambda$$

where $|\lambda|$ denotes the absolute value of the number λ. For example, $\lambda_1 = 1$ is dominant in Example 3.3.12.

Returning to the above discussion, suppose that A has a dominant eigenvalue. By choosing the order in which the columns $\mathbf{x}_i$ are placed in P, we may assume that λ_1 is dominant among the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A (see the discussion following Example 3.3.8). Now recall the exact expression for $\mathbf{v}_k$ in Equation 3.10 above:

$$\mathbf{v}_k = b_1\lambda_1^k\mathbf{x}_1 + b_2\lambda_2^k\mathbf{x}_2 + \cdots + b_n\lambda_n^k\mathbf{x}_n$$

Take λ_1^k out as a common factor in this equation to get

$$\mathbf{v}_k = \lambda_1^k \left[b_1\mathbf{x}_1 + b_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k \mathbf{x}_2 + \cdots + b_n \left(\frac{\lambda_n}{\lambda_1}\right)^k \mathbf{x}_n \right]$$

for each $k \geq 0$. Since λ_1 is dominant, we have $|\lambda_i| < |\lambda_1|$ for each $i \geq 2$, so each of the numbers $(\lambda_i/\lambda_1)^k$ become small in absolute value as k increases. Hence $\mathbf{v}_k$ is approximately equal to the first term $\lambda_1^k b_1\mathbf{x}_1$, and we write this as $\mathbf{v}_k \approx \lambda_1^k b_1\mathbf{x}_1$. These observations are summarized in the following theorem (together with the above exact formula for $\mathbf{v}_k$).

Theorem 3.3.7

Consider the dynamical system $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \dots$ with matrix recurrence

$$\mathbf{v}_{k+1} = A\mathbf{v}_k \text{ for } k \geq 0$$

where A and $\mathbf{v}_0$ are given. Assume that A is a diagonalizable $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and corresponding basic eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, and let $P = \left[\begin{array}{cccc} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_n \end{array} \right]$ be the diagonalizing matrix. Then an exact formula for $\mathbf{v}_k$ is

$$\mathbf{v}_k = b_1\lambda_1^k\mathbf{x}_1 + b_2\lambda_2^k\mathbf{x}_2 + \cdots + b_n\lambda_n^k\mathbf{x}_n \text{ for each } k \geq 0$$

where the coefficients b_i come from

$$\mathbf{b} = P^{-1}\mathbf{v}_0 = \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_n \end{array} \right]$$

Moreover, if A has dominant[12] eigenvalue λ_1, then $\mathbf{v}_k$ is approximated by

$$\mathbf{v}_k = b_1\lambda_1^k\mathbf{x}_1 \text{ for sufficiently large } k.$$

Example 3.3.13

Returning to Example 3.3.12, we see that $\lambda_1 = 1$ is the dominant eigenvalue, with eigenvector $\mathbf{x}_1 = \left[\begin{array}{c} 1 \\ 2 \end{array} \right]$. Here $P = \left[\begin{array}{rr} 1 & -1 \\ 2 & 4 \end{array} \right]$ and $\mathbf{v}_0 = \left[\begin{array}{c} 100 \\ 40 \end{array} \right]$ so $P^{-1}\mathbf{v}_0 = \frac{1}{3}\left[\begin{array}{r} 220 \\ -80 \end{array} \right]$. Hence $b_1 = \frac{220}{3}$ in

[12] Similar results can be found in other situations. If for example, eigenvalues λ_1 and λ_2 (possibly equal) satisfy $|\lambda_1| = |\lambda_2| > |\lambda_i|$ for all $i > 2$, then we obtain $\mathbf{v}_k \approx b_1\lambda_1^k x_1 + b_2\lambda_2^k x_2$ for large k.

the notation of Theorem 3.3.7, so

$$\begin{bmatrix} a_k \\ j_k \end{bmatrix} = \mathbf{v}_k \approx b_1 \lambda_1^k \mathbf{x}_1 = \frac{220}{3} 1^k \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

where k is large. Hence $a_k \approx \frac{220}{3}$ and $j_k \approx \frac{440}{3}$ as in Example 3.3.12.

This next example uses Theorem 3.3.7 to solve a "linear recurrence." See also Section 3.4.

Example 3.3.14

Suppose a sequence $x_0, x_1, x_2, \dots$ is determined by insisting that

$$x_0 = 1, \; x_1 = -1, \text{ and } x_{k+2} = 2x_k - x_{k+1} \text{ for every } k \geq 0$$

Find a formula for x_k in terms of k.

Solution. Using the linear recurrence $x_{k+2} = 2x_k - x_{k+1}$ repeatedly gives

$$x_2 = 2x_0 - x_1 = 3, \quad x_3 = 2x_1 - x_2 = -5, \quad x_4 = 11, \quad x_5 = -21, \dots$$

so the x_i are determined but no pattern is apparent. The idea is to find $\mathbf{v}_k = \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix}$ for each k instead, and then retrieve x_k as the top component of $\mathbf{v}_k$. The reason this works is that the linear recurrence guarantees that these $\mathbf{v}_k$ are a dynamical system:

$$\mathbf{v}_{k+1} = \begin{bmatrix} x_{k+1} \\ x_{k+2} \end{bmatrix} = \begin{bmatrix} x_{k+1} \\ 2x_k - x_{k+1} \end{bmatrix} = A\mathbf{v}_k \text{ where } A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$$

The eigenvalues of A are $\lambda_1 = -2$ and $\lambda_2 = 1$ with eigenvectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, so the diagonalizing matrix is $P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$.

Moreover, $\mathbf{b} = P_0^{-1}\mathbf{v}_0 = \frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ so the exact formula for $\mathbf{v}_k$ is

$$\begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix} = \mathbf{v}_k = b_1 \lambda_1^k \mathbf{x}_1 + b_2 \lambda_2^k \mathbf{x}_2 = \frac{2}{3}(-2)^k \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \frac{1}{3} 1^k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Equating top entries gives the desired formula for x_k:

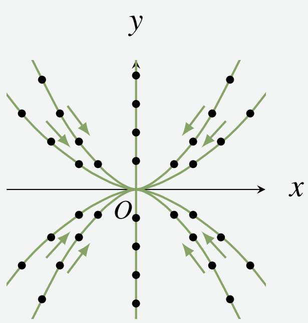
$$x_k = \frac{1}{3}\left[2(-2)^k + 1\right] \text{ for all } k = 0, 1, 2, \dots$$

The reader should check this for the first few values of k.

Graphical Description of Dynamical Systems

If a dynamical system $\mathbf{v}_{k+1} = A\mathbf{v}_k$ is given, the sequence $\mathbf{v}_0,\ \mathbf{v}_1,\ \mathbf{v}_2,\ \dots$ is called the **trajectory** of the system starting at $\mathbf{v}_0$. It is instructive to obtain a graphical plot of the system by writing $\mathbf{v}_k = \left[\begin{array}{c} x_k \\ y_k \end{array}\right]$ and plotting the successive values as points in the plane, identifying $\mathbf{v}_k$ with the point $(x_k,\ y_k)$ in the plane. We give several examples which illustrate properties of dynamical systems. For ease of calculation we assume that the matrix A is simple, usually diagonal.

Example 3.3.15

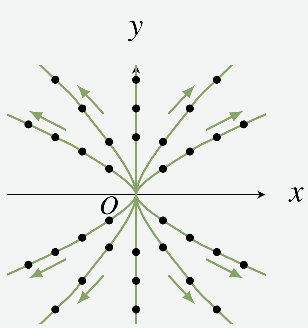


Let $A = \left[\begin{array}{cc} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{array}\right]$ Then the eigenvalues are $\frac{1}{2}$ and $\frac{1}{3}$, with corresponding eigenvectors $\mathbf{x}_1 = \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ and $\mathbf{x}_2 = \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$. The exact formula is

$$\mathbf{v}_k = b_1 \left(\tfrac{1}{2}\right)^k \left[\begin{array}{c} 1 \\ 0 \end{array}\right] + b_2 \left(\tfrac{1}{3}\right)^k \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$$

for $k = 0,\ 1,\ 2,\ \dots$ by Theorem 3.3.7, where the coefficients b_1 and b_2 depend on the initial point $\mathbf{v}_0$. Several trajectories are plotted in the diagram and, for each choice of $\mathbf{v}_0$, the trajectories converge toward the origin because both eigenvalues are less than 1 in absolute value. For this reason, the origin is called an **attractor** for the system.

Example 3.3.16

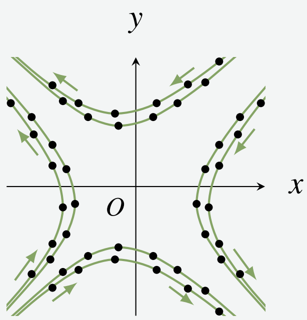


Let $A = \left[\begin{array}{cc} \frac{3}{2} & 0 \\ 0 & \frac{4}{3} \end{array}\right]$. Here the eigenvalues are $\frac{3}{2}$ and $\frac{4}{3}$, with corresponding eigenvectors $\mathbf{x}_1 = \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ and $\mathbf{x}_2 = \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$ as before. The exact formula is

$$\mathbf{v}_k = b_1 \left(\tfrac{3}{2}\right)^k \left[\begin{array}{c} 1 \\ 0 \end{array}\right] + b_2 \left(\tfrac{4}{3}\right)^k \left[\begin{array}{c} 0 \\ 1 \end{array}\right]$$

for $k = 0,\ 1,\ 2,\ \dots$. Since both eigenvalues are greater than 1 in absolute value, the trajectories diverge away from the origin for every choice of initial point V_0. For this reason, the origin is called a **repellor** for the system.

Example 3.3.17



Let $A = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}$. Now the eigenvalues are $\frac{3}{2}$ and $\frac{1}{2}$, with corresponding eigenvectors $\mathbf{x}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ The exact formula is

$$\mathbf{v}_k = b_1 \left(\frac{3}{2}\right)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix} + b_2 \left(\frac{1}{2}\right)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

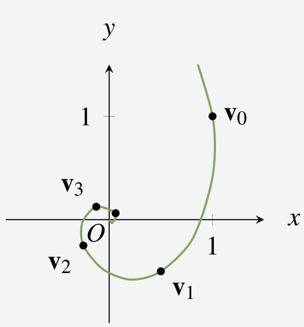
for $k = 0, 1, 2, \dots$. In this case $\frac{3}{2}$ is the dominant eigenvalue so, if $b_1 \neq 0$, we have $\mathbf{v}_k \approx b_1 \left(\frac{3}{2}\right)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ for large k and $\mathbf{v}_k$ is approaching the line $y = -x$.

However, if $b_1 = 0$, then $\mathbf{v}_k = b_2 \left(\frac{1}{2}\right)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and so approaches the origin along the line $y = x$. In general the trajectories appear as in the diagram, and the origin is called a **saddle point** for the dynamical system in this case.

Example 3.3.18

Let $A = \begin{bmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$. Now the characteristic polynomial is $c_A(x) = x^2 + \frac{1}{4}$, so the eigenvalues are the complex numbers $\frac{i}{2}$ and $-\frac{i}{2}$ where $i^2 = -1$. Hence A is not diagonalizable as a real matrix. However, the trajectories are not difficult to describe. If we start with $\mathbf{v}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ then the trajectory begins as

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -\frac{1}{8} \\ \frac{1}{8} \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} \frac{1}{16} \\ \frac{1}{16} \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} \frac{1}{32} \\ -\frac{1}{32} \end{bmatrix}, \mathbf{v}_6 = \begin{bmatrix} -\frac{1}{64} \\ -\frac{1}{64} \end{bmatrix}, \dots$$



The first five of these points are plotted in the diagram. Here each trajectory spirals in toward the origin, so the origin is an attractor. Note that the two (complex) eigenvalues have absolute value less than 1 here. If they had absolute value greater than 1, the trajectories would spiral out from the origin.

Google PageRank

Dominant eigenvalues are useful to the Google search engine for finding information on the Web. If an information query comes in from a client, Google has a sophisticated method of establishing the "relevance" of each site to that query. When the relevant sites have been determined, they are placed in order of importance using a ranking of *all* sites called the PageRank. The relevant sites with the highest PageRank are the ones presented to the client. It is the construction of the PageRank that is our interest here.

The Web contains many links from one site to another. Google interprets a link from site j to site i as a "vote" for the importance of site i. Hence if site i has more links to it than does site j, then i is regarded as more "important" and assigned a higher PageRank. One way to look at this is to view the sites as vertices in a huge directed graph (see Section 2.2). Then if site j links to site i there is an edge from j to i, and hence the $(i,\ j)$-entry is a 1 in the associated adjacency matrix (called the *connectivity* matrix in this context). Thus a large number of 1s in row i of this matrix is a measure of the PageRank of site i.[13]

However this does not take into account the PageRank of the sites that link to i. Intuitively, the higher the rank of these sites, the higher the rank of site i. One approach is to compute a dominant eigenvector $\mathbf{x}$ for the connectivity matrix. In most cases the entries of $\mathbf{x}$ can be chosen to be positive with sum 1. Each site corresponds to an entry of $\mathbf{x}$, so the sum of the entries of sites linking to a given site i is a measure of the rank of site i. In fact, Google chooses the PageRank of a site so that it is proportional to this sum.[14]

Exercises for 3.3

Exercise 3.3.1 In each case find the characteristic polynomial, eigenvalues, eigenvectors, and (if possible) an invertible matrix P such that $P^{-1}AP$ is diagonal.

a. $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

b. $A = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix}$

c. $A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 1 & -3 \\ 2 & 0 & 6 \\ 1 & -1 & 5 \end{bmatrix}$

e. $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 6 & -6 \\ 1 & 2 & -1 \end{bmatrix}$

f. $A = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$

g. $A = \begin{bmatrix} 3 & 1 & 1 \\ -4 & -2 & -5 \\ 2 & 2 & 5 \end{bmatrix}$

h. $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & 2 \end{bmatrix}$

i. $A = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \mu \end{bmatrix}, \lambda \neq \mu$

Exercise 3.3.2 Consider a linear dynamical system $\mathbf{v}_{k+1} = A\mathbf{v}_k$ for $k \geq 0$. In each case approximate $\mathbf{v}_k$ using Theorem 3.3.7.

a. $A = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}, \mathbf{v}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

b. $A = \begin{bmatrix} 3 & -2 \\ 2 & -2 \end{bmatrix}, \mathbf{v}_0 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 3 \\ 1 & 4 & 1 \end{bmatrix}, \mathbf{v}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 3 & 2 \\ -1 & 2 & 1 \\ 4 & -1 & -1 \end{bmatrix}, \mathbf{v}_0 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

Exercise 3.3.3 Show that A has $\lambda = 0$ as an eigenvalue if and only if A is not invertible.

Exercise 3.3.4 Let A denote an $n \times n$ matrix and put $A_1 = A - \alpha I$, α in $\mathbb{R}$. Show that λ is an eigenvalue of A if and only if $\lambda - \alpha$ is an eigenvalue of A_1. (Hence,

[13]For more on PageRank, visit https://en.wikipedia.org/wiki/PageRank.

[14]See the articles "Searching the web with eigenvectors" by Herbert S. Wilf, UMAP Journal 23(2), 2002, pages 101–103, and "The worlds largest matrix computation: Google's PageRank is an eigenvector of a matrix of order 2.7 billion" by Cleve Moler, Matlab News and Notes, October 2002, pages 12–13.

the eigenvalues of A_1 are just those of A "shifted" by α.) How do the eigenvectors compare?

Exercise 3.3.5 Show that the eigenvalues of $\left[\begin{array}{cc} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{array}\right]$ are $e^{i\theta}$ and $e^{-i\theta}$. (See Appendix A)

Exercise 3.3.6 Find the characteristic polynomial of the $n \times n$ identity matrix I. Show that I has exactly one eigenvalue and find the eigenvectors.

Exercise 3.3.7 Given $A = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$ show that:

a. $c_A(x) = x^2 - \text{tr } Ax + \det A$, where $\text{tr } A = a + d$ is called the **trace** of A.

b. The eigenvalues are $\frac{1}{2}\left[(a+d) \pm \sqrt{(a-b)^2 + 4bc}\right]$.

Exercise 3.3.8 In each case, find $P^{-1}AP$ and then compute A^n.

a. $A = \left[\begin{array}{rr} 6 & -5 \\ 2 & -1 \end{array}\right]$, $P = \left[\begin{array}{rr} 1 & 5 \\ 1 & 2 \end{array}\right]$

b. $A = \left[\begin{array}{rr} -7 & -12 \\ 6 & -10 \end{array}\right]$, $P = \left[\begin{array}{rr} -3 & 4 \\ 2 & -3 \end{array}\right]$

[*Hint*: $(PDP^{-1})^n = PD^nP^{-1}$ for each $n = 1, 2, \dots$.]

Exercise 3.3.9

a. If $A = \left[\begin{array}{rr} 1 & 3 \\ 0 & 2 \end{array}\right]$ and $B = \left[\begin{array}{rr} 2 & 0 \\ 0 & 1 \end{array}\right]$ verify that A and B are diagonalizable, but AB is not.

b. If $D = \left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array}\right]$ find a diagonalizable matrix A such that $D + A$ is not diagonalizable.

Exercise 3.3.10 If A is an $n \times n$ matrix, show that A is diagonalizable if and only if A^T is diagonalizable.

Exercise 3.3.11 If A is diagonalizable, show that each of the following is also diagonalizable.

a. A^n, $n \geq 1$

b. kA, k any scalar.

c. $p(A)$, $p(x)$ any polynomial (Theorem 3.3.1)

d. $U^{-1}AU$ for any invertible matrix U.

e. $kI + A$ for any scalar k.

Exercise 3.3.12 Give an example of two diagonalizable matrices A and B whose sum $A + B$ is not diagonalizable.

Exercise 3.3.13 If A is diagonalizable and 1 and -1 are the only eigenvalues, show that $A^{-1} = A$.

Exercise 3.3.14 If A is diagonalizable and 0 and 1 are the only eigenvalues, show that $A^2 = A$.

Exercise 3.3.15 If A is diagonalizable and $\lambda \geq 0$ for each eigenvalue of A, show that $A = B^2$ for some matrix B.

Exercise 3.3.16 If $P^{-1}AP$ and $P^{-1}BP$ are both diagonal, show that $AB = BA$. [*Hint*: Diagonal matrices commute.]

Exercise 3.3.17 A square matrix A is called **nilpotent** if $A^n = 0$ for some $n \geq 1$. Find all nilpotent diagonalizable matrices. [*Hint*: Theorem 3.3.1.]

Exercise 3.3.18 Let A be any $n \times n$ matrix and $r \neq 0$ a real number.

a. Show that the eigenvalues of rA are precisely the numbers $r\lambda$, where λ is an eigenvalue of A.

b. Show that $c_{rA}(x) = r^n c_A\left(\frac{x}{r}\right)$.

Exercise 3.3.19

a. If all rows of A have the same sum s, show that s is an eigenvalue.

b. If all columns of A have the same sum s, show that s is an eigenvalue.

Exercise 3.3.20 Let A be an invertible $n \times n$ matrix.

a. Show that the eigenvalues of A are nonzero.

b. Show that the eigenvalues of A^{-1} are precisely the numbers $1/\lambda$, where λ is an eigenvalue of A.

c. Show that $c_{A^{-1}}(x) = \frac{(-x)^n}{\det A} c_A\left(\frac{1}{x}\right)$.

Exercise 3.3.21 Suppose λ is an eigenvalue of a square matrix A with eigenvector $\mathbf{x} \neq \mathbf{0}$.

a. Show that λ^2 is an eigenvalue of A^2 (with the same $\mathbf{x}$).

b. Show that $\lambda^3 - 2\lambda + 3$ is an eigenvalue of $A^3 - 2A + 3I$.

c. Show that $p(\lambda)$ is an eigenvalue of $p(A)$ for any nonzero polynomial $p(x)$.

Exercise 3.3.22 If A is an $n \times n$ matrix, show that $c_{A^2}(x^2) = (-1)^n c_A(x) c_A(-x)$.

Exercise 3.3.23 An $n \times n$ matrix A is called nilpotent if $A^m = 0$ for some $m \geq 1$.

a. Show that every triangular matrix with zeros on the main diagonal is nilpotent.

b. If A is nilpotent, show that $\lambda = 0$ is the only eigenvalue (even complex) of A.

c. Deduce that $c_A(x) = x^n$, if A is $n \times n$ and nilpotent.

Exercise 3.3.24 Let A be diagonalizable with real eigenvalues and assume that $A^m = I$ for some $m \geq 1$.

a. Show that $A^2 = I$.

b. If m is odd, show that $A = I$.

[*Hint*: Theorem A.3]

Exercise 3.3.25 Let $A^2 = I$, and assume that $A \neq I$ and $A \neq -I$.

a. Show that the only eigenvalues of A are $\lambda = 1$ and $\lambda = -1$.

b. Show that A is diagonalizable. [*Hint*: Verify that $A(A+I) = A+I$ and $A(A-I) = -(A-I)$, and then look at nonzero columns of $A+I$ and of $A-I$.]

c. If $Q_m : \mathbb{R}^2 \to \mathbb{R}^2$ is reflection in the line $y = mx$ where $m \neq 0$, use (b) to show that the matrix of Q_m is diagonalizable for each m.

d. Now prove (c) geometrically using Theorem 3.3.3.

Exercise 3.3.26 Let $A = \left[\begin{array}{rrr} 2 & 3 & -3 \\ 1 & 0 & -1 \\ 1 & 1 & -2 \end{array} \right]$ and $B = \left[\begin{array}{rrr} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 2 & 0 & 0 \end{array} \right]$. Show that $c_A(x) = c_B(x) = (x+1)^2(x-2)$, but A is diagonalizable and B is not.

Exercise 3.3.27

a. Show that the only diagonalizable matrix A that has only one eigenvalue λ is the scalar matrix $A = \lambda I$.

b. Is $\left[\begin{array}{rr} 3 & -2 \\ 2 & -1 \end{array} \right]$ diagonalizable?

Exercise 3.3.28 Characterize the diagonalizable $n \times n$ matrices A such that $A^2 - 3A + 2I = 0$ in terms of their eigenvalues. [*Hint*: Theorem 3.3.1.]

Exercise 3.3.31 Referring to the model in Example 3.3.1, determine if the population stabilizes, becomes extinct, or becomes large in each case. Denote the adult and juvenile survival rates as A and J, and the reproduction rate as R.

	R	A	J
$a.$	2	$\frac{1}{2}$	$\frac{1}{2}$
$b.$	3	$\frac{1}{4}$	$\frac{1}{4}$
$c.$	2	$\frac{1}{4}$	$\frac{1}{3}$
$d.$	3	$\frac{3}{5}$	$\frac{1}{5}$

Exercise 3.3.32 In the model of Example 3.3.1, does the final outcome depend on the initial population of adult and juvenile females? Support your answer.

Exercise 3.3.33 In Example 3.3.1, keep the same reproduction rate of 2 and the same adult survival rate of $\frac{1}{2}$, but suppose that the juvenile survival rate is ρ. Determine which values of ρ cause the population to become extinct or to become large.

Exercise 3.3.34 In Example 3.3.1, let the juvenile survival rate be $\frac{2}{5}$ and let the reproduction rate be 2. What values of the adult survival rate α will ensure that the population stabilizes?

Chapter 4

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Vector Geometry

4.1 Vectors and Lines

In this chapter we study the geometry of 3-dimensional space. We view a point in 3-space as an arrow from the origin to that point. Doing so provides a "picture" of the point that is truly worth a thousand words. We used this idea earlier, in Section 2.6, to describe rotations, reflections, and projections of the plane $\mathbb{R}^2$. We now apply the same techniques to 3-space to examine similar transformations of $\mathbb{R}^3$. Moreover, the method enables us to completely describe all lines and planes in space.

Vectors in $\mathbb{R}^3$

Introduce a coordinate system in 3-dimensional space in the usual way. First choose a point O called the *origin*, then choose three mutually perpendicular lines through O, called the x, y, and z *axes*, and establish a number scale on each axis with zero at the origin. Given a point P in 3-space we associate three numbers x, y, and z with P, as described in Figure 4.1.1. These numbers are called the *coordinates* of P, and we denote the point as (x, y, z), or $P(x, y, z)$ to emphasize the label P. The result is called a *cartesian*[1] coordinate system for 3-space, and the resulting description of 3-space is called *cartesian geometry*.

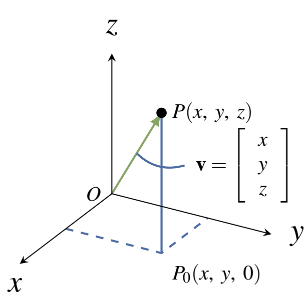


Figure 4.1.1

As in the plane, we introduce vectors by identifying each point $P(x, y, z)$ with the vector $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in $\mathbb{R}^3$, represented by the **arrow** from the origin to P as in Figure 4.1.1. Informally, we say that the point P *has vector* $\mathbf{v}$, and that vector $\mathbf{v}$ *has point* P. In this way 3-space is identified with $\mathbb{R}^3$, and this identification will be made throughout this chapter, often without comment. In particular, the terms "vector" and "point" are interchangeable.[2] The resulting description of 3-space is called **vector geometry**. Note that the origin is $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Length and Direction

We are going to discuss two fundamental geometric properties of vectors in $\mathbb{R}^3$: length and direction. First, if $\mathbf{v}$ is a vector with point P, the **length** $\|\mathbf{v}\|$ of vector $\mathbf{v}$ is defined to be the distance from the origin to P, that is the length of the arrow representing $\mathbf{v}$. The following properties of length will be used frequently.

[1]Named after René Descartes who introduced the idea in 1637.

[2]Recall that we defined $\mathbb{R}^n$ as the set of all ordered n-tuples of real numbers, and reserved the right to denote them as rows or as columns.

Theorem 4.1.1

Let $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ *be a vector.*

1. $\|\mathbf{v}\| = \sqrt{x^2+y^2+z^2}$. [3]
2. $\mathbf{v} = \mathbf{0}$ *if and only if* $\|\mathbf{v}\| = 0$
3. $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$ *for all scalars* a. [4]

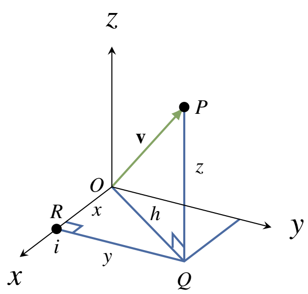


Figure 4.1.2

Proof. Let $\mathbf{v}$ have point $P(x,\ y,\ z)$.

1. In Figure 4.1.2, $\|\mathbf{v}\|$ is the hypotenuse of the right triangle OQP, and so $\|\mathbf{v}\|^2 = h^2 + z^2$ by Pythagoras' theorem.[5] But h is the hypotenuse of the right triangle ORQ, so $h^2 = x^2 + y^2$. Now (1) follows by eliminating h^2 and taking positive square roots.

2. If $\|\mathbf{v}\| = 0$, then $x^2+y^2+z^2 = 0$ by (1). Because squares of real numbers are nonnegative, it follows that $x = y = z = 0$, and hence that $\mathbf{v} = \mathbf{0}$. The converse is because $\|\mathbf{0}\| = 0$.

3. We have $a\mathbf{v} = \begin{bmatrix} ax & ay & az \end{bmatrix}^T$ so (1) gives
$$\|a\mathbf{v}\|^2 = (ax)^2 + (ay)^2 + (az)^2 = a^2\|\mathbf{v}\|^2$$

Hence $\|a\mathbf{v}\| = \sqrt{a^2}\|\mathbf{v}\|$, and we are done because $\sqrt{a^2} = |a|$ for any real number a. □

Of course the $\mathbb{R}^2$-version of Theorem 4.1.1 also holds.

[3]When we write $\sqrt{p}$ we mean the positive square root of p.

[4]Recall that the absolute value $|a|$ of a real number is defined by $|a| = \begin{cases} a \text{ if } a \geq 0 \\ -a \text{ if } a < 0 \end{cases}$.

[5]Pythagoras' theorem states that if a and b are sides of right triangle with hypotenuse c, then $a^2+b^2=c^2$. A proof is given at the end of this section.

Example 4.1.1

If $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ then $\|\mathbf{v}\| = \sqrt{4+1+9} = \sqrt{14}$. Similarly if $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$ in 2-space then $\|\mathbf{v}\| = \sqrt{9+16} = 5$.

When we view two nonzero vectors as arrows emanating from the origin, it is clear geometrically what we mean by saying that they have the same or opposite **direction**. This leads to a fundamental new description of vectors.

Theorem 4.1.2

Let $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$ be vectors in $\mathbb{R}^3$. Then $\mathbf{v} = \mathbf{w}$ as matrices if and only if $\mathbf{v}$ and $\mathbf{w}$ have the same direction and the same length.[6]

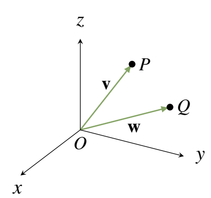


Figure 4.1.3

Proof. If $\mathbf{v} = \mathbf{w}$, they clearly have the same direction and length. Conversely, let $\mathbf{v}$ and $\mathbf{w}$ be vectors with points $P(x, y, z)$ and $Q(x_1, y_1, z_1)$ respectively. If $\mathbf{v}$ and $\mathbf{w}$ have the same length and direction then, geometrically, P and Q must be the same point (see Figure 4.1.3). Hence $x = x_1$, $y = y_1$, and $z = z_1$, that is $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \mathbf{w}$. □

A characterization of a vector in terms of its length and direction only is called an **intrinsic** description of the vector. The point to note is that such a description does *not* depend on the choice of coordinate system in $\mathbb{R}^3$. Such descriptions are important in applications because physical laws are often stated in terms of vectors, and these laws cannot depend on the particular coordinate system used to describe the situation.

Geometric Vectors

If A and B are distinct points in space, the arrow from A to B has length and direction.

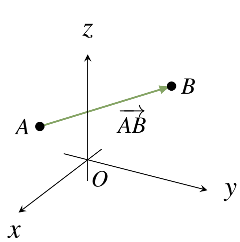


Figure 4.1.4

[6]It is Theorem 4.1.2 that gives vectors their power in science and engineering because many physical quantities are determined by their length and magnitude (and are called **vector quantities**). For example, saying that an airplane is flying at 200 km/h does not describe where it is going; the direction must also be specified. The speed and direction comprise the **velocity** of the airplane, a vector quantity.

Hence:

Definition 4.1 Geometric Vectors

Suppose that A and B are any two points in $\mathbb{R}^3$. In Figure 4.1.4 the line segment from A to B is denoted $\overrightarrow{AB}$ and is called the **geometric vector** *from A to B. Point A is called the* **tail** *of $\overrightarrow{AB}$, B is called the* **tip** *of $\overrightarrow{AB}$, and the* **length** *of $\overrightarrow{AB}$ is denoted $\|\overrightarrow{AB}\|$.*

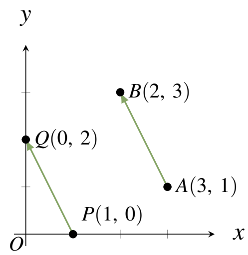


Figure 4.1.5

Note that if $\mathbf{v}$ is any vector in $\mathbb{R}^3$ with point P then $\mathbf{v} = \overrightarrow{OP}$ is itself a geometric vector where O is the origin. Referring to $\overrightarrow{AB}$ as a "vector" seems justified by Theorem 4.1.2 because it has a direction (from A to B) and a length $\|\overrightarrow{AB}\|$. However there appears to be a problem because two geometric vectors can have the same length and direction even if the tips and tails are different. For example $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ in Figure 4.1.5 have the same length $\sqrt{5}$ and the same direction (1 unit left and 2 units up) so, by Theorem 4.1.2, they are the same vector! The best way to understand this apparent paradox is to see $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ as different *representations* of the same[7] underlying vector $\left[\begin{array}{r} -1 \\ 2 \end{array}\right]$. Once it is clarified, this phenomenon is a great benefit because, thanks to Theorem 4.1.2, it means that the same geometric vector can be positioned anywhere in space; what is important is the length and direction, not the location of the tip and tail. This ability to move geometric vectors about is very useful as we shall soon see.

The Parallelogram Law

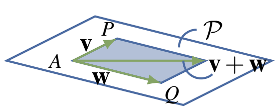


Figure 4.1.6

We now give an intrinsic description of the sum of two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, that is a description that depends only on the lengths and directions of $\mathbf{v}$ and $\mathbf{w}$ and not on the choice of coordinate system. Using Theorem 4.1.2 we can think of these vectors as having a common tail A. If their tips are P and Q respectively, then they both lie in a plane $\mathcal{P}$ containing A, P, and Q, as shown in Figure 4.1.6. The vectors $\mathbf{v}$ and $\mathbf{w}$ create a parallelogram[8] in $\mathcal{P}$, shaded in Figure 4.1.6, called the parallelogram **determined** by $\mathbf{v}$ and $\mathbf{w}$.

If we now choose a coordinate system in the plane $\mathcal{P}$ with A as origin, then the parallelogram law in the plane (Section 2.6) shows that their sum $\mathbf{v}+\mathbf{w}$ is the diagonal of the parallelogram they determine with tail A. This is an intrinsic description of the sum $\mathbf{v}+\mathbf{w}$ because it makes no reference to coordinates. This discussion proves:

The Parallelogram Law

In the parallelogram determined by two vectors $\mathbf{v}$ and $\mathbf{w}$, the vector $\mathbf{v}+\mathbf{w}$ is the diagonal with the same tail as $\mathbf{v}$ and $\mathbf{w}$.

[7]Fractions provide another example of quantities that can be the same but look different. For example $\frac{6}{9}$ and $\frac{14}{21}$ certainly appear different, but they are equal fractions—both equal $\frac{2}{3}$ in "lowest terms".

[8]Recall that a parallelogram is a four-sided figure whose opposite sides are parallel and of equal length.

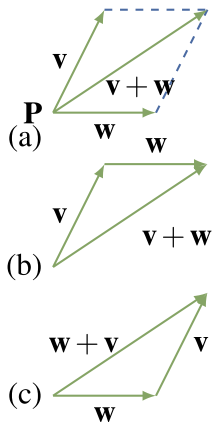


Figure 4.1.7

Because a vector can be positioned with its tail at any point, the parallelogram law leads to another way to view vector addition. In Figure 4.1.7(a) the sum $\mathbf{v}+\mathbf{w}$ of two vectors $\mathbf{v}$ and $\mathbf{w}$ is shown as given by the parallelogram law. If $\mathbf{w}$ is moved so its tail coincides with the tip of $\mathbf{v}$ (Figure 4.1.7(b)) then the sum $\mathbf{v}+\mathbf{w}$ is seen as "first $\mathbf{v}$ and then $\mathbf{w}$. Similarly, moving the tail of $\mathbf{v}$ to the tip of $\mathbf{w}$ shows in Figure 4.1.7(c) that $\mathbf{v}+\mathbf{w}$ is "first $\mathbf{w}$ and then $\mathbf{v}$." This will be referred to as the **tip-to-tail rule**, and it gives a graphic illustration of why $\mathbf{v}+\mathbf{w}=\mathbf{w}+\mathbf{v}$.

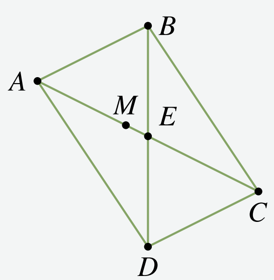
Since $\overrightarrow{AB}$ denotes the vector from a point A to a point B, the tip-to-tail rule takes the easily remembered form

$$\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$$

for any points A, B, and C. The next example uses this to derive a theorem in geometry without using coordinates.

Example 4.1.2

Show that the diagonals of a parallelogram bisect each other.



Solution. Let the parallelogram have vertices A, B, C, and D, as shown; let E denote the intersection of the two diagonals; and let M denote the midpoint of diagonal AC. We must show that $M=E$ and that this is the midpoint of diagonal BD. This is accomplished by showing that $\overrightarrow{BM}=\overrightarrow{MD}$. (Then the fact that these vectors have the same direction means that $M=E$, and the fact that they have the same length means that $M=E$ is the midpoint of BD.) Now $\overrightarrow{AM}=\overrightarrow{MC}$ because M is the midpoint of AC, and $\overrightarrow{BA}=\overrightarrow{CD}$ because the figure is a parallelogram. Hence

$$\overrightarrow{BM}=\overrightarrow{BA}+\overrightarrow{AM}=\overrightarrow{CD}+\overrightarrow{MC}=\overrightarrow{MC}+\overrightarrow{CD}=\overrightarrow{MD}$$

where the first and last equalities use the tip-to-tail rule of vector addition.

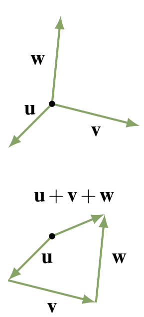


Figure 4.1.8

One reason for the importance of the tip-to-tail rule is that it means two or more vectors can be added by placing them tip-to-tail in sequence. This gives a useful "picture" of the sum of several vectors, and is illustrated for three vectors in Figure 4.1.8 where $\mathbf{u}+\mathbf{v}+\mathbf{w}$ is viewed as first $\mathbf{u}$, then $\mathbf{v}$, then $\mathbf{w}$.

There is a simple geometrical way to visualize the (matrix) **difference** $\mathbf{v}-\mathbf{w}$ of two vectors. If $\mathbf{v}$ and $\mathbf{w}$ are positioned so that they have a common tail A (see Figure 4.1.9), and if B and C are their respective tips, then the tip-to-tail rule gives $\mathbf{w}+\overrightarrow{CB}=\mathbf{v}$. Hence $\mathbf{v}-\mathbf{w}=\overrightarrow{CB}$ is the vector from the tip of $\mathbf{w}$ to the tip of $\mathbf{v}$. Thus both $\mathbf{v}-\mathbf{w}$ and $\mathbf{v}+\mathbf{w}$ appear as diagonals in the parallelogram determined by $\mathbf{v}$ and $\mathbf{w}$ (see Figure 4.1.9). We record this for reference.

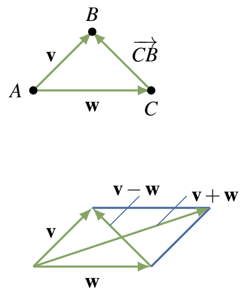


Figure 4.1.9

> **Theorem 4.1.3**
>
> *If $\mathbf{v}$ and $\mathbf{w}$ have a common tail, then $\mathbf{v}-\mathbf{w}$ is the vector from the tip of $\mathbf{w}$ to the tip of $\mathbf{v}$.*

One of the most useful applications of vector subtraction is that it gives a simple formula for the vector from one point to another, and for the distance between the points.

> **Theorem 4.1.4**
>
> *Let $P_1(x_1,\ y_1,\ z_1)$ and $P_2(x_2,\ y_2,\ z_2)$ be two points. Then:*
>
> 1. $\overrightarrow{P_1P_2} = \left[\begin{array}{c} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{array} \right]$.
>
> 2. *The distance between P_1 and P_2 is* $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$.

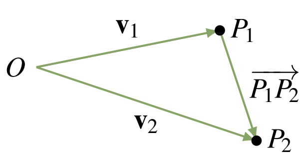


Figure 4.1.10

Proof. If O is the origin, write

$$\mathbf{v}_1 = \overrightarrow{OP}_1 = \left[\begin{array}{c} x_1 \\ y_1 \\ z_1 \end{array} \right] \text{ and } \mathbf{v}_2 = \overrightarrow{OP}_2 = \left[\begin{array}{c} x_2 \\ y_2 \\ z_2 \end{array} \right]$$

as in Figure 4.1.10.

Then Theorem 4.1.3 gives $\overrightarrow{P_1P_2} = \mathbf{v}_2 - \mathbf{v}_1$, and (1) follows. But the distance between P_1 and P_2 is $\|\overrightarrow{P_1P_2}\|$, so (2) follows from (1) and Theorem 4.1.1. □

Of course the $\mathbb{R}^2$-version of Theorem 4.1.4 is also valid: If $P_1(x_1,\ y_1)$ and $P_2(x_2,\ y_2)$ are points in $\mathbb{R}^2$, then $\overrightarrow{P_1P_2} = \left[\begin{array}{c} x_2 - x_1 \\ y_2 - y_1 \end{array} \right]$, and the distance between P_1 and P_2 is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$.

> **Example 4.1.3**
>
> The distance between $P_1(2,\ -1,\ 3)$ and $P_2(1,\ 1,\ 4)$ is $\sqrt{(-1)^2+(2)^2+(1)^2} = \sqrt{6}$, and the vector from P_1 to P_2 is $\overrightarrow{P_1P_2} = \left[\begin{array}{r} -1 \\ 2 \\ 1 \end{array} \right]$.

As for the parallelogram law, the intrinsic rule for finding the length and direction of a scalar multiple of a vector in $\mathbb{R}^3$ follows easily from the same situation in $\mathbb{R}^2$.

Scalar Multiple Law

If a is a real number and $\mathbf{v} \neq \mathbf{0}$ is a vector then:

1. *The length of $a\mathbf{v}$ is $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$.*
2. *If[9] $a\mathbf{v} \neq \mathbf{0}$, the direction of $a\mathbf{v}$ is* $\begin{cases} \text{the same as } \mathbf{v} \text{ if } a > 0, \\ \text{opposite to } \mathbf{v} \text{ if } a < 0. \end{cases}$

Proof.

1. This is part of Theorem 4.1.1.
2. Let O denote the origin in $\mathbb{R}^3$, let $\mathbf{v}$ have point P, and choose any plane containing O and P. If we set up a coordinate system in this plane with O as origin, then $\mathbf{v} = \overrightarrow{OP}$ so the result in (2) follows from the scalar multiple law in the plane (Section 2.6). □

Figure 4.1.11 gives several examples of scalar multiples of a vector $\mathbf{v}$.

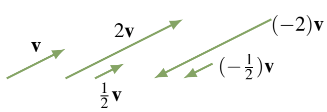


Figure 4.1.11

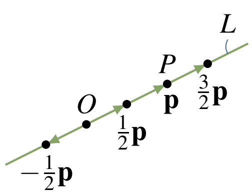


Figure 4.1.12

Consider a line L through the origin, let P be any point on L other than the origin O, and let $\mathbf{p} = \overrightarrow{OP}$. If $t \neq 0$, then $t\mathbf{p}$ is a point on L because it has direction the same or opposite as that of $\mathbf{p}$. Moreover $t > 0$ or $t < 0$ according as the point $t\mathbf{p}$ lies on the same or opposite side of the origin as P. This is illustrated in Figure 4.1.12.

A vector $\mathbf{u}$ is called a **unit vector** if $\|\mathbf{u}\| = 1$. Then $\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are unit vectors, called the **coordinate** vectors. We discuss them in more detail in Section 4.2.

Example 4.1.4

If $\mathbf{v} \neq \mathbf{0}$ show that $\frac{1}{\|\mathbf{v}\|}\mathbf{v}$ is the unique unit vector in the same direction as $\mathbf{v}$.

Solution. The vectors in the same direction as $\mathbf{v}$ are the scalar multiples $a\mathbf{v}$ where $a > 0$. But $\|a\mathbf{v}\| = |a|\|\mathbf{v}\| = a\|\mathbf{v}\|$ when $a > 0$, so $a\mathbf{v}$ is a unit vector if and only if $a = \frac{1}{\|\mathbf{v}\|}$.

The next example shows how to find the coordinates of a point on the line segment between two given points. The technique is important and will be used again below.

[9] Since the zero vector has no direction, we deal only with the case $a\mathbf{v} \neq \mathbf{0}$.

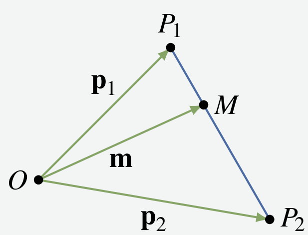
Example 4.1.5

Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be the vectors of two points P_1 and P_2. If M is the point one third the way from P_1 to P_2, show that the vector $\mathbf{m}$ of M is given by

$$\mathbf{m} = \frac{2}{3}\mathbf{p}_1 + \frac{1}{3}\mathbf{p}_2$$

Conclude that if $P_1 = P_1(x_1,\ y_1,\ z_1)$ and $P_2 = P_2(x_2,\ y_2,\ z_2)$, then M has coordinates

$$M = M\left(\frac{2}{3}x_1 + \frac{1}{3}x_2,\ \frac{2}{3}y_1 + \frac{1}{3}y_2,\ \frac{2}{3}z_1 + \frac{1}{3}z_2\right)$$



Solution. The vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{m}$ are shown in the diagram. We have $\overrightarrow{P_1M} = \frac{1}{3}\overrightarrow{P_1P_2}$ because $\overrightarrow{P_1M}$ is in the same direction as $\overrightarrow{P_1P_2}$ and $\frac{1}{3}$ as long. By Theorem 4.1.3 we have $\overrightarrow{P_1P_2} = \mathbf{p}_2 - \mathbf{p}_1$, so tip-to-tail addition gives

$$\mathbf{m} = \mathbf{p}_1 + \overrightarrow{P_1M} = \mathbf{p}_1 + \frac{1}{3}(\mathbf{p}_2 - \mathbf{p}_1) = \frac{2}{3}\mathbf{p}_1 + \frac{1}{3}\mathbf{p}_2$$

as required. For the coordinates, we have $\mathbf{p}_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\mathbf{p}_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, so

$$\mathbf{m} = \frac{2}{3}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \frac{1}{3}\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3}x_1 + \frac{1}{3}x_2 \\ \frac{2}{3}y_1 + \frac{1}{3}y_2 \\ \frac{2}{3}z_1 + \frac{1}{3}z_2 \end{bmatrix}$$

by matrix addition. The last statement follows.

Note that in Example 4.1.5 $\mathbf{m} = \frac{2}{3}\mathbf{p}_1 + \frac{1}{3}\mathbf{p}_2$ is a "weighted average" of $\mathbf{p}_1$ and $\mathbf{p}_2$ with more weight on $\mathbf{p}_1$ because $\mathbf{m}$ is closer to $\mathbf{p}_1$.

The point M halfway between points P_1 and P_2 is called the **midpoint** between these points. In the same way, the vector $\mathbf{m}$ of M is

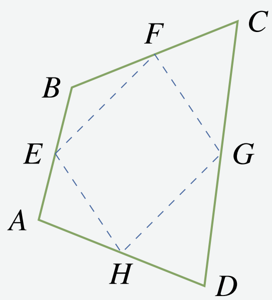
$$\mathbf{m} = \frac{1}{2}\mathbf{p}_1 + \frac{1}{2}\mathbf{p}_2 = \frac{1}{2}(\mathbf{p}_1 + \mathbf{p}_2)$$

as the reader can verify, so $\mathbf{m}$ is the "average" of $\mathbf{p}_1$ and $\mathbf{p}_2$ in this case.

Example 4.1.6

Show that the midpoints of the four sides of any quadrilateral are the vertices of a parallelogram. Here a quadrilateral is any figure with four vertices and straight sides.

Solution. Suppose that the vertices of the quadrilateral are A, B, C, and D (in that order) and that E, F, G, and H are the midpoints of the sides as shown in the diagram. It suffices to show $\overrightarrow{EF} = \overrightarrow{HG}$ (because then sides EF and HG are parallel and of equal length).



Now the fact that E is the midpoint of AB means that $\overrightarrow{EB} = \frac{1}{2}\overrightarrow{AB}$. Similarly, $\overrightarrow{BF} = \frac{1}{2}\overrightarrow{BC}$, so

$$\overrightarrow{EF} = \overrightarrow{EB} + \overrightarrow{BF} = \tfrac{1}{2}\overrightarrow{AB} + \tfrac{1}{2}\overrightarrow{BC} = \tfrac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC}) = \tfrac{1}{2}\overrightarrow{AC}$$

A similar argument shows that $\overrightarrow{HG} = \frac{1}{2}\overrightarrow{AC}$ too, so $\overrightarrow{EF} = \overrightarrow{HG}$ as required.

Definition 4.2 Parallel Vectors in $\mathbb{R}^3$

Two nonzero vectors are called **parallel** *if they have the same or opposite direction.*

Many geometrical propositions involve this notion, so the following theorem will be referred to repeatedly.

Theorem 4.1.5

Two nonzero vectors $\mathbf{v}$ *and* $\mathbf{w}$ *are parallel if and only if one is a scalar multiple of the other.*

Proof. If one of them is a scalar multiple of the other, they are parallel by the scalar multiple law.

Conversely, assume that $\mathbf{v}$ and $\mathbf{w}$ are parallel and write $d = \frac{\|\mathbf{v}\|}{\|\mathbf{w}\|}$ for convenience. Then $\mathbf{v}$ and $\mathbf{w}$ have the same or opposite direction. If they have the same direction we show that $\mathbf{v} = d\mathbf{w}$ by showing that $\mathbf{v}$ and $d\mathbf{w}$ have the same length and direction. In fact, $\|d\mathbf{w}\| = |d|\|\mathbf{w}\| = \|\mathbf{v}\|$ by Theorem 4.1.1; as to the direction, $d\mathbf{w}$ and $\mathbf{w}$ have the same direction because $d > 0$, and this is the direction of $\mathbf{v}$ by assumption. Hence $\mathbf{v} = d\mathbf{w}$ in this case by Theorem 4.1.2. In the other case, $\mathbf{v}$ and $\mathbf{w}$ have opposite direction and a similar argument shows that $\mathbf{v} = -d\mathbf{w}$. We leave the details to the reader. □

Example 4.1.7

Given points $P(2, -1, 4)$, $Q(3, -1, 3)$, $A(0, 2, 1)$, and $B(1, 3, 0)$, determine if $\overrightarrow{PQ}$ and $\overrightarrow{AB}$ are parallel.

Solution. By Theorem 4.1.3, $\overrightarrow{PQ} = (1, 0, -1)$ and $\overrightarrow{AB} = (1, 1, -1)$. If $\overrightarrow{PQ} = t\overrightarrow{AB}$ then $(1, 0, -1) = (t, t, -t)$, so $1 = t$ and $0 = t$, which is impossible. Hence $\overrightarrow{PQ}$ is *not* a scalar multiple of $\overrightarrow{AB}$, so these vectors are not parallel by Theorem 4.1.5.

Lines in Space

These vector techniques can be used to give a very simple way of describing straight lines in space. In order to do this, we first need a way to specify the orientation of such a line, much as the slope does in the plane.

Definition 4.3 Direction Vector of a Line

With this in mind, we call a nonzero vector $\mathbf{d} \neq \mathbf{0}$ *a* ***direction vector*** *for the line if it is parallel to* $\overrightarrow{AB}$ *for some pair of distinct points* A *and* B *on the line.*

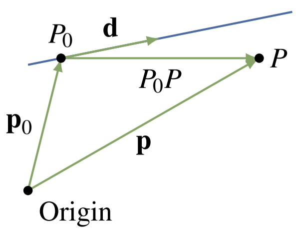


Figure 4.1.13

Of course it is then parallel to $\overrightarrow{CD}$ for *any* distinct points C and D on the line. In particular, any nonzero scalar multiple of $\mathbf{d}$ will also serve as a direction vector of the line.

We use the fact that there is exactly one line that passes through a particular point $P_0(x_0, y_0, z_0)$ and has a given direction vector $\mathbf{d} = \left[\begin{array}{c} a \\ b \\ c \end{array}\right]$. We want to describe this line by giving a condition on x, y, and z that the point $P(x, y, z)$ lies on this line. Let $\mathbf{p}_0 = \left[\begin{array}{c} x_0 \\ y_0 \\ z_0 \end{array}\right]$ and $\mathbf{p} = \left[\begin{array}{c} x \\ y \\ z \end{array}\right]$ denote the vectors of P_0 and P, respectively (see Figure 4.1.13). Then

$$\mathbf{p} = \mathbf{p}_0 + \overrightarrow{P_0P}$$

Hence P lies on the line if and only if $\overrightarrow{P_0P}$ is parallel to $\mathbf{d}$—that is, if and only if $\overrightarrow{P_0P} = t\mathbf{d}$ for some scalar t by Theorem 4.1.5. Thus $\mathbf{p}$ is the vector of a point on the line if and only if $\mathbf{p} = \mathbf{p}_0 + t\mathbf{d}$ for some scalar t. This discussion is summed up as follows.

Vector Equation of a Line

The line parallel to $\mathbf{d} \neq \mathbf{0}$ *through the point with vector* $\mathbf{p}_0$ *is given by*

$$\mathbf{p} = \mathbf{p}_0 + t\mathbf{d} \quad t \text{ any scalar}$$

In other words, the point P *with vector* $\mathbf{p}$ *is on this line if and only if a real number* t *exists such that* $\mathbf{p} = \mathbf{p}_0 + t\mathbf{d}$.

In component form the vector equation becomes

$$\left[\begin{array}{c} x \\ y \\ z \end{array}\right] = \left[\begin{array}{c} x_0 \\ y_0 \\ z_0 \end{array}\right] + t\left[\begin{array}{c} a \\ b \\ c \end{array}\right]$$

Equating components gives a different description of the line.

Parametric Equations of a Line

The line through $P_0(x_0, y_0, z_0)$ *with direction vector* $\mathbf{d} = \left[\begin{array}{c} a \\ b \\ c \end{array} \right] \neq \mathbf{0}$ *is given by*

$$\begin{array}{l} x = x_0 + ta \\ y = y_0 + tb \quad t \text{ any scalar} \\ z = z_0 + tc \end{array}$$

In other words, the point $P(x, y, z)$ *is on this line if and only if a real number* t *exists such that* $x = x_0 + ta$, $y = y_0 + tb$, *and* $z = z_0 + tc$.

Example 4.1.8

Find the equations of the line through the points $P_0(2, 0, 1)$ and $P_1(4, -1, 1)$.

Solution. Let $\mathbf{d} = \overrightarrow{P_0P_1} = \left[\begin{array}{r} 2 \\ -1 \\ 0 \end{array} \right]$ denote the vector from P_0 to P_1. Then $\mathbf{d}$ is parallel to the line (P_0 and P_1 are *on* the line), so $\mathbf{d}$ serves as a direction vector for the line. Using P_0 as the point on the line leads to the parametric equations

$$\begin{array}{l} x = 2 + 2t \\ y = -t \qquad t \text{ a parameter} \\ z = 1 \end{array}$$

Note that if P_1 is used (rather than P_0), the equations are

$$\begin{array}{l} x = 4 + 2s \\ y = -1 - s \quad s \text{ a parameter} \\ z = 1 \end{array}$$

These are different from the preceding equations, but this is merely the result of a change of parameter. In fact, $s = t - 1$.

Example 4.1.9

Find the equations of the line through $P_0(3, -1, 2)$ parallel to the line with equations

$$\begin{array}{l} x = -1 + 2t \\ y = 1 + t \\ z = -3 + 4t \end{array}$$

Solution. The coefficients of t give a direction vector $\mathbf{d} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$ of the given line. Because the line we seek is parallel to this line, $\mathbf{d}$ also serves as a direction vector for the new line. It passes through P_0, so the parametric equations are

$$\begin{aligned} x &= 3 + 2t \\ y &= -1 + t \\ z &= 2 + 4t \end{aligned}$$

Example 4.1.10

Determine whether the following lines intersect and, if so, find the point of intersection.

$$\begin{array}{ll} x = 1 - 3t & x = -1 + s \\ y = 2 + 5t & y = 3 - 4s \\ z = 1 + t & z = 1 - s \end{array}$$

Solution. Suppose $P(x,\ y,\ z)$ with vector $\mathbf{p}$ lies on both lines. Then

$$\begin{bmatrix} 1 - 3t \\ 2 + 5t \\ 1 + t \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 + s \\ 3 - 4s \\ 1 - s \end{bmatrix} \text{ for some } t \text{ and } s,$$

where the first (second) equation is because P lies on the first (second) line. Hence the lines intersect if and only if the three equations

$$\begin{aligned} 1 - 3t &= -1 + s \\ 2 + 5t &= 3 - 4s \\ 1 + t &= 1 - s \end{aligned}$$

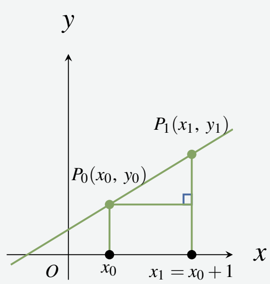
have a solution. In this case, $t = 1$ and $s = -1$ satisfy all three equations, so the lines *do* intersect and the point of intersection is

$$\mathbf{p} = \begin{bmatrix} 1 - 3t \\ 2 + 5t \\ 1 + t \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \\ 2 \end{bmatrix}$$

using $t = 1$. Of course, this point can also be found from $\mathbf{p} = \begin{bmatrix} -1 + s \\ 3 - 4s \\ 1 - s \end{bmatrix}$ using $s = -1$.

Example 4.1.11

Show that the line through $P_0(x_0, y_0)$ with slope m has direction vector $\mathbf{d} = \begin{bmatrix} 1 \\ m \end{bmatrix}$ and equation $y - y_0 = m(x - x_0)$. This equation is called the *point-slope* formula.



Solution. Let $P_1(x_1, y_1)$ be the point on the line one unit to the right of P_0 (see the diagram). Hence $x_1 = x_0 + 1$. Then $\mathbf{d} = \overrightarrow{P_0P_1}$ serves as direction vector of the line, and $\mathbf{d} = \begin{bmatrix} x_1 - x_0 \\ y_1 - y_0 \end{bmatrix} = \begin{bmatrix} 1 \\ y_1 - y_0 \end{bmatrix}$. But the slope m can be computed as follows:

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{y_1 - y_0}{1} = y_1 - y_0$$

Hence $\mathbf{d} = \begin{bmatrix} 1 \\ m \end{bmatrix}$ and the parametric equations are $x = x_0 + t$, $y = y_0 + mt$. Eliminating t gives $y - y_0 = mt = m(x - x_0)$, as asserted.

Note that the vertical line through $P_0(x_0, y_0)$ has a direction vector $\mathbf{d} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ that is *not* of the form $\begin{bmatrix} 1 \\ m \end{bmatrix}$ for any m. This result confirms that the notion of slope makes no sense in this case. However, the vector method gives parametric equations for the line:

$$\begin{aligned} x &= x_0 \\ y &= y_0 + t \end{aligned}$$

Because y is arbitrary here (t is arbitrary), this is usually written simply as $x = x_0$.

Pythagoras' Theorem

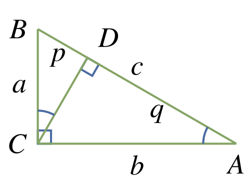


Figure 4.1.14

The Pythagorean theorem was known earlier, but Pythagoras (c. 550 B.C.) is credited with giving the first rigorous, logical, deductive proof of the result. The proof we give depends on a basic property of similar triangles: ratios of corresponding sides are equal.

Theorem 4.1.6: Pythagoras' Theorem

Given a right-angled triangle with hypotenuse c and sides a and b, then $a^2 + b^2 = c^2$.

Proof. Let A, B, and C be the vertices of the triangle as in Figure 4.1.14. Draw a perpendicular line from C to the point D on the hypotenuse, and let p and q be the lengths of BD and DA respectively. Then DBC and CBA are similar triangles so $\frac{p}{a} = \frac{a}{c}$. This means $a^2 = pc$. In the same way, the similarity of DCA and CBA gives $\frac{q}{b} = \frac{b}{c}$, whence $b^2 = qc$. But then

$$a^2 + b^2 = pc + qc = (p + q)c = c^2$$

because $p+q=c$. This proves Pythagoras' theorem[10]. $\square$

Exercises for 4.1

Exercise 4.1.1 Compute $\|\mathbf{v}\|$ if $\mathbf{v}$ equals:

a. $\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$ b. $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

c. $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ d. $\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

e. $2\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ f. $-3\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

Exercise 4.1.2 Find a unit vector in the direction of:

a. $\begin{bmatrix} 7 \\ -1 \\ 5 \end{bmatrix}$ b. $\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$

Exercise 4.1.3

a. Find a unit vector in the direction from $\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$.

b. If $\mathbf{u} \neq \mathbf{0}$, for which values of a is $a\mathbf{u}$ a unit vector?

Exercise 4.1.4 Find the distance between the following pairs of points.

a. $\begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ b. $\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

c. $\begin{bmatrix} -3 \\ 5 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$ d. $\begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

Exercise 4.1.5 Use vectors to show that the line joining the midpoints of two sides of a triangle is parallel to the third side and half as long.

Exercise 4.1.6 Let A, B, and C denote the three vertices of a triangle.

a. If E is the midpoint of side BC, show that

$$\overrightarrow{AE} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC})$$

b. If F is the midpoint of side AC, show that

$$\overrightarrow{FE} = \frac{1}{2}\overrightarrow{AB}$$

Exercise 4.1.7 Determine whether $\mathbf{u}$ and $\mathbf{v}$ are parallel in each of the following cases.

a. $\mathbf{u} = \begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix}; \mathbf{v} = \begin{bmatrix} 5 \\ 10 \\ -5 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}; \mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}; \mathbf{v} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}; \mathbf{v} = \begin{bmatrix} -8 \\ 0 \\ 4 \end{bmatrix}$

Exercise 4.1.8 Let $\mathbf{p}$ and $\mathbf{q}$ be the vectors of points P and Q, respectively, and let R be the point whose vector is $\mathbf{p}+\mathbf{q}$. Express the following in terms of $\mathbf{p}$ and $\mathbf{q}$.

a. $\overrightarrow{QP}$ b. $\overrightarrow{QR}$

c. $\overrightarrow{RP}$ d. $\overrightarrow{RO}$ where O is the origin

Exercise 4.1.9 In each case, find $\overrightarrow{PQ}$ and $\|\overrightarrow{PQ}\|$.

a. $P(1, -1, 3)$, $Q(3, 1, 0)$

b. $P(2, 0, 1)$, $Q(1, -1, 6)$

c. $P(1, 0, 1)$, $Q(1, 0, -3)$

d. $P(1, -1, 2)$, $Q(1, -1, 2)$

[10] There is an intuitive geometrical proof of Pythagoras' theorem in Example B.3.

e. $P(1,\ 0,\ -3)$, $Q(-1,\ 0,\ 3)$

f. $P(3,\ -1,\ 6)$, $Q(1,\ 1,\ 4)$

Exercise 4.1.10 In each case, find a point Q such that $\overrightarrow{PQ}$ has (i) the same direction as $\mathbf{v}$; (ii) the opposite direction to $\mathbf{v}$.

a. $P(-1,\ 2,\ 2)$, $\mathbf{v} = \left[\begin{array}{r} 1 \\ 3 \\ 1 \end{array}\right]$

b. $P(3,\ 0,\ -1)$, $\mathbf{v} = \left[\begin{array}{r} 2 \\ -1 \\ 3 \end{array}\right]$

Exercise 4.1.11 Let $\mathbf{u} = \left[\begin{array}{r} 3 \\ -1 \\ 0 \end{array}\right]$, $\mathbf{v} = \left[\begin{array}{r} 4 \\ 0 \\ 1 \end{array}\right]$, and $\mathbf{w} = \left[\begin{array}{r} -1 \\ 1 \\ 5 \end{array}\right]$. In each case, find $\mathbf{x}$ such that:

a. $3(2\mathbf{u}+\mathbf{x})+\mathbf{w} = 2\mathbf{x}-\mathbf{v}$

b. $2(3\mathbf{v}-\mathbf{x}) = 5\mathbf{w}+\mathbf{u}-3\mathbf{x}$

Exercise 4.1.12 Let $\mathbf{u} = \left[\begin{array}{r} 1 \\ 1 \\ 2 \end{array}\right]$, $\mathbf{v} = \left[\begin{array}{r} 0 \\ 1 \\ 2 \end{array}\right]$, and $\mathbf{w} = \left[\begin{array}{r} 1 \\ 0 \\ -1 \end{array}\right]$. In each case, find numbers a, b, and c such that $\mathbf{x} = a\mathbf{u}+b\mathbf{v}+c\mathbf{w}$.

a. $\mathbf{x} = \left[\begin{array}{r} 2 \\ -1 \\ 6 \end{array}\right]$

b. $\mathbf{x} = \left[\begin{array}{r} 1 \\ 3 \\ 0 \end{array}\right]$

Exercise 4.1.13 Let $\mathbf{u} = \left[\begin{array}{r} 3 \\ -1 \\ 0 \end{array}\right]$, $\mathbf{v} = \left[\begin{array}{r} 4 \\ 0 \\ 1 \end{array}\right]$, and $\mathbf{z} = \left[\begin{array}{r} 1 \\ 1 \\ 1 \end{array}\right]$. In each case, show that there are no numbers a, b, and c such that:

a. $a\mathbf{u}+b\mathbf{v}+c\mathbf{z} = \left[\begin{array}{r} 1 \\ 2 \\ 1 \end{array}\right]$

b. $a\mathbf{u}+b\mathbf{v}+c\mathbf{z} = \left[\begin{array}{r} 5 \\ 6 \\ -1 \end{array}\right]$

Exercise 4.1.14 Given $P_1(2,\ 1,\ -2)$ and $P_2(1,\ -2,\ 0)$. Find the coordinates of the point P:

a. $\frac{1}{5}$ the way from P_1 to P_2

b. $\frac{1}{4}$ the way from P_2 to P_1

Exercise 4.1.15 Find the two points trisecting the segment between $P(2,\ 3,\ 5)$ and $Q(8,\ -6,\ 2)$.

Exercise 4.1.16 Let $P_1(x_1,\ y_1,\ z_1)$ and $P_2(x_2,\ y_2,\ z_2)$ be two points with vectors $\mathbf{p}_1$ and $\mathbf{p}_2$, respectively. If r and s are positive integers, show that the point P lying $\frac{r}{r+s}$ the way from P_1 to P_2 has vector

$$\mathbf{p} = \left(\tfrac{s}{r+s}\right)\mathbf{p}_1 + \left(\tfrac{r}{r+s}\right)\mathbf{p}_2$$

Exercise 4.1.17 In each case, find the point Q:

a. $\overrightarrow{PQ} = \left[\begin{array}{r} 2 \\ 0 \\ -3 \end{array}\right]$ and $P = P(2,\ -3,\ 1)$

b. $\overrightarrow{PQ} = \left[\begin{array}{r} -1 \\ 4 \\ 7 \end{array}\right]$ and $P = P(1,\ 3,\ -4)$

Exercise 4.1.18 Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. In each case find $\mathbf{x}$:

a. $2\mathbf{u} - \|\mathbf{v}\|\mathbf{v} = \frac{3}{2}(\mathbf{u} - 2\mathbf{x})$

b. $3\mathbf{u} + 7\mathbf{v} = \|\mathbf{u}\|^2(2\mathbf{x} + \mathbf{v})$

Exercise 4.1.19 Find all vectors $\mathbf{u}$ that are parallel to $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and satisfy $\|\mathbf{u}\| = 3\|\mathbf{v}\|$.

Exercise 4.1.20 Let P, Q, and R be the vertices of a parallelogram with adjacent sides PQ and PR. In each case, find the other vertex S.

a. $P(3, -1, -1)$, $Q(1, -2, 0)$, $R(1, -1, 2)$

b. $P(2, 0, -1)$, $Q(-2, 4, 1)$, $R(3, -1, 0)$

Exercise 4.1.21 In each case either prove the statement or give an example showing that it is false.

a. The zero vector $\mathbf{0}$ is the only vector of length 0.

b. If $\|\mathbf{v} - \mathbf{w}\| = 0$, then $\mathbf{v} = \mathbf{w}$.

c. If $\mathbf{v} = -\mathbf{v}$, then $\mathbf{v} = \mathbf{0}$.

d. If $\|\mathbf{v}\| = \|\mathbf{w}\|$, then $\mathbf{v} = \mathbf{w}$.

e. If $\|\mathbf{v}\| = \|\mathbf{w}\|$, then $\mathbf{v} = \pm\mathbf{w}$.

f. If $\mathbf{v} = t\mathbf{w}$ for some scalar t, then $\mathbf{v}$ and $\mathbf{w}$ have the same direction.

g. If $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{v} + \mathbf{w}$ are nonzero, and $\mathbf{v}$ and $\mathbf{v} + \mathbf{w}$ parallel, then $\mathbf{v}$ and $\mathbf{w}$ are parallel.

h. $\|-5\mathbf{v}\| = -5\|\mathbf{v}\|$, for all $\mathbf{v}$.

i. If $\|\mathbf{v}\| = \|2\mathbf{v}\|$, then $\mathbf{v} = \mathbf{0}$.

j. $\|\mathbf{v} + \mathbf{w}\| = \|\mathbf{v}\| + \|\mathbf{w}\|$, for all $\mathbf{v}$ and $\mathbf{w}$.

Exercise 4.1.22 Find the vector and parametric equations of the following lines.

a. The line parallel to $\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ and passing through $P(1, -1, 3)$.

b. The line passing through $P(3, -1, 4)$ and $Q(1, 0, -1)$.

c. The line passing through $P(3, -1, 4)$ and $Q(3, -1, 5)$.

d. The line parallel to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and passing through $P(1, 1, 1)$.

e. The line passing through $P(1, 0, -3)$ and parallel to the line with parametric equations $x = -1 + 2t$, $y = 2 - t$, and $z = 3 + 3t$.

f. The line passing through $P(2, -1, 1)$ and parallel to the line with parametric equations $x = 2 - t$, $y = 1$, and $z = t$.

g. The lines through $P(1, 0, 1)$ that meet the line with vector equation $\mathbf{p} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$ at points at distance 3 from $P_0(1, 2, 0)$.

Exercise 4.1.23 In each case, verify that the points P and Q lie on the line.

a. $\begin{aligned} x &= 3 - 4t \\ y &= 2 + t \\ z &= 1 - t \end{aligned}$ $\quad P(-1, 3, 0)$, $Q(11, 0, 3)$

b. $\begin{aligned} x &= 4 - t \\ y &= 3 \\ z &= 1 - 2t \end{aligned}$ $\quad P(2, 3, -3)$, $Q(-1, 3, -9)$

Exercise 4.1.24 Find the point of intersection (if any) of the following pairs of lines.

a. $\begin{aligned} x &= 3 + t \\ y &= 1 - 2t \\ z &= 3 + 3t \end{aligned}$ $\quad \begin{aligned} x &= 4 + 2s \\ y &= 6 + 3s \\ z &= 1 + s \end{aligned}$

b. $\begin{aligned} x &= 1 - t \\ y &= 2 + 2t \\ z &= -1 + 3t \end{aligned}$ $\quad \begin{aligned} x &= 2s \\ y &= 1 + s \\ z &= 3 \end{aligned}$

c. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} + t\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + s\begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$

d. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \\ 12 \end{bmatrix} + s \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix}$

Exercise 4.1.25 Show that if a line passes through the origin, the vectors of points on the line are all scalar multiples of some fixed nonzero vector.

Exercise 4.1.26 Show that every line parallel to the z axis has parametric equations $x = x_0$, $y = y_0$, $z = t$ for some fixed numbers x_0 and y_0.

Exercise 4.1.27 Let $\mathbf{d} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ be a vector where a, b, and c are *all* nonzero. Show that the equations of the line through $P_0(x_0, y_0, z_0)$ with direction vector $\mathbf{d}$ can be written in the form

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$$

This is called the **symmetric form** of the equations.

Exercise 4.1.28 A parallelogram has sides AB, BC, CD, and DA. Given $A(1, -1, 2)$, $C(2, 1, 0)$, and the midpoint $M(1, 0, -3)$ of AB, find $\overrightarrow{BD}$.

Exercise 4.1.29 Find all points C on the line through $A(1, -1, 2)$ and $B = (2, 0, 1)$ such that $\|\overrightarrow{AC}\| = 2\|\overrightarrow{BC}\|$.

Exercise 4.1.30 Let A, B, C, D, E, and F be the vertices of a regular hexagon, taken in order. Show that $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = 3\overrightarrow{AD}$.

Exercise 4.1.31

a. Let P_1, P_2, P_3, P_4, P_5, and P_6 be six points equally spaced on a circle with centre C. Show that

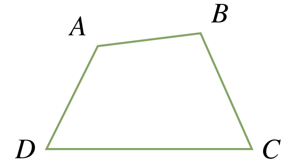
$$\overrightarrow{CP}_1 + \overrightarrow{CP}_2 + \overrightarrow{CP}_3 + \overrightarrow{CP}_4 + \overrightarrow{CP}_5 + \overrightarrow{CP}_6 = \mathbf{0}$$

b. Show that the conclusion in part (a) holds for any *even* set of points evenly spaced on the circle.

c. Show that the conclusion in part (a) holds for *three* points.

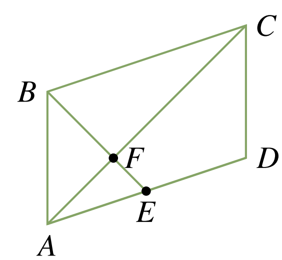
d. Do you think it works for *any* finite set of points evenly spaced around the circle?

Exercise 4.1.32 Consider a quadrilateral with vertices A, B, C, and D in order (as shown in the diagram).



If the diagonals AC and BD bisect each other, show that the quadrilateral is a parallelogram. (This is the converse of Example 4.1.2.) [*Hint*: Let E be the intersection of the diagonals. Show that $\overrightarrow{AB} = \overrightarrow{DC}$ by writing $\overrightarrow{AB} = \overrightarrow{AE} + \overrightarrow{EB}$.]

Exercise 4.1.33 Consider the parallelogram $ABCD$ (see diagram), and let E be the midpoint of side AD.



Show that BE and AC trisect each other; that is, show that the intersection point is one-third of the way from E to B and from A to C. [*Hint*: If F is one-third of the way from A to C, show that $2\overrightarrow{EF} = \overrightarrow{FB}$ and argue as in Example 4.1.2.]

Exercise 4.1.34 The line from a vertex of a triangle to the midpoint of the opposite side is called a **median** of the triangle. If the vertices of a triangle have vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, show that the point on each median that is $\frac{1}{3}$ the way from the midpoint to the vertex has vector $\frac{1}{3}(\mathbf{u}+\mathbf{v}+\mathbf{w})$. Conclude that the point C with vector $\frac{1}{3}(\mathbf{u}+\mathbf{v}+\mathbf{w})$ lies on all three medians. This point C is called the **centroid** of the triangle.

Exercise 4.1.35 Given four noncoplanar points in space, the figure with these points as vertices is called a **tetrahedron**. The line from a vertex through the centroid (see previous exercise) of the triangle formed by the remaining vertices is called a **median** of the tetrahedron. If $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{x}$ are the vectors of the four vertices, show that the point on a median one-fourth the way from the centroid to the vertex has vector $\frac{1}{4}(\mathbf{u}+\mathbf{v}+\mathbf{w}+\mathbf{x})$. Conclude that the four medians are concurrent.

4.2 Projections and Planes

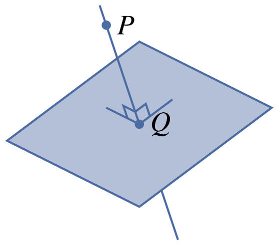


Figure 4.2.1

Any student of geometry soon realizes that the notion of perpendicular lines is fundamental. As an illustration, suppose a point P and a plane are given and it is desired to find the point Q that lies in the plane and is closest to P, as shown in Figure 4.2.1. Clearly, what is required is to find the line through P that is perpendicular to the plane and then to obtain Q as the point of intersection of this line with the plane. Finding the line *perpendicular* to the plane requires a way to determine when two vectors are perpendicular. This can be done using the idea of the dot product of two vectors.

The Dot Product and Angles

Definition 4.4 Dot Product in $\mathbb{R}^3$

Given vectors $\mathbf{v} = \left[\begin{array}{c} x_1 \\ y_1 \\ z_1 \end{array}\right]$ *and* $\mathbf{w} = \left[\begin{array}{c} x_2 \\ y_2 \\ z_2 \end{array}\right]$, *their* **dot product** $\mathbf{v} \cdot \mathbf{w}$ *is a number defined*

$$\mathbf{v} \cdot \mathbf{w} = x_1x_2 + y_1y_2 + z_1z_2 = \mathbf{v}^T\mathbf{w}$$

Because $\mathbf{v} \cdot \mathbf{w}$ is a number, it is sometimes called the **scalar product** of $\mathbf{v}$ and $\mathbf{w}$.[11]

Example 4.2.1

If $\mathbf{v} = \left[\begin{array}{r} 2 \\ -1 \\ 3 \end{array}\right]$ and $\mathbf{w} = \left[\begin{array}{r} 1 \\ 4 \\ -1 \end{array}\right]$, then $\mathbf{v} \cdot \mathbf{w} = 2 \cdot 1 + (-1) \cdot 4 + 3 \cdot (-1) = -5$.

The next theorem lists several basic properties of the dot product.

Theorem 4.2.1

Let $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *denote vectors in* $\mathbb{R}^3$ *(or* $\mathbb{R}^2$*).*

1. $\mathbf{v} \cdot \mathbf{w}$ *is a real number.*
2. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$.
3. $\mathbf{v} \cdot \mathbf{0} = 0 = \mathbf{0} \cdot \mathbf{v}$.
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$.
5. $(k\mathbf{v}) \cdot \mathbf{w} = k(\mathbf{w} \cdot \mathbf{v}) = \mathbf{v} \cdot (k\mathbf{w})$ *for all scalars* k.
6. $\mathbf{u} \cdot (\mathbf{v} \pm \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} \pm \mathbf{u} \cdot \mathbf{w}$

[11] Similarly, if $\mathbf{v} = \left[\begin{array}{c} x_1 \\ y_1 \end{array}\right]$ and $\mathbf{w} = \left[\begin{array}{c} x_2 \\ y_2 \end{array}\right]$ in $\mathbb{R}^2$, then $\mathbf{v} \cdot \mathbf{w} = x_1x_2 + y_1y_2$.

Proof. (1), (2), and (3) are easily verified, and (4) comes from Theorem 4.1.1. The rest are properties of matrix arithmetic (because $\mathbf{w} \cdot \mathbf{v} = \mathbf{v}^T \mathbf{w}$), and are left to the reader. □

The properties in Theorem 4.2.1 enable us to do calculations like

$$3\mathbf{u} \cdot (2\mathbf{v} - 3\mathbf{w} + 4\mathbf{z}) = 6(\mathbf{u} \cdot \mathbf{v}) - 9(\mathbf{u} \cdot \mathbf{w}) + 12(\mathbf{u} \cdot \mathbf{z})$$

and such computations will be used without comment below. Here is an example.

Example 4.2.2

Verify that $\|\mathbf{v} - 3\mathbf{w}\|^2 = 1$ when $\|\mathbf{v}\| = 2$, $\|\mathbf{w}\| = 1$, and $\mathbf{v} \cdot \mathbf{w} = 2$.

Solution. We apply Theorem 4.2.1 several times:

$$\begin{aligned}
\|\mathbf{v} - 3\mathbf{w}\|^2 &= (\mathbf{v} - 3\mathbf{w}) \cdot (\mathbf{v} - 3\mathbf{w}) \\
&= \mathbf{v} \cdot (\mathbf{v} - 3\mathbf{w}) - 3\mathbf{w} \cdot (\mathbf{v} - 3\mathbf{w}) \\
&= \mathbf{v} \cdot \mathbf{v} - 3(\mathbf{v} \cdot \mathbf{w}) - 3(\mathbf{w} \cdot \mathbf{v}) + 9(\mathbf{w} \cdot \mathbf{w}) \\
&= \|\mathbf{v}\|^2 - 6(\mathbf{v} \cdot \mathbf{w}) + 9\|\mathbf{w}\|^2 \\
&= 4 - 12 + 9 = 1
\end{aligned}$$

Theorem 4.2.3

Two vectors $\mathbf{v}$ *and* $\mathbf{w}$ *are orthogonal if and only if* $\mathbf{v} \cdot \mathbf{w} = 0$.

Example 4.2.4

Show that the points $P(3,\ -1,\ 1)$, $Q(4,\ 1,\ 4)$, and $R(6,\ 0,\ 4)$ are the vertices of a right triangle.

Solution. The vectors along the sides of the triangle are

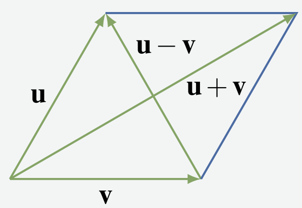
$$\overrightarrow{PQ} = \left[\begin{array}{r} 1 \\ 2 \\ 3 \end{array}\right],\ \overrightarrow{PR} = \left[\begin{array}{r} 3 \\ 1 \\ 3 \end{array}\right],\ \text{and } \overrightarrow{QR} = \left[\begin{array}{r} 2 \\ -1 \\ 0 \end{array}\right]$$

Evidently $\overrightarrow{PQ} \cdot \overrightarrow{QR} = 2 - 2 + 0 = 0$, so $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ are orthogonal vectors. This means sides PQ and QR are perpendicular—that is, the angle at Q is a right angle.

Example 4.2.5 demonstrates how the dot product can be used to verify geometrical theorems involving perpendicular lines.

Example 4.2.5

A parallelogram with sides of equal length is called a **rhombus**. Show that the diagonals of a rhombus are perpendicular.



Solution. Let $\mathbf{u}$ and $\mathbf{v}$ denote vectors along two adjacent sides of a rhombus, as shown in the diagram. Then the diagonals are $\mathbf{u}-\mathbf{v}$ and $\mathbf{u}+\mathbf{v}$, and we compute

$$\begin{aligned}(\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) &= \mathbf{u}\cdot(\mathbf{u}+\mathbf{v})-\mathbf{v}\cdot(\mathbf{u}+\mathbf{v})\\ &= \mathbf{u}\cdot\mathbf{u}+\mathbf{u}\cdot\mathbf{v}-\mathbf{v}\cdot\mathbf{u}-\mathbf{v}\cdot\mathbf{v}\\ &= \|\mathbf{u}\|^2-\|\mathbf{v}\|^2\\ &= 0\end{aligned}$$

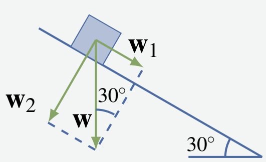
because $\|\mathbf{u}\| = \|\mathbf{v}\|$ (it is a rhombus). Hence $\mathbf{u}-\mathbf{v}$ and $\mathbf{u}+\mathbf{v}$ are orthogonal.

Projections

In applications of vectors, it is frequently useful to write a vector as the sum of two orthogonal vectors. Here is an example.

Example 4.2.6

Suppose a ten-kilogram block is placed on a flat surface inclined 30° to the horizontal as in the diagram. Neglecting friction, how much force is required to keep the block from sliding down the surface?



Solution. Let $\mathbf{w}$ denote the weight (force due to gravity) exerted on the block. Then $\|\mathbf{w}\| = 10$ kilograms and the direction of $\mathbf{w}$ is vertically down as in the diagram. The idea is to write $\mathbf{w}$ as a sum $\mathbf{w} = \mathbf{w}_1+\mathbf{w}_2$ where $\mathbf{w}_1$ is parallel to the inclined surface and $\mathbf{w}_2$ is perpendicular to the surface. Since there is no friction, the force required is $-\mathbf{w}_1$ because the force $\mathbf{w}_2$ has no effect parallel to the surface. As the angle between $\mathbf{w}$ and $\mathbf{w}_2$ is 30° in the diagram, we have $\frac{\|\mathbf{w}_1\|}{\|\mathbf{w}\|} = \sin 30^\circ = \frac{1}{2}$. Hence $\|\mathbf{w}_1\| = \frac{1}{2}\|\mathbf{w}\| = \frac{1}{2}10 = 5$. Thus the required force has a magnitude of 5 kilograms weight directed up the surface.

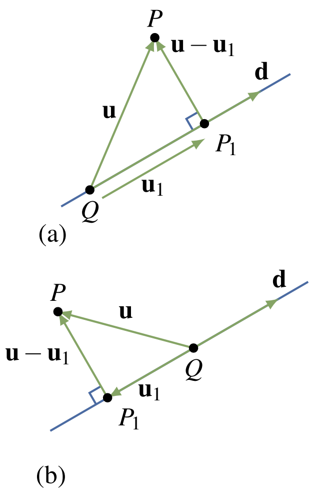


Figure 4.2.5

If a nonzero vector $\mathbf{d}$ is specified, the key idea in Example 4.2.6 is to be able to write an arbitrary vector $\mathbf{u}$ as a sum of two vectors,

$$\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$$

where $\mathbf{u}_1$ is parallel to $\mathbf{d}$ and $\mathbf{u}_2 = \mathbf{u} - \mathbf{u}_1$ is orthogonal to $\mathbf{d}$. Suppose that $\mathbf{u}$ and $\mathbf{d} \neq \mathbf{0}$ emanate from a common tail Q (see Figure 4.2.5). Let P be the tip of $\mathbf{u}$, and let P_1 denote the foot of the perpendicular from P to the line through Q parallel to $\mathbf{d}$.

Then $\mathbf{u}_1 = \overrightarrow{QP}_1$ has the required properties:

1. $\mathbf{u}_1$ *is parallel to* $\mathbf{d}$.
2. $\mathbf{u}_2 = \mathbf{u} - \mathbf{u}_1$ *is orthogonal to* $\mathbf{d}$.
3. $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$.

Definition 4.6 Projection in $\mathbb{R}^3$

*The vector $\boldsymbol{u}_1 = \overrightarrow{QP}_1$ in Figure 4.2.5 is called **the projection** of $\boldsymbol{u}$ on $\boldsymbol{d}$. It is denoted*

$$\boldsymbol{u}_1 = \operatorname{proj}_{\boldsymbol{d}} \boldsymbol{u}$$

In Figure 4.2.5(a) the vector $\mathbf{u}_1 = \operatorname{proj}_{\mathbf{d}} \mathbf{u}$ has the same direction as $\mathbf{d}$; however, $\mathbf{u}_1$ and $\mathbf{d}$ have opposite directions if the angle between $\mathbf{u}$ and $\mathbf{d}$ is greater than $\frac{\pi}{2}$ (Figure 4.2.5(b)). Note that the projection $\mathbf{u}_1 = \operatorname{proj}_{\mathbf{d}} \mathbf{u}$ is zero if and only if $\mathbf{u}$ and $\mathbf{d}$ are orthogonal.

Calculating the projection of $\mathbf{u}$ on $\mathbf{d} \neq \mathbf{0}$ is remarkably easy.

Theorem 4.2.4

Let $\boldsymbol{u}$ and $\boldsymbol{d} \neq \boldsymbol{0}$ be vectors.

1. *The projection of $\boldsymbol{u}$ on $\boldsymbol{d}$ is given by $\operatorname{proj}_{\boldsymbol{d}} \boldsymbol{u} = \frac{\boldsymbol{u} \cdot \boldsymbol{d}}{\|\boldsymbol{d}\|^2} \boldsymbol{d}$.*
2. *The vector $\boldsymbol{u} - \operatorname{proj}_{\boldsymbol{d}} \boldsymbol{u}$ is orthogonal to $\boldsymbol{d}$.*

Proof. The vector $\mathbf{u}_1 = \operatorname{proj}_{\mathbf{d}} \mathbf{u}$ is parallel to $\mathbf{d}$ and so has the form $\mathbf{u}_1 = t\mathbf{d}$ for some scalar t. The requirement that $\mathbf{u} - \mathbf{u}_1$ and $\mathbf{d}$ are orthogonal determines t. In fact, it means that $(\mathbf{u} - \mathbf{u}_1) \cdot \mathbf{d} = 0$ by Theorem 4.2.3. If $\mathbf{u}_1 = t\mathbf{d}$ is substituted here, the condition is

$$0 = (\mathbf{u} - t\mathbf{d}) \cdot \mathbf{d} = \mathbf{u} \cdot \mathbf{d} - t(\mathbf{d} \cdot \mathbf{d}) = \mathbf{u} \cdot \mathbf{d} - t\|\mathbf{d}\|^2$$

It follows that $t = \frac{\mathbf{u} \cdot \mathbf{d}}{\|\mathbf{d}\|^2}$, where the assumption that $\mathbf{d} \neq \mathbf{0}$ guarantees that $\|\mathbf{d}\|^2 \neq 0$. □

Example 4.2.7

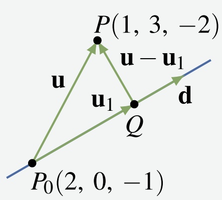
Find the projection of $\mathbf{u} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ on $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and express $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$ where $\mathbf{u}_1$ is parallel to $\mathbf{d}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{d}$.

Solution. The projection $\mathbf{u}_1$ of $\mathbf{u}$ on $\mathbf{d}$ is

$$\mathbf{u}_1 = \text{proj}_{\mathbf{d}}\, \mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d} = \frac{2+3+3}{1^2+(-1)^2+3^2}\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} = \frac{8}{11}\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

Hence $\mathbf{u}_2 = \mathbf{u} - \mathbf{u}_1 = \frac{1}{11}\begin{bmatrix} 14 \\ -25 \\ -13 \end{bmatrix}$, and this is orthogonal to $\mathbf{d}$ by Theorem 4.2.4 (alternatively, observe that $\mathbf{d}\cdot\mathbf{u}_2 = 0$). Since $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$, we are done.

Example 4.2.8



Find the shortest distance (see diagram) from the point $P(1,\ 3,\ -2)$ to the line through $P_0(2,\ 0,\ -1)$ with direction vector $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Also find the point Q that lies on the line and is closest to P.

Solution. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}$ denote the vector from P_0 to P, and let $\mathbf{u}_1$ denote the projection of $\mathbf{u}$ on $\mathbf{d}$. Thus

$$\mathbf{u}_1 = \frac{\mathbf{u}\cdot\mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d} = \frac{-1-3+0}{1^2+(-1)^2+0^2}\mathbf{d} = -2\mathbf{d} = \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}$$

by Theorem 4.2.4. We see geometrically that the point Q on the line is closest to P, so the distance is

$$\|\overrightarrow{QP}\| = \|\mathbf{u} - \mathbf{u}_1\| = \left\| \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\| = \sqrt{3}$$

To find the coordinates of Q, let $\mathbf{p}_0$ and $\mathbf{q}$ denote the vectors of P_0 and Q, respectively. Then $\mathbf{p}_0 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ and $\mathbf{q} = \mathbf{p}_0 + \mathbf{u}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$. Hence $Q(0,\ 2,\ -1)$ is the required point. It can be checked that the distance from Q to P is $\sqrt{3}$, as expected.

Planes

It is evident geometrically that among all planes that are perpendicular to a given straight line there is exactly one containing any given point. This fact can be used to give a very simple description of a plane. To do this, it is necessary to introduce the following notion:

Definition 4.7 Normal Vector in a Plane

A nonzero vector $\mathbf{n}$ *is called a* ***normal*** *for a plane if it is orthogonal to every vector in the plane.*

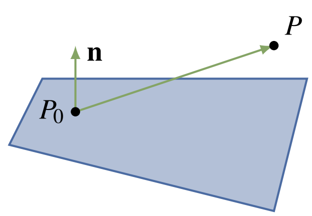


Figure 4.2.6

For example, the coordinate vector $\mathbf{k}$ is a normal for the x-y plane.

Given a point $P_0 = P_0(x_0,\ y_0,\ z_0)$ and a nonzero vector $\mathbf{n}$, there is a unique plane through P_0 with normal $\mathbf{n}$, shaded in Figure 4.2.6. A point $P = P(x,\ y,\ z)$ lies on this plane if and only if the vector $\overrightarrow{P_0P}$ is orthogonal to $\mathbf{n}$—that is, if and only if $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$. Because $\overrightarrow{P_0P} = \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix}$ this gives the following result:

Scalar Equation of a Plane

The plane through $P_0(x_0,\ y_0,\ z_0)$ *with normal* $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq \mathbf{0}$ *as a normal vector is given by*

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

In other words, a point $P(x,\ y,\ z)$ *is on this plane if and only if* x, y, *and* z *satisfy this equation.*

Example 4.2.9

Find an equation of the plane through $P_0(1,\ -1,\ 3)$ with $\mathbf{n} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ as normal.

Solution. Here the general scalar equation becomes

$$3(x - 1) - (y + 1) + 2(z - 3) = 0$$

This simplifies to $3x - y + 2z = 10$.

If we write $d = ax_0 + by_0 + cz_0$, the scalar equation shows that every plane with normal $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ has a linear equation of the form

$$ax + by + cz = d \tag{4.2}$$

for some constant d. Conversely, the graph of this equation is a plane with $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ as a normal vector (assuming that a, b, and c are not all zero).

Example 4.2.10

Find an equation of the plane through $P_0(3,\ -1,\ 2)$ that is parallel to the plane with equation $2x - 3y = 6$.

Solution. The plane with equation $2x - 3y = 6$ has normal $\mathbf{n} = \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}$. Because the two planes are parallel, $\mathbf{n}$ serves as a normal for the plane we seek, so the equation is $2x - 3y = d$ for some d by Equation 4.2. Insisting that $P_0(3,\ -1,\ 2)$ lies on the plane determines d; that is, $d = 2 \cdot 3 - 3(-1) = 9$. Hence, the equation is $2x - 3y = 9$.

Consider points $P_0(x_0,\ y_0,\ z_0)$ and $P(x,\ y,\ z)$ with vectors $\mathbf{p}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$ and $\mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Given a nonzero vector $\mathbf{n}$, the scalar equation of the plane through $P_0(x_0,\ y_0,\ z_0)$ with normal $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ takes the vector form:

Vector Equation of a Plane

The plane with normal $\mathbf{n} \neq \mathbf{0}$ through the point with vector $\mathbf{p}_0$ is given by

$$\mathbf{n} \cdot (\mathbf{p} - \mathbf{p}_0) = 0$$

In other words, the point with vector $\mathbf{p}$ is on the plane if and only if $\mathbf{p}$ satisfies this condition.

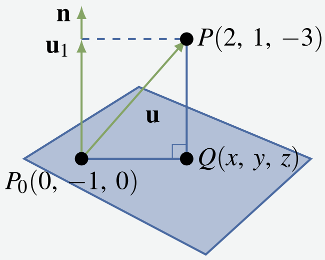
Moreover, Equation 4.2 translates as follows:

Every plane with normal $\mathbf{n}$ has vector equation $\mathbf{n} \cdot \mathbf{p} = d$ for some number d.

This is useful in the second solution of Example 4.2.11.

Example 4.2.11

Find the shortest distance from the point $P(2,\ 1,\ -3)$ to the plane with equation $3x - y + 4z = 1$. Also find the point Q on this plane closest to P.



Solution 1. The plane in question has normal $\mathbf{n} = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}$.

Choose any point P_0 on the plane—say $P_0(0,\ -1,\ 0)$—and let $Q(x,\ y,\ z)$ be the point on the plane closest to P (see the diagram).

The vector from P_0 to P is $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \\ -3 \end{bmatrix}$. Now erect $\mathbf{n}$ with its tail at P_0. Then $\overrightarrow{QP} = \mathbf{u}_1$ and $\mathbf{u}_1$ is the projection of $\mathbf{u}$ on $\mathbf{n}$:

$$\mathbf{u}_1 = \frac{\mathbf{n}\cdot\mathbf{u}}{\|\mathbf{n}\|^2}\mathbf{n} = \frac{-8}{26}\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \frac{-4}{13}\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}$$

Hence the distance is $\|\overrightarrow{QP}\| = \|\mathbf{u}_1\| = \frac{4\sqrt{26}}{13}$. To calculate the point Q, let $\mathbf{q} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\mathbf{p}_0 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$ be the vectors of Q and P_0. Then

$$\mathbf{q} = \mathbf{p}_0 + \mathbf{u} - \mathbf{u}_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ -3 \end{bmatrix} + \frac{4}{13}\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{38}{13} \\ \frac{9}{13} \\ \frac{-23}{13} \end{bmatrix}$$

This gives the coordinates of $Q(\frac{38}{13},\ \frac{9}{13},\ \frac{-23}{13})$.

Solution 2. Let $\mathbf{q} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\mathbf{p} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ be the vectors of Q and P. Then Q is on the line through P with direction vector $\mathbf{n}$, so $\mathbf{q} = \mathbf{p} + t\mathbf{n}$ for some scalar t. In addition, Q lies on the plane, so $\mathbf{n}\cdot\mathbf{q} = 1$. This determines t:

$$1 = \mathbf{n}\cdot\mathbf{q} = \mathbf{n}\cdot(\mathbf{p} + t\mathbf{n}) = \mathbf{n}\cdot\mathbf{p} + t\|\mathbf{n}\|^2 = -7 + t(26)$$

This gives $t = \frac{8}{26} = \frac{4}{13}$, so

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{q} = \mathbf{p} + t\mathbf{n} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} + \frac{4}{13}\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} 38 \\ 9 \\ -23 \end{bmatrix}$$

as before. This determines Q (in the diagram), and the reader can verify that the required distance is $\|\overrightarrow{QP}\| = \frac{4}{13}\sqrt{26}$, as before.

Exercises for 4.2

Exercise 4.2.1 Compute $\mathbf{u} \cdot \mathbf{v}$ where:

a. $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v} = \mathbf{u}$

c. $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 6 \\ -7 \\ -5 \end{bmatrix}$

e. $\mathbf{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

f. $\mathbf{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \mathbf{v} = \mathbf{0}$

Exercise 4.2.2 Find the angle between the following pairs of vectors.

a. $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -6 \\ 2 \\ 0 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 7 \\ -1 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}$

e. $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

f. $\mathbf{u} = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 5\sqrt{2} \\ -7 \\ -1 \end{bmatrix}$

Exercise 4.2.3 Find all real numbers x such that:

a. $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} x \\ -2 \\ 1 \end{bmatrix}$ are orthogonal.

b. $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ x \\ 2 \end{bmatrix}$ are at an angle of $\frac{\pi}{3}$.

Exercise 4.2.4 Find all vectors $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ orthogonal to both:

a. $\mathbf{u}_1 = \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

b. $\mathbf{u}_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

c. $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -4 \\ 0 \\ 2 \end{bmatrix}$

d. $\mathbf{u}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Exercise 4.2.5 Find two orthogonal vectors that are both orthogonal to $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$.

Exercise 4.2.6 Consider the triangle with vertices $P(2, 0, -3)$, $Q(5, -2, 1)$, and $R(7, 5, 3)$.

a. Show that it is a right-angled triangle.

b. Find the lengths of the three sides and verify the Pythagorean theorem.

Exercise 4.2.7 Show that the triangle with vertices $A(4,\ -7,\ 9)$, $B(6,\ 4,\ 4)$, and $C(7,\ 10,\ -6)$ is not a right-angled triangle.

Exercise 4.2.8 Find the three internal angles of the triangle with vertices:

a. $A(3,\ 1,\ -2)$, $B(3,\ 0,\ -1)$, and $C(5,\ 2,\ -1)$

b. $A(3,\ 1,\ -2)$, $B(5,\ 2,\ -1)$, and $C(4,\ 3,\ -3)$

Exercise 4.2.9 Show that the line through $P_0(3,\ 1,\ 4)$ and $P_1(2,\ 1,\ 3)$ is perpendicular to the line through $P_2(1,\ -1,\ 2)$ and $P_3(0,\ 5,\ 3)$.

Exercise 4.2.10 In each case, compute the projection of **u** on **v**.

a. $\mathbf{u} = \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -6 \\ 4 \\ 2 \end{bmatrix}$

Exercise 4.2.11 In each case, write $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$, where $\mathbf{u}_1$ is parallel to **v** and $\mathbf{u}_2$ is orthogonal to **v**.

a. $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -6 \\ 4 \\ -1 \end{bmatrix}$

Exercise 4.2.12 Calculate the distance from the point P to the line in each case and find the point Q on the line closest to P.

a. $P(3,\ 2-1)$

line: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$

b. $P(1,\ -1,\ 3)$

line: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$

Exercise 4.2.13 Compute $\mathbf{u} \times \mathbf{v}$ where:

a. $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -6 \\ 2 \\ 0 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$

Exercise 4.2.14 Find an equation of each of the following planes.

a. Passing through $A(2,\ 1,\ 3)$, $B(3,\ -1,\ 5)$, and $C(1,\ 2,\ -3)$.

b. Passing through $A(1,\ -1,\ 6)$, $B(0,\ 0,\ 1)$, and $C(4,\ 7,\ -11)$.

c. Passing through $P(2,\ -3,\ 5)$ and parallel to the plane with equation $3x-2y-z=0$.

d. Passing through $P(3,\ 0,\ -1)$ and parallel to the plane with equation $2x-y+z=3$.

e. Containing $P(3,\ 0,\ -1)$ and the line
$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

f. Containing $P(2, 1, 0)$ and the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

g. Containing the lines $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$.

h. Containing the lines $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$.

i. Each point of which is equidistant from $P(2, -1, 3)$ and $Q(1, 1, -1)$.

j. Each point of which is equidistant from $P(0, 1, -1)$ and $Q(2, -1, -3)$.

Exercise 4.2.15 In each case, find a vector equation of the line.

a. Passing through $P(3, -1, 4)$ and perpendicular to the plane $3x - 2y - z = 0$.

b. Passing through $P(2, -1, 3)$ and perpendicular to the plane $2x + y = 1$.

c. Passing through $P(0, 0, 0)$ and perpendicular to the lines $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}$.

d. Passing through $P(1, 1, -1)$, and perpendicular to the lines $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ -2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$.

e. Passing through $P(2, 1, -1)$, intersecting the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$, and perpendicular to that line.

f. Passing through $P(1, 1, 2)$, intersecting the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and perpendicular to that line.

Exercise 4.2.16 In each case, find the shortest distance from the point P to the plane and find the point Q on the plane closest to P.

a. $P(2, 3, 0)$; plane with equation $5x + y + z = 1$.

b. $P(3, 1, -1)$; plane with equation $2x + y - z = 6$.

Exercise 4.2.17

a. Does the line through $P(1, 2, -3)$ with direction vector $\mathbf{d} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ lie in the plane $2x - y - z = 3$? Explain.

b. Does the plane through $P(4, 0, 5)$, $Q(2, 2, 1)$, and $R(1, -1, 2)$ pass through the origin? Explain.

Exercise 4.2.18 Show that every plane containing $P(1, 2, -1)$ and $Q(2, 0, 1)$ must also contain $R(-1, 6, -5)$.

Exercise 4.2.19 Find the equations of the line of intersection of the following planes.

a. $2x - 3y + 2z = 5$ and $x + 2y - z = 4$.

b. $3x + y - 2z = 1$ and $x + y + z = 5$.

Exercise 4.2.20 In each case, find all points of intersection of the given plane and the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$.

a. $x - 3y + 2z = 4$

b. $2x - y - z = 5$

c. $3x - y + z = 8$

d. $-x - 4y - 3z = 6$

Exercise 4.2.21 Find the equation of *all* planes:

a. Perpendicular to the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$.

b. Perpendicular to the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$.

c. Containing the origin.

d. Containing $P(3, 2, -4)$.

e. Containing $P(1, 1, -1)$ and $Q(0, 1, 1)$.

f. Containing $P(2, -1, 1)$ and $Q(1, 0, 0)$.

g. Containing the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$.

h. Containing the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$.

Exercise 4.2.22 If a plane contains two distinct points P_1 and P_2, show that it contains every point on the line through P_1 and P_2.

Exercise 4.2.23 Find the shortest distance between the following pairs of parallel lines.

a. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$

b. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

Exercise 4.2.24 Find the shortest distance between the following pairs of nonparallel lines and find the points on the lines that are closest together.

a. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

b. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

c. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

d. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + s \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$; $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Exercise 4.2.25 Show that two lines in the plane with slopes m_1 and m_2 are perpendicular if and only if $m_1m_2 = -1$. [*Hint*: Example 4.1.11.]

Exercise 4.2.26

a. Show that, of the four diagonals of a cube, no pair is perpendicular.

b. Show that each diagonal is perpendicular to the face diagonals it does not meet.

Exercise 4.2.27 Given a rectangular solid with sides of lengths 1, 1, and $\sqrt{2}$, find the angle between a diagonal and one of the longest sides.

Exercise 4.2.28 Consider a rectangular solid with sides of lengths a, b, and c. Show that it has two orthogonal diagonals if and only if the sum of two of a^2, b^2, and c^2 equals the third.

Exercise 4.2.29 Let A, B, and $C(2, -1, 1)$ be the vertices of a triangle where $\overrightarrow{AB}$ is parallel to $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, $\overrightarrow{AC}$ is parallel to $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$, and angle $C = 90°$. Find the equation of the line through B and C.

Exercise 4.2.30 If the diagonals of a parallelogram have equal length, show that the parallelogram is a rectangle.

Exercise 4.2.31 Given $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in component form, show that the projections of $\mathbf{v}$ on $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are $x\mathbf{i}$, $y\mathbf{j}$, and $z\mathbf{k}$, respectively.

Exercise 4.2.32

a. Can $\mathbf{u} \cdot \mathbf{v} = -7$ if $\|\mathbf{u}\| = 3$ and $\|\mathbf{v}\| = 2$? Defend your answer.

b. Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $\|\mathbf{v}\| = 6$, and the angle between $\mathbf{u}$ and $\mathbf{v}$ is $\frac{2\pi}{3}$.

Exercise 4.2.33 Show $(\mathbf{u}+\mathbf{v}) \cdot (\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ for any vectors $\mathbf{u}$ and $\mathbf{v}$.

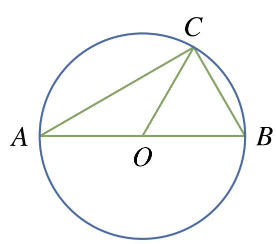
Exercise 4.2.34

a. Show $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$ for any vectors $\mathbf{u}$ and $\mathbf{v}$.

b. What does this say about parallelograms?

Exercise 4.2.35 Show that if the diagonals of a parallelogram are perpendicular, it is necessarily a rhombus. [*Hint*: Example 4.2.5.]

Exercise 4.2.36 Let A and B be the end points of a diameter of a circle (see the diagram). If C is any point on the circle, show that AC and BC are perpendicular. [*Hint*: Express $\overrightarrow{AB} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) = 0$ and $\overrightarrow{BC}$ in terms of $\mathbf{u} = \overrightarrow{OA}$ and $\mathbf{v} = \overrightarrow{OC}$, where O is the centre.]



Exercise 4.2.37 Show that $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, if and only if $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Exercise 4.2.38 Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be pairwise orthogonal vectors.

a. Show that $\|\mathbf{u}+\mathbf{v}+\mathbf{w}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$.

b. If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are all the same length, show that they all make the same angle with $\mathbf{u}+\mathbf{v}+\mathbf{w}$.

Exercise 4.2.39

a. Show that $\mathbf{n} = \begin{bmatrix} a \\ b \end{bmatrix}$ is orthogonal to every vector along the line $ax+by+c=0$.

b. Show that the shortest distance from $P_0(x_0,\ y_0)$ to the line is $\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$.

[*Hint*: If P_1 is on the line, project $\mathbf{u} = \overrightarrow{P_1P_0}$ on $\mathbf{n}$.]

Exercise 4.2.40 Assume $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors that are not parallel. Show that $\mathbf{w} = \|\mathbf{u}\|\mathbf{v} + \|\mathbf{v}\|\mathbf{u}$ is a nonzero vector that bisects the angle between $\mathbf{u}$ and $\mathbf{v}$.

Exercise 4.2.41 Let α, β, and γ be the angles a vector $\mathbf{v} \neq \mathbf{0}$ makes with the positive x, y, and z axes, respectively. Then $\cos\alpha$, $\cos\beta$, and $\cos\gamma$ are called the **direction cosines** of the vector $\mathbf{v}$.

a. If $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, show that $\cos\alpha = \frac{a}{\|\mathbf{v}\|}$, $\cos\beta = \frac{b}{\|\mathbf{v}\|}$, and $\cos\gamma = \frac{c}{\|\mathbf{v}\|}$.

b. Show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

Exercise 4.2.42 Let $\mathbf{v} \neq \mathbf{0}$ be any nonzero vector and suppose that a vector $\mathbf{u}$ can be written as $\mathbf{u} = \mathbf{p}+\mathbf{q}$, where $\mathbf{p}$ is parallel to $\mathbf{v}$ and $\mathbf{q}$ is orthogonal to $\mathbf{v}$. Show that $\mathbf{p}$ must equal the projection of $\mathbf{u}$ on $\mathbf{v}$. [*Hint*: Argue as in the proof of Theorem 4.2.4.]

Exercise 4.2.43 Let $\mathbf{v} \neq \mathbf{0}$ be a nonzero vector and let $a \neq 0$ be a scalar. If $\mathbf{u}$ is any vector, show that the projection of $\mathbf{u}$ on $\mathbf{v}$ equals the projection of $\mathbf{u}$ on $a\mathbf{v}$.

Exercise 4.2.44

a. Show that the **Cauchy-Schwarz inequality** $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ holds for all vectors $\mathbf{u}$ and $\mathbf{v}$. [*Hint*: $|\cos\theta| \leq 1$ for all angles θ.]

b. Show that $|\mathbf{u} \cdot \mathbf{v}| = \|\mathbf{u}\|\|\mathbf{v}\|$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel.

[*Hint*: When is $\cos\theta = \pm 1$?]

c. Show that $|x_1x_2 + y_1y_2 + z_1z_2| \leq \sqrt{x_1^2 + y_1^2 + z_1^2}\sqrt{x_2^2 + y_2^2 + z_2^2}$ holds for all numbers x_1, x_2, y_1, y_2, z_1, and z_2.

d. Show that $|xy + yz + zx| \leq x^2 + y^2 + z^2$ for all x, y, and z.

e. Show that $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ holds for all x, y, and z.

Exercise 4.2.45 Prove that the **triangle inequality** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ holds for all vectors $\mathbf{u}$ and $\mathbf{v}$. [*Hint*: Consider the triangle with $\mathbf{u}$ and $\mathbf{v}$ as two sides.]

4.5 An Application to Computer Graphics

Computer graphics deals with images displayed on a computer screen, and so arises in a variety of applications, ranging from word processors, to *Star Wars* animations, to video games, to wire-frame images of an airplane. These images consist of a number of points on the screen, together with instructions on how to fill in areas bounded by lines and curves. Often curves are approximated by a set of short straight-line segments, so that the curve is specified by a series of points on the screen at the end of these segments. Matrix transformations are important here because matrix images of straight line segments are again line segments.[14] Note that a colour image requires that three images are sent, one to each of the red, green, and blue phosphorus dots on the screen, in varying intensities.

[14] If $\mathbf{v}_0$ and $\mathbf{v}_1$ are vectors, the vector from $\mathbf{v}_0$ to $\mathbf{v}_1$ is $\mathbf{d} = \mathbf{v}_1 - \mathbf{v}_0$. So a vector $\mathbf{v}$ lies on the line segment between $\mathbf{v}_0$ and $\mathbf{v}_1$ if and only if $\mathbf{v} = \mathbf{v}_0 + t\mathbf{d}$ for some number t in the range $0 \leq t \leq 1$. Thus the image of this segment is the set of vectors $A\mathbf{v} = A\mathbf{v}_0 + tA\mathbf{d}$ with $0 \leq t \leq 1$, that is the image is the segment between $A\mathbf{v}_0$ and $A\mathbf{v}_1$.

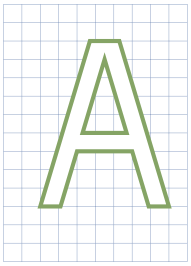


Figure 4.5.1

Consider displaying the letter A. In reality, it is depicted on the screen, as in Figure 4.5.1, by specifying the coordinates of the 11 corners and filling in the interior.

For simplicity, we will disregard the thickness of the letter, so we require only five coordinates as in Figure 4.5.2.

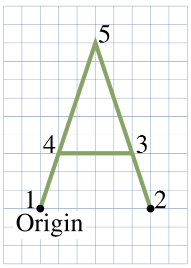


Figure 4.5.2

This simplified letter can then be stored as a data matrix

$$\begin{array}{cc} \textit{Vertex} & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ D = & \left[\begin{array}{ccccc} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right] \end{array}$$

where the columns are the coordinates of the vertices in order. Then if we want to transform the letter by a 2×2 matrix A, we left-multiply this data matrix by A (the effect is to multiply each column by A and so transform each vertex).

For example, we can slant the letter to the right by multiplying by an x-shear matrix $A = \left[\begin{array}{cc} 1 & 0.2 \\ 0 & 1 \end{array}\right]$—see Section 2.2. The result is the letter with data matrix

$$A = \left[\begin{array}{cc} 1 & 0.2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ccccc} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right] = \left[\begin{array}{ccccc} 0 & 6 & 5.6 & 1.6 & 4.8 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right]$$

which is shown in Figure 4.5.3.

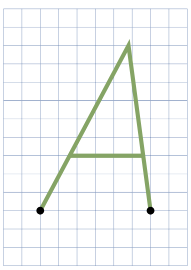


Figure 4.5.3

If we want to make this slanted matrix narrower, we can now apply an x-scale matrix $B = \left[\begin{array}{cc} 0.8 & 0 \\ 0 & 1 \end{array}\right]$ that shrinks the x-coordinate by 0.8. The result is the composite transformation

$$\begin{aligned} BAD &= \left[\begin{array}{cc} 0.8 & 0 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0.2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ccccc} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right] \\ &= \left[\begin{array}{ccccc} 0 & 4.8 & 4.48 & 1.28 & 3.84 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right] \end{aligned}$$

which is drawn in Figure 4.5.4.

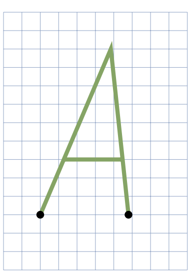


Figure 4.5.4

On the other hand, we can rotate the letter about the origin through $\frac{\pi}{6}$ (or $30°$) by multiplying by the matrix $R_{\frac{\pi}{2}} = \left[\begin{array}{cc} \cos(\frac{\pi}{6}) & -\sin(\frac{\pi}{6}) \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) \end{array}\right] = \left[\begin{array}{cc} 0.866 & -0.5 \\ 0.5 & 0.866 \end{array}\right]$.

This gives

$$\begin{aligned} R_{\frac{\pi}{2}} &= \left[\begin{array}{cc} 0.866 & -0.5 \\ 0.5 & 0.866 \end{array}\right]\left[\begin{array}{ccccc} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \end{array}\right] \\ &= \left[\begin{array}{ccccc} 0 & 5.196 & 2.83 & -0.634 & -1.902 \\ 0 & 3 & 5.098 & 3.098 & 9.294 \end{array}\right] \end{aligned}$$

and is plotted in Figure 4.5.5.

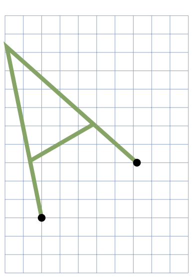


Figure 4.5.5

This poses a problem: How do we rotate at a point other than the origin? It turns out that we can do this when we have solved another more basic problem.

It is clearly important to be able to translate a screen image by a fixed vector $\mathbf{w}$, that is apply the transformation $T_w : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T_w(\mathbf{v}) = \mathbf{v} + \mathbf{w}$ for all $\mathbf{v}$ in $\mathbb{R}^2$. The problem is that these translations are not matrix transformations $\mathbb{R}^2 \to \mathbb{R}^2$ because they do not carry $\mathbf{0}$ to $\mathbf{0}$ (unless $\mathbf{w} = \mathbf{0}$). However, there is a clever way around this.

The idea is to represent a point $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ as a 3×1 column $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, called the **homogeneous coordinates** of $\mathbf{v}$. Then translation by $\mathbf{w} = \begin{bmatrix} p \\ q \end{bmatrix}$ can be achieved by multiplying by a 3×3 matrix:

$$\begin{bmatrix} 1 & 0 & p \\ 0 & 1 & q \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x+p \\ y+q \\ 1 \end{bmatrix} = \begin{bmatrix} T_{\mathbf{w}}(\mathbf{v}) \\ 1 \end{bmatrix}$$

Thus, by using homogeneous coordinates we can implement the translation T_w in the top two coordinates. On the other hand, the matrix transformation induced by $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is also given by a 3×3 matrix:

$$\begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \\ 1 \end{bmatrix} = \begin{bmatrix} A\mathbf{v} \\ 1 \end{bmatrix}$$

So everything can be accomplished at the expense of using 3×3 matrices and homogeneous coordinates.

Example 4.5.1

Rotate the letter A in Figure 4.5.2 through $\frac{\pi}{6}$ about the point $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$.

Solution. Using homogeneous coordinates for the vertices of the letter results in a data matrix with three rows:

$$K_d = \begin{bmatrix} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

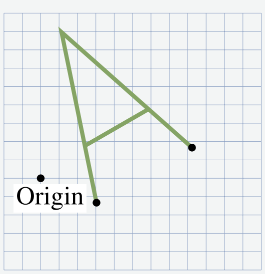


Figure 4.5.6

If we write $\mathbf{w} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, the idea is to use a composite of transformations: First translate the letter by $-\mathbf{w}$ so that the point $\mathbf{w}$ moves to the origin, then rotate this translated letter, and then translate it by $\mathbf{w}$ back to its original position. The matrix arithmetic is as follows (remember the order of composition!):

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 6 & 5 & 1 & 3 \\ 0 & 0 & 3 & 3 & 9 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 3.036 & 8.232 & 5.866 & 2.402 & 1.134 \\ -1.33 & 1.67 & 3.768 & 1.768 & 7.964 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

This is plotted in Figure 4.5.6.

This discussion merely touches the surface of computer graphics, and the reader is referred to specialized books on the subject. Realistic graphic rendering requires an enormous number of matrix calculations. In fact, matrix multiplication algorithms are now embedded in microchip circuits, and can perform over 100 million matrix multiplications per second. This is particularly important in the field of three-dimensional graphics where the homogeneous coordinates have four components and 4×4 matrices are required.

Exercises for 4.5

Exercise 4.5.1 Consider the letter A described in Figure 4.5.2. Find the data matrix for the letter obtained by:

a. Rotating the letter through $\frac{\pi}{4}$ about the origin.

b. Rotating the letter through $\frac{\pi}{4}$ about the point $\left[\begin{array}{c} 1 \\ 2 \end{array}\right]$.

Exercise 4.5.2 Find the matrix for turning the letter A in Figure 4.5.2 upside-down in place.

Exercise 4.5.3 Find the 3×3 matrix for reflecting in the line $y = mx + b$. Use $\left[\begin{array}{c} 1 \\ m \end{array}\right]$ as direction vector for the line.

Exercise 4.5.4 Find the 3×3 matrix for rotating through the angle θ about the point $P(a, b)$.

Exercise 4.5.5 Find the reflection of the point P in the line $y = 1 + 2x$ in $\mathbb{R}^2$ if:

a. $P = P(1, 1)$

b. $P = P(1, 4)$

c. What about $P = P(1, 3)$? Explain. [*Hint*: Example 4.5.1 and Section 4.4.]

Chapter 5

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Vector Space $\mathbb{R}^n$

5.1 Subspaces and Spanning

In Section 2.2 we introduced the set $\mathbb{R}^n$ of all n-tuples (called *vectors*), and began our investigation of the matrix transformations $\mathbb{R}^n \to \mathbb{R}^m$ given by matrix multiplication by an $m \times n$ matrix. Particular attention was paid to the euclidean plane $\mathbb{R}^2$ where certain simple geometric transformations were seen to be matrix transformations. Then in Section 2.6 we introduced linear transformations, showed that they are all matrix transformations, and found the matrices of rotations and reflections in $\mathbb{R}^2$. We returned to this in Section 4.4 where we showed that projections, reflections, and rotations of $\mathbb{R}^2$ and $\mathbb{R}^3$ were all linear, and where we related areas and volumes to determinants.

In this chapter we investigate $\mathbb{R}^n$ in full generality, and introduce some of the most important concepts and methods in linear algebra. The n-tuples in $\mathbb{R}^n$ will continue to be denoted $\mathbf{x}$, $\mathbf{y}$, and so on, and will be written as rows or columns depending on the context.

Subspaces of $\mathbb{R}^n$

Definition 5.1 Subspace of $\mathbb{R}^n$

A set[1] U *of vectors in* $\mathbb{R}^n$ *is called a* **subspace** *of* $\mathbb{R}^n$ *if it satisfies the following properties:*

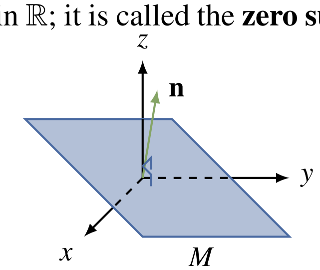
S1. The zero vector $\mathbf{0} \in U$.

S2. If $\mathbf{x} \in U$ *and* $\mathbf{y} \in U$, *then* $\mathbf{x} + \mathbf{y} \in U$.

S3. If $\mathbf{x} \in U$, *then* $a\mathbf{x} \in U$ *for every real number* a.

We say that the subset U is **closed under addition** if S2 holds, and that U is **closed under scalar multiplication** if S3 holds.

Clearly $\mathbb{R}^n$ is a subspace of itself, and this chapter is about these subspaces and their properties. The set $U = \{\mathbf{0}\}$, consisting of only the zero vector, is also a subspace because $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $a\mathbf{0} = \mathbf{0}$ for each a in $\mathbb{R}$; it is called the **zero subspace**. Any subspace of $\mathbb{R}^n$ other than $\{\mathbf{0}\}$ or $\mathbb{R}^n$ is called a **proper** subspace.



We saw in Section 4.2 that every plane M through the origin in $\mathbb{R}^3$ has equation $ax + by + cz = 0$ where a, b, and c are not all zero. Here $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is a normal for the plane and

$$M = \{\mathbf{v} \text{ in } \mathbb{R}^3 \mid \mathbf{n} \cdot \mathbf{v} = 0\}$$

[1]We use the language of sets. Informally, a **set** X is a collection of objects, called the **elements** of the set. The fact that x is an element of X is denoted $x \in X$. Two sets X and Y are called equal (written $X = Y$) if they have the same elements. If every element of X is in the set Y, we say that X is a **subset** of Y, and write $X \subseteq Y$. Hence $X \subseteq Y$ and $Y \subseteq X$ both hold if and only if $X = Y$.

where $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\mathbf{n} \cdot \mathbf{v}$ denotes the dot product introduced in Section 2.2 (see the diagram).[2] Then M is a subspace of $\mathbb{R}^3$. Indeed we show that M satisfies S1, S2, and S3 as follows:

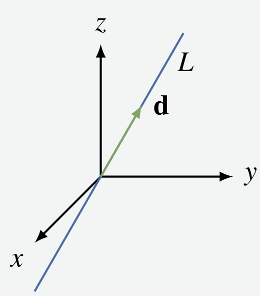
S1. $\mathbf{0} \in M$ *because* $\mathbf{n} \cdot \mathbf{0} = 0$*;*

S2. If $\mathbf{v} \in M$ *and* $\mathbf{v}_1 \in M$ *, then* $\mathbf{n} \cdot (\mathbf{v} + \mathbf{v}_1) = \mathbf{n} \cdot \mathbf{v} + \mathbf{n} \cdot \mathbf{v}_1 = 0 + 0 = 0$ *, so* $\mathbf{v} + \mathbf{v}_1 \in M$*;*

S3. If $\mathbf{v} \in M$ *, then* $\mathbf{n} \cdot (a\mathbf{v}) = a(\mathbf{n} \cdot \mathbf{v}) = a(0) = 0$ *, so* $a\mathbf{v} \in M$*.*

This proves the first part of

Example 5.1.1



Planes and lines through the origin in $\mathbb{R}^3$ are all subspaces of $\mathbb{R}^3$.

Solution. We dealt with planes above. If L is a line through the origin with direction vector $\mathbf{d}$, then $L = \{t\mathbf{d} \mid t \in \mathbb{R}\}$ (see the diagram). We leave it as an exercise to verify that L satisfies S1, S2, and S3.

Example 5.1.1 shows that lines through the origin in $\mathbb{R}^2$ are subspaces; in fact, they are the *only* proper subspaces of $\mathbb{R}^2$ (Exercise 5.1.24). Indeed, we shall see in Example 5.2.14 that lines and planes through the origin in $\mathbb{R}^3$ are the only proper subspaces of $\mathbb{R}^3$. Thus the geometry of lines and planes through the origin is captured by the subspace concept. (Note that *every* line or plane is just a translation of one of these.)

Subspaces can also be used to describe important features of an $m \times n$ matrix A. The **null space** of A, denoted $\operatorname{null} A$, and the **image space** of A, denoted $\operatorname{im} A$, are defined by

$$\operatorname{null} A = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\} \quad \text{and} \quad \operatorname{im} A = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}$$

In the language of Chapter 2, $\operatorname{null} A$ consists of all solutions $\mathbf{x}$ in $\mathbb{R}^n$ of the homogeneous system $A\mathbf{x} = \mathbf{0}$, and $\operatorname{im} A$ is the set of all vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ *has* a solution $\mathbf{x}$. Note that $\mathbf{x}$ is in $\operatorname{null} A$ if it satisfies the *condition* $A\mathbf{x} = \mathbf{0}$, while $\operatorname{im} A$ consists of vectors of the *form* $A\mathbf{x}$ for some $\mathbf{x}$ in $\mathbb{R}^n$. These two ways to describe subsets occur frequently.

Example 5.1.2

If A is an $m \times n$ matrix, then:

1. $\operatorname{null} A$ is a subspace of $\mathbb{R}^n$.

[2]We are using set notation here. In general $\{q \mid p\}$ means the set of all objects q with property p.

2. $\operatorname{im} A$ is a subspace of $\mathbb{R}^m$.

Solution.

1. The zero vector $\mathbf{0} \in \mathbb{R}^n$ lies in $\operatorname{null} A$ because $A\mathbf{0} = \mathbf{0}$.[3] If $\mathbf{x}$ and $\mathbf{x}_1$ are in $\operatorname{null} A$, then $\mathbf{x} + \mathbf{x}_1$ and $a\mathbf{x}$ are in $\operatorname{null} A$ because they satisfy the required condition:

$$A(\mathbf{x} + \mathbf{x}_1) = A\mathbf{x} + A\mathbf{x}_1 = \mathbf{0} + \mathbf{0} = \mathbf{0} \quad \text{and} \quad A(a\mathbf{x}) = a(A\mathbf{x}) = a\mathbf{0} = \mathbf{0}$$

Hence $\operatorname{null} A$ satisfies S1, S2, and S3, and so is a subspace of $\mathbb{R}^n$.

2. The zero vector $\mathbf{0} \in \mathbb{R}^m$ lies in $\operatorname{im} A$ because $\mathbf{0} = A\mathbf{0}$. Suppose that $\mathbf{y}$ and $\mathbf{y}_1$ are in $\operatorname{im} A$, say $\mathbf{y} = A\mathbf{x}$ and $\mathbf{y}_1 = A\mathbf{x}_1$ where $\mathbf{x}$ and $\mathbf{x}_1$ are in $\mathbb{R}^n$. Then

$$\mathbf{y} + \mathbf{y}_1 = A\mathbf{x} + A\mathbf{x}_1 = A(\mathbf{x} + \mathbf{x}_1) \quad \text{and} \quad a\mathbf{y} = a(A\mathbf{x}) = A(a\mathbf{x})$$

show that $\mathbf{y} + \mathbf{y}_1$ and $a\mathbf{y}$ are both in $\operatorname{im} A$ (they have the required form). Hence $\operatorname{im} A$ is a subspace of $\mathbb{R}^m$.

There are other important subspaces associated with a matrix A that clarify basic properties of A. If A is an $n \times n$ matrix and λ is any number, let

$$E_\lambda(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$$

A vector $\mathbf{x}$ is in $E_\lambda(A)$ if and only if $(\lambda I - A)\mathbf{x} = \mathbf{0}$, so Example 5.1.2 gives:

Example 5.1.3

$E_\lambda(A) = \operatorname{null}(\lambda I - A)$ is a subspace of $\mathbb{R}^n$ for each $n \times n$ matrix A and number λ.

$E_\lambda(A)$ is called the **eigenspace** of A corresponding to λ. The reason for the name is that, in the terminology of Section 3.3, λ is an **eigenvalue** of A if $E_\lambda(A) \neq \{\mathbf{0}\}$. In this case the nonzero vectors in $E_\lambda(A)$ are called the **eigenvectors** of A corresponding to λ.

The reader should not get the impression that *every* subset of $\mathbb{R}^n$ is a subspace. For example:

$$U_1 = \left\{ \left[\begin{array}{c} x \\ y \end{array} \right] \middle| \, x \geq 0 \right\} \text{ satisfies S1 and S2, but not S3;}$$

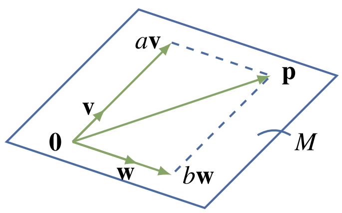
$$U_2 = \left\{ \left[\begin{array}{c} x \\ y \end{array} \right] \middle| \, x^2 = y^2 \right\} \text{ satisfies S1 and S3, but not S2;}$$

Hence neither U_1 nor U_2 is a subspace of $\mathbb{R}^2$. (However, see Exercise 5.1.20.)

[3] We are using $\mathbf{0}$ to represent the zero vector in both $\mathbb{R}^m$ and $\mathbb{R}^n$. This abuse of notation is common and causes no confusion once everybody knows what is going on.

Spanning Sets

Let $\mathbf{v}$ and $\mathbf{w}$ be two nonzero, nonparallel vectors in $\mathbb{R}^3$ with their tails at the origin. The plane M through the origin containing these vectors is described in Section 4.2 by saying that $\mathbf{n} = \mathbf{v} \times \mathbf{w}$ is a *normal* for M, and that M consists of all vectors $\mathbf{p}$ such that $\mathbf{n} \cdot \mathbf{p} = 0$.[4] While this is a very useful way to look at planes, there is another approach that is at least as useful in $\mathbb{R}^3$ and, more importantly, works for all subspaces of $\mathbb{R}^n$ for any $n \geq 1$.



The idea is as follows: Observe that, by the diagram, a vector $\mathbf{p}$ is in M if and only if it has the form

$$\mathbf{p} = a\mathbf{v} + b\mathbf{w}$$

for certain real numbers a and b (we say that $\mathbf{p}$ is a *linear combination* of $\mathbf{v}$ and $\mathbf{w}$). Hence we can describe M as

$$M = \{a\mathbf{x} + b\mathbf{w} \mid a,\ b \in \mathbb{R}\}.^5$$

and we say that $\{\mathbf{v},\ \mathbf{w}\}$ is a *spanning set* for M. It is this notion of a spanning set that provides a way to describe all subspaces of $\mathbb{R}^n$.

As in Section 1.3, given vectors $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$ in $\mathbb{R}^n$, a vector of the form

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k \quad \text{where the } t_i \text{ are scalars}$$

is called a **linear combination** of the $\mathbf{x}_i$, and t_i is called the **coefficient** of $\mathbf{x}_i$ in the linear combination.

Definition 5.2 Linear Combinations and Span in $\mathbb{R}^n$

The set of all such linear combinations is called the ***span*** *of the $\mathbf{x}_i$ and is denoted*

$$\operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k \mid t_i \text{ in } \mathbb{R}\}$$

If $V = \operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$, we say that V is ***spanned*** *by the vectors $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$, and that the vectors $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$* ***span*** *the space V.*

Here are two examples:

$$\operatorname{span}\{\mathbf{x}\} = \{t\mathbf{x} \mid t \in \mathbb{R}\}$$

which we write as $\operatorname{span}\{\mathbf{x}\} = \mathbb{R}\mathbf{x}$ for simplicity.

$$\operatorname{span}\{\mathbf{x},\ \mathbf{y}\} = \{r\mathbf{x} + s\mathbf{y} \mid r,\ s \in \mathbb{R}\}$$

In particular, the above discussion shows that, if $\mathbf{v}$ and $\mathbf{w}$ are two nonzero, nonparallel vectors in $\mathbb{R}^3$, then

$$M = \operatorname{span}\{\mathbf{v},\ \mathbf{w}\}$$

is the plane in $\mathbb{R}^3$ containing $\mathbf{v}$ and $\mathbf{w}$. Moreover, if $\mathbf{d}$ is any nonzero vector in $\mathbb{R}^3$ (or $\mathbb{R}^2$), then

$$L = \operatorname{span}\{\mathbf{v}\} = \{t\mathbf{d} \mid t \in \mathbb{R}\} = \mathbb{R}\mathbf{d}$$

is the line with direction vector $\mathbf{d}$. Hence lines and planes can both be described in terms of spanning sets.

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[4] The vector $\mathbf{n} = \mathbf{v} \times \mathbf{w}$ is nonzero because $\mathbf{v}$ and $\mathbf{w}$ are not parallel.

[5] In particular, this implies that any vector $\mathbf{p}$ orthogonal to $\mathbf{v} \times \mathbf{w}$ must be a linear combination $\mathbf{p} = a\mathbf{v} + b\mathbf{w}$ of $\mathbf{v}$ and $\mathbf{w}$ for some a and b. Can you prove this directly?

Example 5.1.4

Let $\mathbf{x} = (2, -1, 2, 1)$ and $\mathbf{y} = (3, 4, -1, 1)$ in $\mathbb{R}^4$. Determine whether $\mathbf{p} = (0, -11, 8, 1)$ or $\mathbf{q} = (2, 3, 1, 2)$ are in $U = \text{span}\{\mathbf{x}, \mathbf{y}\}$.

Solution. The vector $\mathbf{p}$ is in U if and only if $\mathbf{p} = s\mathbf{x} + t\mathbf{y}$ for scalars s and t. Equating components gives equations

$$2s + 3t = 0, \quad -s + 4t = -11, \quad 2s - t = 8, \quad \text{and} \quad s + t = 1$$

This linear system has solution $s = 3$ and $t = -2$, so $\mathbf{p}$ is in U. On the other hand, asking that $\mathbf{q} = s\mathbf{x} + t\mathbf{y}$ leads to equations

$$2s + 3t = 2, \quad -s + 4t = 3, \quad 2s - t = 1, \quad \text{and} \quad s + t = 2$$

and this system has *no* solution. So $\mathbf{q}$ does *not* lie in U.

Theorem 5.1.1: Span Theorem

Let $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ *in* $\mathbb{R}^n$. *Then:*

1. U *is a subspace of* $\mathbb{R}^n$ *containing each* $\mathbf{x}_i$.
2. *If* W *is a subspace of* $\mathbb{R}^n$ *and each* $\mathbf{x}_i \in W$, *then* $U \subseteq W$.

Proof.

1. The zero vector $\mathbf{0}$ is in U because $\mathbf{0} = 0\mathbf{x}_1 + 0\mathbf{x}_2 + \cdots + 0\mathbf{x}_k$ is a linear combination of the $\mathbf{x}_i$. If $\mathbf{x} = t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k$ and $\mathbf{y} = s_1\mathbf{x}_1 + s_2\mathbf{x}_2 + \cdots + s_k\mathbf{x}_k$ are in U, then $\mathbf{x} + \mathbf{y}$ and $a\mathbf{x}$ are in U because

$$\begin{aligned} \mathbf{x} + \mathbf{y} &= (t_1 + s_1)\mathbf{x}_1 + (t_2 + s_2)\mathbf{x}_2 + \cdots + (t_k + s_k)\mathbf{x}_k, \text{ and} \\ a\mathbf{x} &= (at_1)\mathbf{x}_1 + (at_2)\mathbf{x}_2 + \cdots + (at_k)\mathbf{x}_k \end{aligned}$$

   Finally each $\mathbf{x}_i$ is in U (for example, $\mathbf{x}_2 = 0\mathbf{x}_1 + 1\mathbf{x}_2 + \cdots + 0\mathbf{x}_k$) so S1, S2, and S3 are satisfied for U, proving (1).

2. Let $\mathbf{x} = t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k$ where the t_i are scalars and each $\mathbf{x}_i \in W$. Then each $t_i\mathbf{x}_i \in W$ because W satisfies S3. But then $\mathbf{x} \in W$ because W satisfies S2 (verify). This proves (2). □

Condition (2) in Theorem 5.1.1 can be expressed by saying that $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is the *smallest* subspace of $\mathbb{R}^n$ that contains each $\mathbf{x}_i$. This is useful for showing that two subspaces U and W are equal, since this amounts to showing that both $U \subseteq W$ and $W \subseteq U$. Here is an example of how it is used.

Example 5.1.5

If $\mathbf{x}$ and $\mathbf{y}$ are in $\mathbb{R}^n$, show that $\text{span}\{\mathbf{x}, \mathbf{y}\} = \text{span}\{\mathbf{x} + \mathbf{y}, \mathbf{x} - \mathbf{y}\}$.

Solution. Since both $\mathbf{x}+\mathbf{y}$ and $\mathbf{x}-\mathbf{y}$ are in $\operatorname{span}\{\mathbf{x},\ \mathbf{y}\}$, Theorem 5.1.1 gives

$$\operatorname{span}\{\mathbf{x}+\mathbf{y},\ \mathbf{x}-\mathbf{y}\} \subseteq \operatorname{span}\{\mathbf{x},\ \mathbf{y}\}$$

But $\mathbf{x} = \frac{1}{2}(\mathbf{x}+\mathbf{y}) + \frac{1}{2}(\mathbf{x}-\mathbf{y})$ and $\mathbf{y} = \frac{1}{2}(\mathbf{x}+\mathbf{y}) - \frac{1}{2}(\mathbf{x}-\mathbf{y})$ are both in $\operatorname{span}\{\mathbf{x}+\mathbf{y},\ \mathbf{x}-\mathbf{y}\}$, so

$$\operatorname{span}\{\mathbf{x},\ \mathbf{y}\} \subseteq \operatorname{span}\{\mathbf{x}+\mathbf{y},\ \mathbf{x}-\mathbf{y}\}$$

again by Theorem 5.1.1. Thus $\operatorname{span}\{\mathbf{x},\ \mathbf{y}\} = \operatorname{span}\{\mathbf{x}+\mathbf{y},\ \mathbf{x}-\mathbf{y}\}$, as desired.

It turns out that many important subspaces are best described by giving a spanning set. Here are three examples, beginning with an important spanning set for $\mathbb{R}^n$ itself. Column j of the $n \times n$ identity matrix I_n is denoted $\mathbf{e}_j$ and called the jth **coordinate vector** in $\mathbb{R}^n$, and the set $\{\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_n\}$ is called the **standard basis** of $\mathbb{R}^n$. If $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]$ is any vector in $\mathbb{R}^n$, then $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$, as the reader can verify. This proves:

Example 5.1.6

$\mathbb{R}^n = \operatorname{span}\{\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_n\}$ where $\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_n$ are the columns of I_n.

If A is an $m \times n$ matrix A, the next two examples show that it is a routine matter to find spanning sets for $\operatorname{null} A$ and $\operatorname{im} A$.

Example 5.1.7

Given an $m \times n$ matrix A, let $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$ denote the basic solutions to the system $A\mathbf{x} = \mathbf{0}$ given by the gaussian algorithm. Then

$$\operatorname{null} A = \operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$$

Solution. If $\mathbf{x} \in \operatorname{null} A$, then $A\mathbf{x} = \mathbf{0}$ so Theorem 1.3.2 shows that $\mathbf{x}$ is a linear combination of the basic solutions; that is, $\operatorname{null} A \subseteq \operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$. On the other hand, if $\mathbf{x}$ is in $\operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$, then $\mathbf{x} = t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k$ for scalars t_i, so

$$A\mathbf{x} = t_1A\mathbf{x}_1 + t_2A\mathbf{x}_2 + \cdots + t_kA\mathbf{x}_k = t_1\mathbf{0} + t_2\mathbf{0} + \cdots + t_k\mathbf{0} = \mathbf{0}$$

This shows that $\mathbf{x} \in \operatorname{null} A$, and hence that $\operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\} \subseteq \operatorname{null} A$. Thus we have equality.

Example 5.1.8

Let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ denote the columns of the $m \times n$ matrix A. Then

$$\operatorname{im} A = \operatorname{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$$

Solution. If $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$, observe that

$$\left[\begin{array}{cccc} A\mathbf{e}_1 & A\mathbf{e}_2 & \cdots & A\mathbf{e}_n \end{array}\right] = A\left[\begin{array}{cccc} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{array}\right] = AI_n = A = \left[\begin{array}{ccc} \mathbf{c}_1 & \mathbf{c}_2 & \cdots \mathbf{c}_n \end{array}\right].$$

Hence $\mathbf{c}_i = A\mathbf{e}_i$ is in $\operatorname{im} A$ for each i, so $\operatorname{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\} \subseteq \operatorname{im} A$.

Conversely, let $\mathbf{y}$ be in $\operatorname{im} A$, say $\mathbf{y} = A\mathbf{x}$ for some $\mathbf{x}$ in $\mathbb{R}^n$. If $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]$, then Definition 2.5 gives

$$\mathbf{y} = A\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n \text{ is in } \operatorname{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$$

This shows that $\operatorname{im} A \subseteq \operatorname{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$, and the result follows.

Exercises for 5.1

We often write vectors in $\mathbb{R}^n$ as rows.

Exercise 5.1.1 In each case determine whether U is a subspace of $\mathbb{R}^3$. Support your answer.

a. $U = \{(1, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}$.

b. $U = \{(0, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}$.

c. $U = \{(r, s, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}, -r + 3s + 2t = 0\}$.

d. $U = \{(r, 3s, r-2) \mid r \text{ and } s \text{ in } \mathbb{R}\}$.

e. $U = \{(r, 0, s) \mid r^2 + s^2 = 0, r \text{ and } s \text{ in } \mathbb{R}\}$.

f. $U = \{(2r, -s^2, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}\}$.

Exercise 5.1.2 In each case determine if $\mathbf{x}$ lies in $U = \operatorname{span}\{\mathbf{y}, \mathbf{z}\}$. If $\mathbf{x}$ is in U, write it as a linear combination of $\mathbf{y}$ and $\mathbf{z}$; if $\mathbf{x}$ is not in U, show why not.

a. $\mathbf{x} = (2, -1, 0, 1)$, $\mathbf{y} = (1, 0, 0, 1)$, and $\mathbf{z} = (0, 1, 0, 1)$.

b. $\mathbf{x} = (1, 2, 15, 11)$, $\mathbf{y} = (2, -1, 0, 2)$, and $\mathbf{z} = (1, -1, -3, 1)$.

c. $\mathbf{x} = (8, 3, -13, 20)$, $\mathbf{y} = (2, 1, -3, 5)$, and $\mathbf{z} = (-1, 0, 2, -3)$.

d. $\mathbf{x} = (2, 5, 8, 3)$, $\mathbf{y} = (2, -1, 0, 5)$, and $\mathbf{z} = (-1, 2, 2, -3)$.

Exercise 5.1.3 In each case determine if the given vectors span $\mathbb{R}^4$. Support your answer.

a. $\{(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1)\}$.

b. $\{(1, 3, -5, 0), (-2, 1, 0, 0), (0, 2, 1, -1), (1, -4, 5, 0)\}$.

Exercise 5.1.4 Is it possible that $\{(1, 2, 0), (2, 0, 3)\}$ can span the subspace $U = \{(r, s, 0) \mid r \text{ and } s \text{ in } \mathbb{R}\}$? Defend your answer.

Exercise 5.1.5 Give a spanning set for the zero subspace $\{\mathbf{0}\}$ of $\mathbb{R}^n$.

Exercise 5.1.6 Is $\mathbb{R}^2$ a subspace of $\mathbb{R}^3$? Defend your answer.

Exercise 5.1.7 If $U = \text{span}\,\{\mathbf{x},\ \mathbf{y},\ \mathbf{z}\}$ in $\mathbb{R}^n$, show that $U = \text{span}\,\{\mathbf{x}+t\mathbf{z},\ \mathbf{y},\ \mathbf{z}\}$ for every t in $\mathbb{R}$.

Exercise 5.1.8 If $U = \text{span}\,\{\mathbf{x},\ \mathbf{y},\ \mathbf{z}\}$ in $\mathbb{R}^n$, show that $U = \text{span}\,\{\mathbf{x}+\mathbf{y},\ \mathbf{y}+\mathbf{z},\ \mathbf{z}+\mathbf{x}\}$.

Exercise 5.1.9 If $a \neq 0$ is a scalar, show that $\text{span}\,\{a\mathbf{x}\} = \text{span}\,\{\mathbf{x}\}$ for every vector $\mathbf{x}$ in $\mathbb{R}^n$.

Exercise 5.1.10 If $a_1,\ a_2,\ \dots,\ a_k$ are nonzero scalars, show that $\text{span}\,\{a_1\mathbf{x}_1,\ a_2\mathbf{x}_2,\ \dots,\ a_k\mathbf{x}_k\} = \text{span}\,\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ for any vectors $\mathbf{x}_i$ in $\mathbb{R}^n$.

Exercise 5.1.11 If $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$, determine all subspaces of $\text{span}\,\{\mathbf{x}\}$.

Exercise 5.1.12 Suppose that $U = \text{span}\,\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ where each $\mathbf{x}_i$ is in $\mathbb{R}^n$. If A is an $m \times n$ matrix and $A\mathbf{x}_i = \mathbf{0}$ for each i, show that $A\mathbf{y} = \mathbf{0}$ for every vector $\mathbf{y}$ in U.

Exercise 5.1.13 If A is an $m \times n$ matrix, show that, for each invertible $m \times m$ matrix U, $\text{null}\,(A) = \text{null}\,(UA)$.

Exercise 5.1.14 If A is an $m \times n$ matrix, show that, for each invertible $n \times n$ matrix V, $\text{im}\,(A) = \text{im}\,(AV)$.

Exercise 5.1.15 Let U be a subspace of $\mathbb{R}^n$, and let $\mathbf{x}$ be a vector in $\mathbb{R}^n$.

a. If $a\mathbf{x}$ is in U where $a \neq 0$ is a number, show that $\mathbf{x}$ is in U.

b. If $\mathbf{y}$ and $\mathbf{x}+\mathbf{y}$ are in U where $\mathbf{y}$ is a vector in $\mathbb{R}^n$, show that $\mathbf{x}$ is in U.

Exercise 5.1.16 In each case either show that the statement is true or give an example showing that it is false.

a. If $U \neq \mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ and $\mathbf{x}+\mathbf{y}$ is in U, then $\mathbf{x}$ and $\mathbf{y}$ are both in U.

b. If U is a subspace of $\mathbb{R}^n$ and $r\mathbf{x}$ is in U for all r in $\mathbb{R}$, then $\mathbf{x}$ is in U.

c. If U is a subspace of $\mathbb{R}^n$ and $\mathbf{x}$ is in U, then $-\mathbf{x}$ is also in U.

d. If $\mathbf{x}$ is in U and $U = \text{span}\,\{\mathbf{y},\ \mathbf{z}\}$, then $U = \text{span}\,\{\mathbf{x},\ \mathbf{y},\ \mathbf{z}\}$.

e. The empty set of vectors in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$.

f. $\left[\begin{array}{c} 0 \\ 1 \end{array}\right]$ is in span $\left\{\left[\begin{array}{c} 1 \\ 0 \end{array}\right],\ \left[\begin{array}{c} 2 \\ 0 \end{array}\right]\right\}$.

Exercise 5.1.17

a. If A and B are $m \times n$ matrices, show that $U = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid A\mathbf{x} = B\mathbf{x}\}$ is a subspace of $\mathbb{R}^n$.

b. What if A is $m \times n$, B is $k \times n$, and $m \neq k$?

Exercise 5.1.18 Suppose that $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$ are vectors in $\mathbb{R}^n$. If $\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k$ where $a_1 \neq 0$, show that $\text{span}\,\{\mathbf{x}_1\,\mathbf{x}_2,\ \dots,\ \mathbf{x}_k\} = \text{span}\,\{\mathbf{y}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$.

Exercise 5.1.19 If $U \neq \{\mathbf{0}\}$ is a subspace of $\mathbb{R}$, show that $U = \mathbb{R}$.

Exercise 5.1.20 Let U be a nonempty subset of $\mathbb{R}^n$. Show that U is a subspace if and only if S2 and S3 hold.

Exercise 5.1.21 If S and T are nonempty sets of vectors in $\mathbb{R}^n$, and if $S \subseteq T$, show that $\text{span}\,\{S\} \subseteq \text{span}\,\{T\}$.

Exercise 5.1.22 Let U and W be subspaces of $\mathbb{R}^n$. Define their **intersection** $U \cap W$ and their **sum** $U + W$ as follows:

$U \cap W = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \text{ belongs to both } U \text{ and } W\}$.

$U + W = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \text{ is a sum of a vector in } U$ and a vector in $W\}$.

a. Show that $U \cap W$ is a subspace of $\mathbb{R}^n$.

b. Show that $U + W$ is a subspace of $\mathbb{R}^n$.

Exercise 5.1.23 Let P denote an invertible $n \times n$ matrix. If λ is a number, show that

$$E_\lambda(PAP^{-1}) = \{P\mathbf{x} \mid \mathbf{x} \text{ is in } E_\lambda(A)\}$$

for each $n \times n$ matrix A.

Exercise 5.1.24 Show that every proper subspace U of $\mathbb{R}^2$ is a line through the origin. [*Hint*: If $\mathbf{d}$ is a nonzero vector in U, let $L = \mathbb{R}\mathbf{d} = \{r\mathbf{d} \mid r \text{ in } \mathbb{R}\}$ denote the line with direction vector $\mathbf{d}$. If $\mathbf{u}$ is in U but not in L, argue geometrically that every vector $\mathbf{v}$ in $\mathbb{R}^2$ is a linear combination of $\mathbf{u}$ and $\mathbf{d}$.]

5.2 Independence and Dimension

Some spanning sets are better than others. If $U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a subspace of $\mathbb{R}^n$, then every vector in U can be written as a linear combination of the $\mathbf{x}_i$ in at least one way. Our interest here is in spanning sets where each vector in U has *exactly one* representation as a linear combination of these vectors.

Linear Independence

Given $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ in $\mathbb{R}^n$, suppose that two linear combinations are equal:

$$r_1\mathbf{x}_1 + r_2\mathbf{x}_2 + \cdots + r_k\mathbf{x}_k = s_1\mathbf{x}_1 + s_2\mathbf{x}_2 + \cdots + s_k\mathbf{x}_k$$

We are looking for a condition on the set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors that guarantees that this representation is *unique*; that is, $r_i = s_i$ for each i. Taking all terms to the left side gives

$$(r_1 - s_1)\mathbf{x}_1 + (r_2 - s_2)\mathbf{x}_2 + \cdots + (r_k - s_k)\mathbf{x}_k = \mathbf{0}$$

so the required condition is that this equation forces all the coefficients $r_i - s_i$ to be zero.

Definition 5.3 Linear Independence in $\mathbb{R}^n$

*With this in mind, we call a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors **linearly independent** (or simply **independent**) if it satisfies the following condition:*

$$\text{If } t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k = \mathbf{0} \text{ then } t_1 = t_2 = \cdots = t_k = 0$$

We record the result of the above discussion for reference.

Theorem 5.2.1

*If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is an independent set of vectors in $\mathbb{R}^n$, then every vector in $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ has a **unique** representation as a linear combination of the $\mathbf{x}_i$.*

It is useful to state the definition of independence in different language. Let us say that a linear combination **vanishes** if it equals the zero vector, and call a linear combination **trivial** if every coefficient is zero. Then the definition of independence can be compactly stated as follows:

> A set of vectors is independent if and only if the only linear combination that vanishes is the trivial one.

Hence we have a procedure for checking that a set of vectors is independent:

Independence Test

To verify that a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is independent, proceed as follows:

1. *Set a linear combination equal to zero: $t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$.*

2. *Show that $t_i = 0$ for each i (that is, the linear combination is trivial).*

Of course, if some nontrivial linear combination vanishes, the vectors are not independent.

Example 5.2.1

Determine whether $\{(1,\ 0,\ -2,\ 5),\ (2,\ 1,\ 0,\ -1),\ (1,\ 1,\ 2,\ 1)\}$ is independent in $\mathbb{R}^4$.

Solution. Suppose a linear combination vanishes:

$$r(1,\ 0,\ -2,\ 5) + s(2,\ 1,\ 0,\ -1) + t(1,\ 1,\ 2,\ 1) = (0,\ 0,\ 0,\ 0)$$

Equating corresponding entries gives a system of four equations:

$$r + 2s + t = 0,\ s + t = 0,\ -2r + 2t = 0, \text{ and } 5r - s + t = 0$$

The only solution is the trivial one $r = s = t = 0$ (verify), so these vectors are independent by the independence test.

Example 5.2.2

Show that the standard basis $\{\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_n\}$ of $\mathbb{R}^n$ is independent.

Solution. The components of $t_1\mathbf{e}_1 + t_2\mathbf{e}_2 + \cdots + t_n\mathbf{e}_n$ are $t_1,\ t_2,\ \dots,\ t_n$ (see the discussion preceding Example 5.1.6) So the linear combination vanishes if and only if each $t_i = 0$. Hence the independence test applies.

Example 5.2.3

If $\{\mathbf{x},\ \mathbf{y}\}$ is independent, show that $\{2\mathbf{x} + 3\mathbf{y},\ \mathbf{x} - 5\mathbf{y}\}$ is also independent.

Solution. If $s(2\mathbf{x} + 3\mathbf{y}) + t(\mathbf{x} - 5\mathbf{y}) = \mathbf{0}$, collect terms to get $(2s + t)\mathbf{x} + (3s - 5t)\mathbf{y} = \mathbf{0}$. Since $\{\mathbf{x},\ \mathbf{y}\}$ is independent this combination must be trivial; that is, $2s + t = 0$ and $3s - 5t = 0$. These equations have only the trivial solution $s = t = 0$, as required.

Example 5.2.4

Show that the zero vector in $\mathbb{R}^n$ does not belong to any independent set.

Solution. No set $\{\mathbf{0},\ \mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ of vectors is independent because we have a vanishing, nontrivial linear combination $1 \cdot \mathbf{0} + 0\mathbf{x}_1 + 0\mathbf{x}_2 + \cdots + 0\mathbf{x}_k = \mathbf{0}$.

Example 5.2.5

Given $\mathbf{x}$ in $\mathbb{R}^n$, show that $\{\mathbf{x}\}$ is independent if and only if $\mathbf{x} \neq \mathbf{0}$.

Solution. A vanishing linear combination from $\{\mathbf{x}\}$ takes the form $t\mathbf{x} = \mathbf{0}$, t in $\mathbb{R}$. This implies that $t = 0$ because $\mathbf{x} \neq \mathbf{0}$.

The next example will be needed later.

Example 5.2.6

Show that the nonzero rows of a row-echelon matrix R are independent.

Solution. We illustrate the case with 3 leading 1s; the general case is analogous. Suppose R has the form $R = \left[\begin{array}{cccccc} 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$ where $*$ indicates a nonspecified number. Let R_1, R_2, and R_3 denote the nonzero rows of R. If $t_1R_1 + t_2R_2 + t_3R_3 = 0$ we show that $t_1 = 0$, then $t_2 = 0$, and finally $t_3 = 0$. The condition $t_1R_1 + t_2R_2 + t_3R_3 = 0$ becomes

$$(0,\ t_1,\ *,\ *,\ *,\ *) + (0,\ 0,\ 0,\ t_2,\ *,\ *) + (0,\ 0,\ 0,\ 0,\ t_3,\ *) = (0,\ 0,\ 0,\ 0,\ 0,\ 0)$$

Equating second entries show that $t_1 = 0$, so the condition becomes $t_2R_2 + t_3R_3 = 0$. Now the same argument shows that $t_2 = 0$. Finally, this gives $t_3R_3 = 0$ and we obtain $t_3 = 0$.

A set of vectors in $\mathbb{R}^n$ is called **linearly dependent** (or simply **dependent**) if it is *not* linearly independent, equivalently if some nontrivial linear combination vanishes.

Example 5.2.7

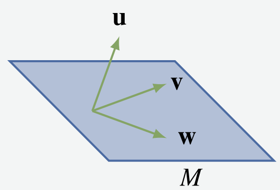
If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v},\ \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel.

Solution. If $\mathbf{v}$ and $\mathbf{w}$ are parallel, then one is a scalar multiple of the other (Theorem 4.1.5), say $\mathbf{v} = a\mathbf{w}$ for some scalar a. Then the nontrivial linear combination $\mathbf{v} - a\mathbf{w} = \mathbf{0}$ vanishes, so $\{\mathbf{v},\ \mathbf{w}\}$ is dependent.

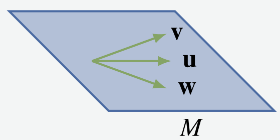
Conversely, if $\{\mathbf{v},\ \mathbf{w}\}$ is dependent, let $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$ be nontrivial, say $s \neq 0$. Then $\mathbf{v} = -\frac{t}{s}\mathbf{w}$ so $\mathbf{v}$ and $\mathbf{w}$ are parallel (by Theorem 4.1.5). A similar argument works if $t \neq 0$.

With this we can give a geometric description of what it means for a set $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ in $\mathbb{R}^3$ to be independent. Note that this requirement means that $\{\mathbf{v},\ \mathbf{w}\}$ is also independent ($a\mathbf{v} + b\mathbf{w} = \mathbf{0}$ means that $0\mathbf{u} + a\mathbf{v} + b\mathbf{w} = \mathbf{0}$), so $M = \text{span}\,\{\mathbf{v},\ \mathbf{w}\}$ is the plane containing $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{0}$ (see the discussion preceding Example 5.1.4). So we assume that $\{\mathbf{v},\ \mathbf{w}\}$ is independent in the following example.

Example 5.2.8



$\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ independent



$\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ not independent

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be nonzero vectors in $\mathbb{R}^3$ where $\{\mathbf{v},\ \mathbf{w}\}$ independent. Show that $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the plane $M = \text{span}\{\mathbf{v},\ \mathbf{w}\}$. This is illustrated in the diagrams.

Solution. If $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent, suppose $\mathbf{u}$ is in the plane $M = \text{span}\{\mathbf{v},\ \mathbf{w}\}$, say $\mathbf{u} = a\mathbf{v} + b\mathbf{w}$, where a and b are in $\mathbb{R}$. Then $1\mathbf{u} - a\mathbf{v} - b\mathbf{w} = \mathbf{0}$, contradicting the independence of $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$.
On the other hand, suppose that $\mathbf{u}$ is not in M; we must show that $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent. If $r\mathbf{u} + s\mathbf{v} + t\mathbf{w} = \mathbf{0}$ where r, s, and t are in $\mathbb{R}^3$, then $r = 0$ since otherwise $\mathbf{u} = -\frac{s}{r}\mathbf{v} + \frac{-t}{r}\mathbf{w}$ is in M. But then $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$, so $s = t = 0$ by our assumption. This shows that $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent, as required.

By the inverse theorem, the following conditions are equivalent for an $n \times n$ matrix A:

1. *A is invertible.*
2. *If $A\mathbf{x} = \mathbf{0}$ where $\mathbf{x}$ is in $\mathbb{R}^n$, then $\mathbf{x} = \mathbf{0}$.*
3. *$A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every vector $\mathbf{b}$ in $\mathbb{R}^n$.*

While condition 1 makes no sense if A is not square, conditions 2 and 3 are meaningful for any matrix A and, in fact, are related to independence and spanning. Indeed, if $\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n$ are the columns of A, and if we write $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]$, then

$$A\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n$$

by Definition 2.5. Hence the definitions of independence and spanning show, respectively, that condition 2 is equivalent to the independence of $\{\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n\}$ and condition 3 is equivalent to the requirement that $\text{span}\{\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n\} = \mathbb{R}^m$. This discussion is summarized in the following theorem:

Theorem 5.2.2

If A is an $m \times n$ matrix, let $\{\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n\}$ denote the columns of A.

1. *$\{\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n\}$ is independent in $\mathbb{R}^m$ if and only if $A\mathbf{x} = \mathbf{0}$, $\mathbf{x}$ in $\mathbb{R}^n$, implies $\mathbf{x} = \mathbf{0}$.*
2. *$\mathbb{R}^m = \text{span}\{\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n\}$ if and only if $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every vector $\mathbf{b}$ in $\mathbb{R}^m$.*

For a *square* matrix A, Theorem 5.2.2 characterizes the invertibility of A in terms of the spanning and independence of its columns (see the discussion preceding Theorem 5.2.2). It is important to be able to discuss these notions for *rows*. If $\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k$ are $1 \times n$ rows, we define $\text{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ to be the set of all linear combinations of the $\mathbf{x}_i$ (as matrices), and we say that $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is linearly independent if the only vanishing linear combination is the trivial one (that is, if $\{\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_k^T\}$ is independent in $\mathbb{R}^n$, as the reader can verify).[6]

Theorem 5.2.3

The following are equivalent for an $n \times n$ matrix A:

1. *A is invertible.*
2. *The columns of A are linearly independent.*
3. *The columns of A span $\mathbb{R}^n$.*
4. *The rows of A are linearly independent.*
5. *The rows of A span the set of all $1 \times n$ rows.*

Proof. Let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ denote the columns of A.

(1) $\Leftrightarrow$ (2). By Theorem 2.4.5, A is invertible if and only if $A\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$; this holds if and only if $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ is independent by Theorem 5.2.2.

(1) $\Leftrightarrow$ (3). Again by Theorem 2.4.5, A is invertible if and only if $A\mathbf{x} = \mathbf{b}$ has a solution for every column B in $\mathbb{R}^n$; this holds if and only if $\text{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\} = \mathbb{R}^n$ by Theorem 5.2.2.

(1) $\Leftrightarrow$ (4). The matrix A is invertible if and only if A^T is invertible (by Corollary 2.4.1 to Theorem 2.4.4); this in turn holds if and only if A^T has independent columns (by (1) $\Leftrightarrow$ (2)); finally, this last statement holds if and only if A has independent rows (because the rows of A are the transposes of the columns of A^T).

(1) $\Leftrightarrow$ (5). The proof is similar to (1) $\Leftrightarrow$ (4). □

Example 5.2.9

Show that $S = \{(2, -2, 5), (-3, 1, 1), (2, 7, -4)\}$ is independent in $\mathbb{R}^3$.

Solution. Consider the matrix $A = \begin{bmatrix} 2 & -2 & 5 \\ -3 & 1 & 1 \\ 2 & 7 & -4 \end{bmatrix}$ with the vectors in S as its rows. A routine computation shows that $\det A = -117 \neq 0$, so A is invertible. Hence S is independent by Theorem 5.2.3. Note that Theorem 5.2.3 also shows that $\mathbb{R}^3 = \text{span } S$.

[6] It is best to view columns and rows as just two different notations for ordered n-tuples. This discussion will become redundant in Chapter 6 where we define the general notion of a vector space.

Dimension

It is common geometrical language to say that $\mathbb{R}^3$ is 3-dimensional, that planes are 2-dimensional and that lines are 1-dimensional. The next theorem is a basic tool for clarifying this idea of "dimension". Its importance is difficult to exaggerate.

Theorem 5.2.4: Fundamental Theorem

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

This proof is given in Theorem 6.3.2 in much greater generality.

Definition 5.4 Basis of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$, a set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ of vectors in U is called a **basis** *of U if it satisfies the following two conditions:*

1. *$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is linearly independent.*
2. *$U = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$.*

The most remarkable result about bases[7] is:

Theorem 5.2.5: Invariance Theorem

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ and $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k\}$ are bases of a subspace U of $\mathbb{R}^n$, then $m = k$.

Proof. We have $k \leq m$ by the fundamental theorem because $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ spans U, and $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k\}$ is independent. Similarly, by interchanging $\mathbf{x}$'s and $\mathbf{y}$'s we get $m \leq k$. Hence $m = k$. □

The invariance theorem guarantees that there is no ambiguity in the following definition:

Definition 5.5 Dimension of a Subspace of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$ and $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is any basis of U, the number, m, of vectors in the basis is called the **dimension** *of U, denoted*

$$\dim U = m$$

The importance of the invariance theorem is that the dimension of U can be determined by counting the number of vectors in *any* basis.[8]

Let $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ denote the standard basis of $\mathbb{R}^n$, that is the set of columns of the identity matrix. Then $\mathbb{R}^n = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ by Example 5.1.6, and $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is independent by Example 5.2.2. Hence it is indeed a basis of $\mathbb{R}^n$ in the present terminology, and we have

[7] The plural of "basis" is "bases".

[8] We will show in Theorem 5.2.6 that every subspace of $\mathbb{R}^n$ does indeed have a basis.

Example 5.2.10

$\dim(\mathbb{R}^n) = n$ and $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis.

This agrees with our geometric sense that $\mathbb{R}^2$ is two-dimensional and $\mathbb{R}^3$ is three-dimensional. It also says that $\mathbb{R}^1 = \mathbb{R}$ is one-dimensional, and $\{1\}$ is a basis. Returning to subspaces of $\mathbb{R}^n$, we define

$$\dim\{\mathbf{0}\} = 0$$

This amounts to saying $\{\mathbf{0}\}$ has a basis containing *no* vectors. This makes sense because $\mathbf{0}$ cannot belong to *any* independent set (Example 5.2.4).

Example 5.2.11

Let $U = \left\{ \left[\begin{array}{c} r \\ s \\ r \end{array} \right] \middle| r, s \text{ in } \mathbb{R} \right\}$. Show that U is a subspace of $\mathbb{R}^3$, find a basis, and calculate $\dim U$.

Solution. Clearly, $\left[\begin{array}{c} r \\ s \\ r \end{array} \right] = r\mathbf{u} + s\mathbf{v}$ where $\mathbf{u} = \left[\begin{array}{c} 1 \\ 0 \\ 1 \end{array} \right]$ and $\mathbf{v} = \left[\begin{array}{c} 0 \\ 1 \\ 0 \end{array} \right]$. It follows that $U = \text{span}\{\mathbf{u}, \mathbf{v}\}$, and hence that U is a subspace of $\mathbb{R}^3$. Moreover, if $r\mathbf{u} + s\mathbf{v} = \mathbf{0}$, then $\left[\begin{array}{c} r \\ s \\ r \end{array} \right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$ so $r = s = 0$. Hence $\{\mathbf{u}, \mathbf{v}\}$ is independent, and so a **basis** of U. This means $\dim U = 2$.

Example 5.2.12

Let $B = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a basis of $\mathbb{R}^n$. If A is an invertible $n \times n$ matrix, then $D = \{A\mathbf{x}_1, A\mathbf{x}_2, \dots, A\mathbf{x}_n\}$ is also a basis of $\mathbb{R}^n$.

Solution. Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$. Then $A^{-1}\mathbf{x}$ is in $\mathbb{R}^n$ so, since B is a basis, we have $A^{-1}\mathbf{x} = t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_n\mathbf{x}_n$ for t_i in $\mathbb{R}$. Left multiplication by A gives $\mathbf{x} = t_1(A\mathbf{x}_1) + t_2(A\mathbf{x}_2) + \cdots + t_n(A\mathbf{x}_n)$, and it follows that D spans $\mathbb{R}^n$. To show independence, let $s_1(A\mathbf{x}_1) + s_2(A\mathbf{x}_2) + \cdots + s_n(A\mathbf{x}_n) = \mathbf{0}$, where the s_i are in $\mathbb{R}$. Then $A(s_1\mathbf{x}_1 + s_2\mathbf{x}_2 + \cdots + s_n\mathbf{x}_n) = \mathbf{0}$ so left multiplication by A^{-1} gives $s_1\mathbf{x}_1 + s_2\mathbf{x}_2 + \cdots + s_n\mathbf{x}_n = \mathbf{0}$. Now the independence of B shows that each $s_i = 0$, and so proves the independence of D. Hence D is a basis of $\mathbb{R}^n$.

While we have found bases in many subspaces of $\mathbb{R}^n$, we have not yet shown that *every* subspace *has* a basis. This is part of the next theorem, the proof of which is deferred to Section 6.4 (Theorem 6.4.1) where it will be proved in more generality.

Theorem 5.2.6

Let $U \neq \{\mathbf{0}\}$ be a subspace of $\mathbb{R}^n$. Then:

1. *U has a basis and $\dim U \leq n$.*
2. *Any independent set in U can be enlarged (by adding vectors from any fixed basis of U) to a basis of U.*
3. *Any spanning set for U can be cut down (by deleting vectors) to a basis of U.*

Example 5.2.13

Find a basis of $\mathbb{R}^4$ containing $S = \{\mathbf{u},\ \mathbf{v}\}$ where $\mathbf{u} = (0,\ 1,\ 2,\ 3)$ and $\mathbf{v} = (2,\ -1,\ 0,\ 1)$.

Solution. By Theorem 5.2.6 we can find such a basis by adding vectors from the standard basis of $\mathbb{R}^4$ to S. If we try $\mathbf{e}_1 = (1,\ 0,\ 0,\ 0)$, we find easily that $\{\mathbf{e}_1,\ \mathbf{u},\ \mathbf{v}\}$ is independent. Now add another vector from the standard basis, say $\mathbf{e}_2$.
Again we find that $B = \{\mathbf{e}_1,\ \mathbf{e}_2,\ \mathbf{u},\ \mathbf{v}\}$ is independent. Since B has $4 = \dim \mathbb{R}^4$ vectors, then B must span $\mathbb{R}^4$ by Theorem 5.2.7 below (or simply verify it directly). Hence B is a basis of $\mathbb{R}^4$.

Theorem 5.2.6 has a number of useful consequences. Here is the first.

Theorem 5.2.7

Let U be a subspace of $\mathbb{R}^n$ where $\dim U = m$ and let $B = \{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_m\}$ be a set of m vectors in U. Then B is independent if and only if B spans U.

Proof. Suppose B is independent. If B does not span U then, by Theorem 5.2.6, B can be enlarged to a basis of U containing more than m vectors. This contradicts the invariance theorem because $\dim U = m$, so B spans U. Conversely, if B spans U but is not independent, then B can be cut down to a basis of U containing fewer than m vectors, again a contradiction. So B is independent, as required. $\square$

As we saw in Example 5.2.13, Theorem 5.2.7 is a "labour-saving" result. It asserts that, given a subspace U of dimension m and a set B of exactly m vectors in U, to prove that B is a basis of U it suffices to show either that B spans U or that B is independent. It is not necessary to verify both properties.

Theorem 5.2.8

Let $U \subseteq W$ be subspaces of $\mathbb{R}^n$. Then:

1. *$\dim U \leq \dim W$.*
2. *If $\dim U = \dim W$, then $U = W$.*

Proof. Write $\dim W = k$, and let B be a basis of U.

1. If $\dim U > k$, then B is an independent set in W containing more than k vectors, contradicting the fundamental theorem. So $\dim U \leq k = \dim W$.

2. If $\dim U = k$, then B is an independent set in W containing $k = \dim W$ vectors, so B spans W by Theorem 5.2.7. Hence $W = \text{span } B = U$, proving (2). □

It follows from Theorem 5.2.8 that if U is a subspace of $\mathbb{R}^n$, then $\dim U$ is one of the integers $0, 1, 2, \ldots, n$, and that:

$$\begin{array}{lll} \dim U = 0 & \text{if and only if} & U = \{\mathbf{0}\}, \\ \dim U = n & \text{if and only if} & U = \mathbb{R}^n \end{array}$$

The other subspaces of $\mathbb{R}^n$ are called **proper**. The following example uses Theorem 5.2.8 to show that the proper subspaces of $\mathbb{R}^2$ are the lines through the origin, while the proper subspaces of $\mathbb{R}^3$ are the lines and planes through the origin.

Example 5.2.14

1. If U is a subspace of $\mathbb{R}^2$ or $\mathbb{R}^3$, then $\dim U = 1$ if and only if U is a line through the origin.

2. If U is a subspace of $\mathbb{R}^3$, then $\dim U = 2$ if and only if U is a plane through the origin.

Proof.

1. Since $\dim U = 1$, let $\{\mathbf{u}\}$ be a basis of U. Then $U = \text{span}\{\mathbf{u}\} = \{t\mathbf{u} \mid t \text{ in } \mathbb{R}\}$, so U is the line through the origin with direction vector $\mathbf{u}$. Conversely each line L with direction vector $\mathbf{d} \neq \mathbf{0}$ has the form $L = \{t\mathbf{d} \mid t \text{ in } \mathbb{R}\}$. Hence $\{\mathbf{d}\}$ is a basis of U, so U has dimension 1.

2. If $U \subseteq \mathbb{R}^3$ has dimension 2, let $\{\mathbf{v}, \mathbf{w}\}$ be a basis of U. Then $\mathbf{v}$ and $\mathbf{w}$ are not parallel (by Example 5.2.7) so $\mathbf{n} = \mathbf{v} \times \mathbf{w} \neq \mathbf{0}$. Let $P = \{\mathbf{x} \text{ in } \mathbb{R}^3 \mid \mathbf{n} \cdot \mathbf{x} = 0\}$ denote the plane through the origin with normal $\mathbf{n}$. Then P is a subspace of $\mathbb{R}^3$ (Example 5.1.1) and both $\mathbf{v}$ and $\mathbf{w}$ lie in P (they are orthogonal to $\mathbf{n}$), so $U = \text{span}\{\mathbf{v}, \mathbf{w}\} \subseteq P$ by Theorem 5.1.1. Hence

   $$U \subseteq P \subseteq \mathbb{R}^3$$

   Since $\dim U = 2$ and $\dim(\mathbb{R}^3) = 3$, it follows from Theorem 5.2.8 that $\dim P = 2$ or 3, whence $P = U$ or $\mathbb{R}^3$. But $P \neq \mathbb{R}^3$ (for example, $\mathbf{n}$ is not in P) and so $U = P$ is a plane through the origin.

   Conversely, if U is a plane through the origin, then $\dim U = 0$, 1, 2, or 3 by Theorem 5.2.8. But $\dim U \neq 0$ or 3 because $U \neq \{\mathbf{0}\}$ and $U \neq \mathbb{R}^3$, and $\dim U \neq 1$ by (1). So $\dim U = 2$. □

Note that this proof shows that if $\mathbf{v}$ and $\mathbf{w}$ are nonzero, nonparallel vectors in $\mathbb{R}^3$, then $\text{span}\{\mathbf{v}, \mathbf{w}\}$ is the plane with normal $\mathbf{n} = \mathbf{v} \times \mathbf{w}$. We gave a geometrical verification of this fact in Section 5.1.

Exercises for 5.2

In Exercises 5.2.1-5.2.6 we write vectors $\mathbb{R}^n$ as rows.

Exercise 5.2.1 Which of the following subsets are independent? Support your answer.

a. $\{(1, -1, 0), (3, 2, -1), (3, 5, -2)\}$ in $\mathbb{R}^3$

b. $\{(1, 1, 1), (1, -1, 1), (0, 0, 1)\}$ in $\mathbb{R}^3$

c. $\{(1, -1, 1, -1), (2, 0, 1, 0), (0, -2, 1, -2)\}$ in $\mathbb{R}^4$

d. $\{(1, 1, 0, 0), (1, 0, 1, 0), (0, 0, 1, 1), (0, 1, 0, 1)\}$ in $\mathbb{R}^4$

Exercise 5.2.2 Let $\{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w}\}$ be an independent set in $\mathbb{R}^n$. Which of the following sets is independent? Support your answer.

a. $\{\mathbf{x}-\mathbf{y}, \mathbf{y}-\mathbf{z}, \mathbf{z}-\mathbf{x}\}$

b. $\{\mathbf{x}+\mathbf{y}, \mathbf{y}+\mathbf{z}, \mathbf{z}+\mathbf{x}\}$

c. $\{\mathbf{x}-\mathbf{y}, \mathbf{y}-\mathbf{z}, \mathbf{z}-\mathbf{w}, \mathbf{w}-\mathbf{x}\}$

d. $\{\mathbf{x}+\mathbf{y}, \mathbf{y}+\mathbf{z}, \mathbf{z}+\mathbf{w}, \mathbf{w}+\mathbf{x}\}$

Exercise 5.2.3 Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$.

a. span $\{(1, -1, 2, 0), (2, 3, 0, 3), (1, 9, -6, 6)\}$

b. span $\{(2, 1, 0, -1), (-1, 1, 1, 1), (2, 7, 4, 1)\}$

c. span $\{(-1, 2, 1, 0), (2, 0, 3, -1), (4, 4, 11, -3), (3, -2, 2, -1)\}$

d. span $\{(-2, 0, 3, 1), (1, 2, -1, 0), (-2, 8, 5, 3), (-1, 2, 2, 1)\}$

Exercise 5.2.4 Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$.

a. $U = \left\{ \left[\begin{array}{c} a \\ a+b \\ a-b \\ b \end{array} \right] \middle|\, a \text{ and } b \text{ in } \mathbb{R} \right\}$

b. $U = \left\{ \left[\begin{array}{c} a+b \\ a-b \\ b \\ a \end{array} \right] \middle|\, a \text{ and } b \text{ in } \mathbb{R} \right\}$

c. $U = \left\{ \left[\begin{array}{c} a \\ b \\ c+a \\ c \end{array} \right] \middle|\, a,\ b, \text{ and } c \text{ in } \mathbb{R} \right\}$

d. $U = \left\{ \left[\begin{array}{c} a-b \\ b+c \\ a \\ b+c \end{array} \right] \middle|\, a,\ b, \text{ and } c \text{ in } \mathbb{R} \right\}$

e. $U = \left\{ \left[\begin{array}{c} a \\ b \\ c \\ d \end{array} \right] \middle|\, a+b-c+d=0 \text{ in } \mathbb{R} \right\}$

f. $U = \left\{ \left[\begin{array}{c} a \\ b \\ c \\ d \end{array} \right] \middle|\, a+b=c+d \text{ in } \mathbb{R} \right\}$

Exercise 5.2.5 Suppose that $\{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w}\}$ is a basis of $\mathbb{R}^4$. Show that:

a. $\{\mathbf{x}+a\mathbf{w}, \mathbf{y}, \mathbf{z}, \mathbf{w}\}$ is also a basis of $\mathbb{R}^4$ for any choice of the scalar a.

b. $\{\mathbf{x}+\mathbf{w}, \mathbf{y}+\mathbf{w}, \mathbf{z}+\mathbf{w}, \mathbf{w}\}$ is also a basis of $\mathbb{R}^4$.

c. $\{\mathbf{x}, \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}+\mathbf{z}, \mathbf{x}+\mathbf{y}+\mathbf{z}+\mathbf{w}\}$ is also a basis of $\mathbb{R}^4$.

Exercise 5.2.6 Use Theorem 5.2.3 to determine if the following sets of vectors are a basis of the indicated space.

a. $\{(3, -1), (2, 2)\}$ in $\mathbb{R}^2$

b. $\{(1, 1, -1), (1, -1, 1), (0, 0, 1)\}$ in $\mathbb{R}^3$

c. $\{(-1, 1, -1), (1, -1, 2), (0, 0, 1)\}$ in $\mathbb{R}^3$

d. $\{(5, 2, -1), (1, 0, 1), (3, -1, 0)\}$ in $\mathbb{R}^3$

e. $\{(2, 1, -1, 3), (1, 1, 0, 2), (0, 1, 0, -3), (-1, 2, 3, 1)\}$ in $\mathbb{R}^4$

f. $\{(1, 0, -2, 5), (4, 4, -3, 2), (0, 1, 0, -3), (1, 3, 3, -10)\}$ in $\mathbb{R}^4$

Exercise 5.2.7 In each case show that the statement is true or give an example showing that it is false.

a. If $\{\mathbf{x}, \mathbf{y}\}$ is independent, then $\{\mathbf{x}, \mathbf{y}, \mathbf{x}+\mathbf{y}\}$ is independent.

b. If $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is independent, then $\{\mathbf{y}, \mathbf{z}\}$ is independent.

c. If $\{\mathbf{y}, \mathbf{z}\}$ is dependent, then $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is dependent for any $\mathbf{x}$.

d. If all of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are nonzero, then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is independent.

e. If one of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ is zero, then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is dependent.

f. If $a\mathbf{x}+b\mathbf{y}+c\mathbf{z}=\mathbf{0}$, then $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is independent.

g. If $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is independent, then $a\mathbf{x}+b\mathbf{y}+c\mathbf{z}=\mathbf{0}$ for some a, b, and c in $\mathbb{R}$.

h. If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is dependent, then $t_1\mathbf{x}_1+t_2\mathbf{x}_2+\cdots+t_k\mathbf{x}_k=\mathbf{0}$ for some numbers t_i in $\mathbb{R}$ not all zero.

i. If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is independent, then $t_1\mathbf{x}_1+t_2\mathbf{x}_2+\cdots+t_k\mathbf{x}_k=\mathbf{0}$ for some t_i in $\mathbb{R}$.

j. Every non-empty subset of a linearly independent set is again linearly independent.

k. Every set containing a spanning set is again a spanning set.

Exercise 5.2.8 If A is an $n \times n$ matrix, show that $\det A = 0$ if and only if some column of A is a linear combination of the other columns.

Exercise 5.2.9 Let $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ be a linearly independent set in $\mathbb{R}^4$. Show that $\{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{e}_k\}$ is a basis of $\mathbb{R}^4$ for some $\mathbf{e}_k$ in the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$.

Exercise 5.2.10 If $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\}$ is an independent set of vectors, show that the subset $\{\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_5\}$ is also independent.

Exercise 5.2.11 Let A be any $m \times n$ matrix, and let $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_k$ be columns in $\mathbb{R}^m$ such that the system $A\mathbf{x}=\mathbf{b}_i$ has a solution $\mathbf{x}_i$ for each i. If $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_k\}$ is independent in $\mathbb{R}^m$, show that $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_k\}$ is independent in $\mathbb{R}^n$.

Exercise 5.2.12 If $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_k\}$ is independent, show $\{\mathbf{x}_1, \mathbf{x}_1+\mathbf{x}_2, \mathbf{x}_1+\mathbf{x}_2+\mathbf{x}_3, \dots, \mathbf{x}_1+\mathbf{x}_2+\cdots+\mathbf{x}_k\}$ is also independent.

Exercise 5.2.13 If $\{\mathbf{y}, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_k\}$ is independent, show that $\{\mathbf{y}+\mathbf{x}_1, \mathbf{y}+\mathbf{x}_2, \mathbf{y}+\mathbf{x}_3, \dots, \mathbf{y}+\mathbf{x}_k\}$ is also independent.

Exercise 5.2.14 If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is independent in $\mathbb{R}^n$, and if $\mathbf{y}$ is not in $\operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$, show that $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k, \mathbf{y}\}$ is independent.

Exercise 5.2.15 If A and B are matrices and the columns of AB are independent, show that the columns of B are independent.

Exercise 5.2.16 Suppose that $\{\mathbf{x}, \mathbf{y}\}$ is a basis of $\mathbb{R}^2$, and let $A = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$.

a. If A is invertible, show that $\{a\mathbf{x}+b\mathbf{y}, c\mathbf{x}+d\mathbf{y}\}$ is a basis of $\mathbb{R}^2$.

b. If $\{a\mathbf{x}+b\mathbf{y}, c\mathbf{x}+d\mathbf{y}\}$ is a basis of $\mathbb{R}^2$, show that A is invertible.

Exercise 5.2.17 Let A denote an $m \times n$ matrix.

a. Show that $\operatorname{null} A = \operatorname{null}(UA)$ for every invertible $m \times m$ matrix U.

b. Show that $\dim(\operatorname{null} A) = \dim(\operatorname{null}(AV))$ for every invertible $n \times n$ matrix V. [*Hint*: If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a basis of $\operatorname{null} A$, show that $\{V^{-1}\mathbf{x}_1, V^{-1}\mathbf{x}_2, \dots, V^{-1}\mathbf{x}_k\}$ is a basis of $\operatorname{null}(AV)$.]

Exercise 5.2.18 Let A denote an $m \times n$ matrix.

a. Show that $\operatorname{im} A = \operatorname{im}(AV)$ for every invertible $n \times n$ matrix V.

b. Show that $\dim(\operatorname{im} A) = \dim(\operatorname{im}(UA))$ for every invertible $m \times m$ matrix U. [*Hint*: If $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k\}$ is a basis of $\operatorname{im}(UA)$, show that $\{U^{-1}\mathbf{y}_1, U^{-1}\mathbf{y}_2, \dots, U^{-1}\mathbf{y}_k\}$ is a basis of $\operatorname{im} A$.]

Exercise 5.2.19 Let U and W denote subspaces of $\mathbb{R}^n$, and assume that $U \subseteq W$. If $\dim U = n-1$, show that either $W = U$ or $W = \mathbb{R}^n$.

Exercise 5.2.20 Let U and W denote subspaces of $\mathbb{R}^n$, and assume that $U \subseteq W$. If $\dim W = 1$, show that either $U = \{\mathbf{0}\}$ or $U = W$.

5.3 Orthogonality

Length and orthogonality are basic concepts in geometry and, in $\mathbb{R}^2$ and $\mathbb{R}^3$, they both can be defined using the dot product. In this section we extend the dot product to vectors in $\mathbb{R}^n$, and so endow $\mathbb{R}^n$ with euclidean geometry. We then introduce the idea of an orthogonal basis—one of the most useful concepts in linear algebra, and begin exploring some of its applications.

Dot Product, Length, and Distance

If $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ are two n-tuples in $\mathbb{R}^n$, recall that their **dot product** was defined in Section 2.2 as follows:

$$\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

Observe that if $\mathbf{x}$ and $\mathbf{y}$ are written as columns then $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T\mathbf{y}$ is a matrix product (and $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}\mathbf{y}^T$ if they are written as rows). Here $\mathbf{x} \cdot \mathbf{y}$ is a 1×1 matrix, which we take to be a number.

Definition 5.6 Length in $\mathbb{R}^n$

*As in $\mathbb{R}^3$, the **length** $\|\mathbf{x}\|$ of the vector is defined by*

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Where $\sqrt{(\quad)}$ indicates the positive square root.

A vector $\mathbf{x}$ of length 1 is called a **unit vector**. If $\mathbf{x} \neq \mathbf{0}$, then $\|\mathbf{x}\| \neq 0$ and it follows easily that $\frac{1}{\|\mathbf{x}\|}\mathbf{x}$ is a unit vector (see Theorem 5.3.6 below), a fact that we shall use later.

Example 5.3.1

If $\mathbf{x} = (1, -1, -3, 1)$ and $\mathbf{y} = (2, 1, 1, 0)$ in $\mathbb{R}^4$, then $\mathbf{x} \cdot \mathbf{y} = 2 - 1 - 3 + 0 = -2$ and $\|\mathbf{x}\| = \sqrt{1+1+9+1} = \sqrt{12} = 2\sqrt{3}$. Hence $\frac{1}{2\sqrt{3}}\mathbf{x}$ is a unit vector; similarly $\frac{1}{\sqrt{6}}\mathbf{y}$ is a unit vector.

These definitions agree with those in $\mathbb{R}^2$ and $\mathbb{R}^3$, and many properties carry over to $\mathbb{R}^n$:

Theorem 5.3.1

Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ denote vectors in $\mathbb{R}^n$. Then:

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
2. $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$.
3. $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$ *for all scalars a.*
4. $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$.
5. $\|\mathbf{x}\| \geq 0$, *and* $\|\mathbf{x}\| = 0$ *if and only if* $\mathbf{x} = \mathbf{0}$.

6. $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$ *for all scalars* a.

Proof. (1), (2), and (3) follow from matrix arithmetic because $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T\mathbf{y}$; (4) is clear from the definition; and (6) is a routine verification since $|a| = \sqrt{a^2}$. If $\mathbf{x} = (x_1, x_2, \dots, x_n)$, then $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ so $\|\mathbf{x}\| = 0$ if and only if $x_1^2 + x_2^2 + \cdots + x_n^2 = 0$. Since each x_i is a real number this happens if and only if $x_i = 0$ for each i; that is, if and only if $\mathbf{x} = \mathbf{0}$. This proves (5). □

Because of Theorem 5.3.1, computations with dot products in $\mathbb{R}^n$ are similar to those in $\mathbb{R}^3$. In particular, the dot product

$$(\mathbf{x}_1 + \mathbf{x}_2 + \cdots + \mathbf{x}_m) \cdot (\mathbf{y}_1 + \mathbf{y}_2 + \cdots + \mathbf{y}_k)$$

equals the sum of mk terms, $\mathbf{x}_i \cdot \mathbf{y}_j$, one for each choice of i and j. For example:

$$\begin{aligned}(3\mathbf{x} - 4\mathbf{y}) \cdot (7\mathbf{x} + 2\mathbf{y}) &= 21(\mathbf{x} \cdot \mathbf{x}) + 6(\mathbf{x} \cdot \mathbf{y}) - 28(\mathbf{y} \cdot \mathbf{x}) - 8(\mathbf{y} \cdot \mathbf{y}) \\ &= 21\|\mathbf{x}\|^2 - 22(\mathbf{x} \cdot \mathbf{y}) - 8\|\mathbf{y}\|^2\end{aligned}$$

holds for all vectors $\mathbf{x}$ and $\mathbf{y}$.

Example 5.3.2

Show that $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$ for any $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$.

Solution. Using Theorem 5.3.1 several times:

$$\begin{aligned}\|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} \\ &= \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2\end{aligned}$$

Example 5.3.3

Suppose that $\mathbb{R}^n = \text{span}\,\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ for some vectors $\mathbf{f}_i$. If $\mathbf{x} \cdot \mathbf{f}_i = 0$ for each i where $\mathbf{x}$ is in $\mathbb{R}^n$, show that $\mathbf{x} = \mathbf{0}$.

Solution. We show $\mathbf{x} = \mathbf{0}$ by showing that $\|\mathbf{x}\| = 0$ and using (5) of Theorem 5.3.1. Since the $\mathbf{f}_i$ span $\mathbb{R}^n$, write $\mathbf{x} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k$ where the t_i are in $\mathbb{R}$. Then

$$\begin{aligned}\|\mathbf{x}\|^2 &= \mathbf{x} \cdot \mathbf{x} = \mathbf{x} \cdot (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k) \\ &= t_1(\mathbf{x} \cdot \mathbf{f}_1) + t_2(\mathbf{x} \cdot \mathbf{f}_2) + \cdots + t_k(\mathbf{x} \cdot \mathbf{f}_k) \\ &= t_1(0) + t_2(0) + \cdots + t_k(0) \\ &= 0\end{aligned}$$

Theorem 5.3.2: Cauchy Inequality[9]

If $\mathbf{x}$ *and* $\mathbf{y}$ *are vectors in* $\mathbb{R}^n$*, then*

$$|\mathbf{x}\cdot\mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

Moreover $|\mathbf{x}\cdot\mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$ *if and only if one of* $\mathbf{x}$ *and* $\mathbf{y}$ *is a multiple of the other.*

Proof. The inequality holds if $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$ (in fact it is equality). Otherwise, write $\|\mathbf{x}\| = a > 0$ and $\|\mathbf{y}\| = b > 0$ for convenience. A computation like that preceding Example 5.3.2 gives

$$\|b\mathbf{x} - a\mathbf{y}\|^2 = 2ab(ab - \mathbf{x}\cdot\mathbf{y}) \text{ and } \|b\mathbf{x} + a\mathbf{y}\|^2 = 2ab(ab + \mathbf{x}\cdot\mathbf{y}) \tag{5.1}$$

It follows that $ab - \mathbf{x}\cdot\mathbf{y} \geq 0$ and $ab + \mathbf{x}\cdot\mathbf{y} \geq 0$, and hence that $-ab \leq \mathbf{x}\cdot\mathbf{y} \leq ab$. Hence $|\mathbf{x}\cdot\mathbf{y}| \leq ab = \|\mathbf{x}\|\|\mathbf{y}\|$, proving the Cauchy inequality.

If equality holds, then $|\mathbf{x}\cdot\mathbf{y}| = ab$, so $\mathbf{x}\cdot\mathbf{y} = ab$ or $\mathbf{x}\cdot\mathbf{y} = -ab$. Hence Equation 5.1 shows that $b\mathbf{x} - a\mathbf{y} = 0$ or $b\mathbf{x} + a\mathbf{y} = 0$, so one of $\mathbf{x}$ and $\mathbf{y}$ is a multiple of the other (even if $a = 0$ or $b = 0$). □

The Cauchy inequality is equivalent to $(\mathbf{x}\cdot\mathbf{y})^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$. In $\mathbb{R}^5$ this becomes

$$(x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4 + x_5y_5)^2 \leq (x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2)(y_1^2 + y_2^2 + y_3^2 + y_4^2 + y_5^2)$$

for all x_i and y_i in $\mathbb{R}$.

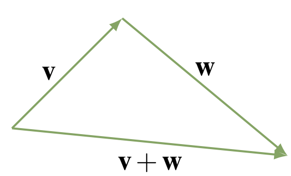
There is an important consequence of the Cauchy inequality. Given $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, use Example 5.3.2 and the fact that $\mathbf{x}\cdot\mathbf{y} \leq \|\mathbf{x}\|\|\mathbf{y}\|$ to compute

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2(\mathbf{x}\cdot\mathbf{y}) + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x} + \mathbf{y}\|)^2$$

Taking positive square roots gives:

Corollary 5.3.1: Triangle Inequality

If $\mathbf{x}$ *and* $\mathbf{y}$ *are vectors in* $\mathbb{R}^n$*, then* $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.



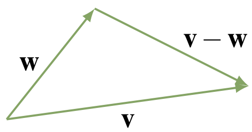
The reason for the name comes from the observation that in $\mathbb{R}^3$ the inequality asserts that the sum of the lengths of two sides of a triangle is not less than the length of the third side. This is illustrated in the diagram.

[9]Augustin Louis Cauchy (1789–1857) was born in Paris and became a professor at the École Polytechnique at the age of 26. He was one of the great mathematicians, producing more than 700 papers, and is best remembered for his work in analysis in which he established new standards of rigour and founded the theory of functions of a complex variable. He was a devout Catholic with a long-term interest in charitable work, and he was a royalist, following King Charles X into exile in Prague after he was deposed in 1830. Theorem 5.3.2 first appeared in his 1812 memoir on determinants.

Definition 5.7 Distance in $\mathbb{R}^n$

If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in $\mathbb{R}^n$, we define the ***distance*** *$d(\mathbf{x},\ \mathbf{y})$ between $\mathbf{x}$ and $\mathbf{y}$ by*

$$d(\mathbf{x},\ \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$



The motivation again comes from $\mathbb{R}^3$ as is clear in the diagram. This distance function has all the intuitive properties of distance in $\mathbb{R}^3$, including another version of the triangle inequality.

Theorem 5.3.3

If $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ are three vectors in $\mathbb{R}^n$ we have:

1. $d(\mathbf{x},\ \mathbf{y}) \geq 0$ *for all* $\mathbf{x}$ *and* $\mathbf{y}$.
2. $d(\mathbf{x},\ \mathbf{y}) = 0$ *if and only if* $\mathbf{x} = \mathbf{y}$.
3. $d(\mathbf{x},\ \mathbf{y}) = d(\mathbf{y},\ \mathbf{x})$ *for all* $\mathbf{x}$ *and* $\mathbf{y}$.
4. $d(\mathbf{x},\ \mathbf{z}) \leq d(\mathbf{x},\ \mathbf{y}) + d(\mathbf{y},\ \mathbf{z})$ *for all* $\mathbf{x}$, $\mathbf{y}$, *and* $\mathbf{z}$. *Triangle inequality.*

Proof. (1) and (2) restate part (5) of Theorem 5.3.1 because $d(\mathbf{x},\ \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, and (3) follows because $\|\mathbf{u}\| = \|-\mathbf{u}\|$ for every vector $\mathbf{u}$ in $\mathbb{R}^n$. To prove (4) use the Corollary to Theorem 5.3.2:

$$\begin{aligned} d(\mathbf{x},\ \mathbf{z}) = \|\mathbf{x} - \mathbf{z}\| &= \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|(\mathbf{x} - \mathbf{y})\| + \|(\mathbf{y} - \mathbf{z})\| = d(\mathbf{x},\ \mathbf{y}) + d(\mathbf{y},\ \mathbf{z}) \end{aligned}$$

□

Orthogonal Sets and the Expansion Theorem

Definition 5.8 Orthogonal and Orthonormal Sets

We say that two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ are ***orthogonal*** *if $\mathbf{x} \cdot \mathbf{y} = 0$, extending the terminology in $\mathbb{R}^3$ (See Theorem 4.2.3). More generally, a set $\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is called an* ***orthogonal set*** *if*

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0 \text{ for all } i \neq j \quad \text{and} \quad \mathbf{x}_i \neq \mathbf{0} \text{ for all } i\text{[10]}$$

Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. A set $\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_k\}$ of vectors in $\mathbb{R}^n$ is called ***orthonormal*** *if it is orthogonal and, in addition, each $\mathbf{x}_i$ is a unit vector:*

$$\|\mathbf{x}_i\| = 1 \text{ for each } i.$$

[10]The reason for insisting that orthogonal sets consist of nonzero vectors is that we will be primarily concerned with orthogonal bases.

Example 5.3.4

The standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is an orthonormal set in $\mathbb{R}^n$.

The routine verification is left to the reader, as is the proof of:

Example 5.3.5

If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is orthogonal, so also is $\{a_1\mathbf{x}_1, a_2\mathbf{x}_2, \dots, a_k\mathbf{x}_k\}$ for any nonzero scalars a_i.

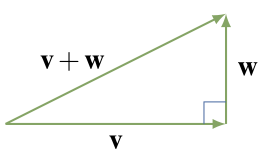
If $\mathbf{x} \neq \mathbf{0}$, it follows from item (6) of Theorem 5.3.1 that $\frac{1}{\|\mathbf{x}\|}\mathbf{x}$ is a unit vector, that is it has length 1.

Definition 5.9 Normalizing an Orthogonal Set

*Hence if $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is an orthogonal set, then $\{\frac{1}{\|\mathbf{x}_1\|}\mathbf{x}_1, \frac{1}{\|\mathbf{x}_2\|}\mathbf{x}_2, \cdots, \frac{1}{\|\mathbf{x}_k\|}\mathbf{x}_k\}$ is an orthonormal set, and we say that it is the result of **normalizing** the orthogonal set $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$.*

Example 5.3.6

If $\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$, $\mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, and $\mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}$ then $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ is an orthogonal set in $\mathbb{R}^4$ as is easily verified. After normalizing, the corresponding orthonormal set is $\{\frac{1}{2}\mathbf{f}_1, \frac{1}{\sqrt{6}}\mathbf{f}_2, \frac{1}{\sqrt{2}}\mathbf{f}_3, \frac{1}{2\sqrt{3}}\mathbf{f}_4\}$



The most important result about orthogonality is Pythagoras' theorem. Given orthogonal vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, it asserts that

$$\|\mathbf{v}+\mathbf{w}\|^2 = \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$$

as in the diagram. In this form the result holds for any orthogonal set in $\mathbb{R}^n$.

Theorem 5.3.5

Every orthogonal set in $\mathbb{R}^n$ is linearly independent.

Proof. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ be an orthogonal set in $\mathbb{R}^n$ and suppose a linear combination vanishes, say: $t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$. Then

$$\begin{aligned}
0 = \mathbf{x}_1 \cdot \mathbf{0} = \mathbf{x}_1 \cdot (t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k) \\
&= t_1(\mathbf{x}_1 \cdot \mathbf{x}_1) + t_2(\mathbf{x}_1 \cdot \mathbf{x}_2) + \cdots + t_k(\mathbf{x}_1 \cdot \mathbf{x}_k) \\
&= t_1\|\mathbf{x}_1\|^2 + t_2(0) + \cdots + t_k(0) \\
&= t_1\|\mathbf{x}_1\|^2
\end{aligned}$$

Since $\|\mathbf{x}_1\|^2 \neq 0$, this implies that $t_1 = 0$. Similarly $t_i = 0$ for each i. □

A natural question arises here: Does every subspace U of $\mathbb{R}^n$ *have* an orthogonal basis? The answer is "yes"; in fact, there is a systematic procedure, called the Gram-Schmidt algorithm, for turning any basis of U into an orthogonal one. This leads to a definition of the projection onto a subspace U that generalizes the projection along a vector used in $\mathbb{R}^2$ and $\mathbb{R}^3$. All this is discussed in Section 8.1.

Exercises for 5.3

We often write vectors in $\mathbb{R}^n$ as row n-tuples.

Exercise 5.3.1 Obtain orthonormal bases of $\mathbb{R}^3$ by normalizing the following.

a. $\{(1,\ -1,\ 2),\ (0,\ 2,\ 1),\ (5,\ 1,\ -2)\}$

b. $\{(1,\ 1,\ 1),\ (4,\ 1,\ -5),\ (2,\ -3,\ 1)\}$

Exercise 5.3.2 In each case, show that the set of vectors is orthogonal in $\mathbb{R}^4$.

a. $\{(1,\ -1,\ 2,\ 5),\ (4,\ 1,\ 1,\ -1),\ (-7,\ 28,\ 5,\ 5)\}$

b. $\{(2,\ -1,\ 4,\ 5),\ (0,\ -1,\ 1,\ -1),\ (0,\ 3,\ 2,\ -1)\}$

Exercise 5.3.3 In each case, show that B is an orthogonal basis of $\mathbb{R}^3$ and use Theorem 5.3.6 to expand $\mathbf{x} = (a,\ b,\ c)$ as a linear combination of the basis vectors.

a. $B = \{(1,\ -1,\ 3),\ (-2,\ 1,\ 1),\ (4,\ 7,\ 1)\}$

b. $B = \{(1,\ 0,\ -1),\ (1,\ 4,\ 1),\ (2,\ -1,\ 2)\}$

c. $B = \{(1,\ 2,\ 3),\ (-1,\ -1,\ 1),\ (5,\ -4,\ 1)\}$

d. $B = \{(1,\ 1,\ 1),\ (1,\ -1,\ 0),\ (1,\ 1,\ -2)\}$

Exercise 5.3.4 In each case, write $\mathbf{x}$ as a linear combination of the orthogonal basis of the subspace U.

a. $\mathbf{x} = (13,\ -20,\ 15); U = \text{span}\,\{(1,\ -2,\ 3),\ (-1,\ 1,\ 1)\}$

b. $\mathbf{x} = (14,\ 1,\ -8,\ 5)$;
$U = \text{span}\,\{(2,\ -1,\ 0,\ 3),\ (2,\ 1,\ -2,\ -1)\}$

Exercise 5.3.5 In each case, find all $(a,\ b,\ c,\ d)$ in $\mathbb{R}^4$ such that the given set is orthogonal.

a. $\{(1,\ 2,\ 1,\ 0),\ (1,\ -1,\ 1,\ 3),\ (2,\ -1,\ 0,\ -1),$
$(a,\ b,\ c,\ d)\}$

b. $\{(1,\ 0,\ -1,\ 1),\ (2,\ 1,\ 1,\ -1),\ (1,\ -3,\ 1,\ 0),\ (a,\ b,\ c,\ d)\}$

Exercise 5.3.6 If $\|\mathbf{x}\| = 3$, $\|\mathbf{y}\| = 1$, and $\mathbf{x}\cdot\mathbf{y} = -2$, compute:

a. $\|3\mathbf{x} - 5\mathbf{y}\|$

b. $\|2\mathbf{x} + 7\mathbf{y}\|$

c. $(3\mathbf{x} - \mathbf{y})\cdot(2\mathbf{y} - \mathbf{x})$

d. $(\mathbf{x} - 2\mathbf{y})\cdot(3\mathbf{x} + 5\mathbf{y})$

Exercise 5.3.7 In each case either show that the statement is true or give an example showing that it is false.

a. Every independent set in $\mathbb{R}^n$ is orthogonal.

b. If $\{\mathbf{x},\ \mathbf{y}\}$ is an orthogonal set in $\mathbb{R}^n$, then $\{\mathbf{x},\ \mathbf{x}+\mathbf{y}\}$ is also orthogonal.

c. If $\{\mathbf{x},\ \mathbf{y}\}$ and $\{\mathbf{z},\ \mathbf{w}\}$ are both orthogonal in $\mathbb{R}^n$, then $\{\mathbf{x},\ \mathbf{y},\ \mathbf{z},\ \mathbf{w}\}$ is also orthogonal.

d. If $\{\mathbf{x}_1,\ \mathbf{x}_2\}$ and $\{\mathbf{y}_1,\ \mathbf{y}_2,\ \mathbf{y}_3\}$ are both orthogonal and $\mathbf{x}_i\cdot\mathbf{y}_j = 0$ for all i and j, then $\{\mathbf{x}_1,\ \mathbf{x}_2,\ \mathbf{y}_1,\ \mathbf{y}_2,\ \mathbf{y}_3\}$ is orthogonal.

e. If $\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_n\}$ is orthogonal in $\mathbb{R}^n$, then $\mathbb{R}^n = \operatorname{span}\{\mathbf{x}_1,\ \mathbf{x}_2,\ \dots,\ \mathbf{x}_n\}$.

f. If $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$, then $\{\mathbf{x}\}$ is an orthogonal set.

Exercise 5.3.8 Let $\mathbf{v}$ denote a nonzero vector in $\mathbb{R}^n$.

a. Show that $P = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x}\cdot\mathbf{v} = 0\}$ is a subspace of $\mathbb{R}^n$.

b. Show that $\mathbb{R}\mathbf{v} = \{t\mathbf{v} \mid t \text{ in } \mathbb{R}\}$ is a subspace of $\mathbb{R}^n$.

c. Describe P and $\mathbb{R}\mathbf{v}$ geometrically when $n = 3$.

Exercise 5.3.9 If A is an $m\times n$ matrix with orthonormal columns, show that $A^TA = I_n$. [*Hint*: If $\mathbf{c}_1,\ \mathbf{c}_2,\ \dots,\ \mathbf{c}_n$ are the columns of A, show that column j of A^TA has entries $\mathbf{c}_1\cdot\mathbf{c}_j,\ \mathbf{c}_2\cdot\mathbf{c}_j,\ \dots,\ \mathbf{c}_n\cdot\mathbf{c}_j$].

Exercise 5.3.10 Use the Cauchy inequality to show that $\sqrt{xy} \le \frac{1}{2}(x+y)$ for all $x \ge 0$ and $y \ge 0$. Here $\sqrt{xy}$ and $\frac{1}{2}(x+y)$ are called, respectively, the *geometric mean* and *arithmetic mean* of x and y.

[*Hint*: Use $\mathbf{x} = \left[\begin{array}{c}\sqrt{x}\\ \sqrt{y}\end{array}\right]$ *and* $\mathbf{y} = \left[\begin{array}{c}\sqrt{y}\\ \sqrt{x}\end{array}\right]$.]

Exercise 5.3.11 Use the Cauchy inequality to prove that:

a. $r_1 + r_2 + \cdots + r_n \le n(r_1^2 + r_2^2 + \cdots + r_n^2)$ for all r_i in $\mathbb{R}$ and all $n \ge 1$.

b. $r_1r_2 + r_1r_3 + r_2r_3 \le r_1^2 + r_2^2 + r_3^2$ for all $r_1,\ r_2,$ and r_3 in $\mathbb{R}$. [*Hint*: See part (a).]

Exercise 5.3.12

a. Show that $\mathbf{x}$ and $\mathbf{y}$ are orthogonal in $\mathbb{R}^n$ if and only if $\|\mathbf{x}+\mathbf{y}\| = \|\mathbf{x}-\mathbf{y}\|$.

b. Show that $\mathbf{x}+\mathbf{y}$ and $\mathbf{x}-\mathbf{y}$ are orthogonal in $\mathbb{R}^n$ if and only if $\|\mathbf{x}\| = \|\mathbf{y}\|$.

Exercise 5.3.13

a. Show that $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$ if and only if $\mathbf{x}$ is orthogonal to $\mathbf{y}$.

b. If $\mathbf{x} = \left[\begin{array}{c}1\\1\end{array}\right]$, $\mathbf{y} = \left[\begin{array}{c}1\\0\end{array}\right]$ and $\mathbf{z} = \left[\begin{array}{r}-2\\3\end{array}\right]$, show that $\|\mathbf{x}+\mathbf{y}+\mathbf{z}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \|\mathbf{z}\|^2$ but $\mathbf{x}\cdot\mathbf{y} \neq 0$, $\mathbf{x}\cdot\mathbf{z} \neq 0$, and $\mathbf{y}\cdot\mathbf{z} \neq 0$.

Exercise 5.3.14

a. Show that $\mathbf{x}\cdot\mathbf{y} = \frac{1}{4}[\|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2]$ for all $\mathbf{x}$, $\mathbf{y}$ in $\mathbb{R}^n$.

b. Show that $\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 = \frac{1}{2}\left[\|\mathbf{x}+\mathbf{y}\|^2 + \|\mathbf{x}-\mathbf{y}\|^2\right]$ for all $\mathbf{x}$, $\mathbf{y}$ in $\mathbb{R}^n$.

Exercise 5.3.15 If A is $n\times n$, show that every eigenvalue of A^TA is nonnegative. [*Hint*: Compute $\|A\mathbf{x}\|^2$ where $\mathbf{x}$ is an eigenvector.]

Exercise 5.3.16 If $\mathbb{R}^n = \operatorname{span}\{\mathbf{x}_1,\ \dots,\ \mathbf{x}_m\}$ and $\mathbf{x}\cdot\mathbf{x}_i = 0$ for all i, show that $\mathbf{x} = 0$. [*Hint*: Show $\|\mathbf{x}\| = 0$.]

Exercise 5.3.17 If $\mathbb{R}^n = \operatorname{span}\{\mathbf{x}_1,\ \dots,\ \mathbf{x}_m\}$ and $\mathbf{x}\cdot\mathbf{x}_i = \mathbf{y}\cdot\mathbf{x}_i$ for all i, show that $\mathbf{x} = \mathbf{y}$. [*Hint*: Exercise 5.3.16]

Exercise 5.3.18 Let $\{\mathbf{e}_1,\ \dots,\ \mathbf{e}_n\}$ be an orthogonal basis of $\mathbb{R}^n$. Given $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, show that

$$\mathbf{x}\cdot\mathbf{y} = \frac{(\mathbf{x}\cdot\mathbf{e}_1)(\mathbf{y}\cdot\mathbf{e}_1)}{\|\mathbf{e}_1\|^2} + \cdots + \frac{(\mathbf{x}\cdot\mathbf{e}_n)(\mathbf{y}\cdot\mathbf{e}_n)}{\|\mathbf{e}_n\|^2}$$

5.4 Rank of a Matrix

In this section we use the concept of dimension to clarify the definition of the rank of a matrix given in Section 1.2, and to study its properties. This requires that we deal with rows and columns in the same way. While it has been our custom to write the n-tuples in $\mathbb{R}^n$ as columns, in this section we will frequently write them as rows. Subspaces, independence, spanning, and dimension are defined for rows using matrix operations, just as for columns. If A is an $m \times n$ matrix, we define:

Definition 5.10 Column and Row Space of a Matrix

*The **column space**, $\operatorname{col} A$, of A is the subspace of $\mathbb{R}^m$ spanned by the columns of A.*
*The **row space**, $\operatorname{row} A$, of A is the subspace of $\mathbb{R}^n$ spanned by the rows of A.*

Much of what we do in this section involves these subspaces. We begin with:

Lemma 5.4.1

Let A and B denote $m \times n$ matrices.

1. *If $A \to B$ by elementary row operations, then $\operatorname{row} A = \operatorname{row} B$.*
2. *If $A \to B$ by elementary column operations, then $\operatorname{col} A = \operatorname{col} B$.*

Proof. We prove (1); the proof of (2) is analogous. It is enough to do it in the case when $A \to B$ by a single row operation. Let $R_1, R_2, \dots, R_m$ denote the rows of A. The row operation $A \to B$ either interchanges two rows, multiplies a row by a nonzero constant, or adds a multiple of a row to a different row. We leave the first two cases to the reader. In the last case, suppose that a times row p is added to row q where $p < q$. Then the rows of B are $R_1, \dots, R_p, \dots, R_q + aR_p, \dots, R_m$, and Theorem 5.1.1 shows that

$$\operatorname{span}\{R_1, \dots, R_p, \dots, R_q, \dots, R_m\} = \operatorname{span}\{R_1, \dots, R_p, \dots, R_q + aR_p, \dots, R_m\}$$

That is, $\operatorname{row} A = \operatorname{row} B$. □

If A is any matrix, we can carry $A \to R$ by elementary row operations where R is a row-echelon matrix. Hence $\operatorname{row} A = \operatorname{row} R$ by Lemma 5.4.1; so the first part of the following result is of interest.

Lemma 5.4.2

If R is a row-echelon matrix, then

1. *The nonzero rows of R are a basis of $\operatorname{row} R$.*
2. *The columns of R containing leading ones are a basis of $\operatorname{col} R$.*

Proof. The rows of R are independent by Example 5.2.6, and they span $\operatorname{row} R$ by definition. This proves (1).

Let $\mathbf{c}_{j_1}, \mathbf{c}_{j_2}, \dots, \mathbf{c}_{j_r}$ denote the columns of R containing leading 1s. Then $\{\mathbf{c}_{j_1}, \mathbf{c}_{j_2}, \dots, \mathbf{c}_{j_r}\}$ is independent because the leading 1s are in different rows (and have zeros below and to the left of them).

Let U denote the subspace of all columns in $\mathbb{R}^m$ in which the last $m-r$ entries are zero. Then $\dim U = r$ (it is just $\mathbb{R}^r$ with extra zeros). Hence the independent set $\{\mathbf{c}_{j_1}, \mathbf{c}_{j_2}, \dots, \mathbf{c}_{j_r}\}$ is a basis of U by Theorem 5.2.7. Since each $\mathbf{c}_{j_i}$ is in $\text{col } R$, it follows that $\text{col } R = U$, proving (2). □

With Lemma 5.4.2 we can fill a gap in the definition of the rank of a matrix given in Chapter 1. Let A be any matrix and suppose A is carried to some row-echelon matrix R by row operations. Note that R is not unique. In Section 1.2 we defined the **rank** of A, denoted $\text{rank } A$, to be the number of leading 1s in R, that is the number of nonzero rows of R. The fact that this number does not depend on the choice of R was not proved in Section 1.2. However part 1 of Lemma 5.4.2 shows that

$$\text{rank } A = \dim(\text{row } A)$$

and hence that $\text{rank } A$ is independent of R.

Lemma 5.4.2 can be used to find bases of subspaces of $\mathbb{R}^n$ (written as rows). Here is an example.

Example 5.4.1

Find a basis of $U = \text{span}\{(1,\ 1,\ 2,\ 3),\ (2,\ 4,\ 1,\ 0),\ (1,\ 5,\ -4,\ -9)\}$.

Solution. U is the row space of $\left[\begin{array}{rrrr} 1 & 1 & 2 & 3 \\ 2 & 4 & 1 & 0 \\ 1 & 5 & -4 & -9 \end{array}\right]$. This matrix has row-echelon form $\left[\begin{array}{rrrr} 1 & 1 & 2 & 3 \\ 0 & 1 & -\frac{3}{2} & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$, so $\{(1,\ 1,\ 2,\ 3),\ (0,\ 1,\ -\frac{3}{2},\ -3)\}$ is basis of U by Lemma 5.4.2.
Note that $\{(1,\ 1,\ 2,\ 3),\ (0,\ 2,\ -3,\ -6)\}$ is another basis that avoids fractions.

Lemmas 5.4.1 and 5.4.2 are enough to prove the following fundamental theorem.

Theorem 5.4.1: Rank Theorem

Let A denote any $m \times n$ matrix of rank r. Then

$$\dim(\text{col } A) = \dim(\text{row } A) = r$$

Moreover, if A is carried to a row-echelon matrix R by row operations, then

1. *The r nonzero rows of R are a basis of $\text{row } A$.*
2. *If the leading 1s lie in columns $j_1, j_2, \dots, j_r$ of R, then columns $j_1, j_2, \dots, j_r$ of A are a basis of $\text{col } A$.*

Proof. We have $\text{row } A = \text{row } R$ by Lemma 5.4.1, so (1) follows from Lemma 5.4.2. Moreover, $R = UA$ for some invertible matrix U by Theorem 2.5.1. Now write $A = \left[\begin{array}{cccc} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n \end{array}\right]$ where $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ are the columns of A. Then

$$R = UA = U\left[\begin{array}{cccc} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{array}\right] = \left[\begin{array}{cccc} U\mathbf{c}_1 & U\mathbf{c}_2 & \cdots & U\mathbf{c}_n \end{array}\right]$$

Thus, in the notation of (2), the set $B = \{U\mathbf{c}_{j_1}, U\mathbf{c}_{j_2}, \dots, U\mathbf{c}_{j_r}\}$ is a basis of $\operatorname{col} R$ by Lemma 5.4.2. So, to prove (2) and the fact that $\dim(\operatorname{col} A) = r$, it is enough to show that $D = \{\mathbf{c}_{j_1}, \mathbf{c}_{j_2}, \dots, \mathbf{c}_{j_r}\}$ is a basis of $\operatorname{col} A$. First, D is linearly independent because U is invertible (verify), so we show that, for each j, column $\mathbf{c}_j$ is a linear combination of the $\mathbf{c}_{j_i}$. But $U\mathbf{c}_j$ is column j of R, and so is a linear combination of the $U\mathbf{c}_{j_i}$, say $U\mathbf{c}_j = a_1 U\mathbf{c}_{j_1} + a_2 U\mathbf{c}_{j_2} + \cdots + a_r U\mathbf{c}_{j_r}$ where each a_i is a real number.

Since U is invertible, it follows that $\mathbf{c}_j = a_1\mathbf{c}_{j_1} + a_2\mathbf{c}_{j_2} + \cdots + a_r\mathbf{c}_{j_r}$ and the proof is complete. □

Example 5.4.2

Compute the rank of $A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$ and find bases for $\operatorname{row} A$ and $\operatorname{col} A$.

Solution. The reduction of A to row-echelon form is as follows:

$$\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence $\operatorname{rank} A = 2$, and $\left\{\begin{bmatrix} 1 & 2 & 2 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & -3 \end{bmatrix}\right\}$ is a basis of $\operatorname{row} A$ by Lemma 5.4.2. Since the leading 1s are in columns 1 and 3 of the row-echelon matrix, Theorem 5.4.1 shows that columns 1 and 3 of A are a basis $\left\{\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}\right\}$ of $\operatorname{col} A$.

Theorem 5.4.1 has several important consequences. The first, Corollary 5.4.1 below, follows because the rows of A are independent (respectively span $\operatorname{row} A$) if and only if their transposes are independent (respectively span $\operatorname{col} A$).

Corollary 5.4.1

If A is any matrix, then $\operatorname{rank} A = \operatorname{rank}(A^T)$.

If A is an $m \times n$ matrix, we have $\operatorname{col} A \subseteq \mathbb{R}^m$ and $\operatorname{row} A \subseteq \mathbb{R}^n$. Hence Theorem 5.2.8 shows that $\dim(\operatorname{col} A) \leq \dim(\mathbb{R}^m) = m$ and $\dim(\operatorname{row} A) \leq \dim(\mathbb{R}^n) = n$. Thus Theorem 5.4.1 gives:

Corollary 5.4.2

If A is an $m \times n$ matrix, then $\operatorname{rank} A \leq m$ *and* $\operatorname{rank} A \leq n$.

Corollary 5.4.3

$\operatorname{rank} A = \operatorname{rank}(UA) = \operatorname{rank}(AV)$ *whenever U and V are invertible.*

Proof. Lemma 5.4.1 gives $\text{rank}\, A = \text{rank}\,(UA)$. Using this and Corollary 5.4.1 we get

$$\text{rank}\,(AV) = \text{rank}\,(AV)^T = \text{rank}\,(V^T A^T) = \text{rank}\,(A^T) = \text{rank}\, A$$

The next corollary requires a preliminary lemma. □

Lemma 5.4.3

Let A, U, and V be matrices of sizes $m \times n$, $p \times m$, and $n \times q$ respectively.

1. $\text{col}\,(AV) \subseteq \text{col}\, A$, *with equality if* $VV' = I_n$ *for some* V'.
2. $\text{row}\,(UA) \subseteq \text{row}\, A$, *with equality if* $U'U = I_m$ *for some* U'.

Proof. For (1), write $V = \begin{bmatrix} \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q \end{bmatrix}$ where $\mathbf{v}_j$ is column j of V. Then we have $AV = \begin{bmatrix} A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_q \end{bmatrix}$, and each $A\mathbf{v}_j$ is in $\text{col}\, A$ by Definition 2.4. It follows that $\text{col}\,(AV) \subseteq \text{col}\, A$. If $VV' = I_n$, we obtain $\text{col}\, A = \text{col}\,[(AV)V'] \subseteq \text{col}\,(AV)$ in the same way. This proves (1).

As to (2), we have $\text{col}\,\left[(UA)^T\right] = \text{col}\,(A^T U^T) \subseteq \text{col}\,(A^T)$ by (1), from which $\text{row}\,(UA) \subseteq \text{row}\, A$. If $U'U = I_m$, this is equality as in the proof of (1). □

Corollary 5.4.4

If A is $m \times n$ and B is $n \times m$, then $\text{rank}\, AB \leq \text{rank}\, A$ *and* $\text{rank}\, AB \leq \text{rank}\, B$.

Proof. By Lemma 5.4.3, $\text{col}\,(AB) \subseteq \text{col}\, A$ and $\text{row}\,(BA) \subseteq \text{row}\, A$, so Theorem 5.4.1 applies. □

In Section 5.1 we discussed two other subspaces associated with an $m \times n$ matrix A: the null space $\text{null}\,(A)$ and the image space $\text{im}\,(A)$

$$\text{null}\,(A) = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\} \text{ and } \text{im}\,(A) = \{A\mathbf{x} \mid \mathbf{x} \text{ in } \mathbb{R}^n\}$$

Using rank, there are simple ways to find bases of these spaces. If A has rank r, we have $\text{im}\,(A) = \text{col}\,(A)$ by Example 5.1.8, so $\dim\,[\text{im}\,(A)] = \dim\,[\text{col}\,(A)] = r$. Hence Theorem 5.4.1 provides a method of finding a basis of $\text{im}\,(A)$. This is recorded as part (2) of the following theorem.

Theorem 5.4.2

Let A denote an $m \times n$ matrix of rank r. Then

1. *The $n - r$ basic solutions to the system $A\mathbf{x} = \mathbf{0}$ provided by the gaussian algorithm are a basis of* $\text{null}\,(A)$, *so* $\dim\,[\text{null}\,(A)] = n - r$.
2. *Theorem 5.4.1 provides a basis of* $\text{im}\,(A) = \text{col}\,(A)$, *and* $\dim\,[\text{im}\,(A)] = r$.

Proof. It remains to prove (1). We already know (Theorem 2.2.1) that $\text{null}\,(A)$ is spanned by the $n - r$ basic solutions of $A\mathbf{x} = \mathbf{0}$. Hence using Theorem 5.2.7, it suffices to show that $\dim\,[\text{null}\,(A)] = n - r$. So let $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ be a basis of $\text{null}\,(A)$, and extend it to a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{x}_{k+1}, \dots, \mathbf{x}_n\}$ of $\mathbb{R}^n$ (by

Theorem 5.2.6). It is enough to show that $\{A\mathbf{x}_{k+1}, \dots, A\mathbf{x}_n\}$ is a basis of $\text{im}(A)$; then $n-k=r$ by the above and so $k=n-r$ as required.

Spanning. Choose $A\mathbf{x}$ in $\text{im}(A)$, $\mathbf{x}$ in $\mathbb{R}^n$, and write $\mathbf{x} = a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + a_{k+1}\mathbf{x}_{k+1} + \cdots + a_n\mathbf{x}_n$ where the a_i are in $\mathbb{R}$. Then $A\mathbf{x} = a_{k+1}A\mathbf{x}_{k+1} + \cdots + a_nA\mathbf{x}_n$ because $\{\mathbf{x}_1, \dots, \mathbf{x}_k\} \subseteq \text{null}(A)$.

Independence. Let $t_{k+1}A\mathbf{x}_{k+1} + \cdots + t_nA\mathbf{x}_n = \mathbf{0}$, t_i in $\mathbb{R}$. Then $t_{k+1}\mathbf{x}_{k+1} + \cdots + t_n\mathbf{x}_n$ is in $\text{null}\,A$, so $t_{k+1}\mathbf{x}_{k+1} + \cdots + t_n\mathbf{x}_n = t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k$ for some $t_1, \dots, t_k$ in $\mathbb{R}$. But then the independence of the $\mathbf{x}_i$ shows that $t_i = 0$ for every i. $\square$

Example 5.4.3

If $A = \left[\begin{array}{rrrr} 1 & -2 & 1 & 1 \\ -1 & 2 & 0 & 1 \\ 2 & -4 & 1 & 0 \end{array}\right]$, find bases of $\text{null}(A)$ and $\text{im}(A)$, and so find their dimensions.

Solution. If $\mathbf{x}$ is in $\text{null}(A)$, then $A\mathbf{x} = \mathbf{0}$, so $\mathbf{x}$ is given by solving the system $A\mathbf{x} = \mathbf{0}$. The reduction of the augmented matrix to reduced form is

$$\left[\begin{array}{rrrr|r} 1 & -2 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 2 & -4 & 1 & 0 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrr|r} 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Hence $r = \text{rank}(A) = 2$. Here, $\text{im}(A) = \text{col}(A)$ has basis $\left\{ \left[\begin{array}{r} 1 \\ -1 \\ 2 \end{array}\right], \left[\begin{array}{r} 1 \\ 0 \\ 1 \end{array}\right] \right\}$ by Theorem 5.4.1 because the leading 1s are in columns 1 and 3. In particular, $\dim[\text{im}(A)] = 2 = r$ as in Theorem 5.4.2.

Turning to $\text{null}(A)$, we use gaussian elimination. The leading variables are x_1 and x_3, so the nonleading variables become parameters: $x_2 = s$ and $x_4 = t$. It follows from the reduced matrix that $x_1 = 2s + t$ and $x_3 = -2t$, so the general solution is

$$\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array}\right] = \left[\begin{array}{c} 2s+t \\ s \\ -2t \\ t \end{array}\right] = s\mathbf{x}_1 + t\mathbf{x}_2 \text{ where } \mathbf{x}_1 = \left[\begin{array}{r} 2 \\ 1 \\ 0 \\ 0 \end{array}\right], \text{ and } \mathbf{x}_2 = \left[\begin{array}{r} 1 \\ 0 \\ -2 \\ 1 \end{array}\right].$$

Hence $\text{null}(A)$. But $\mathbf{x}_1$ and $\mathbf{x}_2$ *are* solutions (basic), so

$$\text{null}(A) = \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$$

However Theorem 5.4.2 asserts that $\{\mathbf{x}_1, \mathbf{x}_2\}$ is a *basis* of $\text{null}(A)$. (In fact it is easy to verify directly that $\{\mathbf{x}_1, \mathbf{x}_2\}$ is independent in this case.) In particular, $\dim[\text{null}(A)] = 2 = n - r$, as Theorem 5.4.2 asserts.

Let A be an $m \times n$ matrix. Corollary 5.4.2 of Theorem 5.4.1 asserts that $\text{rank}\,A \le m$ and $\text{rank}\,A \le n$, and it is natural to ask when these extreme cases arise. If $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ are the columns of A, Theorem 5.2.2 shows that $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ spans $\mathbb{R}^m$ if and only if the system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$, and that $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ is independent if and only if $A\mathbf{x} = \mathbf{0}$, $\mathbf{x}$ in $\mathbb{R}^n$, implies $\mathbf{x} = \mathbf{0}$. The next two useful theorems improve on both these results, and relate them to when the rank of A is n or m.

Theorem 5.4.3

The following are equivalent for an $m \times n$ matrix A:

1. rank $A = n$.
2. *The rows of A span $\mathbb{R}^n$.*
3. *The columns of A are linearly independent in $\mathbb{R}^m$.*
4. *The $n \times n$ matrix A^TA is invertible.*
5. $CA = I_n$ *for some $n \times m$ matrix C.*
6. *If $A\mathbf{x} = \mathbf{0}$, $\mathbf{x}$ in $\mathbb{R}^n$, then $\mathbf{x} = \mathbf{0}$.*

Proof. (1) $\Rightarrow$ (2). We have row $A \subseteq \mathbb{R}^n$, and $\dim(\text{row } A) = n$ by (1), so row $A = \mathbb{R}^n$ by Theorem 5.2.8. This is (2).

(2) $\Rightarrow$ (3). By (2), row $A = \mathbb{R}^n$, so rank $A = n$. This means $\dim(\text{col } A) = n$. Since the n columns of A span col A, they are independent by Theorem 5.2.7.

(3) $\Rightarrow$ (4). If $(A^TA)\mathbf{x} = \mathbf{0}$, $\mathbf{x}$ in $\mathbb{R}^n$, we show that $\mathbf{x} = \mathbf{0}$ (Theorem 2.4.5). We have

$$\|A\mathbf{x}\|^2 = (A\mathbf{x})^T A\mathbf{x} = \mathbf{x}^T A^T A\mathbf{x} = \mathbf{x}^T \mathbf{0} = \mathbf{0}$$

Hence $A\mathbf{x} = \mathbf{0}$, so $\mathbf{x} = \mathbf{0}$ by (3) and Theorem 5.2.2.

(4) $\Rightarrow$ (5). Given (4), take $C = (A^TA)^{-1}A^T$.

(5) $\Rightarrow$ (6). If $A\mathbf{x} = \mathbf{0}$, then left multiplication by C (from (5)) gives $\mathbf{x} = \mathbf{0}$.

(6) $\Rightarrow$ (1). Given (6), the columns of A are independent by Theorem 5.2.2. Hence $\dim(\text{col } A) = n$, and (1) follows. □

Theorem 5.4.4

The following are equivalent for an $m \times n$ matrix A:

1. rank $A = m$.
2. *The columns of A span $\mathbb{R}^m$.*
3. *The rows of A are linearly independent in $\mathbb{R}^n$.*
4. *The $m \times m$ matrix AA^T is invertible.*
5. $AC = I_m$ *for some $n \times m$ matrix C.*
6. *The system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$.*

Proof. (1) $\Rightarrow$ (2). By (1), $\dim(\text{col } A = m$, so $\text{col } A = \mathbb{R}^m$ by Theorem 5.2.8.

(2) $\Rightarrow$ (3). By (2), $\text{col } A = \mathbb{R}^m$, so $\text{rank } A = m$. This means $\dim(\text{row } A) = m$. Since the m rows of A span $\text{row } A$, they are independent by Theorem 5.2.7.

(3) $\Rightarrow$ (4). We have $\text{rank } A = m$ by (3), so the $n \times m$ matrix A^T has rank m. Hence applying Theorem 5.4.3 to A^T in place of A shows that $(A^T)^T A^T$ is invertible, proving (4).

(4) $\Rightarrow$ (5). Given (4), take $C = A^T(AA_T)^{-1}$ in (5).

(5) $\Rightarrow$ (6). Comparing columns in $AC = I_m$ gives $A\mathbf{c}_j = \mathbf{e}_j$ for each j, where $\mathbf{c}_j$ and $\mathbf{e}_j$ denote column j of C and I_m respectively. Given $\mathbf{b}$ in $\mathbb{R}^m$, write $\mathbf{b} = \sum_{j=1}^m r_j\mathbf{e}_j$, r_j in $\mathbb{R}$. Then $A\mathbf{x} = \mathbf{b}$ holds with $\mathbf{x} = \sum_{j=1}^m r_j\mathbf{c}_j$ as the reader can verify.

(6) $\Rightarrow$ (1). Given (6), the columns of A span $\mathbb{R}^m$ by Theorem 5.2.2. Thus $\text{col } A = \mathbb{R}^m$ and (1) follows. □

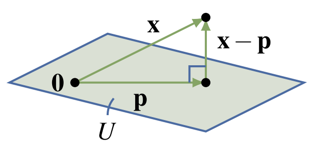
Example 5.4.4

Show that $\left[\begin{array}{cc} 3 & x+y+z \\ x+y+z & x^2+y^2+z^2 \end{array}\right]$ is invertible if x, y, and z are not all equal.

Solution. The given matrix has the form A^TA where $A = \left[\begin{array}{cc} 1 & x \\ 1 & y \\ 1 & z \end{array}\right]$ has independent columns because x, y, and z are not all equal (verify). Hence Theorem 5.4.3 applies.

Theorem 5.4.3 and Theorem 5.4.4 relate several important properties of an $m \times n$ matrix A to the invertibility of the square, symmetric matrices A^TA and AA^T. In fact, even if the columns of A are not independent or do not span $\mathbb{R}^m$, the matrices A^TA and AA^T are both symmetric and, as such, have real eigenvalues as we shall see. We return to this in Chapter 7.

The last topic for this section is the idea of the orthogonal complement of a subspace.



Suppose a point $\mathbf{x}$ and a plane U through the origin in $\mathbb{R}^3$ are given, and we want to find the point $\mathbf{p}$ in the plane that is closest to $\mathbf{x}$. Our geometric intuition assures us that such a point $\mathbf{p}$ exists. In fact (see the diagram), $\mathbf{p}$ must be chosen in such a way that $\mathbf{x}-\mathbf{p}$ is *perpendicular* to the plane.

Now we make two observations: first, the plane U is a *subspace of* $\mathbb{R}^3$ (because U contains the origin); and second, that the condition that $\mathbf{x}-\mathbf{p}$ is perpendicular to the plane U means that $\mathbf{x}-\mathbf{p}$ is *orthogonal* to every vector in U. In these terms the whole discussion makes sense in $\mathbb{R}^n$. Furthermore, the orthogonal lemma provides exactly what is needed to find $\mathbf{p}$ in this more general setting.

Definition 8.1 Orthogonal Complement of a Subspace of $\mathbb{R}^n$

If U is a subspace of $\mathbb{R}^n$, define the ***orthogonal complement*** *$U^\perp$ of U (pronounced "U-perp") by*

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x}\cdot\mathbf{y} = 0 \text{ for all } \mathbf{y} \text{ in } U\}$$

The following lemma collects some useful properties of the orthogonal complement; the proof of (1) and (2) is left as an exercise.

Lemma 8.1.2

Let U be a subspace of $\mathbb{R}^n$.

1. *$U^\perp$ is a subspace of $\mathbb{R}^n$.*
2. *$\{\mathbf{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$.*
3. *If $U = \operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$, then $U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x}\cdot\mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$.*

Proof.

3. Let $U = \operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$; we must show that $U^\perp = \{\mathbf{x} \mid \mathbf{x}\cdot\mathbf{x}_i = 0 \text{ for each } i\}$. If $\mathbf{x}$ is in $U^\perp$ then $\mathbf{x}\cdot\mathbf{x}_i = 0$ for all i because each $\mathbf{x}_i$ is in U. Conversely, suppose that $\mathbf{x}\cdot\mathbf{x}_i = 0$ for all i; we must show that $\mathbf{x}$ is in $U^\perp$, that is, $\mathbf{x}\cdot\mathbf{y} = 0$ for each $\mathbf{y}$ in U. Write $\mathbf{y} = r_1\mathbf{x}_1 + r_2\mathbf{x}_2 + \cdots + r_k\mathbf{x}_k$, where each r_i is in $\mathbb{R}$. Then, using Theorem 5.3.1,

$$\mathbf{x}\cdot\mathbf{y} = r_1(\mathbf{x}\cdot\mathbf{x}_1) + r_2(\mathbf{x}\cdot\mathbf{x}_2) + \cdots + r_k(\mathbf{x}\cdot\mathbf{x}_k) = r_1 0 + r_2 0 + \cdots + r_k 0 = 0$$

as required. $\square$

Fundamental Subspaces

It turns out that any that the orthogonal complement of a subspace is relati el eas to find hen ou recogni e that subspace as being one of the four fundamental subspaces of some matrix.

Definition 8.10

The ***fundamental subspaces*** *of an* $m \times n$ *matrix* A *are:*

$$\text{row } A = \text{span}\,\{\mathbf{x} \mid \mathbf{x} \textit{ is a row of } A\}$$

$$\text{col } A = \text{span}\,\{\mathbf{x} \mid \mathbf{x} \textit{ is a column of } A\}$$

$$\text{null } A = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

$$\text{null } A^T = \{\mathbf{x} \in \mathbb{R}^n \mid A^T\mathbf{x} = \mathbf{0}\}$$

The follo ing lemma sho s a relationship bet een each of these subspaces and their orthogonal complements. of these fundamental subspaces.

Lemma 8.6.4

If A *is any matrix then:*

1. $(\text{row } A)^{\perp} = \text{null } A$ *and* $(\text{col } A)^{\perp} = \text{null } A^T$.
2. *If* U *is any subspace of* $\mathbb{R}^n$ *then* $U^{\perp\perp} = U$.
3. *Let* $\{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ *be an orthonormal basis of* $\mathbb{R}^m$. *If* $U = \text{span}\,\{\mathbf{f}_1, \ldots, \mathbf{f}_k\}$, *then*

$$U^{\perp} = \text{span}\,\{\mathbf{f}_{k+1}, \ldots, \mathbf{f}_m\}$$

Exercises for 5.4

Exercise 5.4.1 In each case find bases for the row and column spaces of A and determine the rank of A.

a. $\begin{bmatrix} 2 & -4 & 6 & 8 \\ 2 & -1 & 3 & 2 \\ 4 & -5 & 9 & 10 \\ 0 & -1 & 1 & 2 \end{bmatrix}$ b. $\begin{bmatrix} 2 & -1 & 1 \\ -2 & 1 & 1 \\ 4 & -2 & 3 \\ -6 & 3 & 0 \end{bmatrix}$

c. $\begin{bmatrix} 1 & -1 & 5 & -2 & 2 \\ 2 & -2 & -2 & 5 & 1 \\ 0 & 0 & -12 & 9 & -3 \\ -1 & 1 & 7 & -7 & 1 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 2 & -1 & 3 \\ -3 & -6 & 3 & -2 \end{bmatrix}$

Exercise 5.4.2 In each case find a basis of the subspace U.

a. $U = \text{span}\,\{(1, -1, 0, 3), (2, 1, 5, 1), (4, -2, 5, 7)\}$

b. $U = \text{span}\,\{(1, -1, 2, 5, 1), (3, 1, 4, 2, 7), (1, 1, 0, 0, 0), (5, 1, 6, 7, 8)\}$

c. $U = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}\right\}$

d.
$$U = \text{span}\left\{\begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -8 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -10 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}\right\}$$

Exercise 5.4.3

a. Can a 3×4 matrix have independent columns? Independent rows? Explain.

b. If A is 4×3 and $\text{rank } A = 2$, can A have independent columns? Independent rows? Explain.

c. If A is an $m \times n$ matrix and $\text{rank } A = m$, show that $m \leq n$.

d. Can a nonsquare matrix have its rows independent and its columns independent? Explain.

e. Can the null space of a 3×6 matrix have dimension 2? Explain.

f. Suppose that A is 5×4 and $\text{null}(A) = \mathbb{R}\mathbf{x}$ for some column $\mathbf{x} \neq \mathbf{0}$. Can $\dim(\text{im } A) = 2$?

Exercise 5.4.4 If A is $m \times n$ show that

$$\text{col}(A) = \{A\mathbf{x} \mid \mathbf{x} \text{ in } \mathbb{R}^n\}$$

Exercise 5.4.5 If A is $m \times n$ and B is $n \times m$, show that $AB = 0$ if and only if $\text{col } B \subseteq \text{null } A$.

Exercise 5.4.6 Show that the rank does not change when an elementary row or column operation is performed on a matrix.

Exercise 5.4.7 In each case find a basis of the null space of A. Then compute $\text{rank } A$ and verify (1) of Theorem 5.4.2.

a. $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & 1 \\ 4 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 3 & 5 & 5 & 2 & 0 \\ 1 & 0 & 2 & 2 & 1 \\ 1 & 1 & 1 & -2 & -2 \\ -2 & 0 & -4 & -4 & -2 \end{bmatrix}$

Exercise 5.4.8 Let $A = \mathbf{cr}$ where $\mathbf{c} \neq \mathbf{0}$ is a column in $\mathbb{R}^m$ and $\mathbf{r} \neq \mathbf{0}$ is a row in $\mathbb{R}^n$.

a. Show that $\text{col } A = \text{span}\{\mathbf{c}\}$ and $\text{row } A = \text{span}\{\mathbf{r}\}$.

b. Find $\dim(\text{null } A)$.

c. Show that $\text{null } A = \text{null } \mathbf{r}$.

Exercise 5.4.9 Let A be $m \times n$ with columns $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$.

a. If $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ is independent, show $\text{null } A = \{\mathbf{0}\}$.

b. If $\text{null } A = \{\mathbf{0}\}$, show that $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ is independent.

Exercise 5.4.10 Let A be an $n \times n$ matrix.

a. Show that $A^2 = 0$ if and only if $\text{col } A \subseteq \text{null } A$.

b. Conclude that if $A^2 = 0$, then $\text{rank } A \leq \frac{n}{2}$.

c. Find a matrix A for which $\text{col } A = \text{null } A$.

Exercise 5.4.11 Let B be $m \times n$ and let AB be $k \times n$. If $\text{rank } B = \text{rank}(AB)$, show that $\text{null } B = \text{null}(AB)$. [*Hint*: Theorem 5.4.1.]

Exercise 5.4.12 Give a careful argument why $\text{rank}(A^T) = \text{rank } A$.

Exercise 5.4.13 Let A be an $m \times n$ matrix with columns $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$. If $\text{rank } A = n$, show that $\{A^T\mathbf{c}_1, A^T\mathbf{c}_2, \dots, A^T\mathbf{c}_n\}$ is a basis of $\mathbb{R}^n$.

Exercise 5.4.14 If A is $m \times n$ and $\mathbf{b}$ is $m \times 1$, show that $\mathbf{b}$ lies in the column space of A if and only if $\text{rank}[A\ \mathbf{b}] = \text{rank } A$.

Exercise 5.4.15

a. Show that $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\text{rank } A = \text{rank}[A\ \mathbf{b}]$. [*Hint*: Exercises 5.4.12 and 5.4.14.]

b. If $A\mathbf{x} = \mathbf{b}$ has no solution, show that $\text{rank}[A\ \mathbf{b}] = 1 + \text{rank } A$.

Exercise 5.4.16 Let X be a $k \times m$ matrix. If I is the $m \times m$ identity matrix, show that $I + X^T X$ is invertible.

[*Hint*: $I+X^TX = A^TA$ where $A = \left[\begin{array}{c} I \\ X \end{array}\right]$ in block form.]

Exercise 5.4.17 If A is $m \times n$ of rank r, show that A can be factored as $A = PQ$ where P is $m \times r$ with r independent columns, and Q is $r \times n$ with r independent rows. [*Hint*: Let $UAV = \left[\begin{array}{cc} I_r & 0 \\ 0 & 0 \end{array}\right]$ by Theorem 2.5.3, and write $U^{-1} = \left[\begin{array}{cc} U_1 & U_2 \\ U_3 & U_4 \end{array}\right]$ and $V^{-1} = \left[\begin{array}{cc} V_1 & V_2 \\ V_3 & V_4 \end{array}\right]$ in block form, where U_1 and V_1 are $r \times r$.]

Exercise 5.4.18

a. Show that if A and B have independent columns, so does AB.

b. Show that if A and B have independent rows, so does AB.

Exercise 5.4.19 A matrix obtained from A by deleting rows and columns is called a **submatrix** of A. If A has an invertible $k \times k$ submatrix, show that $\text{rank } A \geq k$. [*Hint*: Show that row and column operations carry $A \to \left[\begin{array}{cc} I_k & P \\ 0 & Q \end{array}\right]$ in block form.] *Remark*: It can be shown that $\text{rank } A$ is the largest integer r such that A has an invertible $r \times r$ submatrix.

5.6 Best Approximation and Least Squares

Often an exact solution to a problem in applied mathematics is difficult to obtain. However, it is usually just as useful to find arbitrarily close approximations to a solution. In particular, finding "linear approximations" is a potent technique in applied mathematics. One basic case is the situation where a system of linear equations has no solution, and it is desirable to find a "best approximation" to a solution to the system. In this section best approximations are defined and a method for finding them is described. The result is then applied to "least squares" approximation of data.

Suppose A is an $m \times n$ matrix and $\mathbf{b}$ is a column in $\mathbb{R}^m$, and consider the system

$$A\mathbf{x} = \mathbf{b}$$

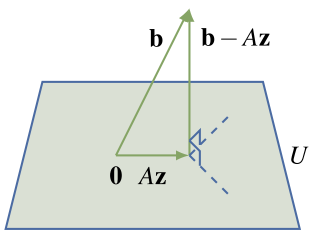
of m linear equations in n variables. This need not have a solution. However, given any column $\mathbf{z} \in \mathbb{R}^n$, the distance $||\mathbf{b} - A\mathbf{z}||$ is a measure of how far $A\mathbf{z}$ is from $\mathbf{b}$. Hence it is natural to ask whether there is a column $\mathbf{z}$ in $\mathbb{R}^n$ that is as close as possible to a solution in the sense that

$$||\mathbf{b} - A\mathbf{z}||$$

is the minimum value of $||\mathbf{b} - A\mathbf{x}||$ as $\mathbf{x}$ ranges over all columns in $\mathbb{R}^n$.

The answer is "yes", and to describe it define

$$U = \{A\mathbf{x} \mid \mathbf{x} \text{ lies in } \mathbb{R}^n\}$$



This is a subspace of $\mathbb{R}^n$ (verify) and we want a vector $A\mathbf{z}$ in U as close as possible to $\mathbf{b}$. That there is such a vector is clear geometrically if $n = 3$ by the diagram. In general such a vector $A\mathbf{z}$ exists by a general result called the *projection theorem* that will be proved in Chapter 8 (Theorem 8.1.3). Moreover, the projection theorem gives a simple way to compute $\mathbf{z}$ because it also shows that the vector $\mathbf{b} - A\mathbf{z}$ is *orthogonal* to every vector $A\mathbf{x}$ in U. Thus, for all $\mathbf{x}$ in $\mathbb{R}^n$,

$$\begin{aligned} 0 = (A\mathbf{x}) \cdot (\mathbf{b} - A\mathbf{z}) = (A\mathbf{x})^T(\mathbf{b} - A\mathbf{z}) &= \mathbf{x}^T A^T(\mathbf{b} - A\mathbf{z}) \\ &= \mathbf{x} \cdot [A^T(\mathbf{b} - A\mathbf{z})] \end{aligned}$$

In other words, the vector $A^T(\mathbf{b} - A\mathbf{z})$ in $\mathbb{R}^n$ is orthogonal to *every* vector in $\mathbb{R}^n$ and so must be zero (being orthogonal to itself). Hence $\mathbf{z}$ satisfies

$$(A^T A)\mathbf{z} = A^T\mathbf{b}$$

Definition 5.14 Normal Equations

This is a system of linear equations called the ***normal equations*** *for* $\mathbf{z}$.

Note that this system can have more than one solution (see Exercise 5.6.5). However, the $n \times n$ matrix $A^T A$ is invertible if (and only if) the columns of A are linearly independent (Theorem 5.4.3); so, in this case, $\mathbf{z}$ is uniquely determined and is given explicitly by $\mathbf{z} = (A^T A)^{-1} A^T \mathbf{b}$. However, the most efficient way to find $\mathbf{z}$ is to apply gaussian elimination to the normal equations.

This discussion is summarized in the following theorem.

Theorem 5.6.1: Best Approximation Theorem

Let A be an $m \times n$ matrix, let $\mathbf{b}$ be any column in $\mathbb{R}^m$, and consider the system

$$A\mathbf{x} = \mathbf{b}$$

of m equations in n variables.

1. *Any solution $\mathbf{z}$ to the normal equations*

$$(A^T A)\mathbf{z} = A^T\mathbf{b}$$

   is a best approximation to a solution to $A\mathbf{x} = \mathbf{b}$ in the sense that $\|\mathbf{b} - A\mathbf{z}\|$ is the minimum value of $\|\mathbf{b} - A\mathbf{x}\|$ as $\mathbf{x}$ ranges over all columns in $\mathbb{R}^n$.

2. *If the columns of A are linearly independent, then $A^T A$ is invertible and $\mathbf{z}$ is given uniquely by $\mathbf{z} = (A^T A)^{-1} A^T \mathbf{b}$.*

We note in passing that if A is $n \times n$ and invertible, then

$$\mathbf{z} = (A^T A)^{-1} A^T \mathbf{b} = A^{-1} \mathbf{b}$$

is the solution to the system of equations, and $\|\mathbf{b} - A\mathbf{z}\| = 0$. Hence if A has independent columns, then $(A^T A)^{-1} A^T$ is playing the role of the inverse of the nonsquare matrix A. The matrix $A^T (AA^T)^{-1}$ plays a similar role when the rows of A are linearly independent. These are both special cases of the **generalized inverse** of a matrix A (see Exercise 5.6.14). However, we shall not pursue this topic here.

Example 5.6.1

The system of linear equations

$$\begin{aligned} 3x - y &= 4 \\ x + 2y &= 0 \\ 2x + y &= 1 \end{aligned}$$

has no solution. Find the vector $\mathbf{z} = \left[\begin{array}{c} x_0 \\ y_0 \end{array} \right]$ that best approximates a solution.

Solution. In this case,

$$A = \left[\begin{array}{rr} 3 & -1 \\ 1 & 2 \\ 2 & 1 \end{array} \right], \text{ so } A^T A = \left[\begin{array}{rrr} 3 & 1 & 2 \\ -1 & 2 & 1 \end{array} \right] \left[\begin{array}{rr} 3 & -1 \\ 1 & 2 \\ 2 & 1 \end{array} \right] = \left[\begin{array}{rr} 14 & 1 \\ 1 & 6 \end{array} \right]$$

is invertible. The normal equations $(A^T A)\mathbf{z} = A^T \mathbf{b}$ are

$$\left[\begin{array}{rr} 14 & 1 \\ 1 & 6 \end{array} \right] \mathbf{z} = \left[\begin{array}{r} 14 \\ -3 \end{array} \right], \text{ so } \mathbf{z} = \frac{1}{83} \left[\begin{array}{r} 87 \\ -56 \end{array} \right]$$

Thus $x_0 = \frac{87}{83}$ and $y_0 = \frac{-56}{83}$. With these values of x and y, the left sides of the equations are, approximately,

$$\begin{aligned} 3x_0 - y_0 &= \tfrac{317}{83} = 3.82 \\ x_0 + 2y_0 &= \tfrac{-25}{83} = -0.30 \\ 2x_0 + y_0 &= \tfrac{118}{83} = 1.42 \end{aligned}$$

This is as close as possible to a solution.

Example 5.6.2

The average number g of goals per game scored by a hockey player seems to be related linearly to two factors: the number x_1 of years of experience and the number x_2 of goals in the preceding 10 games. The data on the following page were collected on four players. Find the linear function

$g = a_0 + a_1x_1 + a_2x_2$ that best fits these data.

g	x_1	x_2
0.8	5	3
0.8	3	4
0.6	1	5
0.4	2	1

Solution. If the relationship is given by $g = r_0 + r_1x_1 + r_2x_2$, then the data can be described as follows:

$$\begin{bmatrix} 1 & 5 & 3 \\ 1 & 3 & 4 \\ 1 & 1 & 5 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.8 \\ 0.6 \\ 0.4 \end{bmatrix}$$

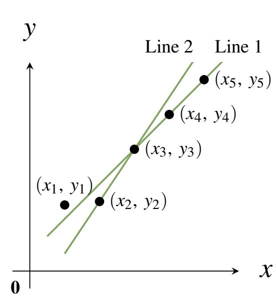
Using the notation in Theorem 5.6.1, we get

$$\begin{aligned} \mathbf{z} &= (A^TA)^{-1}A^T\mathbf{b} \\ &= \frac{1}{42}\begin{bmatrix} 119 & -17 & -19 \\ -17 & 5 & 1 \\ -19 & 1 & 5 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 5 & 3 & 1 & 2 \\ 3 & 4 & 5 & 1 \end{bmatrix}\begin{bmatrix} 0.8 \\ 0.8 \\ 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.09 \\ 0.08 \end{bmatrix} \end{aligned}$$

Hence the best-fitting function is $g = 0.14 + 0.09x_1 + 0.08x_2$. The amount of computation would have been reduced if the normal equations had been constructed and then solved by gaussian elimination.

Least Squares Approximation

In many scientific investigations, data are collected that relate two variables. For example, if x is the number of dollars spent on advertising by a manufacturer and y is the value of sales in the region in question, the manufacturer could generate data by spending $x_1, x_2, \dots, x_n$ dollars at different times and measuring the corresponding sales values $y_1, y_2, \dots, y_n$.

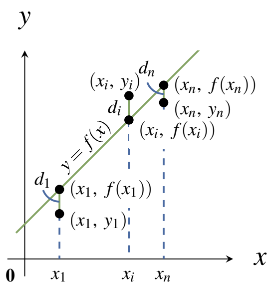


Suppose it is known that a linear relationship exists between the variables x and y—in other words, that $y = a + bx$ for some constants a and b. If the data are plotted, the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ may appear to lie on a straight line and estimating a and b requires finding the "best-fitting" line through these data points. For example, if five data points occur as shown in the diagram, line 1 is clearly a better fit than line 2. In general, the problem is to find the values of the constants a and b such that the line $y = a + bx$ best approximates the data in question. Note that an exact fit would be obtained if a and b were such that $y_i = a + bx_i$ were true for each data point (x_i, y_i). But this is too much to expect. Experimental errors in measurement are bound to occur, so the choice of a and b should be made in such a way that the errors between the observed values y_i and the corresponding fitted values $a + bx_i$ are in some sense minimized. Least squares approximation is a way to do this.

The first thing we must do is explain exactly what we mean by the *best fit* of a line $y = a + bx$ to an observed set of data points $(x_1,\ y_1),\ (x_2,\ y_2),\ \dots,\ (x_n,\ y_n)$. For convenience, write the linear function $r_0 + r_1 x$ as

$$f(x) = r_0 + r_1 x$$

so that the fitted points (on the line) have coordinates $(x_1,\ f(x_1)),\ \dots,\ (x_n,\ f(x_n))$.



The second diagram is a sketch of what the line $y = f(x)$ might look like. For each i the observed data point $(x_i,\ y_i)$ and the fitted point $(x_i,\ f(x_i))$ need not be the same, and the distance d_i between them measures how far the line misses the observed point. For this reason d_i is often called the **error** at x_i, and a natural measure of how close the line $y = f(x)$ is to the observed data points is the sum $d_1 + d_2 + \cdots + d_n$ of all these errors. However, it turns out to be better to use the sum of squares

$$S = d_1^2 + d_2^2 + \cdots + d_n^2$$

as the measure of error, and the line $y = f(x)$ is to be chosen so as to make this sum as small as possible. This line is said to be the **least squares approximating line** for the data points $(x_1,\ y_1),\ (x_2,\ y_2),\ \dots,\ (x_n,\ y_n)$.

The square of the error d_i is given by $d_i^2 = [y_i - f(x_i)]^2$ for each i, so the quantity S to be minimized is the sum:

$$S = [y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \cdots + [y_n - f(x_n)]^2$$

Note that all the numbers x_i and y_i are *given* here; what is required is that the *function* f be chosen in such a way as to minimize S. Because $f(x) = r_0 + r_1 x$, this amounts to choosing r_0 and r_1 to minimize S. This problem can be solved using Theorem 5.6.1. The following notation is convenient.

$$\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right] \quad \mathbf{y} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_n \end{array}\right] \quad \text{and} \quad f(\mathbf{x}) = \left[\begin{array}{c} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{array}\right] = \left[\begin{array}{c} r_0 + r_1 x_1 \\ r_0 + r_1 x_2 \\ \vdots \\ r_0 + r_1 x_n \end{array}\right]$$

Then the problem takes the following form: Choose r_0 and r_1 such that

$$S = [y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \cdots + [y_n - f(x_n)]^2 = \|\mathbf{y} - f(\mathbf{x})\|^2$$

is as small as possible. Now write

$$M = \left[\begin{array}{cc} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{array}\right] \quad \text{and} \quad \mathbf{r} = \left[\begin{array}{c} r_0 \\ r_1 \end{array}\right]$$

Then $M\mathbf{r} = f(\mathbf{x})$, so we are looking for a column $\mathbf{r} = \left[\begin{array}{c} r_0 \\ r_1 \end{array}\right]$ such that $\|\mathbf{y} - M\mathbf{r}\|^2$ is as small as possible. In other words, we are looking for a best approximation $\mathbf{z}$ to the system $M\mathbf{r} = \mathbf{y}$. Hence Theorem 5.6.1 applies directly, and we have

Theorem 5.6.2

Suppose that n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ are given, where at least two of $x_1, x_2, \dots, x_n$ are distinct. Put

$$\mathbf{y} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_n \end{array} \right] \quad M = \left[\begin{array}{cc} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{array} \right]$$

Then the least squares approximating line for these data points has equation

$$y = z_0 + z_1 x$$

where $\mathbf{z} = \left[\begin{array}{c} z_0 \\ z_1 \end{array} \right]$ is found by gaussian elimination from the normal equations

$$(M^T M)\mathbf{z} = M^T \mathbf{y}$$

The condition that at least two of $x_1, x_2, \dots, x_n$ are distinct ensures that $M^T M$ is an invertible matrix, so $\mathbf{z}$ is unique:

$$\mathbf{z} = (M^T M)^{-1} M^T \mathbf{y}$$

Example 5.6.3

Let data points $(x_1, y_1), (x_2, y_2), \dots, (x_5, y_5)$ be given as in the accompanying table. Find the least squares approximating line for these data.

x	y
1	1
3	2
4	3
6	4
7	5

Solution. In this case we have

$$\begin{aligned} M^T M &= \left[\begin{array}{cccc} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_5 \end{array} \right] \left[\begin{array}{cc} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_5 \end{array} \right] \\ &= \left[\begin{array}{cc} 5 & x_1 + \cdots + x_5 \\ x_1 + \cdots + x_5 & x_1^2 + \cdots + x_5^2 \end{array} \right] = \left[\begin{array}{cc} 5 & 21 \\ 21 & 111 \end{array} \right] \end{aligned}$$

$$\text{and } M^T\mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_5 \end{bmatrix}$$

$$= \begin{bmatrix} y_1 + y_2 + \cdots + y_5 \\ x_1y_1 + x_2y_2 + \cdots + x_5y_5 \end{bmatrix} = \begin{bmatrix} 15 \\ 78 \end{bmatrix}$$

so the normal equations $(M^TM)\mathbf{z} = M^T\mathbf{y}$ for $\mathbf{z} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}$ become

$$\begin{bmatrix} 5 & 21 \\ 21 & 111 \end{bmatrix} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix} = \begin{bmatrix} 15 \\ 78 \end{bmatrix}$$

The solution (using gaussian elimination) is $\mathbf{z} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0.24 \\ 0.66 \end{bmatrix}$ to two decimal places, so the least squares approximating line for these data is $y = 0.24 + 0.66x$. Note that M^TM is indeed invertible here (the determinant is 114), and the exact solution is

$$\mathbf{z} = (M^TM)^{-1}M^T\mathbf{y} = \frac{1}{114}\begin{bmatrix} 111 & -21 \\ -21 & 5 \end{bmatrix}\begin{bmatrix} 15 \\ 78 \end{bmatrix} = \frac{1}{114}\begin{bmatrix} 27 \\ 75 \end{bmatrix} = \frac{1}{38}\begin{bmatrix} 9 \\ 25 \end{bmatrix}$$

Least Squares Approximating Polynomials

Suppose now that, rather than a straight line, we want to find a polynomial

$$y = f(x) = r_0 + r_1x + r_2x^2 + \cdots + r_mx^m$$

of degree m that best approximates the data pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. As before, write

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \text{and} \quad f(\mathbf{x}) = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{bmatrix}$$

For each x_i we have two values of the variable y, the observed value y_i, and the computed value $f(x_i)$. The problem is to choose $f(x)$—that is, choose $r_0, r_1, \dots, r_m$ —such that the $f(x_i)$ are as close as possible to the y_i. Again we define "as close as possible" by the least squares condition: We choose the r_i such that

$$\|\mathbf{y} - f(\mathbf{x})\|^2 = [y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \cdots + [y_n - f(x_n)]^2$$

is as small as possible.

Definition 5.15 Least Squares Approximation

A polynomial $f(x)$ satisfying this condition is called a ***least squares approximating polynomial*** *of degree m for the given data pairs.*

If we write

$$M = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix} \quad \text{and} \quad \mathbf{r} = \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_m \end{bmatrix}$$

we see that $f(\mathbf{x}) = M\mathbf{r}$. Hence we want to find $\mathbf{r}$ such that $\|\mathbf{y} - M\mathbf{r}\|^2$ is as small as possible; that is, we want a best approximation $\mathbf{z}$ to the system $M\mathbf{r} = \mathbf{y}$. Theorem 5.6.1 gives the first part of Theorem 5.6.3.

Theorem 5.6.3

Let n data pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be given, and write

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad M = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_m \end{bmatrix}$$

1. *If $\mathbf{z}$ is any solution to the normal equations*

$$(M^T M)\mathbf{z} = M^T \mathbf{y}$$

then the polynomial

$$z_0 + z_1 x + z_2 x^2 + \cdots + z_m x^m$$

is a least squares approximating polynomial of degree m for the given data pairs.

2. *If at least $m+1$ of the numbers $x_1, x_2, \dots, x_n$ are distinct (so $n \geq m+1$), the matrix $M^T M$ is invertible and $\mathbf{z}$ is uniquely determined by*

$$\mathbf{z} = (M^T M)^{-1} M^T \mathbf{y}$$

Proof. It remains to prove (2), and for that we show that the columns of M are linearly independent (Theorem 5.4.3). Suppose a linear combination of the columns vanishes:

$$r_0 \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + r_1 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \cdots + r_m \begin{bmatrix} x_1^m \\ x_2^m \\ \vdots \\ x_n^m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

If we write $q(x) = r_0 + r_1 x + \cdots + r_m x^m$, equating coefficients shows that

$$q(x_1) = q(x_2) = \cdots = q(x_n) = 0$$

Hence $q(x)$ is a polynomial of degree m with at least $m+1$ distinct roots, so $q(x)$ must be the zero polynomial (see Appendix D or Theorem 6.5.4). Thus $r_0 = r_1 = \cdots = r_m = 0$ as required. $\square$

Example 5.6.4

Find the least squares approximating quadratic $y = z_0 + z_1 x + z_2 x^2$ for the following data points.

$$(-3,\ 3),\ (-1,\ 1),\ (0,\ 1),\ (1,\ 2),\ (3,\ 4)$$

Solution. This is an instance of Theorem 5.6.3 with $m = 2$. Here

$$\mathbf{y} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 2 \\ 4 \end{bmatrix} \quad M = \begin{bmatrix} 1 & -3 & 9 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix}$$

Hence,

$$M^T M = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -3 & -1 & 0 & 1 & 3 \\ 9 & 1 & 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} 1 & -3 & 9 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 20 \\ 0 & 20 & 0 \\ 20 & 0 & 164 \end{bmatrix}$$

$$M^T \mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -3 & -1 & 0 & 1 & 3 \\ 9 & 1 & 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ 4 \\ 66 \end{bmatrix}$$

The normal equations for $\mathbf{z}$ are

$$\begin{bmatrix} 5 & 0 & 20 \\ 0 & 20 & 0 \\ 20 & 0 & 164 \end{bmatrix} \mathbf{z} = \begin{bmatrix} 11 \\ 4 \\ 66 \end{bmatrix} \qquad \text{whence } \mathbf{z} = \begin{bmatrix} 1.15 \\ 0.20 \\ 0.26 \end{bmatrix}$$

This means that the least squares approximating quadratic for these data is
$y = 1.15 + 0.20x + 0.26x^2$.

Other Functions

There is an extension of Theorem 5.6.3 that should be mentioned. Given data pairs (x_1, y_1), (x_2, y_2), $\dots$, (x_n, y_n), that theorem shows how to find a polynomial

$$f(x) = r_0 + r_1 x + \cdots + r_m x^m$$

such that $\|\mathbf{y} - f(\mathbf{x})\|^2$ is as small as possible, where $\mathbf{x}$ and $f(\mathbf{x})$ are as before. Choosing the appropriate polynomial $f(x)$ amounts to choosing the coefficients $r_0, r_1, \dots, r_m$, and Theorem 5.6.3 gives a formula for the optimal choices. Here $f(x)$ is a linear combination of the functions $1, x, x^2, \dots, x^m$ where the r_i are the coefficients, and this suggests applying the method to other functions. If $f_0(x), f_1(x), \dots, f_m(x)$ are given functions, write

$$f(x) = r_0 f_0(x) + r_1 f_1(x) + \cdots + r_m f_m(x)$$

where the r_i are real numbers. Then the more general question is whether $r_0, r_1, \dots, r_m$ can be found such that $\|\mathbf{y} - f(\mathbf{x})\|^2$ is as small as possible where

$$f(\mathbf{x}) = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_m) \end{bmatrix}$$

Such a function $f(\mathbf{x})$ is called a **least squares best approximation** for these data pairs of the form $r_0 f_0(x) + r_1 f_1(x) + \cdots + r_m f_m(x)$, r_i in $\mathbb{R}$. The proof of Theorem 5.6.3 goes through to prove

Theorem 5.6.4

Let n data pairs (x_1, y_1), (x_2, y_2), $\dots$, (x_n, y_n) be given, and suppose that $m+1$ functions $f_0(x), f_1(x), \dots, f_m(x)$ are specified. Write

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad M = \begin{bmatrix} f_0(x_1) & f_1(x_1) & \cdots & f_m(x_1) \\ f_0(x_2) & f_1(x_2) & \cdots & f_m(x_2) \\ \vdots & \vdots & & \vdots \\ f_0(x_n) & f_1(x_n) & \cdots & f_m(x_n) \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix}$$

1. *If $\mathbf{z}$ is any solution to the normal equations*

$$(M^T M)\mathbf{z} = M^T \mathbf{y}$$

   then the function

$$z_0 f_0(x) + z_1 f_1(x) + \cdots + z_m f_m(x)$$

   is the best approximation for these data among all functions of the form $r_0 f_0(x) + r_1 f_1(x) + \cdots + r_m f_m(x)$ where the r_i are in $\mathbb{R}$.

2. *If $M^T M$ is invertible (that is, if $\operatorname{rank}(M) = m+1$), then $\mathbf{z}$ is uniquely determined; in fact, $\mathbf{z} = (M^T M)^{-1}(M^T \mathbf{y})$.*

Clearly Theorem 5.6.4 contains Theorem 5.6.3 as a special case, but there is no simple test in general for whether $M^T M$ is invertible. Conditions for this to hold depend on the choice of the functions $f_0(x), f_1(x), \dots, f_m(x)$.

Example 5.6.5

Given the data pairs $(-1, 0)$, $(0, 1)$, and $(1, 4)$, find the least squares approximating function of the form $r_0x + r_1 2^x$.

Solution. The functions are $f_0(x) = x$ and $f_1(x) = 2^x$, so the matrix M is

$$M = \left[\begin{array}{cc} f_0(x_1) & f_1(x_1) \\ f_0(x_2) & f_1(x_2) \\ f_0(x_3) & f_1(x_3) \end{array}\right] = \left[\begin{array}{rr} -1 & 2^{-1} \\ 0 & 2^0 \\ 1 & 2^1 \end{array}\right] = \frac{1}{2}\left[\begin{array}{rr} -2 & 1 \\ 0 & 2 \\ 2 & 4 \end{array}\right]$$

In this case $M^TM = \frac{1}{4}\left[\begin{array}{rr} 8 & 6 \\ 6 & 21 \end{array}\right]$ is invertible, so the normal equations

$$\frac{1}{4}\left[\begin{array}{rr} 8 & 6 \\ 6 & 21 \end{array}\right]\mathbf{z} = \left[\begin{array}{r} 4 \\ 9 \end{array}\right]$$

have a unique solution $\mathbf{z} = \frac{1}{11}\left[\begin{array}{r} 10 \\ 16 \end{array}\right]$. Hence the best-fitting function of the form $r_0x + r_1 2^x$ is $\overline{f}(x) = \frac{10}{11}x + \frac{16}{11}2^x$. Note that $\overline{f}(\mathbf{x}) = \left[\begin{array}{c} \overline{f}(-1) \\ \overline{f}(0) \\ \overline{f}(1) \end{array}\right] = \left[\begin{array}{c} \frac{-2}{11} \\ \frac{16}{11} \\ \frac{42}{11} \end{array}\right]$, compared with $\mathbf{y} = \left[\begin{array}{c} 0 \\ 1 \\ 4 \end{array}\right]$

Exercises for 5.6

Exercise 5.6.1 Find the best approximation to a solution of each of the following systems of equations.

a. $$\begin{aligned} x + y - z &= 5 \\ 2x - y + 6z &= 1 \\ 3x + 2y - z &= 6 \\ -x + 4y + z &= 0 \end{aligned}$$

b. $$\begin{aligned} 3x + y + z &= 6 \\ 2x + 3y - z &= 1 \\ 2x - y + z &= 0 \\ 3x - 3y + 3z &= 8 \end{aligned}$$

Exercise 5.6.2 Find the least squares approximating line $y = z_0 + z_1x$ for each of the following sets of data points.

a. $(1, 1)$, $(3, 2)$, $(4, 3)$, $(6, 4)$

b. $(2, 4)$, $(4, 3)$, $(7, 2)$, $(8, 1)$

c. $(-1, -1)$, $(0, 1)$, $(1, 2)$, $(2, 4)$, $(3, 6)$

d. $(-2, 3)$, $(-1, 1)$, $(0, 0)$, $(1, -2)$, $(2, -4)$

Exercise 5.6.3 Find the least squares approximating quadratic $y = z_0 + z_1x + z_2x^2$ for each of the following sets of data points.

a. $(0, 1)$, $(2, 2)$, $(3, 3)$, $(4, 5)$

b. $(-2, 1)$, $(0, 0)$, $(3, 2)$, $(4, 3)$

Exercise 5.6.4 Find a least squares approximating function of the form $r_0x + r_1x^2 + r_2 2^x$ for each of the following sets of data pairs.

a. $(-1, 1)$, $(0, 3)$, $(1, 1)$, $(2, 0)$

b. $(0, 1)$, $(1, 1)$, $(2, 5)$, $(3, 10)$

Exercise 5.6.5 Find the least squares approximating function of the form $r_0 + r_1x^2 + r_2 \sin\frac{\pi x}{2}$ for each of the following sets of data pairs.

a. $(0, 3)$, $(1, 0)$, $(1, -1)$, $(-1, 2)$

b. $(-1, \frac{1}{2})$, $(0, 1)$, $(2, 5)$, $(3, 9)$

Exercise 5.6.6 If M is a square invertible matrix, show that $\mathbf{z} = M^{-1}\mathbf{y}$ (in the notation of Theorem 5.6.3).

Exercise 5.6.7 Newton's laws of motion imply that an object dropped from rest at a height of 100 metres will be at a height $s = 100 - \frac{1}{2}gt^2$ metres t seconds later, where g is a constant called the acceleration due to gravity. The values of s and t given in the table are observed. Write $x = t^2$, find the least squares approximating line $s = a + bx$ for these data, and use b to estimate g.

Then find the least squares approximating quadratic $s = a_0 + a_1 t + a_2 t^2$ and use the value of a_2 to estimate g.

t	1	2	3
s	95	80	56

Exercise 5.6.8 A naturalist measured the heights y_i (in metres) of several spruce trees with trunk diameters x_i (in centimetres). The data are as given in the table. Find the least squares approximating line for these data and use it to estimate the height of a spruce tree with a trunk of diameter 10 cm.

x_i	5	7	8	12	13	16
y_i	2	3.3	4	7.3	7.9	10.1

Exercise 5.6.9 The yield y of wheat in bushels per acre appears to be a linear function of the number of days x_1 of sunshine, the number of inches x_2 of rain, and the number of pounds x_3 of fertilizer applied per acre. Find the best fit to the data in the table by an equation of the form $y = r_0 + r_1x_1 + r_2x_2 + r_3x_3$. [*Hint*: If a calculator for inverting A^TA is not available, the inverse is given in the answer.]

y	x_1	x_2	x_3
28	50	18	10
30	40	20	16
21	35	14	10
23	40	12	12
23	30	16	14

Exercise 5.6.10

a. Use $m = 0$ in Theorem 5.6.3 to show that the best-fitting horizontal line $y = a_0$ through the data points $(x_1, y_1), \dots, (x_n, y_n)$ is

$$y = \frac{1}{n}(y_1 + y_2 + \cdots + y_n)$$

the average of the y coordinates.

b. Deduce the conclusion in (a) without using Theorem 5.6.3.

Exercise 5.6.11 Assume $n = m+1$ in Theorem 5.6.3 (so M is square). If the x_i are distinct, use Theorem 3.2.6 to show that M is invertible. Deduce that $\mathbf{z} = M^{-1}\mathbf{y}$ and that the least squares polynomial is the interpolating polynomial (Theorem 3.2.6) and actually passes through all the data points.

Exercise 5.6.12 Let A be any $m \times n$ matrix and write $K = \{\mathbf{x} \mid A^TA\mathbf{x} = \mathbf{0}\}$. Let $\mathbf{b}$ be an m-column. Show that, if $\mathbf{z}$ is an n-column such that $\|\mathbf{b} - A\mathbf{z}\|$ is minimal, then *all* such vectors have the form $\mathbf{z} + \mathbf{x}$ for some $\mathbf{x} \in K$. [*Hint*: $\|\mathbf{b} - A\mathbf{y}\|$ is minimal if and only if $A^TA\mathbf{y} = A^T\mathbf{b}$.]

Exercise 5.6.13 Given the situation in Theorem 5.6.4, write

$$f(x) = r_0p_0(x) + r_1p_1(x) + \cdots + r_mp_m(x)$$

Suppose that $f(x)$ has at most k roots for any choice of the coefficients $r_0, r_1, \dots, r_m$, not all zero.

a. Show that M^TM is invertible if at least $k+1$ of the x_i are distinct.

b. If at least two of the x_i are distinct, show that there is always a best approximation of the form $r_0 + r_1e^x$.

c. If at least three of the x_i are distinct, show that there is always a best approximation of the form $r_0 + r_1x + r_2e^x$. [Calculus is needed.]

Exercise 5.6.14 If A is an $m \times n$ matrix, it can be proved that there exists a unique $n \times m$ matrix $A^{\#}$ satisfying the following four conditions: $AA^{\#}A = A$; $A^{\#}AA^{\#} = A^{\#}$; $AA^{\#}$ and $A^{\#}A$ are symmetric. The matrix $A^{\#}$ is called the **generalized inverse** of A, or the **Moore-Penrose** inverse.

a. If A is square and invertible, show that $A^{\#} = A^{-1}$.

b. If rank $A = m$, show that $A^{\#} = A^T(AA^T)^{-1}$.

c. If rank $A = n$, show that $A^{\#} = (A^TA)^{-1}A^T$.

Selected Exercise Answers

Section 1.1

1.1.1 b.
$2(2s+12t+13)+5s+9(-s-3t-3)+3t=-1$;
$(2s+12t+13)+2s+4(-s-3t-3)=1$

1.1.2 b. $x=t$, $y=\frac{1}{3}(1-2t)$ or $x=\frac{1}{2}(1-3s)$, $y=s$

d. $x=1+2s-5t$, $y=s$, $z=t$ or $x=s$, $y=t$, $z=\frac{1}{5}(1-s+2t)$

1.1.4 $x=\frac{1}{4}(3+2s)$, $y=s$, $z=t$

1.1.5 a. No solution if $b \neq 0$. If $b=0$, *any* x is a solution.

b. $x=\frac{b}{a}$

1.1.7 b. $\left[\begin{array}{rr|r} 1 & 2 & 0 \\ 0 & 1 & 1 \end{array}\right]$

d. $\left[\begin{array}{rrr|r} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 2 \end{array}\right]$

1.1.8 b. $\begin{array}{rcrcrcr} 2x & - & y & & & = & -1 \\ -3x & + & 2y & + & z & = & 0 \\ & & y & + & z & = & 3 \end{array}$

or $\begin{array}{rcrcrcr} 2x_1 & - & x_2 & & & = & -1 \\ -3x_1 & + & 2x_2 & + & x_3 & = & 0 \\ & & x_2 & + & x_3 & = & 3 \end{array}$

1.1.9 b. $x=-3$, $y=2$

d. $x=-17$, $y=13$

1.1.10 b. $x=\frac{1}{9}$, $y=\frac{10}{9}$, $z=-\frac{7}{3}$

1.1.11 b. No solution

1.1.14 b. F. $x+y=0$, $x-y=0$ has a unique solution.

d. T. Theorem 1.1.1.

1.1.16 $x'=5$, $y'=1$, so $x=23$, $y=-32$

1.1.17 $a=-\frac{1}{9}$, $b=-\frac{5}{9}$, $c=\frac{11}{9}$

1.1.19 \$4.50, \$5.20

Section 1.2

1.2.1 b. No, no

d. No, yes

f. No, no

1.2.2 b. $\left[\begin{array}{rrrrrrr} 0 & 1 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right]$

1.2.3 b. $x_1=2r-2s-t+1$, $x_2=r$, $x_3=-5s+3t-1$, $x_4=s$, $x_5=-6t+1$, $x_6=t$

d. $x_1=-4s-5t-4$, $x_2=-2s+t-2$, $x_3=s$, $x_4=1$, $x_5=t$

1.2.4 b. $x=-\frac{1}{7}$, $y=-\frac{3}{7}$

d. $x=\frac{1}{3}(t+2)$, $y=t$

f. No solution

1.2.5 b. $x=-15t-21$, $y=-11t-17$, $z=t$

d. No solution

f. $x=-7$, $y=-9$, $z=1$

h. $x=4$, $y=3+2t$, $z=t$

1.2.6 b. Denote the equations as E_1, E_2, and E_3. Apply gaussian elimination to column 1 of the augmented matrix, and observe that $E_3-E_1=-4(E_2-E_1)$. Hence $E_3=5E_1-4E_2$.

1.2.7 b. $x_1=0$, $x_2=-t$, $x_3=0$, $x_4=t$

d. $x_1=1$, $x_2=1-t$, $x_3=1+t$, $x_4=t$

1.2.8 b. If $ab \neq 2$, unique solution $x=\frac{-2-5b}{2-ab}$, $y=\frac{a+5}{2-ab}$. If $ab=2$: no solution if $a \neq -5$; if $a=-5$, the solutions are $x=-1+\frac{2}{5}t$, $y=t$.

d. If $a \neq 2$, unique solution $x=\frac{1-b}{a-2}$, $y=\frac{ab-2}{a-2}$. If $a=2$, no solution if $b \neq 1$; if $b=1$, the solutions are $x=\frac{1}{2}(1-t)$, $y=t$.

1.2.9 b. Unique solution $x=-2a+b+5c$, $y=3a-b-6c$, $z=-2a+b+c$, for any a, b, c.

d. If $abc \neq -1$, unique solution $x = y = z = 0$; if $abc = -1$ the solutions are $x = abt$, $y = -bt$, $z = t$.

f. If $a = 1$, solutions $x = -t$, $y = t$, $z = -1$. If $a = 0$, there is no solution. If $a \neq 1$ and $a \neq 0$, unique solution $x = \frac{a-1}{a}$, $y = 0$, $z = \frac{-1}{a}$.

1.2.10 b. 1

d. 3

f. 1

1.2.11 b. 2

d. 3

f. 2 if $a = 0$ or $a = 2$; 3, otherwise.

1.2.12 b. False. $A = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]$

d. False. $A = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

f. False. $\begin{array}{r} 2x - y = 0 \\ -4x + 2y = 0 \end{array}$ is consistent but $\begin{array}{r} 2x - y = 1 \\ -4x + 2y = 1 \end{array}$ is not.

h. True, A has 3 rows, so there are at most 3 leading 1s.

1.2.14 b. Since one of $b - a$ and $c - a$ is nonzero, then
$$\left[\begin{array}{ccc} 1 & a & b+c \\ 1 & b & c+a \\ 1 & b & c+a \end{array}\right] \to \left[\begin{array}{ccc} 1 & a & b+c \\ 0 & b-a & a-b \\ 0 & c-a & a-c \end{array}\right] \to$$
$$\left[\begin{array}{ccc} 1 & a & b+c \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc} 1 & 0 & b+c+a \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right]$$

1.2.16 b. $x^2 + y^2 - 2x + 6y - 6 = 0$

1.2.18 $\frac{5}{20}$ in A, $\frac{7}{20}$ in B, $\frac{8}{20}$ in C.

Section 1.3

1.3.1 b. False. $A = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$

d. False. $A = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right]$

f. False. $A = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$

h. False. $A = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

1.3.2 b. $a = -3$, $x = 9t$, $y = -5t$, $z = t$

d. $a = 1$, $x = -t$, $y = t$, $z = 0$; or $a = -1$, $x = t$, $y = 0$, $z = t$

1.3.3 b. Not a linear combination.

d. $\mathbf{v} = \mathbf{x} + 2\mathbf{y} - \mathbf{z}$

1.3.4 b. $\mathbf{y} = 2\mathbf{a}_1 - \mathbf{a}_2 + 4\mathbf{a}_3$.

1.3.5 b. $r\left[\begin{array}{r} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{array}\right] + s\left[\begin{array}{r} -2 \\ 0 \\ -1 \\ 1 \\ 0 \end{array}\right] + t\left[\begin{array}{r} -3 \\ 0 \\ -2 \\ 0 \\ 1 \end{array}\right]$

d. $s\left[\begin{array}{r} 0 \\ 2 \\ 1 \\ 0 \\ 0 \end{array}\right] + t\left[\begin{array}{r} -1 \\ 3 \\ 0 \\ 1 \\ 0 \end{array}\right]$

1.3.6 b. The system in (a) has nontrivial solutions.

1.3.7 b. By Theorem 1.2.2, there are $n - r = 6 - 1 = 5$ parameters and thus infinitely many solutions.

d. If R is the row-echelon form of A, then R has a row of zeros and 4 rows in all. Hence R has $r = \text{rank } A = 1$, 2, or 3. Thus there are $n - r = 6 - r = 5$, 4, or 3 parameters and thus infinitely many solutions.

1.3.9 b. That the graph of $ax + by + cz = d$ contains three points leads to 3 linear equations homogeneous in variables a, b, c, and d. Apply Theorem 1.3.1.

1.3.11 There are $n - r$ parameters (Theorem 1.2.2), so there are nontrivial solutions if and only if $n - r > 0$.

Section 1.4

1.4.1 b. $\begin{aligned} f_1 &= 85 - f_4 - f_7 \\ f_2 &= 60 - f_4 - f_7 \\ f_3 &= -75 + f_4 + f_6 \\ f_5 &= 40 - f_6 - f_7 \end{aligned}$

f_4, f_6, f_7 parameters

1.4.2 b. $f_5 = 15$
$25 \leq f_4 \leq 30$

1.4.3 b. CD

Section 1.5

1.5.2 $I_1 = -\frac{1}{5}$, $I_2 = \frac{3}{5}$, $I_3 = \frac{4}{5}$

1.5.4 $I_1 = 2$, $I_2 = 1$, $I_3 = \frac{1}{2}$, $I_4 = \frac{3}{2}$, $I_5 = \frac{3}{2}$, $I_6 = \frac{1}{2}$

Section 1.6

1.6.2 $2NH_3 + 3CuO \to N_2 + 3Cu + 3H_2O$

1.6.4 $15Pb(N_3)_2 + 44Cr(MnO_4)_2 \to$
$22Cr_2O_3 + 88MnO_2 + 5Pb_3O_4 + 90NO$

Supplementary Exercises for Chapter 1

Supplementary Exercise 1.1. b. No. If the corresponding planes are parallel and distinct, there is no solution. Otherwise they either coincide or have a whole common line of solutions, that is, at least one parameter.

Supplementary Exercise 1.2. b. $x_1 = \frac{1}{10}(-6s-6t+16)$, $x_2 = \frac{1}{10}(4s-t+1)$, $x_3 = s$, $x_4 = t$

Supplementary Exercise b.. b. If $a = 1$, no solution. If $a = 2$, $x = 2-2t$, $y = -t$, $z = t$. If $a \neq 1$ and $a \neq 2$, the unique solution is $x = \frac{8-5a}{3(a-1)}$, $y = \frac{-2-a}{3(a-1)}$, $z = \frac{a+2}{3}$

Supplementary Exercise 1.4. $\left[\begin{array}{c} R_1 \\ R_2 \end{array}\right] \to$
$\left[\begin{array}{c} R_1+R_2 \\ R_2 \end{array}\right] \to \left[\begin{array}{c} R_1+R_2 \\ -R_1 \end{array}\right] \to \left[\begin{array}{c} R_2 \\ -R_1 \end{array}\right] \to \left[\begin{array}{c} R_2 \\ R_1 \end{array}\right]$

Supplementary Exercise 1.6. $a = 1$, $b = 2$, $c = -1$

Supplementary Exercise 1.8. The (real) solution is $x = 2$, $y = 3-t$, $z = t$ where t is a parameter. The given complex solution occurs when $t = 3-i$ is complex. If the real system has a unique solution, that solution is real because the coefficients and constants are all real.

Supplementary Exercise 1.9. b. 5 of brand 1, 0 of brand 2, 3 of brand 3

Section 2.1

2.1.1 b. $(a\ b\ c\ d) = (-2, -4, -6, 0) + t(1, 1, 1, 1)$, t arbitrary

d. $a = b = c = d = t$, t arbitrary

2.1.2 b. $\left[\begin{array}{c} -14 \\ -20 \end{array}\right]$

d. $(-12, 4, -12)$

f. $\left[\begin{array}{rrr} 0 & 1 & -2 \\ -1 & 0 & 4 \\ 2 & -4 & 0 \end{array}\right]$

h. $\left[\begin{array}{rr} 4 & -1 \\ -1 & -6 \end{array}\right]$

2.1.3 b. $\left[\begin{array}{rr} 15 & -5 \\ 10 & 0 \end{array}\right]$

d. Impossible

f. $\left[\begin{array}{rr} 5 & 2 \\ 0 & -1 \end{array}\right]$

h. Impossible

2.1.4 b. $\left[\begin{array}{c} 2 \\ -\frac{1}{2} \end{array}\right]$

2.1.5 b. $A = -\frac{11}{3}B$

2.1.6 b. $X = 4A-3B$, $Y = 4B-5A$

2.1.7 b. $Y = (s, t)$, $X = \frac{1}{2}(1+5s, 2+5t)$; s and t arbitrary

2.1.8 b. $20A-7B+2C$

2.1.9 b. If $A = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$, then $(p, q, r, s) = \frac{1}{2}(2d, a+b-c-d, a-b+c-d, -a+b+c+d)$.

2.1.11 b. If $A + A' = 0$ then $-A = -A+0 = -A+(A+A') = (-A+A)+A' = 0+A' = A'$

2.1.13 b. Write $A = \text{diag}(a_1, \dots, a_n)$, where $a_1, \dots, a_n$ are the main diagonal entries. If $B = \text{diag}(b_1, \dots, b_n)$ then $kA = \text{diag}(ka_1, \dots, ka_n)$.

2.1.14 b. $s = 1$ or $t = 0$

d. $s = 0$, and $t = 3$

2.1.15 b. $\left[\begin{array}{rr} 2 & 0 \\ 1 & -1 \end{array}\right]$

d. $\left[\begin{array}{rr} 2 & 7 \\ -\frac{9}{2} & -5 \end{array}\right]$

2.1.16 b. $A = A^T$, so using Theorem 2.1.2, $(kA)^T = kA^T = kA$.

2.1.19 b. False. Take $B = -A$ for any $A \neq 0$.

d. True. Transposing fixes the main diagonal.

f. True.
$(kA+mB)^T = (kA)^T + (mB)^T = kA^T + mB^T = kA + mB$

2.1.20 c. Suppose $A = S+W$, where $S = S^T$ and $W = -W^T$. Then $A^T = S^T + W^T = S - W$, so $A + A^T = 2S$ and $A - A^T = 2W$. Hence $S = \frac{1}{2}(A+A^T)$ and $W = \frac{1}{2}(A-A^T)$ are uniquely determined by A.

2.1.22 b. If $A = [a_{ij}]$ then $(kp)A = [(kp)a_{ij}] = [k(pa_{ij})] = k[pa_{ij}] = k(pA)$.

Section 2.2

2.2.1 b.
$$\begin{aligned} x_1 - 3x_2 - 3x_3 + 3x_4 &= 5 \\ 8x_2 + 2x_4 &= 1 \\ x_1 + 2x_2 + 2x_3 &= 2 \\ x_2 + 2x_3 - 5x_4 &= 0 \end{aligned}$$

2.2.2 $x_1 \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 0 \\ -2 \\ -4 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 7 \\ 9 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ 8 \\ 12 \end{bmatrix}$

2.2.3 b. $A\mathbf{x} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -4 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 + 3x_3 \\ -4x_2 + 5x_3 \end{bmatrix}$

d. $A\mathbf{x} = \begin{bmatrix} 3 & -4 & 1 & 6 \\ 0 & 2 & 1 & 5 \\ -8 & 7 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 \begin{bmatrix} 3 \\ 0 \\ -8 \end{bmatrix} + x_2 \begin{bmatrix} -4 \\ 2 \\ 7 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} + x_4 \begin{bmatrix} 6 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 3x_1 - 4x_2 + x_3 + 6x_4 \\ 2x_2 + x_3 + 5x_4 \\ -8x_1 + 7x_2 - 3x_3 \end{bmatrix}$

2.2.4 b. To solve $A\mathbf{x} = \mathbf{b}$ the reduction is $\left[\begin{array}{rrrr|r} 1 & 3 & 2 & 0 & 4 \\ 1 & 0 & -1 & -3 & 1 \\ -1 & 2 & 3 & 5 & 1 \end{array}\right] \to \left[\begin{array}{rrrr|r} 1 & 0 & -1 & -3 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ so the general solution is $\begin{bmatrix} 1+s+3t \\ 1-s-t \\ s \\ t \end{bmatrix}$.

Hence $(1+s+3t)\mathbf{a}_1 + (1-s-t)\mathbf{a}_2 + s\mathbf{a}_3 + t\mathbf{a}_4 = \mathbf{b}$ for any choice of s and t. If $s = t = 0$, we get $\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{b}$; if $s = 1$ and $t = 0$, we have $2\mathbf{a}_1 + \mathbf{a}_3 = \mathbf{b}$.

2.2.5 b. $\begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$

d. $\begin{bmatrix} 3 \\ -9 \\ -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 4 \\ 1 \\ 1 \end{bmatrix}$

2.2.6 We have $A\mathbf{x}_0 = \mathbf{0}$ and $A\mathbf{x}_1 = \mathbf{0}$ and so $A(s\mathbf{x}_0 + t\mathbf{x}_1) = s(A\mathbf{x}_0) + t(A\mathbf{x}_1) = s \cdot \mathbf{0} + t \cdot \mathbf{0} = \mathbf{0}$.

2.2.8 b. $\mathbf{x} = \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \left(s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -5 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right)$.

2.2.10 b. False. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

d. True. The linear combination $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ equals $A\mathbf{x}$ where $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$ by Theorem 2.2.1.

f. False. If $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 0 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, then $A\mathbf{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \neq s \begin{bmatrix} 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ for any s and t.

h. False. If $A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \end{bmatrix}$, there is a solution for $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ but not for $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

2.2.11 b. Here $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

d. Here $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

2.2.13 b. Here
$$T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$
so the matrix is $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

2.2.16 Write $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ in terms of its columns. If $\mathbf{b} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$ where the x_i are scalars, then $A\mathbf{x} = \mathbf{b}$ by Theorem 2.2.1 where $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^T$. That is, $\mathbf{x}$ is a solution to the system $A\mathbf{x} = \mathbf{b}$.

2.2.18 b. By Theorem 2.2.3, $A(t\mathbf{x}_1) = t(A\mathbf{x}_1) = t \cdot \mathbf{0} = \mathbf{0}$; that is, $t\mathbf{x}_1$ is a solution to $A\mathbf{x} = \mathbf{0}$.

2.2.22 If A is $m \times n$ and $\mathbf{x}$ and $\mathbf{y}$ are n-vectors, we must show that $A(\mathbf{x}+\mathbf{y}) = A\mathbf{x}+A\mathbf{y}$. Denote the columns of A by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and write $\mathbf{x} = \left[\begin{array}{cccc} x_1 & x_2 & \cdots & x_n \end{array}\right]^T$ and $\mathbf{y} = \left[\begin{array}{cccc} y_1 & y_2 & \cdots & y_n \end{array}\right]^T$. Then $\mathbf{x}+\mathbf{y} = \left[\begin{array}{cccc} x_1+y_1 & x_2+y_2 & \cdots & x_n+y_n \end{array}\right]^T$, so Definition 2.1 and Theorem 2.1.1 give $A(\mathbf{x}+\mathbf{y}) = (x_1+y_1)\mathbf{a}_1 + (x_2+y_2)\mathbf{a}_2 + \cdots + (x_n+y_n)\mathbf{a}_n = (x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n) + (y_1\mathbf{a}_1 + y_2\mathbf{a}_2 + \cdots + y_n\mathbf{a}_n) = A\mathbf{x}+A\mathbf{y}$.

Section 2.3

2.3.1 b. $\left[\begin{array}{rrr} -1 & -6 & -2 \\ 0 & 6 & 10 \end{array}\right]$

d. $\left[\begin{array}{rr} -3 & -15 \end{array}\right]$

f. $[-23]$

h. $\left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array}\right]$

j. $\left[\begin{array}{ccc} aa' & 0 & 0 \\ 0 & bb' & 0 \\ 0 & 0 & cc' \end{array}\right]$

2.3.2 b. $BA = \left[\begin{array}{rrr} -1 & 4 & -10 \\ 1 & 2 & 4 \end{array}\right]$, $B^2 = \left[\begin{array}{rr} 7 & -6 \\ -1 & 6 \end{array}\right]$, $CB = \left[\begin{array}{rr} -2 & 12 \\ 2 & -6 \\ 1 & 6 \end{array}\right]$

$AC = \left[\begin{array}{rr} 4 & 10 \\ -2 & -1 \end{array}\right]$, $CA = \left[\begin{array}{rrr} 2 & 4 & 8 \\ -1 & -1 & -5 \\ 1 & 4 & 2 \end{array}\right]$

2.3.3 b. $(a, b, a_1, b_1) = (3, 0, 1, 2)$

2.3.4 b. $A^2 - A - 6I = \left[\begin{array}{rr} 8 & 2 \\ 2 & 5 \end{array}\right] - \left[\begin{array}{rr} 2 & 2 \\ 2 & -1 \end{array}\right] - \left[\begin{array}{rr} 6 & 0 \\ 0 & 6 \end{array}\right] = \left[\begin{array}{rr} 0 & 0 \\ 0 & 0 \end{array}\right]$

2.3.5 b. $A(BC) = \left[\begin{array}{rr} 1 & -1 \\ 0 & 1 \end{array}\right] \left[\begin{array}{rr} -9 & -16 \\ 5 & 1 \end{array}\right] = \left[\begin{array}{rr} -14 & -17 \\ 5 & 1 \end{array}\right] = \left[\begin{array}{rrr} -2 & -1 & -2 \\ 3 & 1 & 0 \end{array}\right] \left[\begin{array}{rr} 1 & 0 \\ 2 & 1 \\ 5 & 8 \end{array}\right] = (AB)C$

2.3.6 b. If $A = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$ and $E = \left[\begin{array}{rr} 0 & 0 \\ 1 & 0 \end{array}\right]$, compare entries an AE and EA.

2.3.7 b. $m \times n$ and $n \times m$ for some m and n

2.3.8 b. i. $\left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array}\right]$, $\left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array}\right]$, $\left[\begin{array}{rr} 1 & 1 \\ 0 & -1 \end{array}\right]$

ii. $\left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array}\right]$, $\left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array}\right]$, $\left[\begin{array}{rr} 1 & 1 \\ 0 & 0 \end{array}\right]$

2.3.12 b. $A^{2k} = \left[\begin{array}{cc|cc} 1 & -2k & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$ for $k = 0, 1, 2, \dots$,

$A^{2k+1} = A^{2k}A = \left[\begin{array}{cc|cc} 1 & -(2k+1) & 2 & -1 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$ for $k = 0, 1, 2, \dots$

2.3.13 b. $\left[\begin{array}{cc} I & 0 \\ 0 & I \end{array}\right] = I_{2k}$

d. 0_k

f. $\left[\begin{array}{cc} X^m & 0 \\ 0 & X^m \end{array}\right]$ if $n = 2m$; $\left[\begin{array}{cc} 0 & X^{m+1} \\ X^m & 0 \end{array}\right]$ if $n = 2m+1$

2.3.14 b. If Y is row i of the identity matrix I, then YA is row i of $IA = A$.

2.3.16 b. $AB - BA$

d. 0

2.3.18 b. $(kA)C = k(AC) = k(CA) = C(kA)$

2.3.20 We have $A^T = A$ and $B^T = B$, so $(AB)^T = B^TA^T = BA$. Hence AB is symmetric if and only if $AB = BA$.

2.3.22 b. $A = 0$

2.3.24 If $BC = I$, then $AB = 0$ gives $0 = 0C = (AB)C = A(BC) = AI = A$, contrary to the assumption that $A \neq 0$.

2.3.26 3 paths $v_1 \to v_4$, 0 paths $v_2 \to v_3$

2.3.27 b. False. If $A = \left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array}\right] = J$, then $AJ = A$ but $J \neq I$.

d. True. Since $A^T = A$, we have $(I + AT = I^T + A^T = I + A$.

f. False. If $A = \left[\begin{array}{rr} 0 & 1 \\ 0 & 0 \end{array}\right]$, then $A \neq 0$ but $A^2 = 0$.

h. True. We have $A(A+B) = (A+B)A$; that is, $A^2 + AB = A^2 + BA$. Subtracting A^2 gives $AB = BA$.

j. False. $A = \left[\begin{array}{rr} 1 & -2 \\ 2 & 4 \end{array}\right]$, $B = \left[\begin{array}{rr} 2 & 4 \\ 1 & 2 \end{array}\right]$

l. False. See **(j)**.

2.3.28 b. If $A = [a_{ij}]$ and $B = [b_{ij}]$ and $\sum_j a_{ij} = 1 = \sum_j b_{ij}$, then the $(i,\ j)$-entry of AB is $c_{ij} = \sum_k a_{ik}b_{kj}$, whence $\sum_j c_{ij} = \sum_j \sum_k a_{ik}b_{kj} = \sum_k a_{ik}(\sum_j b_{kj}) = \sum_k a_{ik} = 1$. Alternatively: If $\mathbf{e} = (1,\ 1,\ \dots,\ 1)$, then the rows of A sum to 1 if and only if $A\mathbf{e} = \mathbf{e}$. If also $B\mathbf{e} = \mathbf{e}$ then $(AB)\mathbf{e} = A(B\mathbf{e}) = A\mathbf{e} = \mathbf{e}$.

2.3.30 b. If $A = [a_{ij}]$, then $\operatorname{tr}(kA) = \operatorname{tr}[ka_{ij}] = \sum_{i=1}^{n} ka_{ii} = k\sum_{i=1}^{n} a_{ii} = k\operatorname{tr}(A)$.

e. Write $A^T = \left[a'_{ij}\right]$, where $a'_{ij} = a_{ji}$. Then $AA^T = \left(\sum_{k=1}^{n} a_{ik}a'_{kj}\right)$, so $\operatorname{tr}(AA^T) = \sum_{i=1}^{n}\left[\sum_{k=1}^{n} a_{ik}a'_{ki}\right] = \sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}^2$.

2.3.32 e. Observe that $PQ = P^2 + PAP - P^2AP = P$, so $Q^2 = PQ + APQ - PAPQ = P + AP - PAP = Q$.

2.3.34 b. $(A+B)(A-B) = A^2 - AB + BA - B^2$, and $(A-B)(A+B) = A^2 + AB - BA - B^2$. These are equal if and only if $-AB + BA = AB - BA$; that is, $2BA = 2AB$; that is, $BA = AB$.

2.3.35 b. $(A+B)(A-B) = A^2 - AB + BA - B^2$ and $(A-B)(A+B) = A^2 - BA + AB - B^2$. These are equal if and only if $-AB + BA = -BA + AB$, that is $2AB = 2BA$, that is $AB = BA$.

Section 2.4

2.4.2 b. $\frac{1}{5}\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$

d. $\begin{bmatrix} 2 & -1 & 3 \\ 3 & 1 & -1 \\ 1 & 1 & -2 \end{bmatrix}$

f. $\frac{1}{10}\begin{bmatrix} 1 & 4 & -1 \\ -2 & 2 & 2 \\ -9 & 14 & -1 \end{bmatrix}$

h. $\frac{1}{4}\begin{bmatrix} 2 & 0 & -2 \\ -5 & 2 & 5 \\ -3 & 2 & -1 \end{bmatrix}$

j. $\begin{bmatrix} 0 & 0 & 1 & -2 \\ -1 & -2 & -1 & -3 \\ 1 & 2 & 1 & 2 \\ 0 & -1 & 0 & 0 \end{bmatrix}$

l. $\begin{bmatrix} 1 & -2 & 6 & -30 & 210 \\ 0 & 1 & -3 & 15 & -105 \\ 0 & 0 & 1 & -5 & 35 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

2.4.3 b. $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} -3 \\ -2 \end{bmatrix}$

d. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 9 & -14 & 6 \\ 4 & -4 & 1 \\ -10 & 15 & -5 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 23 \\ 8 \\ -25 \end{bmatrix}$

2.4.4 b. $B = A^{-1}AB = \begin{bmatrix} 4 & -2 & 1 \\ 7 & -2 & 4 \\ -1 & 2 & -1 \end{bmatrix}$

2.4.5 b. $\frac{1}{10}\begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$

d. $\frac{1}{2}\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$

f. $\frac{1}{2}\begin{bmatrix} 2 & 0 \\ -6 & 1 \end{bmatrix}$

h. $-\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

2.4.6 b. $A = \frac{1}{2}\begin{bmatrix} 2 & -1 & 3 \\ 0 & 1 & -1 \\ -2 & 1 & -1 \end{bmatrix}$

2.4.8 b. A and B are inverses.

2.4.9 b. False. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

d. True. $A^{-1} = \frac{1}{3}A^3$

f. False. $A = B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

h. True. If $(A^2)B = I$, then $A(AB) = I$; use Theorem 2.4.5.

2.4.10 b. $(C^T)^{-1} = (C^{-1})^T = A^T$ because $C^{-1} = (A^{-1})^{-1} = A$.

2.4.11 b. (i) Inconsistent.

(ii) $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

2.4.15 b. $B^4 = I$, so $B^{-1} = B^3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

2.4.16 $\begin{bmatrix} c^2-2 & -c & 1 \\ -c & 1 & 0 \\ 3-c^2 & c & -1 \end{bmatrix}$

2.4.18 b. If column j of A is zero, $A\mathbf{y} = \mathbf{0}$ where $\mathbf{y}$ is column j of the identity matrix. Use Theorem 2.4.5.

d. If each column of A sums to 0, $XA = 0$ where X is the row of 1s. Hence $A^T X^T = 0$ so A has no inverse by Theorem 2.4.5 ($X^T \neq 0$).

2.4.19 b. (ii) $(-1,\ 1,\ 1)A = 0$

2.4.20 b. Each power A^k is invertible by Theorem 2.4.4 (because A is invertible). Hence A^k cannot be 0.

2.4.21 b. By (a), if one has an inverse the other is zero and so has no inverse.

2.4.22 If $A = \left[\begin{array}{cc} a & 0 \\ 0 & 1 \end{array}\right]$, $a > 1$, then $A^{-1} = \left[\begin{array}{cc} \frac{1}{a} & 0 \\ 0 & 1 \end{array}\right]$ is an x-compression because $\frac{1}{a} < 1$.

2.4.24 b. $A^{-1} = \frac{1}{4}(A^3 + 2A^2 - 1)$

2.4.25 b. If $B\mathbf{x} = \mathbf{0}$, then $(AB)\mathbf{x} = (A)B\mathbf{x} = \mathbf{0}$, so $\mathbf{x} = \mathbf{0}$ because AB is invertible. Hence B is invertible by Theorem 2.4.5. But then $A = (AB)B^{-1}$ is invertible by Theorem 2.4.4.

2.4.26 b. $\left[\begin{array}{rr|r} 2 & -1 & 0 \\ -5 & 3 & 0 \\ \hline -13 & 8 & -1 \end{array}\right]$

d. $\left[\begin{array}{rr|rr} 1 & -1 & -14 & 8 \\ -1 & 2 & 16 & -9 \\ \hline 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right]$

2.4.28 d. If $A^n = 0$, $(I-A)^{-1} = I + A + \cdots + A^{n-1}$.

2.4.30 b. $A[B(AB)^{-1}] = I = [(BA)^{-1}B]A$, so A is invertible by Exercise 2.4.10.

2.4.32 a. Have $AC = CA$. Left-multiply by A^{-1} to get $C = A^{-1}CA$. Then right-multiply by A^{-1} to get $CA^{-1} = A^{-1}C$.

2.4.33 b. Given $ABAB = AABB$. Left multiply by A^{-1}, then right multiply by B^{-1}.

2.4.34 If $B\mathbf{x} = \mathbf{0}$ where $\mathbf{x}$ is $n \times 1$, then $AB\mathbf{x} = \mathbf{0}$ so $\mathbf{x} = \mathbf{0}$ as AB is invertible. Hence B is invertible by Theorem 2.4.5, so $A = (AB)B^{-1}$ is invertible.

2.4.35 b. $B\left[\begin{array}{r} -1 \\ 3 \\ -1 \end{array}\right] = 0$ so B is not invertible by Theorem 2.4.5.

2.4.38 b. Write $U = I_n - 2XX^T$. Then $U^T = I_n^T - 2X^{TT}X^T = U$, and $U^2 = I_n^2 - (2XX^T)I_n - I_n(2XX^T) + 4(XX^T)(XX^T) = I_n - 4XX^T + 4XX^T = I_n$.

2.4.39 b. $(I-2P)^2 = I - 4P + 4P^2$, and this equals I if and only if $P^2 = P$.

2.4.41 b. $(A^{-1} + B^{-1})^{-1} = B(A+B)^{-1}A$

Section 2.5

2.5.1 b. Interchange rows 1 and 3 of I. $E^{-1} = E$.

d. Add (-2) times row 1 of I to row 2.
$E^{-1} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$

f. Multiply row 3 of I by 5. $E^{-1} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{5} \end{array}\right]$

2.5.2 b. $\left[\begin{array}{rr} -1 & 0 \\ 0 & 1 \end{array}\right]$

d. $\left[\begin{array}{rr} 1 & -1 \\ 0 & 1 \end{array}\right]$

f. $\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$

2.5.3 b. The only possibilities for E are $\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$, $\left[\begin{array}{cc} k & 0 \\ 0 & 1 \end{array}\right]$, $\left[\begin{array}{cc} 1 & 0 \\ 0 & k \end{array}\right]$, $\left[\begin{array}{cc} 1 & k \\ 0 & 1 \end{array}\right]$, and $\left[\begin{array}{cc} 1 & 0 \\ k & 1 \end{array}\right]$. In each case, EA has a row different from C.

2.5.5 b. No, 0 is not invertible.

2.5.6 b. $\left[\begin{array}{rr} 1 & -2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & \frac{1}{2} \end{array}\right]\left[\begin{array}{rr} 1 & 0 \\ -5 & 1 \end{array}\right]$
$A = \left[\begin{array}{rrr} 1 & 0 & 7 \\ 0 & 1 & -3 \end{array}\right]$. Alternatively,
$\left[\begin{array}{cc} 1 & 0 \\ 0 & \frac{1}{2} \end{array}\right]\left[\begin{array}{rr} 1 & -1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{rr} 1 & 0 \\ -5 & 1 \end{array}\right]$
$A = \left[\begin{array}{rrr} 1 & 0 & 7 \\ 0 & 1 & -3 \end{array}\right]$.

d. $\left[\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & 1 \end{array}\right]\left[\begin{array}{rrr} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{array}\right]$
$\left[\begin{array}{rrr} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{array}\right]\left[\begin{array}{rrr} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] A = \left[\begin{array}{cccc} 1 & 0 & \frac{1}{5} & \frac{1}{5} \\ 0 & 1 & -\frac{7}{5} & -\frac{2}{5} \\ 0 & 0 & 0 & 0 \end{array}\right]$

2.5.7 b. $U = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

2.5.8 b. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

2.5.10 $UA = R$ by Theorem 2.5.1, so $A = U^{-1}R$.

2.5.12 b. $U = A^{-1}$, $V = I^2$; rank $A = 2$

d. $U = \begin{bmatrix} -2 & 1 & 0 \\ 3 & -1 & 0 \\ 2 & -1 & 1 \end{bmatrix}$,
$V = \begin{bmatrix} 1 & 0 & -1 & -3 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; rank $A = 2$

2.5.16 Write $U^{-1} = E_k E_{k-1} \cdots E_2 E_1$, E_i elementary. Then $\begin{bmatrix} I & U^{-1}A \end{bmatrix} = \begin{bmatrix} U^{-1}U & U^{-1}A \end{bmatrix}$ $= U^{-1}\begin{bmatrix} U & A \end{bmatrix} = E_k E_{k-1} \cdots E_2 E_1 \begin{bmatrix} U & A \end{bmatrix}$. So $\begin{bmatrix} U & A \end{bmatrix} \to \begin{bmatrix} I & U^{-1}A \end{bmatrix}$ by row operations (Lemma 2.5.1).

2.5.17 b. (i) $A \overset{r}{\sim} A$ because $A = IA$. (ii) If $A \overset{r}{\sim} B$, then $A = UB$, U invertible, so $B = U^{-1}A$. Thus $B \overset{r}{\sim} A$. (iii) If $A \overset{r}{\sim} B$ and $B \overset{r}{\sim} C$, then $A = UB$ and $B = VC$, U and V invertible. Hence $A = U(VC) = (UV)C$, so $A \overset{r}{\sim} C$.

2.5.19 b. If $B \overset{r}{\sim} A$, let $B = UA$, U invertible. If $U = \begin{bmatrix} d & b \\ -b & d \end{bmatrix}$, $B = UA = \begin{bmatrix} 0 & 0 & b \\ 0 & 0 & d \end{bmatrix}$ where b and d are not both zero (as U is invertible). Every such matrix B arises in this way: Use $U = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$–it is invertible by Example 2.3.5.

2.5.22 b. Multiply column i by $1/k$.

Section 2.6

2.6.1 b. $\begin{bmatrix} 5 \\ 6 \\ -13 \end{bmatrix} = 3\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$, so
$T\begin{bmatrix} 5 \\ 6 \\ -13 \end{bmatrix} = 3T\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} - 2T\begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} =$
$3\begin{bmatrix} 3 \\ 5 \end{bmatrix} - 2\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \end{bmatrix}$

2.6.2 b. As in 1(b), $T\begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -9 \end{bmatrix}$.

2.6.3 b. $T(\mathbf{e}_1) = -\mathbf{e}_2$ and $T(\mathbf{e}_2) = -\mathbf{e}_1$. So $A\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} = \begin{bmatrix} -\mathbf{e}_2 & -\mathbf{e}_1 \end{bmatrix} =$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$.

d. $T(\mathbf{e}_1) = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$ and $T(\mathbf{e}_2) = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$

So $A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} = \frac{\sqrt{2}}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

2.6.4 b. $T(\mathbf{e}_1) = -\mathbf{e}_1$, $T(\mathbf{e}_2) = \mathbf{e}_2$ and $T(\mathbf{e}_3) = \mathbf{e}_3$. Hence Theorem 2.6.2 gives $A\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{bmatrix} = \begin{bmatrix} -\mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \end{bmatrix} =$ $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

2.6.5 b. We have $\mathbf{y}_1 = T(\mathbf{x}_1)$ for some $\mathbf{x}_1$ in $\mathbb{R}^n$, and $\mathbf{y}_2 = T(\mathbf{x}_2)$ for some $\mathbf{x}_2$ in $\mathbb{R}^n$. So $a\mathbf{y}_1 + b\mathbf{y}_2 = aT(\mathbf{x}_1) + bT(\mathbf{x}_2) = T(a\mathbf{x}_1 + b\mathbf{x}_2)$. Hence $a\mathbf{y}_1 + b\mathbf{y}_2$ is also in the image of T.

2.6.7 b. $T\left(2\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \neq 2\begin{bmatrix} 0 \\ -1 \end{bmatrix}$.

2.6.8 b. $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, rotation through $\theta = -\frac{\pi}{4}$.

d. $A = \frac{1}{10}\begin{bmatrix} -8 & -6 \\ -6 & 8 \end{bmatrix}$, reflection in the line $y = -3x$.

2.6.10 b. $\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$

2.6.12 b. Reflection in the y axis

d. Reflection in $y = x$

f. Rotation through $\frac{\pi}{2}$

2.6.13 b. $T(\mathbf{x}) = aR(\mathbf{x}) = a(A\mathbf{x}) = (aA)\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}$. Hence T is induced by aA.

2.6.14 b. If $\mathbf{x}$ is in $\mathbb{R}^n$, then $T(-\mathbf{x}) = T[(-1)\mathbf{x}] = (-1)T(\mathbf{x}) = -T(\mathbf{x})$.

2.6.17 b. If $B^2 = I$ then
$T^2(\mathbf{x}) = T[T(\mathbf{x})] = B(B\mathbf{x}) = B^2\mathbf{x} = I\mathbf{x} = \mathbf{x} = 1_{\mathbb{R}^2}(\mathbf{x})$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Hence $T^2 = 1_{\mathbb{R}^2}$. If $T^2 = 1_{\mathbb{R}^2}$, then $B^2\mathbf{x} = T^2(\mathbf{x}) = 1_{\mathbb{R}^2}(\mathbf{x}) = \mathbf{x} = I\mathbf{x}$ for all $\mathbf{x}$, so $B^2 = I$ by Theorem 2.2.6.

2.6.18 b. The matrix of $Q_1 \circ Q_0$ is
$\left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array}\right] = \left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array}\right]$, which is the matrix of $R_{\frac{\pi}{2}}$.

d. The matrix of $Q_0 \circ R_{\frac{\pi}{2}}$ is
$\left[\begin{array}{rr} 1 & 0 \\ 0 & -1 \end{array}\right]\left[\begin{array}{rr} 0 & -1 \\ 1 & 0 \end{array}\right] = \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array}\right]$, which is the matrix of Q_{-1}.

2.6.20 We have

$$T(\mathbf{x}) = x_1 + x_2 + \cdots + x_n = \left[\begin{array}{cccc} 1 & 1 & \cdots & 1 \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]$$, so T is the matrix transformation induced by the matrix $A = \left[\begin{array}{cccc} 1 & 1 & \cdots & 1 \end{array}\right]$. In particular, T is linear. On the other hand, we can use Theorem 2.6.2 to get A, but to do this we must first show directly that T is linear. If we write

$$\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right] \text{ and } \mathbf{y} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_n \end{array}\right]$$. Then

$$\begin{aligned} T(\mathbf{x}+\mathbf{y}) &= T\left[\begin{array}{c} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{array}\right] \\ &= (x_1 + y_1) + (x_2 + y_2) + \cdots + (x_n + y_n) \\ &= (x_1 + x_2 + \cdots + x_n) + (y_1 + y_2 + \cdots + y_n) \\ &= T(\mathbf{x}) + T(\mathbf{y}) \end{aligned}$$

Similarly, $T(a\mathbf{x}) = aT(\mathbf{x})$ for any scalar a, so T is linear. By Theorem 2.6.2, T has matrix $A = \left[\begin{array}{cccc} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{array}\right] = \left[\begin{array}{cccc} 1 & 1 & \cdots & 1 \end{array}\right]$, as before.

2.6.22 b. If $T : \mathbb{R}^n \to \mathbb{R}$ is linear, write $T(\mathbf{e}_j) = w_j$ for each $j = 1, 2, \dots, n$ where $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$. Since $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$, Theorem 2.6.1 gives

$$\begin{aligned} T(\mathbf{x}) &= T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n) \\ &= x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n) \\ &= x_1w_1 + x_2w_2 + \cdots + x_nw_n \\ &= \mathbf{w}\cdot\mathbf{x} = T_{\mathbf{w}}(\mathbf{x}) \end{aligned}$$

where $\mathbf{w} = \left[\begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_n \end{array}\right]$. Since this holds for all $\mathbf{x}$ in $\mathbb{R}^n$, it shows that $T = T_{\mathbf{W}}$. This also follows from Theorem 2.6.2, but we have first to verify that T is linear. (This comes to showing that $\mathbf{w}\cdot(\mathbf{x}+\mathbf{y}) = \mathbf{w}\cdot\mathbf{s} + \mathbf{w}\cdot\mathbf{y}$ and $\mathbf{w}\cdot(a\mathbf{x}) = a(\mathbf{w}\cdot\mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ and all a in $\mathbb{R}$.) Then T has matrix $A = \left[\begin{array}{cccc} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{array}\right] = \left[\begin{array}{cccc} w_1 & w_2 & \cdots & w_n \end{array}\right]$ by Theorem 2.6.2. Hence if $\mathbf{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]$ in $\mathbb{R}$, then $T(\mathbf{x}) = A\mathbf{x} = \mathbf{w}\cdot\mathbf{x}$, as required.

2.6.23 b. Given $\mathbf{x}$ in $\mathbb{R}$ and a in $\mathbb{R}$, we have

$(S \circ T)(a\mathbf{x})$	$=$	$S[T(a\mathbf{x})]$	Definition of $S \circ T$
	$=$	$S[aT(\mathbf{x})]$	Because T is linear.
	$=$	$a[S[T(\mathbf{x})]]$	Because S is linear.
	$=$	$a[S \circ T(\mathbf{x})]$	Definition of $S \circ T$

Section 2.7

2.7.1 b. $\left[\begin{array}{rrr} 2 & 0 & 0 \\ 1 & -3 & 0 \\ -1 & 9 & 1 \end{array}\right]\left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 \end{array}\right]$

d. $\left[\begin{array}{rrrr} -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & -2 & 0 & 1 \end{array}\right]\left[\begin{array}{rrrrr} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$

f. $\left[\begin{array}{rrrr} 2 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 \\ 3 & -2 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{array}\right]\left[\begin{array}{ccccc} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$

2.7.2 b. $P = \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$

$$\begin{aligned} PA &= \left[\begin{array}{rrr} -1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 4 \end{array}\right] \\ &= \left[\begin{array}{rrr} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{array}\right]\left[\begin{array}{rrr} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right] \end{aligned}$$

d. $P = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right]$

$$PA = \begin{bmatrix} -1 & -2 & 3 & 0 \\ 1 & 1 & -1 & 3 \\ 2 & 5 & -10 & 1 \\ 2 & 4 & -6 & 5 \end{bmatrix}$$
$$= \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 2 & 1 & -2 & 0 \\ 2 & 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.7.3 b. $\mathbf{y} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} -1+2t \\ -t \\ s \\ t \end{bmatrix}$ s and t arbitrary

d. $\mathbf{y} = \begin{bmatrix} 2 \\ 8 \\ -1 \\ 0 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} 8-2t \\ 6-t \\ -1-t \\ t \end{bmatrix}$ t arbitrary

2.7.5 $\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} \to \begin{bmatrix} R_1+R_2 \\ R_2 \end{bmatrix} \to \begin{bmatrix} R_1+R_2 \\ -R_1 \end{bmatrix} \to \begin{bmatrix} R_2 \\ -R_1 \end{bmatrix} \to \begin{bmatrix} R_2 \\ R_1 \end{bmatrix}$

2.7.6 b. Let $A = LU = L_1U_1$ be LU-factorizations of the invertible matrix A. Then U and U_1 have no row of zeros and so (being row-echelon) are upper triangular with 1's on the main diagonal. Thus, using **(a.)**, the diagonal matrix $D = UU_1^{-1}$ has 1's on the main diagonal. Thus $D = I$, $U = U_1$, and $L = L_1$.

2.7.7 If $A = \begin{bmatrix} a & 0 \\ X & A_1 \end{bmatrix}$ and $B = \begin{bmatrix} b & 0 \\ Y & B_1 \end{bmatrix}$ in block form, then $AB = \begin{bmatrix} ab & 0 \\ Xb+A_1Y & A_1B_1 \end{bmatrix}$, and A_1B_1 is lower triangular by induction.

2.7.9 b. Let $A = LU = L_1U_1$ be two such factorizations. Then $UU_1^{-1} = L^{-1}L_1$; write this matrix as $D = UU_1^{-1} = L^{-1}L_1$. Then D is lower triangular (apply Lemma 2.7.1 to $D = L^{-1}L_1$); and D is also upper triangular (consider UU_1^{-1}). Hence D is diagonal, and so $D = I$ because L^{-1} and L_1 are unit triangular. Since $A = LU$; this completes the proof.

Section 2.8

2.8.1 b. $\begin{bmatrix} t \\ 3t \\ t \end{bmatrix}$

d. $\begin{bmatrix} 14t \\ 17t \\ 47t \\ 23t \end{bmatrix}$

2.8.2 $\begin{bmatrix} t \\ t \\ t \end{bmatrix}$

2.8.4 $P = \begin{bmatrix} bt \\ (1-a)t \end{bmatrix}$ is nonzero (for some t) unless $b = 0$ and $a = 1$. In that case, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is a solution. If the entries of E are positive, then $P = \begin{bmatrix} b \\ 1-a \end{bmatrix}$ has positive entries.

2.8.7 b. $\begin{bmatrix} 0.4 & 0.8 \\ 0.7 & 0.2 \end{bmatrix}$

2.8.8 If $E = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $I - E = \begin{bmatrix} 1-a & -b \\ -c & 1-d \end{bmatrix}$, so $\det(I-E) = (1-a)(1-d) - bc = 1 - \operatorname{tr} E + \det E$. If $\det(I-E) \neq 0$, then $(I-E)^{-1} = \frac{1}{\det(I-E)} \begin{bmatrix} 1-d & b \\ c & 1-a \end{bmatrix}$, so $(I-E)^{-1} \geq 0$ if $\det(I-E) > 0$, that is, $\operatorname{tr} E < 1 + \det E$. The converse is now clear.

2.8.9 b. Use $\mathbf{p} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ in Theorem 2.8.2.

d. $\mathbf{p} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$ in Theorem 2.8.2.

Section 2.9

2.9.1 b. Not regular

2.9.2 b. $\frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\frac{3}{8}$

d. $\frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, 0.312

f. $\frac{1}{20}\begin{bmatrix} 5 \\ 7 \\ 8 \end{bmatrix}$, 0.306

2.9.4 b. 50% middle, 25% upper, 25% lower

2.9.6 $\frac{7}{16}$, $\frac{9}{16}$

2.9.8 a. $\frac{7}{75}$

b. He spends most of his time in compartment 3; steady state $\frac{1}{16}\begin{bmatrix} 3 \\ 2 \\ 5 \\ 4 \\ 2 \end{bmatrix}$.

2.9.12 a. Direct verification.

b. Since $0<p<1$ and $0<q<1$ we get $0<p+q<2$ whence $-1<p+q-1<1$. Finally, $-1<1-p-q<1$, so $(1-p-q)^m$ converges to zero as m increases.

Supplementary Exercises for Chapter 2

Supplementary Exercise 2.2. b. $U^{-1}=\frac{1}{4}(U^2-5U+11I)$.

Supplementary Exercise 2.4. b. If $\mathbf{x}_k=\mathbf{x}_m$, then $\mathbf{y}+k(\mathbf{y}-\mathbf{z})=\mathbf{y}+m(\mathbf{y}-\mathbf{z})$. So $(k-m)(\mathbf{y}-\mathbf{z})=\mathbf{0}$. But $\mathbf{y}-\mathbf{z}$ is not zero (because $\mathbf{y}$ and $\mathbf{z}$ are distinct), so $k-m=0$ by Example 2.1.7.

Supplementary Exercise 2.6. d. Using parts **(c)** and **(b)** gives $I_{pq}AI_{rs}=\sum_{i=1}^{n}\sum_{j=1}^{n}a_{ij}I_{pq}I_{ij}I_{rs}$. The only nonzero term occurs when $i=q$ and $j=r$, so $I_{pq}AI_{rs}=a_{qr}I_{ps}$.

Supplementary Exercise 2.7. b. If $A=[a_{ij}]=\sum_{ij}a_{ij}I_{ij}$, then $I_{pq}AI_{rs}=a_{qr}I_{ps}$ by 6(d). But then $a_{qr}I_{ps}=AI_{pq}I_{rs}=0$ if $q\neq r$, so $a_{qr}=0$ if $q\neq r$. If $q=r$, then $a_{qq}I_{ps}=AI_{pq}I_{rs}=AI_{ps}$ is independent of q. Thus $a_{qq}=a_{11}$ for all q.

Section 3.1

3.1.1 b. 0

d. -1

f. -39

h. 0

j. $2abc$

l. 0

n. -56

p. $abcd$

3.1.5 b. -17

d. 106

3.1.6 b. 0

3.1.7 b. 12

3.1.8 b. $\det\begin{bmatrix}2a+p & 2b+q & 2c+r\\ 2p+x & 2q+y & 2r+z\\ 2x+a & 2y+b & 2z+c\end{bmatrix}$

$$=3\det\begin{bmatrix}a+p+x & b+q+y & c+r+z\\ 2p+x & 2q+y & 2r+z\\ 2x+a & 2y+b & 2z+c\end{bmatrix}$$
$$=3\det\begin{bmatrix}a+p+x & b+q+y & c+r+z\\ p-a & q-b & r-c\\ x-p & y-q & z-r\end{bmatrix}$$
$$=3\det\begin{bmatrix}3x & 3y & 3z\\ p-a & q-b & r-c\\ x-p & y-q & z-r\end{bmatrix}\cdots$$

3.1.9 b. False. $A=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$

d. False. $A=\begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}\to R=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

f. False. $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$

h. False. $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

3.1.10 b. 35

3.1.11 b. -6

d. -6

3.1.14 b. $-(x-2)(x^2+2x-12)$

3.1.15 b. -7

3.1.16 b. $\pm\frac{\sqrt{6}}{2}$

d. $x=\pm y$

3.1.21 Let $\mathbf{x}=\begin{bmatrix}x_1\\ x_2\\ \vdots\\ x_n\end{bmatrix}$, $\mathbf{y}=\begin{bmatrix}y_1\\ y_2\\ \vdots\\ y_n\end{bmatrix}$ and $A=\begin{bmatrix}\mathbf{c}_1 & \cdots & \mathbf{x}+\mathbf{y} & \cdots & \mathbf{c}_n\end{bmatrix}$ where $\mathbf{x}+\mathbf{y}$ is in column j. Expanding $\det A$ along column j (the one containing $\mathbf{x}+\mathbf{y}$):

$$\begin{aligned}T(\mathbf{x}+\mathbf{y})=\det A&=\sum_{i=1}^{n}(x_i+y_i)c_{ij}(A)\\ &=\sum_{i=1}^{n}x_ic_{ij}(A)+\sum_{i=1}^{n}y_ic_{ij}(A)\\ &=T(\mathbf{x})+T(\mathbf{y})\end{aligned}$$

Similarly for $T(a\mathbf{x})=aT(\mathbf{x})$.

3.1.24 If A is $n \times n$, then $\det B = (-1)^k \det A$ where $n = 2k$ or $n = 2k+1$.

Section 3.2

3.2.1 b. $\begin{bmatrix} 1 & -1 & -2 \\ -3 & 1 & 6 \\ -3 & 1 & 4 \end{bmatrix}$

d. $\frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} = A$

3.2.2 b. $c \neq 0$

d. any c

f. $c \neq -1$

3.2.3 b. -2

3.2.4 b. 1

3.2.6 b. $\frac{4}{9}$

3.2.7 b. 16

3.2.8 b. $\frac{1}{11}\begin{bmatrix} 5 \\ 21 \end{bmatrix}$

d. $\frac{1}{79}\begin{bmatrix} 12 \\ -37 \\ -2 \end{bmatrix}$

3.2.9 b. $\frac{4}{51}$

3.2.10 b. $\det A = 1, -1$

d. $\det A = 1$

f. $\det A = 0$ if n is odd; nothing can be said if n is even

3.2.15 dA where $d = \det A$

3.2.19 b. $\frac{1}{c}\begin{bmatrix} 1 & 0 & 1 \\ 0 & c & 1 \\ -1 & c & 1 \end{bmatrix}$, $c \neq 0$

d. $\frac{1}{2}\begin{bmatrix} 8-c^2 & -c & c^2-6 \\ c & 1 & -c \\ c^2-10 & c & 8-c^2 \end{bmatrix}$

f. $\frac{1}{c^3+1}\begin{bmatrix} 1-c & c^2+1 & -c-1 \\ c^2 & -c & c+1 \\ -c & 1 & c^2-1 \end{bmatrix}$, $c \neq -1$

3.2.20 b. T. $\det AB = \det A \det B = \det B \det A = \det BA$.

d. T. $\det A \neq 0$ means A^{-1} exists, so $AB = AC$ implies that $B = C$.

f. F. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ then $\operatorname{adj} A = 0$.

h. F. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ then $\operatorname{adj} A = \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$

j. F. If $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ then $\det(I+A) = -1$ but $1 + \det A = 1$.

l. F. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ then $\det A = 1$ but $\operatorname{adj} A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \neq A$

3.2.22 b. $5 - 4x + 2x^2$.

3.2.23 b. $1 - \frac{5}{3}x + \frac{1}{2}x^2 + \frac{7}{6}x^3$

3.2.24 b. $1 - 0.51x + 2.1x^2 - 1.1x^3$; 1.25, so $y = 1.25$

3.2.26 b. Use induction on n where A is $n \times n$. It is clear if $n = 1$. If $n > 1$, write $A = \begin{bmatrix} a & X \\ 0 & B \end{bmatrix}$ in block form where B is $(n-1) \times (n-1)$. Then $A^{-1} = \begin{bmatrix} a^{-1} & -a^{-1}XB^{-1} \\ 0 & B^{-1} \end{bmatrix}$, and this is upper triangular because B is upper triangular by induction.

3.2.28 $-\frac{1}{21}\begin{bmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 3 & 1 & -1 \end{bmatrix}$

3.2.34 b. Have $(\operatorname{adj} A)A = (\det A)I$; so taking inverses, $A^{-1} \cdot (\operatorname{adj} A)^{-1} = \frac{1}{\det A}I$. On the other hand, $A^{-1} \operatorname{adj}(A^{-1}) = \det(A^{-1})I = \frac{1}{\det A}I$. Comparison yields $A^{-1}(\operatorname{adj} A)^{-1} = A^{-1} \operatorname{adj}(A^{-1})$, and part **(b)** follows.

d. Write $\det A = d$, $\det B = e$. By the adjugate formula $AB \operatorname{adj}(AB) = deI$, and $AB \operatorname{adj} B \operatorname{adj} A = A[eI] \operatorname{adj} A = (eI)(dI) = deI$. Done as AB is invertible.

Section 3.3

3.3.1 b. $(x-3)(x+2)$; 3; -2; $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $P = \begin{bmatrix} 4 & 1 \\ -1 & 1 \end{bmatrix}$; $P^{-1}AP = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$.

d. $(x-2)^3$; 2; $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$; No such P; Not diagonalizable.

f. $(x+1)^2(x-2); -1,\ -2; \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$; No such P; Not diagonalizable. Note that this matrix and the matrix in Example 3.3.9 have the same characteristic polynomial, but that matrix is diagonalizable.

h. $(x-1)^2(x-3); 1,\ 3; \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ No such P; Not diagonalizable.

3.3.2 b. $V_k = \frac{7}{3}2^k \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

d. $V_k = \frac{3}{2}3^k \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

3.3.4 $A\mathbf{x} = \lambda\mathbf{x}$ if and only if $(A-\alpha I)\mathbf{x} = (\lambda-\alpha)\mathbf{x}$. Same eigenvectors.

3.3.8 b. $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, so $A^n = P\begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix}P^{-1} = \begin{bmatrix} 9-8\cdot 2^n & 12(1-2^n) \\ 6(2^n-1) & 9\cdot 2^n - 8 \end{bmatrix}$

3.3.9 b. $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$

3.3.11 b. and d. $PAP^{-1} = D$ is diagonal, then b. $P^{-1}(kA)P = kD$ is diagonal, and d. $Q(U^{-1}AU)Q = D$ where $Q = PU$.

3.3.12 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is not diagonalizable by Example 3.3.8. But $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$ where $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$ has diagonalizing matrix $P = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$ is already diagonal.

3.3.14 We have $\lambda^2 = \lambda$ for every eigenvalue λ (as $\lambda = 0,\ 1$) so $D^2 = D$, and so $A^2 = A$ as in Example 3.3.9.

3.3.18 b. $c_{rA}(x) = \det[xI - rA]$ $= r^n \det\left[\frac{x}{r}I - A\right] = r^n c_A\left[\frac{x}{r}\right]$

3.3.20 b. If $\lambda \neq 0$, $A\mathbf{x} = \lambda\mathbf{x}$ if and only if $A^{-1}\mathbf{x} = \frac{1}{\lambda}\mathbf{x}$. The result follows.

3.3.21 b. $(A^3 - 2A - 3I)\mathbf{x} = A^3\mathbf{x} - 2A\mathbf{x} + 3\mathbf{x} =$ $\lambda^3\mathbf{x} - 2\lambda\mathbf{x} + 3\mathbf{x} = (\lambda^3 - 2\lambda - 3)\mathbf{x}$.

3.3.23 b. If $A^m = 0$ and $A\mathbf{x} = \lambda\mathbf{x}$, $\mathbf{x} \neq \mathbf{0}$, then $A^2\mathbf{x} = A(\lambda\mathbf{x}) = \lambda A\mathbf{x} = \lambda^2\mathbf{x}$. In general, $A^k\mathbf{x} = \lambda^k\mathbf{x}$ for all $k \geq 1$. Hence, $\lambda^m\mathbf{x} = A^m\mathbf{x} = 0\mathbf{x} = \mathbf{0}$, so $\lambda = 0$ (because $\mathbf{x} \neq \mathbf{0}$).

3.3.24 a. If $A\mathbf{x} = \lambda\mathbf{x}$, then $A^k\mathbf{x} = \lambda^k\mathbf{x}$ for each k. Hence $\lambda^m\mathbf{x} = A^m\mathbf{x} = \mathbf{x}$, so $\lambda^m = 1$. As λ is real, $\lambda = \pm 1$ by the Hint. So if $P^{-1}AP = D$ is diagonal, then $D^2 = I$ by Theorem 3.3.4. Hence $A^2 = PD^2P = I$.

3.3.27 a. We have $P^{-1}AP = \lambda I$ by the diagonalization algorithm, so $A = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$.

b. No. $\lambda = 1$ is the only eigenvalue.

3.3.31 b. $\lambda_1 = 1$, stabilizes.

d. $\lambda_1 = \frac{1}{24}(3 + \sqrt{69}) = 1.13$, diverges.

3.3.34 Extinct if $\alpha < \frac{1}{5}$, stable if $\alpha = \frac{1}{5}$, diverges if $\alpha > \frac{1}{5}$.

Section 3.4

3.4.1 b. $x_k = \frac{1}{3}\left[4 - (-2)^k\right]$

d. $x_k = \frac{1}{5}\left[2^{k+2} + (-3)^k\right]$

3.4.2 b. $x_k = \frac{1}{2}\left[(-1)^k + 1\right]$

3.4.3 b. $x_{k+4} = x_k + x_{k+2} + x_{k+3}; x_{10} = 169$

3.4.5 $\frac{1}{2\sqrt{5}}\left[3 + \sqrt{5}\right]\lambda_1^k + (-3 + \sqrt{5})\lambda_2^k$ where $\lambda_1 = \frac{1}{2}(1 + \sqrt{5})$ and $\lambda_2 = \frac{1}{2}(1 - \sqrt{5})$.

3.4.7 $\frac{1}{2\sqrt{3}}\left[2 + \sqrt{3}\right]\lambda_1^k + (-2 + \sqrt{3})\lambda_2^k$ where $\lambda_1 = 1 + \sqrt{3}$ and $\lambda_2 = 1 - \sqrt{3}$.

3.4.9 $\frac{34}{3} - \frac{4}{3}\left(-\frac{1}{2}\right)^k$. Long term $11\frac{1}{3}$ million tons.

3.4.11 b. $A\begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \end{bmatrix} = \begin{bmatrix} \lambda \\ \lambda^2 \\ a + b\lambda + c\lambda^2 \end{bmatrix} = \begin{bmatrix} \lambda \\ \lambda^2 \\ \lambda^3 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \end{bmatrix}$

3.4.12 b. $x_k = \frac{11}{10}3^k + \frac{11}{15}(-2)^k - \frac{5}{6}$

3.4.13 a. $p_{k+2} + q_{k+2} = [ap_{k+1} + bp_k + c(k)] + [aq_{k+1} + bq_k] =$ $a(p_{k+1} + q_{k+1}) + b(p_k + q_k) + c(k)$

Section 3.5

3.5.1 b. $c_1\begin{bmatrix}1\\1\end{bmatrix}e^{4x}+c_2\begin{bmatrix}5\\-1\end{bmatrix}e^{-2x}; c_1=-\frac{2}{3},\ c_2=\frac{1}{3}$

d. $c_1\begin{bmatrix}-8\\10\\7\end{bmatrix}e^{-x}+c_2\begin{bmatrix}1\\-2\\1\end{bmatrix}e^{2x}+c_3\begin{bmatrix}1\\0\\1\end{bmatrix}e^{4x}$; $c_1=0,\ c_2=-\frac{1}{2},\ c_3=\frac{3}{2}$

3.5.3 b. The solution to (a) is $m(t)=10\left(\frac{4}{5}\right)^{t/3}$. Hence we want t such that $10\left(\frac{4}{5}\right)^{t/3}=5$. We solve for t by taking natural logarithms:

$$t=\frac{3\ln(\frac{1}{2})}{\ln(\frac{4}{5})}=9.32 \text{ hours.}$$

3.5.5 a. If $\mathbf{g}'=A\mathbf{g}$, put $\mathbf{f}=\mathbf{g}-A^{-1}\mathbf{b}$. Then $\mathbf{f}'=\mathbf{g}'$ and $A\mathbf{f}=A\mathbf{g}-\mathbf{b}$, so $\mathbf{f}'=\mathbf{g}'=A\mathbf{g}=A\mathbf{f}+\mathbf{b}$, as required.

3.5.6 b. Assume that $f_1'=a_1f_1+f_2$ and $f_2'=a_2f_1$. Differentiating gives $f_1''=a_1f_1'+f_2'=a_1f_1'+a_2f_1$, proving that f_1 satisfies Equation 3.15.

Section 3.6

3.6.2 Consider the rows $R_p, R_{p+1}, \dots, R_{q-1}, R_q$. In $q-p$ adjacent interchanges they can be put in the order $R_{p+1}, \dots, R_{q-1}, R_q, R_p$. Then in $q-p-1$ adjacent interchanges we can obtain the order $R_q, R_{p+1}, \dots, R_{q-1}, R_p$. This uses $2(q-p)-1$ adjacent interchanges in all.

Supplementary Exercises for Chapter 3

Supplementary Exercise 3.2. b. If A is 1×1, then $A^T=A$. In general, $\det\left[A_{ij}\right]=\det\left[(A_{ij})^T\right]=\det\left[(A^T)_{ji}\right]$ by (a) and induction. Write $A^T=\left[a'_{ij}\right]$ where $a'_{ij}=a_{ji}$, and expand $\det A^T$ along column 1.

$$\begin{aligned}\det A^T&=\sum_{j=1}^{n}a'_{j1}(-1)^{j+1}\det\left[(A^T)_{j1}\right]\\&=\sum_{j=1}^{n}a_{1j}(-1)^{1+j}\det\left[A_{1j}\right]=\det A\end{aligned}$$

where the last equality is the expansion of $\det A$ along row 1.

Section 4.1

4.1.1 b. $\sqrt{6}$

d. $\sqrt{5}$

f. $3\sqrt{6}$

4.1.2 b. $\frac{1}{3}\begin{bmatrix}-2\\-1\\2\end{bmatrix}$

4.1.4 b. $\sqrt{2}$

d. 3

4.1.6 b. $\overrightarrow{FE}=\overrightarrow{FC}+\overrightarrow{CE}=\frac{1}{2}\overrightarrow{AC}+\frac{1}{2}\overrightarrow{CB}=\frac{1}{2}(\overrightarrow{AC}+\overrightarrow{CB})=\frac{1}{2}\overrightarrow{AB}$

4.1.7 b. Yes

d. Yes

4.1.8 b. $\mathbf{p}$

d. $-(\mathbf{p}+\mathbf{q})$.

4.1.9 b. $\begin{bmatrix}-1\\-1\\5\end{bmatrix}, \sqrt{27}$

d. $\begin{bmatrix}0\\0\\0\end{bmatrix}, 0$

f. $\begin{bmatrix}-2\\2\\2\end{bmatrix}, \sqrt{12}$

4.1.10 b. **(i)** $Q(5,\ -1,\ 2)$ **(ii)** $Q(1,\ 1,\ -4)$.

4.1.11 b. $\mathbf{x}=\mathbf{u}-6\mathbf{v}+5\mathbf{w}=\begin{bmatrix}-26\\4\\19\end{bmatrix}$

4.1.12 b. $\begin{bmatrix}a\\b\\c\end{bmatrix}=\begin{bmatrix}-5\\8\\6\end{bmatrix}$

4.1.13 b. If it holds then $\begin{bmatrix}3a+4b+c\\-a+c\\b+c\end{bmatrix}=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$.

$$\left[\begin{array}{rrrc}3&4&1&x_1\\-1&0&1&x_2\\0&1&1&x_3\end{array}\right]\rightarrow\left[\begin{array}{rrrc}0&4&4&x_1+3x_2\\-1&0&1&x_2\\0&1&1&x_3\end{array}\right]$$

If there is to be a solution then $x_1+3x_2=4x_3$ must hold. This is not satisfied.

4.1.14 b. $\frac{1}{4}\begin{bmatrix}5\\-5\\-2\end{bmatrix}$

4.1.17 b. $Q(0,\ 7,\ 3)$.

4.1.18 b. $\mathbf{x}=\frac{1}{40}\begin{bmatrix}-20\\-13\\14\end{bmatrix}$

4.1.20 b. $S(-1,\ 3,\ 2)$.

4.1.21 b. T. $\|\mathbf{v}-\mathbf{w}\| = 0$ implies that $\mathbf{v}-\mathbf{w} = \mathbf{0}$.

d. F. $\|\mathbf{v}\| = \|-\mathbf{v}\|$ for all $\mathbf{v}$ but $\mathbf{v} = -\mathbf{v}$ only holds if $\mathbf{v} = \mathbf{0}$.

f. F. If $t < 0$ they have the *opposite* direction.

h. F. $\|-5\mathbf{v}\| = 5\|\mathbf{v}\|$ for all $\mathbf{v}$, so it fails if $\mathbf{v} \neq \mathbf{0}$.

j. F. Take $\mathbf{w} = -\mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$.

4.1.22 b. $\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$; $x = 3+2t$, $y = -1-t$, $z = 4+5t$

d. $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$; $x = y = z = 1+t$

f. $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$; $x = 2-t$, $y = -1$, $z = 1+t$

4.1.23 b. P corresponds to $t = 2$; Q corresponds to $t = 5$.

4.1.24 b. No intersection

d. $P(2,\ -1,\ 3)$; $t = -2$, $s = -3$

4.1.29 $P(3,\ 1,\ 0)$ or $P(\frac{5}{3},\ \frac{-1}{3},\ \frac{4}{3})$

4.1.31 b. $\overrightarrow{CP}_k = -\overrightarrow{CP}_{n+k}$ if $1 \leq k \leq n$, where there are $2n$ points.

4.1.33 $\overrightarrow{DA} = 2\overrightarrow{EA}$ and $2\overrightarrow{AF} = \overrightarrow{FC}$, so $2\overrightarrow{EF} = 2(\overrightarrow{EF}+\overrightarrow{AF}) = \overrightarrow{DA}+\overrightarrow{FC} = \overrightarrow{CB}+\overrightarrow{FC} = \overrightarrow{FC}+\overrightarrow{CB} = \overrightarrow{FB}$. Hence $\overrightarrow{EF} = \frac{1}{2}\overrightarrow{FB}$. So F is the trisection point of both AC and EB.

Section 4.2

4.2.1 b. 6

d. 0

f. 0

4.2.2 b. π or 180°

d. $\frac{\pi}{3}$ or 60°

f. $\frac{2\pi}{3}$ or 120°

4.2.3 b. 1 or -17

4.2.4 b. $t \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$

d. $s \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$

4.2.6 b. $29+57 = 86$

4.2.8 b. $A = B = C = \frac{\pi}{3}$ or 60°

4.2.10 b. $\frac{11}{18}\mathbf{v}$

d. $-\frac{1}{2}\mathbf{v}$

4.2.11 b. $\frac{5}{21} \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix} + \frac{1}{21} \begin{bmatrix} 53 \\ 26 \\ 20 \end{bmatrix}$

d. $\frac{27}{53} \begin{bmatrix} 6 \\ -4 \\ 1 \end{bmatrix} + \frac{1}{53} \begin{bmatrix} -3 \\ 2 \\ 26 \end{bmatrix}$

4.2.12 b. $\frac{1}{26}\sqrt{5642}$, $Q(\frac{71}{26},\ \frac{15}{26},\ \frac{34}{26})$

4.2.13 b. $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

b. $\begin{bmatrix} 4 \\ -15 \\ 8 \end{bmatrix}$

4.2.14 b. $-23x+32y+11z = 11$

d. $2x-y+z = 5$

f. $2x+3y+2z = 7$

h. $2x-7y-3z = -1$

j. $x-y-z = 3$

4.2.15 b. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

d. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

f. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 4 \\ 1 \\ -5 \end{bmatrix}$

4.2.16 b. $\frac{\sqrt{6}}{3}$, $Q(\frac{7}{3},\ \frac{2}{3},\ \frac{-2}{3})$

4.2.17 b. Yes. The equation is $5x-3y-4z = 0$.

4.2.19 b. $(-2,\ 7,\ 0)+t(3,\ -5,\ 2)$

4.2.20 b. None

d. $P(\frac{13}{19}, \frac{-78}{19}, \frac{65}{19})$

4.2.21 b. $3x+2z=d$, d arbitrary

d. $a(x-3)+b(y-2)+c(z+4)=0$; a, b, and c not all zero

f. $ax+by+(b-a)z=a$; a and b not both zero

h. $ax+by+(a-2b)z=5a-4b$; a and b not both zero

4.2.23 b. $\sqrt{10}$

4.2.24 b. $\frac{\sqrt{14}}{2}$, $A(3, 1, 2)$, $B(\frac{7}{2}, -\frac{1}{2}, 3)$

d. $\frac{\sqrt{6}}{6}$, $A(\frac{19}{3}, 2, \frac{1}{3})$, $B(\frac{37}{6}, \frac{13}{6}, 0)$

4.2.26 b. Consider the diagonal $\mathbf{d} = \left[\begin{array}{c} a \\ a \\ a \end{array}\right]$ The six face diagonals in question are $\pm\left[\begin{array}{c} a \\ 0 \\ -a \end{array}\right]$, $\pm\left[\begin{array}{c} 0 \\ a \\ -a \end{array}\right]$, $\pm\left[\begin{array}{c} a \\ -a \\ 0 \end{array}\right]$. All of these are orthogonal to $\mathbf{d}$. The result works for the other diagonals by symmetry.

4.2.28 The four diagonals are (a, b, c), $(-a, b, c)$, $(a, -b, c)$ and $(a, b, -c)$ or their negatives. The dot products are $\pm(-a^2+b^2+c^2)$, $\pm(a^2-b^2+c^2)$, and $\pm(a^2+b^2-c^2)$.

4.2.34 b. The sum of the squares of the lengths of the diagonals equals the sum of the squares of the lengths of the four sides.

4.2.38 b. The angle θ between $\mathbf{u}$ and $(\mathbf{u}+\mathbf{v}+\mathbf{w})$ is given by $\cos\theta = \frac{\mathbf{u}\cdot(\mathbf{u}+\mathbf{v}+\mathbf{w})}{\|\mathbf{u}\|\|\mathbf{u}+\mathbf{v}+\mathbf{w}\|} = \frac{\|\mathbf{u}\|}{\sqrt{\|\mathbf{u}\|^2+\|\mathbf{v}\|^2+\|\mathbf{w}\|^2}} = \frac{1}{\sqrt{3}}$ because $\|\mathbf{u}\| = \|\mathbf{v}\| = \|\mathbf{w}\|$. Similar remarks apply to the other angles.

4.2.39 b. Let $\mathbf{p}_0$, $\mathbf{p}_1$ be the vectors of P_0, P_1, so $\mathbf{u} = \mathbf{p}_0 - \mathbf{p}_1$. Then $\mathbf{u}\cdot\mathbf{n} = \mathbf{p}_0\cdot\mathbf{n} - \mathbf{p}_1\cdot\mathbf{n} = (ax_0+by_0)-(ax_1+by_1) = ax_0+by_0+c$. Hence the distance is

$$\left\|\left(\frac{\mathbf{u}\cdot\mathbf{n}}{\|\mathbf{n}\|^2}\right)\mathbf{n}\right\| = \frac{|\mathbf{u}\cdot\mathbf{n}|}{\|\mathbf{n}\|}$$

as required.

4.2.41 b. This follows from **(a)** because $\|\mathbf{v}\|^2 = a^2+b^2+c^2$.

4.2.44 d. Take $\left[\begin{array}{c} x_1 \\ y_1 \\ z_1 \end{array}\right] = \left[\begin{array}{c} x \\ y \\ z \end{array}\right]$ and $\left[\begin{array}{c} x_2 \\ y_2 \\ z_2 \end{array}\right] = \left[\begin{array}{c} y \\ z \\ x \end{array}\right]$ in (**c**).

Section 4.3

4.3.3 b. $\pm\frac{\sqrt{3}}{3}\left[\begin{array}{r} 1 \\ -1 \\ -1 \end{array}\right]$.

4.3.4 b. 0

d. $\sqrt{5}$

4.3.5 b. 7

4.3.6 b. The distance is $\|\mathbf{p}-\mathbf{p}_0\|$; use part (a.).

4.3.10 $\|\overrightarrow{AB}\times\overrightarrow{AC}\|$ is the area of the parallelogram determined by A, B, and C.

4.3.12 Because $\mathbf{u}$ and $\mathbf{v}\times\mathbf{w}$ are parallel, the angle θ between them is 0 or π. Hence $\cos(\theta) = \pm 1$, so the volume is $|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})| = \|\mathbf{u}\|\|\mathbf{v}\times\mathbf{w}\|\cos(\theta) = \|\mathbf{u}\|\|(\mathbf{v}\times\mathbf{w})\|$. But the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\frac{\pi}{2}$ so $\|\mathbf{v}\times\mathbf{w}\| = \|\mathbf{v}\|\|\mathbf{w}\|\cos(\frac{\pi}{2}) = \|\mathbf{v}\|\|\mathbf{w}\|$. The result follows.

4.3.15 b. If $\mathbf{u} = \left[\begin{array}{c} u_1 \\ u_2 \\ u_3 \end{array}\right]$, $\mathbf{v} = \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array}\right]$ and $\mathbf{w} = \left[\begin{array}{c} w_1 \\ w_2 \\ w_3 \end{array}\right]$, then $\mathbf{u}\times(\mathbf{v}+\mathbf{w}) = \det\left[\begin{array}{ccc} \mathbf{i} & u_1 & v_1+w_1 \\ \mathbf{j} & u_2 & v_2+w_2 \\ \mathbf{k} & u_3 & v_3+w_3 \end{array}\right]$
$= \det\left[\begin{array}{ccc} \mathbf{i} & u_1 & v_1 \\ \mathbf{j} & u_2 & v_2 \\ \mathbf{k} & u_3 & v_3 \end{array}\right] + \det\left[\begin{array}{ccc} \mathbf{i} & u_1 & w_1 \\ \mathbf{j} & u_2 & w_2 \\ \mathbf{k} & u_3 & w_3 \end{array}\right]$
$= (\mathbf{u}\times\mathbf{v}) + (\mathbf{u}\times\mathbf{w})$ where we used Exercise 4.3.21.

4.3.16 b. $(\mathbf{v}-\mathbf{w})\cdot[(\mathbf{u}\times\mathbf{v})+(\mathbf{v}\times\mathbf{w})+(\mathbf{w}\times\mathbf{u})] = (\mathbf{v}-\mathbf{w})\cdot(\mathbf{u}\times\mathbf{v})+(\mathbf{v}-\mathbf{w})\cdot(\mathbf{v}\times\mathbf{w})+(\mathbf{v}-\mathbf{w})\cdot(\mathbf{w}\times\mathbf{u}) = -\mathbf{w}\cdot(\mathbf{u}\times\mathbf{v})+0+\mathbf{v}\cdot(\mathbf{w}\times\mathbf{u}) = 0$.

4.3.22 Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be vectors of points in the planes, so $\mathbf{p}_1\cdot\mathbf{n} = d_1$ and $\mathbf{p}_2\cdot\mathbf{n} = d_2$. The distance is the length of the projection of $\mathbf{p}_2-\mathbf{p}_1$ along $\mathbf{n}$; that is $\frac{|(\mathbf{p}_2-\mathbf{p}_1)\cdot\mathbf{n}|}{\|\mathbf{n}\|} = \frac{|d_1-d_2|}{\|\mathbf{n}\|}$.

Section 4.4

4.4.1 b. $A = \left[\begin{array}{rr} 1 & -1 \\ -1 & 1 \end{array}\right]$, projection on $y=-x$.

d. $A = \frac{1}{5}\left[\begin{array}{rr} -3 & 4 \\ 4 & 3 \end{array}\right]$, reflection in $y=2x$.

f. $A = \frac{1}{2}\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$, rotation through $\frac{\pi}{3}$.

4.4.2 b. The zero transformation.

4.4.3 b. $\frac{1}{21}\left[\begin{array}{rrr} 17 & 2 & -8 \\ 2 & 20 & 4 \\ -8 & 4 & 5 \end{array}\right]\left[\begin{array}{r} 0 \\ 1 \\ -3 \end{array}\right]$

d. $\frac{1}{30}\left[\begin{array}{rrr} 22 & -4 & 20 \\ -4 & 28 & 10 \\ 20 & 10 & -20 \end{array}\right]\left[\begin{array}{r} 0 \\ 1 \\ -3 \end{array}\right]$

f. $\frac{1}{25}\left[\begin{array}{rrr} 9 & 0 & 12 \\ 0 & 0 & 0 \\ 12 & 0 & 16 \end{array}\right]\left[\begin{array}{r} 1 \\ -1 \\ 7 \end{array}\right]$

h. $\frac{1}{11}\left[\begin{array}{rrr} -9 & 2 & -6 \\ 2 & -9 & -6 \\ -6 & -6 & 7 \end{array}\right]\left[\begin{array}{r} 2 \\ -5 \\ 0 \end{array}\right]$

4.4.4 b. $\frac{1}{2}\left[\begin{array}{ccc} \sqrt{3} & -1 & 0 \\ 1 & \sqrt{3} & 0 \\ 0 & 0 & 1 \end{array}\right]\left[\begin{array}{c} 1 \\ 0 \\ 3 \end{array}\right]$

4.4.6 $\left[\begin{array}{ccc} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{array}\right]$

4.4.9 a. Write $\mathbf{v} = \left[\begin{array}{c} x \\ y \end{array}\right]$.

$$\begin{aligned} P_L(\mathbf{v}) = \left(\frac{\mathbf{v}\cdot\mathbf{d}}{\|\mathbf{d}\|^2}\right)\mathbf{d} &= \frac{ax+by}{a^2+b^2}\left[\begin{array}{c} a \\ b \end{array}\right] \\ &= \frac{1}{a^2+b^2}\left[\begin{array}{c} a^2x+aby \\ abx+b^2y \end{array}\right] \\ &= \frac{1}{a^2+b^2}\left[\begin{array}{cc} a^2+ab \\ ab+b^2 \end{array}\right]\left[\begin{array}{c} x \\ y \end{array}\right] \end{aligned}$$

Section 4.5

4.5.1 b. $\frac{1}{2}\left[\begin{array}{ccccc} \sqrt{2}+2 & 7\sqrt{2}+2 & 3\sqrt{2}+2 & -\sqrt{2}+2 & -5\sqrt{2}+2 \\ -3\sqrt{2}+4 & 3\sqrt{2}+4 & 5\sqrt{2}+4 & \sqrt{2}+4 & 9\sqrt{2}+4 \\ 2 & 2 & 2 & 2 & 2 \end{array}\right]$

4.5.5 b. $P(\frac{9}{5}, \frac{18}{5})$

Supplementary Exercises for Chapter 4

Supplementary Exercise 4.4. 125 knots in a direction θ degrees east of north, where $\cos\theta = 0.6$ ($\theta = 53°$ or 0.93 radians).

Supplementary Exercise 4.6. $(12, 5)$. Actual speed 12 knots.

Section 5.1

5.1.1 b. Yes

d. No

f. No.

5.1.2 b. No

d. Yes, $\mathbf{x} = 3\mathbf{y} + 4\mathbf{z}$.

5.1.3 b. No

5.1.10 $\text{span}\{a_1\mathbf{x}_1, a_2\mathbf{x}_2, \dots, a_k\mathbf{x}_k\} \subseteq \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ by Theorem 5.1.1 because, for each i, $a_i\mathbf{x}_i$ is in $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$. Similarly, the fact that $\mathbf{x}_i = a_i^{-1}(a_i\mathbf{x}_i)$ is in $\text{span}\{a_1\mathbf{x}_1, a_2\mathbf{x}_2, \dots, a_k\mathbf{x}_k\}$ for each i shows that $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\} \subseteq \text{span}\{a_1\mathbf{x}_1, a_2\mathbf{x}_2, \dots, a_k\mathbf{x}_k\}$, again by Theorem 5.1.1.

5.1.12 If $\mathbf{y} = r_1\mathbf{x}_1 + \cdots + r_k\mathbf{x}_k$ then $A\mathbf{y} = r_1(A\mathbf{x}_1) + \cdots + r_k(A\mathbf{x}_k) = 0$.

5.1.15 b. $\mathbf{x} = (\mathbf{x}+\mathbf{y}) - \mathbf{y} = (\mathbf{x}+\mathbf{y}) + (-\mathbf{y})$ is in U because U is a subspace and both $\mathbf{x}+\mathbf{y}$ and $-\mathbf{y} = (-1)\mathbf{y}$ are in U.

5.1.16 b. True. $\mathbf{x} = 1\mathbf{x}$ is in U.

d. True. Always $\text{span}\{\mathbf{y}, \mathbf{z}\} \subseteq \text{span}\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ by Theorem 5.1.1. Since $\mathbf{x}$ is in $\text{span}\{\mathbf{x}, \mathbf{y}\}$ we have $\text{span}\{\mathbf{x}, \mathbf{y}, \mathbf{z}\} \subseteq \text{span}\{\mathbf{y}, \mathbf{z}\}$, again by Theorem 5.1.1.

f. False. $a\left[\begin{array}{c} 1 \\ 0 \end{array}\right] + b\left[\begin{array}{c} 2 \\ 0 \end{array}\right] = \left[\begin{array}{c} a+2b \\ 0 \end{array}\right]$ cannot equal $\left[\begin{array}{c} 0 \\ 1 \end{array}\right]$.

5.1.20 If U is a subspace, then S2 and S3 certainly hold. Conversely, assume that S2 and S3 hold for U. Since U is nonempty, choose $\mathbf{x}$ in U. Then $\mathbf{0} = 0\mathbf{x}$ is in U by S3, so S1 also holds. This means that U is a subspace.

5.1.22 b. The zero vector $\mathbf{0}$ is in $U+W$ because $\mathbf{0} = \mathbf{0}+\mathbf{0}$. Let $\mathbf{p}$ and $\mathbf{q}$ be vectors in $U+W$, say $\mathbf{p} = \mathbf{x}_1 + \mathbf{y}_1$ and $\mathbf{q} = \mathbf{x}_2 + \mathbf{y}_2$ where $\mathbf{x}_1$ and $\mathbf{x}_2$ are in U, and $\mathbf{y}_1$ and $\mathbf{y}_2$ are in W. Then $\mathbf{p}+\mathbf{q} = (\mathbf{x}_1+\mathbf{x}_2) + (\mathbf{y}_1+\mathbf{y}_2)$ is in $U+W$ because $\mathbf{x}_1+\mathbf{x}_2$ is in U and $\mathbf{y}_1+\mathbf{y}_2$ is in W. Similarly, $a(\mathbf{p}+\mathbf{q}) = a\mathbf{p}+a\mathbf{q}$ is in $U+W$ for any scalar a because $a\mathbf{p}$ is in U and $a\mathbf{q}$ is in W. Hence $U+W$ is indeed a subspace of $\mathbb{R}^n$.

Section 5.2

5.2.1 b. Yes. If $r\begin{bmatrix}1\\1\\1\end{bmatrix}+s\begin{bmatrix}1\\1\\1\end{bmatrix}+t\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$, then $r+s=0$, $r-s=0$, and $r+s+t=0$. These equations give $r=s=t=0$.

d. No. Indeed:
$$\begin{bmatrix}1\\1\\0\\0\end{bmatrix}-\begin{bmatrix}1\\0\\1\\0\end{bmatrix}+\begin{bmatrix}0\\0\\1\\1\end{bmatrix}-\begin{bmatrix}0\\1\\0\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}.$$

5.2.2 b. Yes. If $r(\mathbf{x}+\mathbf{y})+s(\mathbf{y}+\mathbf{z})+t(\mathbf{z}+\mathbf{x})=\mathbf{0}$, then $(r+t)\mathbf{x}+(r+s)\mathbf{y}+(s+t)\mathbf{z}=\mathbf{0}$. Since $\{\mathbf{x},\ \mathbf{y},\ \mathbf{z}\}$ is independent, this implies that $r+t=0$, $r+s=0$, and $s+t=0$. The only solution is $r=s=t=0$.

d. No. In fact, $(\mathbf{x}+\mathbf{y})-(\mathbf{y}+\mathbf{z})+(\mathbf{z}+\mathbf{w})-(\mathbf{w}+\mathbf{x})=\mathbf{0}$.

5.2.3 b. $\left\{\begin{bmatrix}2\\1\\0\\-1\end{bmatrix},\begin{bmatrix}-1\\1\\1\\1\end{bmatrix}\right\}$; dimension 2.

d. $\left\{\begin{bmatrix}-2\\0\\3\\1\end{bmatrix},\begin{bmatrix}1\\2\\-1\\0\end{bmatrix}\right\}$; dimension 2.

5.2.4 b. $\left\{\begin{bmatrix}1\\1\\0\\1\end{bmatrix},\begin{bmatrix}1\\-1\\1\\0\end{bmatrix}\right\}$; dimension 2.

d. $\left\{\begin{bmatrix}1\\0\\1\\0\end{bmatrix},\begin{bmatrix}-1\\1\\0\\1\end{bmatrix},\begin{bmatrix}0\\1\\0\\1\end{bmatrix}\right\}$; dimension 3.

f. $\left\{\begin{bmatrix}-1\\1\\0\\0\end{bmatrix},\begin{bmatrix}1\\0\\1\\0\end{bmatrix},\begin{bmatrix}1\\0\\0\\1\end{bmatrix}\right\}$; dimension 3.

5.2.5 b. If $r(\mathbf{x}+\mathbf{w})+s(\mathbf{y}+\mathbf{w})+t(\mathbf{z}+\mathbf{w})+u(\mathbf{w})=\mathbf{0}$, then $r\mathbf{x}+s\mathbf{y}+t\mathbf{z}+(r+s+t+u)\mathbf{w}=\mathbf{0}$, so $r=0$, $s=0$, $t=0$, and $r+s+t+u=0$. The only solution is $r=s=t=u=0$, so the set is independent. Since $\dim\mathbb{R}^4=4$, the set is a basis by Theorem 5.2.7.

5.2.6 b. Yes

d. Yes

f. No.

5.2.7 b. T. If $r\mathbf{y}+s\mathbf{z}=\mathbf{0}$, then $0\mathbf{x}+r\mathbf{y}+s\mathbf{z}=\mathbf{0}$ so $r=s=0$ because $\{\mathbf{x},\ \mathbf{y},\ \mathbf{z}\}$ is independent.

d. F. If $\mathbf{x}\neq\mathbf{0}$, take $k=2$, $\mathbf{x}_1=\mathbf{x}$ and $\mathbf{x}_2=-\mathbf{x}$.

f. F. If $\mathbf{y}=-\mathbf{x}$ and $\mathbf{z}=\mathbf{0}$, then $1\mathbf{x}+1\mathbf{y}+1\mathbf{z}=\mathbf{0}$.

h. T. This is a nontrivial, vanishing linear combination, so the $\mathbf{x}_i$ cannot be independent.

5.2.10 If $r\mathbf{x}_2+s\mathbf{x}_3+t\mathbf{x}_5=\mathbf{0}$ then $0\mathbf{x}_1+r\mathbf{x}_2+s\mathbf{x}_3+0\mathbf{x}_4+t\mathbf{x}_5+0\mathbf{x}_6=\mathbf{0}$ so $r=s=t=0$.

5.2.12 If $t_1\mathbf{x}_1+t_2(\mathbf{x}_1+\mathbf{x}_2)+\cdots+t_k(\mathbf{x}_1+\mathbf{x}_2+\cdots+\mathbf{x}_k)=\mathbf{0}$, then $(t_1+t_2+\cdots+t_k)\mathbf{x}_1+(t_2+\cdots+t_k)\mathbf{x}_2+\cdots+(t_{k-1}+t_k)\mathbf{x}_{k-1}+(t_k)\mathbf{x}_k=\mathbf{0}$. Hence all these coefficients are zero, so we obtain successively $t_k=0,\ t_{k-1}=0,\ \dots,\ t_2=0,\ t_1=0$.

5.2.16 b. We show A^T is invertible (then A is invertible). Let $A^T\mathbf{x}=\mathbf{0}$ where $\mathbf{x}=[s\ t]^T$. This means $as+ct=0$ and $bs+dt=0$, so $s(a\mathbf{x}+b\mathbf{y})+t(c\mathbf{x}+d\mathbf{y})=(sa+tc)\mathbf{x}+(sb+td)\mathbf{y}=\mathbf{0}$. Hence $s=t=0$ by hypothesis.

5.2.17 b. Each $V^{-1}\mathbf{x}_i$ is in $\operatorname{null}(AV)$ because $AV(V^{-1}\mathbf{x}_i)=A\mathbf{x}_i=\mathbf{0}$. The set $\{V^{-1}\mathbf{x}_1,\ \dots,\ V^{-1}\mathbf{x}_k\}$ is independent as V^{-1} is invertible. If $\mathbf{y}$ is in $\operatorname{null}(AV)$, then $V\mathbf{y}$ is in $\operatorname{null}(A)$ so let $V\mathbf{y}=t_1\mathbf{x}_1+\cdots+t_k\mathbf{x}_k$ where each t_k is in $\mathbb{R}$. Thus $\mathbf{y}=t_1V^{-1}\mathbf{x}_1+\cdots+t_kV^{-1}\mathbf{x}_k$ is in $\operatorname{span}\{V^{-1}\mathbf{x}_1,\ \dots,\ V^{-1}\mathbf{x}_k\}$.

5.2.20 We have $\{\mathbf{0}\}\subseteq U\subseteq W$ where $\dim\{\mathbf{0}\}=0$ and $\dim W=1$. Hence $\dim U=0$ or $\dim U=1$ by Theorem 5.2.8, that is $U=0$ or $U=W$, again by Theorem 5.2.8.

Section 5.3

5.3.1 b. $\left\{\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix},\ \frac{1}{\sqrt{42}}\begin{bmatrix}4\\1\\-5\end{bmatrix},\ \frac{1}{\sqrt{14}}\begin{bmatrix}2\\-3\\1\end{bmatrix}\right\}$.

5.3.3 b. $\begin{bmatrix}a\\b\\c\end{bmatrix}=\frac{1}{2}(a-c)\begin{bmatrix}1\\0\\-1\end{bmatrix}+\frac{1}{18}(a+4b+c)\begin{bmatrix}1\\4\\1\end{bmatrix}+\frac{1}{9}(2a-b+2c)\begin{bmatrix}2\\-1\\2\end{bmatrix}$.

d. $\begin{bmatrix}a\\b\\c\end{bmatrix}=\frac{1}{3}(a+b+c)\begin{bmatrix}1\\1\\1\end{bmatrix}+\frac{1}{2}(a-b)\begin{bmatrix}1\\-1\\0\end{bmatrix}+\frac{1}{6}(a+b-2c)\begin{bmatrix}1\\1\\-2\end{bmatrix}$.

5.3.4 b. $\begin{bmatrix}14\\1\\-8\\5\end{bmatrix}=3\begin{bmatrix}2\\-1\\0\\3\end{bmatrix}+4\begin{bmatrix}2\\1\\-2\\-1\end{bmatrix}$.

5.3.5 b. $t\begin{bmatrix} -1 \\ 3 \\ 10 \\ 11 \end{bmatrix}$, in $\mathbb{R}$

5.3.6 b. $\sqrt{29}$

d. 19

5.3.7 b. F. $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

d. T. Every $\mathbf{x}_i \cdot \mathbf{y}_j = 0$ by assumption, every $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ if $i \neq j$ because the $\mathbf{x}_i$ are orthogonal, and every $\mathbf{y}_i \cdot \mathbf{y}_j = 0$ if $i \neq j$ because the $\mathbf{y}_i$ are orthogonal. As all the vectors are nonzero, this does it.

f. T. Every pair of *distinct* vectors in the set $\{\mathbf{x}\}$ has dot product zero (there are no such pairs).

5.3.9 Let $\mathbf{c}_1, \dots, \mathbf{c}_n$ be the columns of A. Then row i of A^T is $\mathbf{c}_i^T$, so the (i, j)-entry of A^TA is $\mathbf{c}_i^T\mathbf{c}_j = \mathbf{c}_i \cdot \mathbf{c}_j = 0, 1$ according as $i \neq j$, $i = j$. So $A^TA = I$.

5.3.11 b. Take $n = 3$ in **(a)**, expand, and simplify.

5.3.12 b. We have $(\mathbf{x}+\mathbf{y}) \cdot (\mathbf{x}-\mathbf{y}) = \|\mathbf{x}\|^2 - \|\mathbf{y}\|^2$. Hence $(\mathbf{x}+\mathbf{y}) \cdot (\mathbf{x}-\mathbf{y}) = 0$ if and only if $\|\mathbf{x}\|^2 = \|\mathbf{y}\|^2$; if and only if $\|\mathbf{x}\| = \|\mathbf{y}\|$—where we used the fact that $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{y}\| \geq 0$.

5.3.15 If $A^TA\mathbf{x} = \lambda\mathbf{x}$, then $\|A\mathbf{x}\|^2 = (A\mathbf{x}) \cdot (A\mathbf{x}) = \mathbf{x}^TA^TA\mathbf{x} = \mathbf{x}^T(\lambda\mathbf{x}) = \lambda\|\mathbf{x}\|^2$.

Section 5.4

5.4.1 b.
$\left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}; \left\{ \begin{bmatrix} 2 \\ -2 \\ 4 \\ -6 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \\ 0 \end{bmatrix} \right\}; 2$

d. $\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}; \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \end{bmatrix} \right\}; 2$

5.4.2 b. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 2 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ -3 \\ 6 \end{bmatrix} \right\}$

d. $\left\{ \begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

5.4.3 b. No; no

d. No

f. Otherwise, if A is $m \times n$, we have $m = \dim(\text{row } A) = \text{rank } A = \dim(\text{col } A) = n$

5.4.4 Let $A = \begin{bmatrix} \mathbf{c}_1 & \dots & \mathbf{c}_n \end{bmatrix}$. Then $\text{col } A = \text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \{x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n \mid x_i \text{ in } \mathbb{R}\} = \{A\mathbf{x} \mid \mathbf{x} \text{ in } \mathbb{R}^n\}$.

5.4.7 b. The basis is $\left\{ \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ so the dimension is 2.

Have rank $A = 3$ and $n - 3 = 2$.

5.4.8 b. $n - 1$

5.4.9 b. If $r_1\mathbf{c}_1 + \cdots + r_n\mathbf{c}_n = \mathbf{0}$, let $\mathbf{x} = [r_1, \dots, r_n]^T$. Then $C\mathbf{x} = r_1\mathbf{c}_1 + \cdots + r_n\mathbf{c}_n = \mathbf{0}$, so $\mathbf{x}$ is in null $A = 0$. Hence each $r_i = 0$.

5.4.10 b. Write $r = \text{rank } A$. Then **(a)** gives $r = \dim(\text{col } A \leq \dim(\text{null } A) = n - r$.

5.4.12 We have $\text{rank}(A) = \dim[\text{col}(A)]$ and $\text{rank}(A^T) = \dim[\text{row}(A^T)]$. Let $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k\}$ be a basis of $\text{col}(A)$; it suffices to show that $\{\mathbf{c}_1^T, \mathbf{c}_2^T, \dots, \mathbf{c}_k^T\}$ is a basis of $\text{row}(A^T)$. But if $t_1\mathbf{c}_1^T + t_2\mathbf{c}_2^T + \cdots + t_k\mathbf{c}_k^T = \mathbf{0}$, t_j in $\mathbb{R}$, then (taking transposes) $t_1\mathbf{c}_1 + t_2\mathbf{c}_2 + \cdots + t_k\mathbf{c}_k = \mathbf{0}$ so each $t_j = 0$. Hence $\{\mathbf{c}_1^T, \mathbf{c}_2^T, \dots, \mathbf{c}_k^T\}$ is independent. Given $\mathbf{v}$ in $\text{row}(A^T)$ then $\mathbf{v}^T$ is in $\text{col}(A)$; say $\mathbf{v}^T = s_1\mathbf{c}_1 + s_2\mathbf{c}_2 + \cdots + s_k\mathbf{c}_k$, s_j in $\mathbb{R}$: Hence $\mathbf{v} = s_1\mathbf{c}_1^T + s_2\mathbf{c}_2^T + \cdots + s_k\mathbf{c}_k^T$, so $\{\mathbf{c}_1^T, \mathbf{c}_2^T, \dots, \mathbf{c}_k^T\}$ spans $\text{row}(A^T)$, as required.

5.4.15 b. Let $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ be a basis of $\text{col}(A)$. Then $\mathbf{b}$ is *not* in $\text{col}(A)$, so $\{\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{b}\}$ is linearly independent. Show that $\text{col}[A\ \mathbf{b}] = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{b}\}$.

Section 5.5

5.5.1 b. traces $= 2$, ranks $= 2$, but $\det A = -5$, $\det B = -1$

d. ranks $= 2$, determinants $= 7$, but $\text{tr } A = 5$, $\text{tr } B = 4$

f. traces $= -5$, determinants $= 0$, but $\text{rank } A = 2$, $\text{rank } B = 1$

5.5.3 b. If $B = P^{-1}AP$, then $B^{-1} = P^{-1}A^{-1}(P^{-1})^{-1} = P^{-1}A^{-1}P$.

5.5.4 b. Yes, $P = \begin{bmatrix} -1 & 0 & 6 \\ 0 & 1 & 0 \\ 1 & 0 & 5 \end{bmatrix}$, $P^{-1}AP = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

d. No, $c_A(x) = (x+1)(x-4)^2$ so $\lambda = 4$ has multiplicity 2. But $\dim(E_4) = 1$ so Theorem 5.5.6 applies.

5.5.8 b. If $B = P^{-1}AP$ and $A^k = 0$, then $B^k = (P^{-1}AP)^k = P^{-1}A^kP = P^{-1}0P = 0$.

5.5.9 b. The eigenvalues of A are all equal (they are the diagonal elements), so if $P^{-1}AP = D$ is diagonal, then $D = \lambda I$. Hence $A = P^{-1}(\lambda I)P = \lambda I$.

5.5.10 b. A is similar to $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ so (Theorem 5.5.1) $\text{tr}\, A = \text{tr}\, D = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.

5.5.12 b. $T_P(A)T_P(B) = (P^{-1}AP)(P^{-1}BP) = P^{-1}(AB)P = T_P(AB)$.

5.5.13 b. If A is diagonalizable, so is A^T, and they have the same eigenvalues. Use **(a)**.

5.5.17 b. $c_B(x) = [x-(a+b+c)][x^2-k]$ where $k = a^2+b^2+c^2-[ab+ac+bc]$. Use Theorem 5.5.7.

Section 5.6

5.6.1 b. $\frac{1}{12}\begin{bmatrix} -20 \\ 46 \\ 95 \end{bmatrix}$, $(A^TA)^{-1}$ $= \frac{1}{12}\begin{bmatrix} 8 & -10 & -18 \\ -10 & 14 & 24 \\ -18 & 24 & 43 \end{bmatrix}$

5.6.2 b. $\frac{64}{13} - \frac{6}{13}x$

d. $-\frac{4}{10} - \frac{17}{10}x$

5.6.3 b. $y = 0.127 - 0.024x + 0.194x^2$, $(M^TM)^{-1} = \frac{1}{4248}\begin{bmatrix} 3348 & 642 & -426 \\ 642 & 571 & -187 \\ -426 & -187 & 91 \end{bmatrix}$

5.6.4 b. $\frac{1}{92}(-46x + 66x^2 + 60 \cdot 2^x)$, $(M^TM)^{-1} = \frac{1}{46}\begin{bmatrix} 115 & 0 & -46 \\ 0 & 17 & -18 \\ -46 & -18 & 38 \end{bmatrix}$

5.6.5 b. $\frac{1}{20}[18 + 21x^2 + 28\sin(\frac{\pi x}{2})]$, $(M^TM)^{-1} = \frac{1}{40}\begin{bmatrix} 24 & -2 & 14 \\ -2 & 1 & 3 \\ 14 & 3 & 49 \end{bmatrix}$

5.6.7 $s = 99.71 - 4.87x$; the estimate of g is 9.74. [The true value of g is 9.81]. If a quadratic in s is fit, the result is $s = 101 - \frac{3}{2}t - \frac{9}{2}t^2$ giving $g = 9$;
$(M^TM)^{-1} = \frac{1}{2}\begin{bmatrix} 38 & -42 & 10 \\ -42 & 49 & -12 \\ 10 & -12 & 3 \end{bmatrix}$.

5.6.9 $y = -5.19 + 0.34x_1 + 0.51x_2 + 0.71x_3$, $(A^TA)^{-1}$
$= \frac{1}{25080}\begin{bmatrix} 517860 & -8016 & 5040 & -22650 \\ -8016 & 208 & -316 & 400 \\ 5040 & -316 & 1300 & -1090 \\ -22650 & 400 & -1090 & 1975 \end{bmatrix}$

5.6.10 b. $f(x) = a_0$ here, so the sum of squares is $S = \sum(y_i - a_0)^2 = na_0^2 - 2a_0\sum y_i + \sum y_i^2$. Completing the square gives $S = n[a_0 - \frac{1}{n}\sum y_i]^2 + [\sum y_i^2 - \frac{1}{n}(\sum y_i)^2]$ This is minimal when $a_0 = \frac{1}{n}\sum y_i$.

5.6.13 b. Here $f(x) = r_0 + r_1e^x$. If $f(x_1) = 0 = f(x_2)$ where $x_1 \neq x_2$, then $r_0 + r_1 \cdot e^{x_1} = 0 = r_0 + r_1 \cdot e^{x_2}$ so $r_1(e^{x_1} - e^{x_2}) = 0$. Hence $r_1 = 0 = r_0$.

Section 5.7

5.7.2 Let X denote the number of years of education, and let Y denote the yearly income (in 1000's). Then $\overline{x} = 15.3$, $s_x^2 = 9.12$ and $s_x = 3.02$, while $\overline{y} = 40.3$, $s_y^2 = 114.23$ and $s_y = 10.69$. The correlation is $r(X, Y) = 0.599$.

5.7.4 b. Given the sample vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, let $\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ where $z_i = a + bx_i$ for each i. By **(a)** we have $\overline{z} = a + b\overline{x}$, so

$$\begin{aligned} s_z^2 &= \frac{1}{n-1}\sum_i (z_i - \overline{z})^2 \\ &= \frac{1}{n-1}\sum_i [(a+bx_i) - (a+b\overline{x})]^2 \\ &= \frac{1}{n-1}\sum_i b^2(x_i - \overline{x})^2 \\ &= b^2 s_x^2. \end{aligned}$$

Now **(b)** follows because $\sqrt{b^2} = |b|$.

Supplementary Exercises for Chapter 5

Supplementary Exercise 5.1. b. F